OS AXIOMS FOR THE FIELD STRENGTH TENSOR IN THREE-DIMENSIONAL ELECTRODYNAMICS

Abstract. We construct subsequential continuum laws of the field strength tensor for noncompact lattice quantum electrodynamics in three dimensions with a fermion discretization on the chambers of the cubic reflection group. The volume is taken to infinity at each fixed lattice spacing, and the spacing subsequently tends to zero. For every positive charge, every Wilson parameter strictly between zero and one, and every sequence of real lattice masses, these limits exist. Their Schwinger functions satisfy the Osterwalder–Schrader axioms of regularity, Euclidean invariance, reflection positivity and symmetry; almost every translation-ergodic component also satisfies clustering. The argument passes positivity at diagonal reflection planes through the infinite-volume limit and uses the resulting spectral bound to prove rotation invariance of all moments. The construction concerns the field strength tensor and supplies no lower bound excluding the zero-field law.

1. Introduction

The observable constructed in this paper is the field strength tensor $F = (F_{\mu\nu})_{1\le\mu<\nu\le 3}$ of a three-dimensional lattice gauge theory. In the continuum it is a random tempered distribution, and its moments are tested against smooth antisymmetric tensor fields. Our main result is that a nonempty family of its continuum laws satisfies the Osterwalder–Schrader axioms. Clustering holds in almost every translation-ergodic component of each law.

There are two limiting operations. We first send the volume to infinity while keeping the lattice spacing fixed. We then send the spacing to zero, passing to subsequences at both stages when necessary. This order is part of the theorem. The lattice fermions enter through their determinant, which changes the probability distribution of the field strength tensor. The observables in the continuum theorem are moments of that tensor; they do not contain fermionic insertions.

The main difficulty is continuous rotation invariance. A cubic lattice has only finitely many orthogonal symmetries, and taking a limit does not by itself enlarge that group. The fermion discretization used here is designed to preserve reflection positivity at the diagonal planes of the cubic reflection group as well as at coordinate planes. These additional reflections give a spectral bound with its exact constant. Combined with Gaussian domination and the identity $dF = 0$, that bound forces every moment to be invariant under continuous rotations.

We state the result informally before defining its terms precisely. For any charge $e > 0$, Wilson parameter $0 < \rho < 1$, and sequence of real bare masses, the volume-first field strength laws have continuum limit points. Each such law is invariant under every Euclidean motion and is reflection positive at every plane. Its Schwinger functions satisfy regularity and permutation symmetry. Almost every ergodic component satisfies all these properties and the factorizing cluster property. Neither uniqueness of the limiting law nor a positive lower bound on a correlation function is required.

1.1. **The lattice model and the order of limits.** The photon variables are real one-cochains on the links of $\mathbb{Z}^3$. The discrete field strength is

$$u = dA + \frac{2\pi}{q} z, \qquad dz = 0,$$

where z is an integer two-cochain. All closed integer flux sectors are summed. The photon weight is $\exp(-\|u\|^2/2)$. Quotienting the redundancy between A and integer shifts gives the finite-volume integral defined in Section 2.

The group of Euclidean motions preserving $\mathbb{Z}^3$ is

$$W = \mathbb{Z}^3 \rtimes O_{\mathrm{cub}}, \qquad O_{\mathrm{cub}} = O(3) \cap GL(3,\mathbb{Z}).$$

Its reflection planes divide space into chambers. Choose a point v_0 of the fundamental chamber

$$C_0 = \{0 < x_3 < x_2 < x_1 < \tfrac{1}{2}\}.$$

The fermion sites are the orbit Wv_0, with one site in each chamber. Sites in neighboring chambers are joined by a bond. Thus a bond crossing a reflection plane is perpendicular to that plane and joins two reflected sites. There are four types of bonds. Their weights are $w_i = 48|\Pi_i \cap \overline{C}_0|$, where Π_i are the four walls of C_0. The four-component hopping matrices use $\gamma_\mu = \mathrm{diag}(\sigma_\mu, -\sigma_\mu)$ and the Wilson parameter ρ. Their precise definition is Definition 2.8.

On a finite torus the photon density is multiplied by $\det D_A$. The two Pauli blocks of D_A are adjoints, so this determinant is nonnegative. The strict positivity of its integral also has to be proved: nonnegativity alone does not define a normalized probability law. The finite-volume sections prove both assertions for every real mass.

Fix an odd integer $L \geq 3$. At cutoff index N the spacing and charge are

$$\eta_N = L^{-N}, \qquad q_N = e\eta_N^{1/2}.$$

Choose $0 < \bar{m} \leq 1$ and any sequence $(m_{0,N})$ of real numbers; the mass in lattice units is $m_N = \bar{m}\eta_N + m_{0,N}$. The diagonal parameter in the chamber operator is $M_N = m_N + (\rho/2)\sum_i w_i$. Because the sequence is arbitrary, this convention imposes no restriction on the real lattice masses. The finite cubic tori have lattice side lengths $n_{N,T} = 8L^{\lfloor \log_L \bar{m}^{-1} \rfloor} L^{N+T}$, with antiperiodic fermion boundary conditions in each coordinate direction. At fixed N their physical side lengths $\eta_N n_{N,T}$ tend to infinity. Their normalized field strength laws, their interpolation into distributions on $\mathbb{R}^3$, and the cutoff sequence are specified in Sections 2 and 16.

At each fixed N, let ν_N be a subsequential infinite-volume law of the lattice field strength u. Interpolate it into physical units by

$$(\sharp_{\eta_N} u)_{\mu\nu}(y) = \eta_N^{-3/2} u(x;\mu\nu) \quad \text{on} \quad Q_{\eta_N}(x;\mu\nu) = \eta_N\big(x + \tfrac{1}{2}(e_\mu + e_\nu) + [-\tfrac{1}{2},\tfrac{1}{2})^3\big).$$

For each component these cubes partition space. A *volume-first continuum law* is a weak subsequential limit of $(\sharp_{\eta_N})_*\nu_N$ as $N \to \infty$. The topology is that of the weighted Hilbert space H_{-4} defined in Section 15. This fixes both the observable and the limiting procedure in the theorem below.

1.2. **Field strength laws and Schwinger functions.** We regard an antisymmetric tensor as a two-form, using the orthonormal basis $dx_1 \wedge dx_2$, $dx_1 \wedge dx_3$, $dx_2 \wedge dx_3$. Write

$$\mathcal{S} = \mathcal{S}(\mathbb{R}^3;\Lambda^2), \qquad \mathcal{S}' = \mathcal{S}'(\mathbb{R}^3;\Lambda^2).$$

A real distribution F acts on a complex test by complex linearity. Let Σ be the σ-algebra generated by its evaluations $F(f)$. For an open set O, let Σ_O be generated by evaluations with tests in $C_c^\infty(O;\Lambda^2)$. A law means a probability measure on $(\mathcal{S}',\Sigma)$.

A Euclidean motion $g(x) = Rx + a$, $R \in O(3)$, acts on tests and distributions by

$$(g \cdot f)(x) = (\wedge^2 R) f(g^{-1}x), \qquad (gF)(f) = F(g^{-1} \cdot f).$$

These formulas specify the tensor transformation of the observable. They include orientation-reversing transformations. If $r_{n,c}$ is reflection in the plane $n \cdot x = c$, $|n| = 1$, then its action on observables is

$$(\widehat{r}_{n,c}\Phi)(F) = \Phi(r_{n,c}F), \qquad \Theta_{n,c}\Phi = \overline{\widehat{r}_{n,c}\Phi}.$$

The second operator is antilinear. Reflection positivity at this plane means

$$\mu((\Theta_{n,c}\Phi)\Phi) \geq 0 \qquad (\Phi \in L^2(\Sigma_{\{n\cdot x > c\}},\mu)). \tag{1.1}$$

Reflection invariance of the law is stated separately. Initially the observations in (1.1) are supported on the open side; positivity for tensor tests touching the plane will follow by approximation.

For a law with Gaussian domination,

$$\mu(e^{F(f)}) \le e^{\|f\|_2^2/2} \qquad (f \in \mathcal{S} \text{ real}), \tag{1.2}$$

the field extends continuously to all L^2 tests and has moments of every order. On simple tensor products the Schwinger functions are

$$S_n(f_1 \otimes \cdots \otimes f_n) = \mu\big(F(f_1)\cdots F(f_n)\big), \qquad S_0 = 1.$$

They extend to the tensor Schwartz spaces

$$\mathscr{S}_n = \mathcal{S}((\mathbb{R}^3)^n; (\Lambda^2)_{\mathbb{C}}^{\otimes n}), \qquad \mathscr{S}_0 = \mathbb{C}.$$

The extension and its growth estimate are proved in Lemma 15.5. In particular, the definition does not exclude coincident arguments.

For clarity, the following theorem lists the version of the OS axioms used here. For $f \in \mathscr{S}_n$, put

$$q_m(f) = \max_{|\alpha|\le m} \sup_y (1+|y|^2)^{m/2} |\partial^\alpha f(y)|,$$

where the maximum includes tensor components. For a unit vector v, define $\Theta_k^v f = (r_{v,0}\cdot)^{\otimes k}\overline{f}$. A permutation of the arguments includes the same permutation of their tensor factors. Translation of all arguments in the second tensor factor is denoted by $\tau_a^{\otimes n}$.

Theorem 1.1 (Field strength OS theorem)**.** *Fix $L \ge 3$ odd, $0 < \rho < 1$, $v_0 \in C_0$, the weights $w_i = 48|\Pi_i \cap \overline{C}_0|$, $e > 0$, $0 < \bar{m} \le 1$, and an arbitrary real sequence $(m_{0,N})$. The corresponding family of volume-first continuum laws is nonempty. For every such law μ, its Schwinger functions satisfy the following axioms:*

OS0. There is a numerical constant C such that, for $n \ge 1$,

$$|S_n(f)| \le C^n \sqrt{n!}\, q_{6n}(f) \qquad (f \in \mathscr{S}_n).$$

OS1. For every Euclidean motion g, $S_n((g\cdot)^{\otimes n} f) = S_n(f)$.

OS2. For every unit vector v, every $K \ge 0$, and tensor tests $f_k \in \mathscr{S}_k$ supported where $v \cdot y_i \ge 0$ for each argument, with $f_0 \in \mathbb{C}$,

$$\sum_{k,l=0}^{K} S_{k+l}(\Theta_k^v f_k \otimes f_l) \ge 0.$$

OS3. $S_n(f^\pi) = S_n(f)$ for every permutation π.

Let $\mu = \int \mu_\xi \, d\nu(\xi)$ be its translation-ergodic decomposition. For ν-almost every ξ, the Schwinger functions of μ_ξ satisfy OS0–OS3 with the same constant and also:

OS4. For $f \in \mathscr{S}_m$ and $f' \in \mathscr{S}_n$,

$$S_{m+n}(f \otimes \tau_a^{\otimes n} f') \longrightarrow S_m(f) S_n(f') \qquad \text{as } |a| \to \infty.$$

All these statements allow coincident arguments. The laws μ and the indicated components satisfy (1.2), are carried by $dF = 0$, and are invariant and reflection positive at every plane.

The growth estimate is the linear-growth version of OS regularity in [1, 2]. It is stronger than merely requiring each S_n to be a tempered distribution. OS1 is covariance with the tensor action already defined, and OS2 is positivity of a finite sum of polynomial observables. OS4 expresses decorrelation of two groups of observations when their separation tends to infinity in any direction.

The ergodic qualification is necessary. A translation-invariant mixture of ergodic laws need not have factorizing correlations: a common component label can correlate observations at arbitrarily large separation. The full limit law instead clusters to the projection onto translation-invariant observables, as stated in Corollary 20.2.

1.3. **An abstract rotation theorem.** The part of the proof that produces continuous rotations is independent of the lattice construction. Its assumptions are properties of a single continuum probability law. The distinction matters because reflection positivity of another law, or of a torus with different boundary conditions, cannot be substituted for positivity of this law.

Theorem 1.2 (Diagonal reflections and rotations)**.** *Let μ be a law of a real tempered field strength tensor on $\mathbb{R}^3$. Suppose that it satisfies (1.2), is carried by $dF = 0$, and is invariant under continuous translations and O_{cub}. Suppose that it is reflection positive at every translated plane from the nine families*

$$\begin{gathered} x_j = c \quad (1 \le j \le 3), \\ x_j - x_i = c, \qquad x_j + x_i = c \quad (1 \le j < i \le 3), \end{gathered} \qquad c \in \mathbb{R}.$$

Then μ is invariant under $\mathbb{R}^3 \rtimes O(3)$, and all its moments on L^2 tests are covariant under that group. It is reflection positive at every plane.

For coordinate reflection positivity, reconstruction gives a Hilbert space and commuting generators $\mathsf{H}_j \ge 0$ and P_i for translations. The diagonal reflections imply

$$\mathsf{H}_j \ge |\mathsf{P}_i| \qquad (i \ne j)$$

on the entire reconstructed Hilbert space. The constant one is essential: it specifies the common domain of holomorphy needed to compare different time directions. The proof then analyzes all moments, not only the two-point function. Gaussian domination bounds their growth; positivity rules out nonzero angular coefficients; and the identity $dF = 0$ removes a possible contact contribution. Moment determinacy finally identifies the rotated law with the original law.

The finite-volume comparison and the probability arguments supply the hypotheses of Theorem 1.2 for every volume-first limit and almost every one of its ergodic components. The OS theorem then follows from that rotation theorem and the probability estimates. This separates the regulator-specific comparison argument from the general continuum argument.

1.4. **Why the finite-volume comparison is needed.** The chamber geometry allows reflection-positive factorization at a coordinate or diagonal plane when the torus periods are compatible with that plane. A cubic torus is compatible with the coordinate factorizations used here, whereas a diagonal factorization uses a different period lattice. Consequently the diagonal positivity inequality cannot simply be written for the cubic finite-volume law. We have to transfer it through the infinite-volume limit.

Equality of limiting pressures is one part of this transfer, but it does not imply equality of normalized local expectations. To compare those expectations we introduce local perturbations, use reflected copies of them in compatible tori, and control the additional layers. The resulting comparison states are proved translation invariant. A separation argument then yields diagonal reflection positivity of the original infinite-volume law. The continuum limit preserves these inequalities and their common law.

This argument is longer than the compactness argument. We give the transfer identities, the bounds with insertions, and the elimination of the extra layers before taking limits. Their role is to ensure that the law to which the rotation theorem is applied is the law obtained from the specified cubic cutoff sequence.

1.5. **Scope and plan.** The theorem constructs a field strength sector with the stated OS properties. It does not construct continuum fermionic or mixed correlation functions. It also does not prove that a limiting field strength law is nonzero or non-Gaussian. The point mass at $F = 0$ satisfies every OS axiom and every upper bound used above. Establishing noncollapse therefore requires a lower bound of a different kind. The result gives subsequential limits rather than a unique continuum law.

The paper follows the order of the construction. Section 2 defines the two lattices, their fields and their finite-volume measures. The subsequent finite-volume sections establish strict normalization,

reflection-positive factorizations, and comparison of tori with local insertions. The volume comparison ends with diagonal positivity of the infinite-volume laws. Section 15 introduces the distribution spaces, moments and test functions; Section 16 proves compactness and passage to the continuum. Section 18 proves Theorem 1.2, including the argument for higher moments and coincident supports. The sections on ergodic components and clustering finish the proof of Theorem 1.1.

For a first reading, the finite model, the infinite-volume comparison statement, the continuum compactness proof and the opening of the rotation section give the principal narrative. The detailed transfer and harmonic arguments can then be read separately, returning to the notation index as needed. Every construction used in those arguments is defined before its first theorem.

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2. The discrete field strength and the chamber fermion

The fermion sites are chosen so that every bond crossing one of the required reflection planes is perpendicular to that plane. This is the geometric feature used in the positivity argument. The photon variables remain on the ordinary cubic lattice. We begin with both lattices, and then define the normalized finite-volume measure.

2.1. Cochains and reflection planes.

Definition 2.1. The oriented cells of $\mathbb{Z}^3$ are its sites, links $(x,\mu)=[x,x+e_\mu]$, plaquettes $(x;\mu\nu)$ with $\mu<\nu$, and cubes $x+[0,1]^3$. Their orientations are given by the indicated coordinate vectors. A real k-cochain is a real function on the oriented k-cells, extended with opposite sign to the opposite orientation. We write C^k, C^k_{c} and $C^k(\mathbb{Z})$ for all, finitely supported and integer-valued cochains. The pairing is $\langle u,v\rangle=\sum_\sigma u_\sigma v_\sigma$ whenever it converges absolutely, and $\|u\|^2=\langle u,u\rangle$.

The coboundary is

$$\begin{aligned}(d\lambda)_\mu(x)&=\lambda(x+e_\mu)-\lambda(x),\\ (dA)_{\mu\nu}(x)&=A_\mu(x)+A_\nu(x+e_\mu)-A_\mu(x+e_\nu)-A_\nu(x),\\ (dF)(x)&=F_{23}(x+e_1)-F_{23}(x)-F_{13}(x+e_2)+F_{13}(x)+F_{12}(x+e_3)-F_{12}(x).\end{aligned}$$

It satisfies $d^2=0$. Its adjoint on finitely supported cochains is denoted by d^*. On ordered nearest-neighbor pairs we use $A(x,x+e_\mu)=A_\mu(x)$, $A(x+e_\mu,x)=-A_\mu(x)$ and $A(x,x)=0$. An isometry g preserving $\mathbb{Z}^3$ acts by oriented pullback $(g^*u)(\sigma)=\pm u(g\sigma)$. This action preserves the pairing and commutes with d, since it preserves the signed boundary of each cell. We use $\tau_v(x)=x+v$ and $O_{\mathrm{cub}}=O(3)\cap GL(3,\mathbb{Z})$.

Definition 2.2. Let $\mathcal{L}$ consist of the nine linear forms

$$x_\mu\quad(1\le\mu\le3),\qquad x_\mu-x_\nu,\ x_\mu+x_\nu\quad(\mu<\nu).$$

Write $\ell(x)=n_\ell\cdot x$ and $\hat n_\ell=n_\ell/|n_\ell|$. A lattice reflection plane is $\Pi=(\ell,c)$, meaning $\{\ell=c\}$, where $c\in\mathrm{Adm}(\ell)=\frac12\mathbb{Z}$ for coordinate planes and $c\in\mathbb{Z}$ for diagonal planes. Its reflection is

$$\theta_{\ell,c}x=x-2\frac{\ell(x)-c}{|n_\ell|^2}n_\ell.$$

Coordinate planes at integer positions are site planes, and those at half-integer positions are link planes. The group generated by these reflections is denoted by W. Put $e_\pm=e_1\pm e_2$ and $\ell_\pm=x_1\pm x_2$.

The following elementary geometry identifies the sites and the four possible bonds. In particular, the nearest photon site does not change when a fermion crosses a diagonal reflection plane.

Lemma 2.3. *The group W is $\mathbb{Z}^3\rtimes O_{\mathrm{cub}}$, and acts simply transitively on the chambers of the arrangement of lattice reflection planes. A fundamental chamber is*

$$C_0=\{0<x_3<x_2<x_1<\tfrac12\}.$$

Its four walls and inward normals are

$$\begin{aligned}\Pi_0&=\{x_3=0\},&&n_0=e_3,\\ \Pi_1&=\{x_2=x_3\},&&n_1=(e_2-e_3)/\sqrt2,\\ \Pi_2&=\{x_1=x_2\},&&n_2=(e_1-e_2)/\sqrt2,\\ \Pi_3&=\{x_1=\tfrac12\},&&n_3=-e_1.\end{aligned}$$

Let s_i be reflection in Π_i. The faces have areas $(1,\sqrt2,\sqrt2,1)/8$, and $|C_0|=1/48$.

Proof. The coordinate reflections at 0 and $1/2$ generate unit translations and coordinate sign changes. The diagonal reflections generate coordinate permutations. Conversely every listed reflection is an integer translation followed by a signed permutation, proving the group identity. For a point outside the reflection planes, first subtract its unique nearest integer vector, then change the signs of its three nonzero coordinates, and finally sort their distinct absolute values. This places the point in C_0, and all three choices are unique. On each connected component those choices are constant. It follows that the images of C_0 are exactly the chambers, with trivial stabilizers. The area and volume formulas follow from the vertices 0, $e_1/2$, $(e_1+e_2)/2$ and $(e_1+e_2+e_3)/2$. □

Definition 2.4. Fix $v_0=(v_1,v_2,v_3)\in C_0$ and let $\mathbb{X}=Wv_0$. For $x=gv_0$ define $x^{(i)}=gs_iv_0$. The unordered pairs $\{x,x^{(i)}\}$ are the bonds of type i. Every site has a unique representation $x=k+hv_0$ with $k\in\mathbb{Z}^3$ and $h\in O_{\mathrm{cub}}$. Define its nearest photon site by $\pi(x)=k$ and its displacement by $y(x)=hv_0$.

Lemma 2.5. *There are* 48 *fermion sites per photon unit cell. No fermion site lies on a lattice reflection plane. Every bond meets exactly one such plane in its open segment; that plane bisects the bond perpendicularly. For $i=0,1,2$, $\pi(x^{(i)})=\pi(x)$, whereas $\pi(x^{(3)})-\pi(x)$ is a coordinate unit vector with either sign. The latter bond crosses the link plane bisecting the corresponding photon link. Every other reflection plane either contains that photon link or places its open segment and the fermion bond on the same side.*

Proof. The simple transitivity in Lemma 2.3 gives the site count and excludes sites on walls. Two neighboring chambers are related by reflection in their common wall. Their two sites are reflected points, so their connecting segment is perpendicular to that wall and is bisected there. In C_0 the four midpoints are

$$(v_1,v_2,0),\quad (v_1,(v_2+v_3)/2,(v_2+v_3)/2),\quad ((v_1+v_2)/2,(v_1+v_2)/2,v_3),\quad (1/2,v_2,v_3).$$

Each satisfies its defining wall equality and all the other chamber inequalities strictly. Each open half-segment therefore lies in the chamber interior. The midpoint belongs to no other wall, so no other reflection plane meets the open bond.

The reflections s_0,s_1,s_2 are linear signed permutations, while $s_3=\tau_{e_1}\operatorname{diag}(-1,1,1)$. Thus for $x=k+hv_0$ its first three neighbors have nearest site k, and the fourth has nearest site $k+he_1$. In terms of y, the first neighbor changes the sign of the coordinate of modulus v_3; the next two interchange respectively the moduli v_2,v_3 and v_1,v_2 without changing signs. The last changes $\pm v_1$ to $\pm(1-v_1)$ before changing the nearest integer site. To check the assertion about the photon link it suffices, by signed permutations and translations, to use $[0,e_1]$ and the bond from (v_1,v_2,v_3) to $(1-v_1,v_2,v_3)$. For $j=2,3$, every point of this bond satisfies $x_1-x_j>0$ and $x_1+x_j<1$, since $0<v_j<v_1<1/2$. The remaining diagonal heights are constant and lie strictly between consecutive integers. The bond crosses only the coordinate plane $x_1=1/2$. On the photon link each diagonal height is constant or ranges between consecutive integers; its open segment meets no admissible diagonal plane except those containing the link. All other coordinate planes put the open link and this bond on the same side. This proves the final claims. □

2.2. **Photon variables, spinors and the normalized functional.**

Definition 2.6. A period lattice is a rank-three subgroup $\Gamma\subset\mathbb{Z}^3$ with $\operatorname{sys}(\Gamma)=\min_{g\in\Gamma\setminus\{0\}}|g|_\infty\geq 4$. Periodic cochains form $C^k(\Gamma)$. For $q>0$ the configuration space is

$$\operatorname{Conf}(\Gamma)=\{(A,z)\in C^1(\Gamma)\times C^2(\Gamma;\mathbb{Z}) : dz=0\}/\sim,\qquad (A,z)\sim(A+2\pi k/q,z-dk).$$

The quotient measure is Lebesgue measure on $A(b)\in[0,2\pi/q)$ at all links, summed over closed integer z. Its field strength tensor is

$$F=dA+2\pi z/q.$$

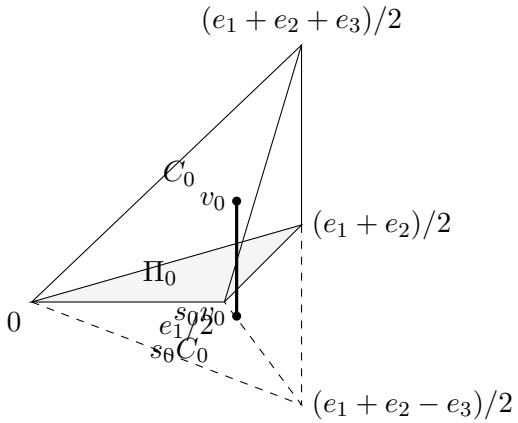


FIGURE 1. A projection of two neighboring chambers. Their common face lies in $\Pi_0 = \{x_3 = 0\}$. The two fermion sites are reflected points, and the bond between them is perpendicular to that face in $\mathbb{R}^3$. The photon site 0 is the nearest integer site to both. The projection illustrates the geometry and does not preserve angles.

It is closed and independent of the representative. The link phases $\exp(iqA(b))$ are likewise independent of the representative. An oriented lattice isometry carries this quotient measure to that on the transformed period lattice, since it permutes the link coordinates up to sign and bijects the closed integer cochains.

Definition 2.7. Let σ_μ be the Pauli matrices and set $\gamma_\mu = \mathrm{diag}(\sigma_\mu, -\sigma_\mu)$. Then $\gamma_\mu^* = \gamma_\mu$ and $\gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2\delta_{\mu\nu}$. For a unit vector n write $\gamma_n = \gamma \cdot n$ and $P_n^\pm = (1 \pm \gamma_n)/2$, and fix orthonormal bases $e_{n,k}^\pm$, $k = 1, 2$, of their ranges with $e_{-n,k}^\pm = e_{n,k}^\mp$. At each site there are eight Grassmann generators: the column $\psi(x)$ and row $\bar\psi(x)$. Their exterior algebra is denoted by Gr. The Berezin integral is the coefficient of $\prod_{x,\alpha} \psi_\alpha(x)\bar\psi_\alpha(x)$; the order of these even pairs is immaterial. In particular

$$\int \exp(-\bar\psi D\psi) = \det D. \tag{2.1}$$

Definition 2.8. Fix $0 < \rho < 1$, $w_i = 48|\Pi_i \cap \overline{C}_0|$, and $M \in \mathbb{R}$. For a bond of type i set

$$w(x,x') = \frac{w_i}{2}\left(\rho - \gamma \cdot \frac{x'-x}{|x'-x|}\right), \qquad U_A(x,x') = \exp(iqA(\pi(x),\pi(x'))).$$

The chamber operator is

$$(D_A\psi)(x) = M\psi(x) - \sum_{i=0}^{3} w(x,x^{(i)})U_A(x,x^{(i)})\psi(x^{(i)}).$$

A torus $\mathbb{T} = (\Gamma, \chi)$ includes a character $\chi : \Gamma \to \{\pm 1\}$, and its spinors obey $\psi(x+g) = \chi(g)\psi(x)$. Its volume is $|\mathbb{T}| = [\mathbb{Z}^3 : \Gamma]$ and its finite matrix is $D_A^{\mathbb{T}}$. The action is $S_{\mathbb{T}}(A) = \sum_{x \in \mathbb{X}/\Gamma} \bar\psi(x)(D_A\psi)(x)$, interpreted using representatives and the stated boundary character. Changing representatives unitarily conjugates the matrix and preserves its determinant. The mass in lattice units is $m = M - \frac{\rho}{2}\sum_i w_i$.

Definition 2.9. Write $\mathbb{X}_{\mathbb{T}} = \mathbb{X}/\Gamma$ and let $\mathrm{Gr}_{\mathbb{T}}$ be the Grassmann algebra with relations $\psi(x+g) = \chi(g)\psi(x)$ and $\bar\psi(x+g) = \chi(g)\bar\psi(x)$. Its quotient map is $r_{\mathbb{T}} : \mathrm{Gr}(\mathbb{X}) \to \mathrm{Gr}_{\mathbb{T}}$. Three families of tori will be used. The coordinate torus $C(n_1, n_2, n_3)$ has periods $n_\mu e_\mu$ and character -1 on all three generators. Write $C(n) = C(n, n, n)$. The torus $D(a, b, c)$ has periods ae_-, be_+, ce_3 and character -1 on all three. The torus $D'(a, b, c)$ has the same periods and character $(-1, +1, -1)$ on those

generators. All the named parameters are integers at least four. Their volumes are $n_1n_2n_3$, $2abc$, and $2abc$, respectively. Their systoles are $\min n_\mu$ and $\min(a,b,c)$: for the second lattice a vector is $(ak+bl,-ak+bl,cm)$, and $\max(|ak+bl|,|-ak+bl|)=|ak|+|bl|$. For a lattice isometry with linear part R set $g\mathbb{T}=(R\Gamma,\chi\circ R^{-1})$.

Lemma 2.10. *For real A, M, $\det D_A^{\mathbb{T}} \geq 0$. Moreover $\mathrm{Re}\langle u, D_A^{\mathbb{T}} u\rangle \geq m\|u\|^2$.*

Proof. The two Pauli blocks of $D_A^{\mathbb{T}}$ are adjoints. Indeed, reversal of a bond conjugates its phase and reverses its unit direction; the scalar hopping part is self-adjoint and the Pauli hopping part is skew-adjoint. Thus their determinants are complex conjugates, and the four-component determinant is the squared modulus of either determinant. For each i, the map

$$(T_i u)(x) = U_A(x, x^{(i)})u(x^{(i)})$$

is a self-adjoint unitary on the finite spinor space. The real part of $D_A^{\mathbb{T}}$ is $M - \frac{\rho}{2}\sum_i w_i T_i$. Since $\|T_i\| = 1$ and $\rho, w_i > 0$, its quadratic form is bounded below by m. □

Definition 2.11. For an integrable insertion G put

$$Z_{\mathbb{T}}[G] = \int_{\mathrm{Conf}(\Gamma)} e^{-\|F\|^2/2} \int_{\mathbb{T}} e^{-S_{\mathbb{T}}(A)} G(A,z), \qquad Z_{\mathbb{T}} = Z_{\mathbb{T}}[1].$$

After Theorem 3.4 proves $Z_{\mathbb{T}} > 0$, the normalized functional is $\mathbb{E}_{\mathbb{T}}[G] = Z_{\mathbb{T}}[G]/Z_{\mathbb{T}}$. Its field strength marginal is a probability measure on closed two-cochains, with density proportional to $e^{-\|F\|^2/2}\det D_A^{\mathbb{T}}$. The periodic extension of F is written $\widetilde{F}$.

Lemma 2.12. *The finite functional is absolutely convergent coefficient by coefficient for a bounded photon insertion times a Grassmann polynomial. It is also convergent for finitely many local factors whose Gaussian growth constants are strictly less than $1/2$ on pairwise disjoint sets of plaquettes.*

Proof. On the compact link box every coefficient of the fermion exponential is bounded. Also $\|dA\|$ is bounded there, and for each $\varepsilon > 0$,

$$\|dA + 2\pi z/q\|^2 \geq (1-\varepsilon)(2\pi/q)^2\|z\|^2 - (\varepsilon^{-1}-1)\|dA\|^2.$$

The sum over closed integer z is bounded by a finite-dimensional Gaussian sum over all integer plaquette values. The same argument applies to local Gaussian growth: disjoint supports leave a strictly positive quadratic coefficient on every plaquette. There are finitely many Grassmann coefficients, so all the coefficient integrals converge absolutely. □

2.3. **Local observables and reflections.** The field strength alone is enough to state the main theorem. The finite-volume comparisons also use local fermion polynomials. Their definition must be independent of a choice of link potential: a closed integer flux has a primitive on a box, but generally has no periodic primitive on a torus.

Definition 2.13. A box is a product of compact intervals. Write $B^{(k)}$ for the oriented lattice k-cells entirely contained in it and let $\mathbb{X}(B)$ contain those fermion sites x for which $\pi(x)+[-1/2,1/2]^3 \subset B$. A local integrand on B is a map

$$G(\widehat{A}) = \sum_{\Psi} a_\Psi(\widehat{A})\Psi \in \mathrm{Gr}(\mathbb{X}(B)), \qquad \widehat{A} \in \mathbb{R}^{B^{(1)}},$$

whose coefficients are Borel functions. The sum ranges over the ordered Grassmann monomials at these sites. If $c_\Psi(k) = \mathfrak{q}(\Psi)(k)$ is the number of ψ factors minus the number of $\bar\psi$ factors based at the photon site k, require

$$a_\Psi(\widehat{A} - d\lambda)e^{iq\sum_k \lambda(k)c_\Psi(k)} = a_\Psi(\widehat{A})$$

for every real zero-cochain λ on B. Require also, for some $C < \infty$ and $\varepsilon > 0$,

$$|a_\Psi(\widehat{A})| \leq C\exp\big((\tfrac{1}{2}-\varepsilon)\|d\widehat{A}\|^2_{B^{(2)}}\big).$$

These integrands form $\mathcal{G}(B)$; its even subspace is $\mathcal{G}^{\mathrm{ev}}(B)$.

For $U \subset \mathbb{R}^3$, the space $\mathcal{A}(U)$ is the linear span, as maps of a full-lattice real one-cochain $\widehat{A}$ to $\mathrm{Gr}(\mathbb{X})$, of products $\Phi(d\widehat{A})G_1(\widehat{A}|_{B_1})\cdots G_r(\widehat{A}|_{B_r})$. We write these products more briefly as $\Phi(F)G_1\cdots G_r$. Here Φ is a bounded Borel function of finitely many field strength plaquettes contained in U, and $G_j \in \mathcal{G}^{\mathrm{ev}}(B_j)$ for pairwise disjoint boxes $B_j \subset U$. A box embeds in a period lattice if the translates of its enlarged box $B + [-1,1]^3$ are pairwise disjoint. A family of boxes embeds disjointly if each box embeds and the translates of the boxes in the entire family are pairwise disjoint.

Lemma 2.14. *Every configuration has a real potential $\widehat{A}$ on a box B such that $d\widehat{A} = F$ there and $e^{iq\widehat{A}} = e^{iqA}$ on its links. The value of a local integrand at this potential is independent of its choice and of the representative (A, z). These values are Borel functions of the configuration. The same statement holds on the full lattice for a configuration without a periodicity requirement.*

Proof. We give the primitive explicitly. A box with no lattice vertices has no cells, and the assertion is vacuous. Otherwise its vertices form $I_1 \times I_2 \times I_3$, where $I_j = [a_j, b_j] \cap \mathbb{Z}$ and $a_j, b_j \in \mathbb{Z}$ are its least and greatest lattice coordinates in direction j. For a closed integer two-cochain z, put, on the links that exist in the box,

$$\begin{aligned} k_1(x) &= 0, \\ k_2(x) &= \sum_{s=a_1}^{x_1-1} z_{12}(s, x_2, x_3), \\ k_3(x) &= \sum_{s=a_1}^{x_1-1} z_{13}(s, x_2, x_3) + \sum_{s=a_2}^{x_2-1} z_{23}(a_1, s, x_3). \end{aligned} \tag{2.2}$$

Empty sums are zero; the formula therefore also covers a singleton coordinate interval. The 12 and 13 components of $dk = z$ follow by telescoping. For the remaining component,

$$(dk)_{23}(x) = z_{23}(a_1, x_2, x_3) + \sum_{s=a_1}^{x_1-1} (\Delta_2 z_{13} - \Delta_3 z_{12})(s, x_2, x_3).$$

Closedness says that the summand is $\Delta_1 z_{23}$, so the sum is $z_{23}(x) - z_{23}(a_1, x_2, x_3)$. This proves $dk = z$ using only integer additions and subtractions. Hence $\widehat{A} = A + 2\pi k/q$ has the required field strength and link phases.

We will also need the corresponding statement for one-cochains. If $dh = 0$, define

$$\begin{aligned} \lambda(x) = &\sum_{s=a_1}^{x_1-1} h_1(s, a_2, a_3) + \sum_{s=a_2}^{x_2-1} h_2(x_1, s, a_3) \\ &+ \sum_{s=a_3}^{x_3-1} h_3(x_1, x_2, s). \end{aligned}$$

For the third difference the last sum telescopes directly. For the second, use $\Delta_2 h_3 = \Delta_3 h_2$ in the last sum, which telescopes from $h_2(x_1, x_2, a_3)$ to $h_2(x_1, x_2, x_3)$. For the first, use $\Delta_1 h_2 = \Delta_2 h_1$ and $\Delta_1 h_3 = \Delta_3 h_1$ in the last two sums. Together they give $\Delta_i \lambda = h_i$ for every existing link. Integer h gives integer λ; the same proof works for real cochains.

The difference of two lifts is $2\pi/q$ times a closed integer one-cochain and therefore equals $(2\pi/q)d\lambda$ with integer λ. The gauge transformation in Definition 2.13 has multiplier one, because all charges c_Ψ are integers. Thus every coefficient is unchanged. Changing the representative (A, z) leaves F and all link phases unchanged and gives the same conclusion.

To make Borel measurability explicit, recover a representative on the box from the link phases by

$$A^0(b) = q^{-1} \operatorname{Arg}_{[0,2\pi)}(e^{iqA(b)}), \qquad z^0 = \frac{q}{2\pi}(F - dA^0).$$

For a configuration, z^0 is a closed integer two-cochain. Apply (2.2) to z^0 and set $\widehat{A}^0 = A^0 + 2\pi k^0/q$. This is a Borel function of finitely many field strength values and link phases. Composing the Borel coefficients a_Ψ with this lift proves the assertion, with exactly the growth bound already specified because $d\widehat{A}^0 = F$.

On the full lattice use the same formulas with $a_1 = a_2 = a_3 = 0$, interpreting an oriented sum with a negative endpoint by

$$\sum_{s=0}^{t-1} u(s) = -\sum_{s=t}^{-1} u(s), \qquad t < 0.$$

Every coordinate of the primitive uses finitely many terms; the telescoping identities hold for either sign of t. This gives a global primitive for every closed integer two-cochain and the analogous global primitive for a closed one-cochain. For periodic z the primitive may be nonperiodic. No assertion about torus cohomology is needed. □

Proposition 2.15. *Suppose a finite sum of products defining $G \in \mathcal{A}(U)$ has local boxes that embed disjointly in $\mathbb{T}$. Its evaluation in $\mathrm{Gr}_\mathbb{T}$ is independent of the local lifts and of the representation of G as a sum of products. This evaluation is linear and Borel. It respects products whenever the combined local boxes embed disjointly.*

Proof. Extend a representative (A, z) periodically to $\mathbb{Z}^3$ and choose one global, possibly nonperiodic primitive k of its integer flux by Lemma 2.14. Then $\widehat{A} = \widetilde{A} + 2\pi k/q$ has field strength $\widetilde{F}$ and the prescribed phases on every link. For a defining product $H = \Phi(F)G_1 \cdots G_r$ its evaluation can be written

$$r_\mathbb{T}(H(\widehat{A})) = \Phi(\widetilde{F}) \prod_{j=1}^{r} r_\mathbb{T}(G_j(\widehat{A}|_{B_j})).$$

The quotient map is an algebra homomorphism. By the local-lift lemma each restricted lift can be replaced by the one used to define that local factor on the torus. If two finite sums represent the same map G, they agree at this single $\widehat{A}$ before applying $r_\mathbb{T}$, hence also afterwards. This proves representation independence and linearity. Borel measurability follows from the finite-box Borel constructions in the lemma. For products the same identity proves compatibility, and the disjointness assumption ensures absolute convergence by Lemma 2.12. □

Definition 2.16. For $\Pi = (\ell, c)$ put $n = \hat{n}_\ell$, $\theta = \theta_{\ell,c}$. The reflected Grassmann algebra is defined by the antilinear anti-automorphism

$$\Theta_\Pi \psi(x) = -\bar{\psi}(\theta x)\gamma_n, \qquad \Theta_\Pi \bar{\psi}(x) = -\gamma_n \psi(\theta x). \tag{2.3}$$

Here the first identity is read component by component; ψ is a column and $\bar{\psi}$ a row. On a local integrand put

$$\Theta_\Pi G(\widehat{A}) = \sum_\Psi \overline{a_\Psi(\theta^* \widehat{A})} \Theta_\Pi \Psi.$$

For a photon function this is $\Theta_\Pi \Phi(F) = \overline{\Phi(\theta^* F)}$.

Lemma 2.17. *The reflection maps $\mathcal{G}(B)$ onto $\mathcal{G}(\theta B)$, preserves its even subspace, and has square one. It reverses products. On an even product of local integrands it depends only on the induced lattice isometry when reflections are composed; an odd number of reflections acts antilinearly, and an even number acts linearly.*

Proof. The Clifford relation gives $\gamma_n^2 = 1$, and hence the square of the component transformation is one. Reversing the order twice has the same property on a monomial. The reflected field strength norm is unchanged. A zero-cochain gauge transformation at x becomes one at θx; conjugation and interchange of ψ with $\bar{\psi}$ reverse both the phase and its charge, proving the coefficient transformation required in Definition 2.13.

For completeness, two successive reflections act by

$$\Theta_\Pi\Theta_{\Pi'}\psi(x) = \gamma_{n'}\gamma_n\psi(\theta_\Pi\theta_{\Pi'}x), \qquad \Theta_\Pi\Theta_{\Pi'}\bar\psi(x) = \bar\psi(\theta_\Pi\theta_{\Pi'}x)\gamma_n\gamma_{n'}.$$

The order of the matrices is thus the reverse of the order of the reflections acting on sites. Their Clifford action is fixed by

$$-\gamma_n\gamma_v\gamma_n = \gamma_{v-2(n\cdot v)n}.$$

Compare two reflection words inducing the same affine isometry by composing the first with the inverse of the second. Their lengths have the same parity, so the resulting word has even length $2r$, fixes all sites, and acts on an unbarred spinor through

$$U = \gamma_{n_{2r}}\cdots\gamma_{n_1} = \operatorname{diag}(u,u), \qquad u = \sigma_{n_{2r}}\cdots\sigma_{n_1}.$$

The Clifford identity shows that U commutes with every γ_v. Thus u commutes with the three Pauli matrices and is scalar. Since

$$\det u = \prod_{a=1}^{2r}\det\sigma_{n_a} = (-1)^{2r} = 1,$$

this scalar is $+1$ or -1. Barred generators acquire its inverse, which is the same sign. A monomial of degree d therefore acquires the factor $(\pm1)^d$, equal to one for even d. The two words also have the same linear or antilinear action on coefficients. Integer translations are included in this argument: the quotient word fixes every site and its linear part is the identity. This proves word independence on the even subspace. □

Proposition 2.18. *The finite photon law is invariant under lattice translations and covariant under the signed permutation group. On the coupled functional, a lattice reflection satisfies*

$$Z_{\theta\mathbb{T}}[\Theta_\Pi G] = \overline{Z_{\mathbb{T}}[G]}.$$

Products of reflections give the corresponding covariance for even local integrands whenever they embed on the tori.

Proof. The reflected bilinear has the exact component formula

$$\Theta_\Pi(\bar\psi(x)B\psi(x')) = \bar\psi(\theta x')\gamma_n B^*\gamma_n\psi(\theta x).$$

It follows by conjugating coefficients, reversing the product, and using the two generator formulas in (2.3). The Clifford identity $\gamma_n\gamma_v\gamma_n = -\gamma_{\theta v}$ transforms the hopping matrix into that of the reversed reflected bond. Conjugating its phase reverses its orientation in the same way. The real mass term is fixed. Thus reflection takes the full action to that on the reflected torus. It preserves the Gaussian photon norm, bijects closed integer fluxes, and preserves the quotient link measure. It also takes a site top monomial to the reflected top monomial: the two matrix determinants and the eight-generator reversal sign are one. A change of variables therefore gives the displayed functional identity. Translations act by relabeling; products of reflections generate all lattice isometries, and their word independence on even integrands was proved above. This proves the remaining assertions. □

2.4. **The polynomial algebra used in the comparisons.**

Lemma 2.19. *Every finitely supported real one-cochain δ with $d^*\delta = 0$ has a finitely supported real two-cochain β with $\delta = d^*\beta$. If δ is integer, β may be chosen integer.*

Proof. Interpret δ as a finite link chain with zero boundary. Choose a finite connected graph containing its support and a spanning tree. Subtract from the chain the coefficient of each non-tree edge times the cycle formed by that edge and its tree path. What remains is a cycle on a tree and is zero, by deleting leaves. Thus the original chain is a finite linear combination of finite lattice loops, with integer coefficients when the chain was integer. A loop in $\mathbb{Z}^3$ can be filled by plaquettes: write it as a word in coordinate unit steps and commute adjacent steps in different directions, adding

the intervening oriented unit square to the filling. After sorting the directions, cancel opposite steps within each direction. The word becomes empty since the loop is closed. These finitely many squares fill the loop. Summing them with their cycle coefficients gives the asserted β. □

Definition 2.20. Let $\mathcal{P}$ be the span of the local expressions

$$P(\widehat{A}) = e^{i\langle\delta,\widehat{A}\rangle}\Psi, \qquad \delta \in C^1_{\mathrm{c}}, \qquad d^*\delta = qc_\Psi,$$

where Ψ is a finite Grassmann monomial and c_Ψ is its photon-site charge from Definition 2.13. The relation is precisely gauge invariance. Summing it over photon sites shows that Ψ has equally many ψ and $\bar{\psi}$ factors, so these expressions are even. Each belongs to $\mathcal{G}(B)$ for a sufficiently large box and has bounded coefficients there. Their coefficients extend entirely to complex link variables.

The bosonic subspace of $\mathcal{P}$ consists of finite sums of $e^{i\langle\beta,d\widehat{A}\rangle}$ for finite two-cochains β. Indeed gauge invariance gives $d^*\delta = 0$, and Lemma 2.19 then provides $\delta = d^*\beta$. The algebra $\mathcal{P}$ is preserved by lattice isometries, with the antilinear reflection rule already defined.

An entire bosonic factor on a torus is an entire function of its finitely many complex field strength coordinates that is bounded on every tube $|\operatorname{Im} F_\sigma| \le r$. Exponentials of finite real trigonometric polynomials in F have this property. It is this tube bound that is used when contours are moved in the layer elimination proof.

For an even local expression, write G^g for the action of a lattice isometry obtained by composing its reflections and translations. A translation replaces fields at x by fields at $x + v$ and pulls the potential back by τ_v. For a reflection it is $\Theta_\Pi G$. The word independence proved above makes this notation unambiguous. On a real photon observable it is the ordinary oriented pullback.

3. Normalization at arbitrary real mass

The mass need not make the fermion matrix accretive. Positivity of its determinant then gives only $Z_{\mathbb{T}} \ge 0$. To normalize the finite-volume functional and compare pressures, we need a strictly positive lower bound exponential in the volume. A coefficient of the determinant polynomial provides such a bound independently of the mass.

Definition 3.1. In this section only, a torus may have any rank-three subgroup $\Gamma \subset \mathbb{Z}^3$ and any unitary character $\chi : \Gamma \to S^1$. Its sites are still $\mathbb{X}/\Gamma$ and its volume is $V = [\mathbb{Z}^3 : \Gamma]$. Choose one representative of each site orbit. If $x^{(i)} = \widehat{x^{(i)}} + g_i(x)$, the matrix block of the chamber operator is

$$D[x,x'] = \begin{cases} M1, & x' = x, \\ -\chi(g_i(x))U_A(x,x^{(i)})w(x,x^{(i)}), & x' = \widehat{x^{(i)}}, \\ 0, & \text{otherwise.} \end{cases} \tag{3.1}$$

Changing representatives conjugates D by a diagonal unitary. The five classes $x, x^{(0)}, \dots, x^{(3)}$ are distinct: their displacements y differ either in their coordinate moduli or in the sign of one of two different moduli, by Lemma 2.5. Thus the cases in the formula do not overlap, even on these smaller tori.

Let $u_b = e^{iqA(b)}$ for each photon link orbit. Only bonds of type 3 carry a variable phase. The Leibniz expansion of D is a Laurent polynomial $P_{\mathbb{T}}(u) = \sum_\lambda a_\lambda u^\lambda$. This polynomial is independent of the chosen representatives. For real M it is nonnegative on the phase torus: the adjoint Pauli-block argument of Lemma 2.10 applies also to a unitary boundary character.

Lemma 3.2. *Fix a coordinate μ. Order the six possible values of $y_\mu(x)$ as $-v_1, -v_2, -v_3, v_3, v_2, v_1$, and let $k_\mu(x) \in \{1, \dots, 6\}$ be its position in this list, read cyclically. Each layer has $8V$ site classes. Each site has a unique neighbor $x^\uparrow$ in the next layer and a unique neighbor $x^\downarrow$ in the preceding layer;*

the other two neighbors stay in its layer. The upward map is a bijection on the torus sites, with inverse the downward map. Its bond types in layers $1,\dots,6$ *are*

$$(i_1,\dots,i_6)=(2,1,0,1,2,3).$$

Its phase is one in layers $1,\dots,5$ *and is* u_b *in layer* 6*, for* $b=(\pi(x),\mu)$ *modulo* Γ*. There are eight such layer-*6 *bonds over each link in direction* μ*; the opposite orientation carries* u_b^{-1}*.*

Proof. There are eight signed permutations of v_0 with a prescribed μ-coordinate: the two remaining moduli can be placed in two orders with four sign choices. This gives the layer count. The explicit neighbor transformations in Lemma 2.5 give the following possibilities. At modulus v_3, type 0 changes its sign and type 1 changes its modulus to v_2. At modulus v_2, types $1,2$ change it to v_3, v_1, keeping its sign. At modulus v_1, type 2 changes it to v_2, whereas type 3 changes its sign and its photon coordinate by one. These give precisely one successor and one predecessor and the displayed type list. The neighbor relation is symmetric, so successor and predecessor are inverse. They commute with translations and descend to the torus. Only the step from v_1 to $-v_1$ in the next photon cell carries a positive coordinate link phase. The layer count at a fixed photon site gives the eight bonds over that link. □

Proposition 3.3. *Let* $\lambda_b^{(\mu)}=32$ *for all links in direction* μ *and zero for the other links. Then*

$$|a_{\lambda^{(\mu)}}|=C_*^V,\qquad C_*=(w_0/2)^{32}(w_1/2)^{64}(w_2/2)^{64}(w_3/2)^{32}(1-\rho^2)^{96}>0.$$

For a character taking only the values ±1*, the coefficient itself equals* C_*^V*.*

Proof. In a nonzero term of the Leibniz expansion, every row maps to a column at its own site or at one of its four neighbors. For a given positive-direction link, only the 32 rows at its eight lower sites can contribute a positive exponent, and only the corresponding 32 upper rows can contribute a negative exponent. Thus each exponent is at most 32. If it is 32 at every μ-link, all rows in layer 6 must map upwards to layer 1, and no row in layer 1 may map downwards. These rows exhaust the layer-1 columns. A row in layer 1 therefore cannot stay in its layer and must map upwards to layer 2. These exhaust layer-2 columns. Inductively, a row in layer k cannot stay there or go downwards, since those columns have already been exhausted by earlier layers. It must map upwards. This determines the site permutation throughout all six layers. Conversely, choosing arbitrary spin permutations in each upward 4-by-4 block gives precisely the exponent $\lambda^{(\mu)}$, so there are no omitted terms contributing to this coefficient.

The site permutation is repeated four times in spin, and therefore has sign one. Summing its internal spin permutations by the determinant formula gives the product over all sites of the upward hopping-block determinants. Their phase product is $\prod_{b\|e_\mu}u_b^{32}$. Each block contributes

$$\det(-\chi(g_i(x))w(x,x^{\uparrow}))=\chi(g_i(x))^4(w_{i_k}/2)^4(1-\rho^2)^2.$$

The character factors have modulus one and are one for a sign character. Each layer has $8V$ sites. The type multiplicities $1,2,2,1$ in the six-layer cycle then give exactly C_*^V. □

Theorem 3.4. *For every real mass* M *and* $0<\rho<1$*,*

$$Z_{\mathbb{T}}\ge(2\pi/q)^{3V}e^{-6(2\pi/q)^2V}C_*^V>0, \tag{3.2}$$

$$Z_{\mathbb{T}}\le\left((2\pi/q)C_q\right)^{3V}\left(|M|+(1+\rho)\textstyle\sum_i w_i/2\right)^{192V}<\infty. \tag{3.3}$$

The determinant is positive outside a Lebesgue null set of link potentials. Every closed integer flux sector has positive weight. The coupled photon measure and the Gaussian photon measure have the same null sets.

Proof. The constant Fourier coefficient a_0 of a nonnegative trigonometric polynomial bounds the modulus of every Fourier coefficient, since $|a_\lambda|=|\int P(u)u^{-\lambda}du|\le\int P(u)du=a_0$. Consequently

the mean of $\det D_A$ on its compact link box is at least C_*^V. On that box each plaquette satisfies $|(dA)_\sigma| < 2(2\pi/q)$, so $\|dA\|^2 \le 12V(2\pi/q)^2$. Keeping just the zero flux sector proves the lower bound.

The finite matrix has dimension $192V$. Each neighbor map with its link phase and boundary character is unitary, and its hopping matrix has norm at most $w_i(1+\rho)/2$. The operator norm of D_A is therefore at most the parenthesis in (3.3). Bound its determinant by that norm to the power $192V$, drop the integer closedness constraint, and apply the one-plaquette Gaussian sum bound C_q. This proves the upper bound.

The extreme coefficient shows that the determinant is a nonzero trigonometric polynomial. Its zero set has Lebesgue measure zero: induct on the number of variables, separating the one-variable polynomial from its coefficient polynomials; outside the common zero set of a nonzero coefficient it has finitely many roots in that variable. Fubini's theorem completes the induction. For any fixed closed flux, its Gaussian density is strictly positive everywhere on the link box. Thus that sector has positive mass. Removing the determinant changes no null set in any sector, and then in their countable sum. □

Lemma 3.5. *For every bounded real Borel insertion V_0, $Z_{\mathbb{T}}[e^{V_0}] \ge e^{-\|V_0\|_\infty} Z_{\mathbb{T}} > 0$. For each real plaquette cochain β, $Z_{\mathbb{T}}[e^{-\langle\beta,F\rangle - \|\beta\|^2/2}]$ is also finite and strictly positive.*

Proof. The first assertion is pointwise monotonicity of the positive measure. In the second the Gaussian exponent completes to $-\|F+\beta\|^2/2$. The shifted Gaussian sum has the same uniform bound C_q; its density is positive everywhere. The determinant has already been proved positive almost everywhere, giving both convergence and strict positivity. □

4. Reflection positivity of the finite measure

The fermion crossing coefficients are positive, but this alone does not prove positivity of the coupled measure. The integer fluxes must remain closed at cubes cut by a reflection plane. We first enforce that constraint by Fourier integration. Poisson summation then expresses the photon crossing as a sum of positive terms. At a link plane, integration over the crossing links imposes Gauss' law, including the charges of the fermion modes.

4.1. The crossing modes and the two halves.

Proposition 4.1. *For the modes of Definition 6.11, the terms of one bond crossing Π satisfy*

$$-S_\Pi^\times = \sum_i \omega_i u_i \Theta_\Pi(\zeta_i)\zeta_i, \qquad e^{-S_\Pi^\times} = \sum_I \omega_I u_I \Theta_\Pi(\zeta_I)\zeta_I.$$

All ω_i are positive when $0<\rho<1$. A bond entirely above Π uses only link phases from the closed upper half-space.

Proof. Fix a crossing bond of type j, with lower endpoint x, upper endpoint θx, unit normal $n=\hat n_\ell$, and phase $u = U_A(x,\theta x)$. The reversed bond has normal $-n$ and phase $\bar u$. The two hopping terms in the negative action are

$$\begin{aligned}-S_b = {}& \frac{w_j}{2} u\big((1+\rho)\bar\psi(x)P_n^-\psi(\theta x) - (1-\rho)\bar\psi(x)P_n^+\psi(\theta x)\big)\\ &+ \frac{w_j}{2}\bar u\big((1+\rho)\bar\psi(\theta x)P_n^+\psi(x) - (1-\rho)\bar\psi(\theta x)P_n^-\psi(x)\big).\end{aligned}$$

For any $e\in\mathbb{C}^4$, antilinearity and (2.3) give

$$\Theta(e^*\psi(\theta x)) = -\bar\psi(x)\gamma_n e, \qquad \Theta(\bar\psi(\theta x)e) = -e^*\gamma_n\psi(x).$$

Since $\gamma_n e^{\pm}_{n,k} = \pm e^{\pm}_{n,k}$, the products $\Theta\zeta_i\zeta_i$ for the four groups, in their listed order, are

$$\bar\psi(x)e^-_{n,k}(e^-_{n,k})^*\psi(\theta x), \qquad -\bar\psi(x)e^+_{n,k}(e^+_{n,k})^*\psi(\theta x),$$
$$\bar\psi(\theta x)e^+_{n,k}(e^+_{n,k})^*\psi(x), \qquad -\bar\psi(\theta x)e^-_{n,k}(e^-_{n,k})^*\psi(x).$$

In the last two expressions, moving $\psi(x)$ past $\bar\psi(\theta x)$ supplies the extra minus sign. Summing over $k = 1, 2$ uses $\sum_k e^{\pm}_{n,k}(e^{\pm}_{n,k})^* = P^{\pm}_n$ and recovers exactly $-S_b$. This proves the first identity, with the positive coefficients in Definition 6.11.

Each $\Theta\zeta_i\zeta_i$ is even and has square zero. The factors therefore commute, and their exponential is the product of $1+\omega_i u_i \Theta\zeta_i\zeta_i$. For $I = \{i_1 < \cdots < i_d\}$, collecting all reflected factors on the left gives the sign $(-1)^{\binom{d}{2}}$. Reversing their order to form $\Theta(\zeta_I)$ gives the same sign, so $\prod_{i\in I}\Theta\zeta_i\zeta_i = \Theta(\zeta_I)\zeta_I$. This proves the second identity. The coefficients are positive in the stated range. For the last assertion, Lemma 2.5 shows that a photon link associated with an upper bond cannot cross the plane; links on the plane are allowed. □

Definition 4.2. For a family ℓ, a torus is split if

$$\Gamma = \Gamma_\parallel + \mathbb{Z}g_0, \qquad \Gamma_\parallel = \Gamma \cap \{\ell = 0\}, \qquad g_0 = \frac{p}{|n_\ell|^2} n_\ell \in \mathbb{Z}^3,$$

where $p > 0$. It is adapted if also $\chi(g_0) = -1$. Given $c \in \mathrm{Adm}(\ell)$, write

$$c' = c + p/2, \quad \Pi = (\ell, c), \quad \Pi' = (\ell, c'), \quad U_+ = \{c < \ell < c'\}, \quad U_- = \{c - p/2 < \ell < c\}.$$

The reflection is $\theta = \theta_\Pi$ and $\mathcal{M} = \bigcup_{j\in\mathbb{Z}}\{\ell = c + jp/2\}$.

Lemma 4.3. *On a split torus, $p \geq 4|n_\ell|^2$, $c' \in \mathrm{Adm}(\ell)$, and $\theta_{\Pi'} = \tau_{g_0}\theta_\Pi$. Both reflections induce the same involution on the photon torus, preserve its character, and exchange its two open halves. No fermion site lies on their boundaries. Choose representatives $\mathbb{X}_+$ of the sites in U_+ modulo $\Gamma_\parallel$ and put $\mathbb{X}_- = \theta\mathbb{X}_+$. These two sets together represent all fermion sites of the torus.*

If local products G, G' in U_+ embed disjointly modulo $\Gamma_\parallel$, then $\Theta_\Pi G' G$ embeds disjointly modulo Γ. On the Grassmann algebra,

$$\Theta_{\Pi'}\xi = \chi(g_0)^{\deg\xi}\Theta_\Pi\xi \tag{4.1}$$

for a homogeneous element ξ.

Proof. The systole bound applied to g_0 gives $p \geq 4$ for a coordinate normal and $p \geq 8$ for a diagonal normal. In the latter case p is even because g_0 is integer. The stated reflection formula follows by subtraction of the two affine reflection formulas. Its linear part fixes $\Gamma_\parallel$ and sends g_0 to $-g_0$, so it preserves Γ and χ. The translated open slabs partition $\mathbb{R}^3 \setminus \mathcal{M}$. Every site orbit therefore has one representative in either half.

A box in U_+ enlarged by $[-1, 1]^3$ has heights in an open interval of length at most $p/2 + 2|n_\ell|^2 \leq p$. Its translates by nonzero multiples of g_0 cannot meet it; the remaining disjointness is the assumed tangential embedding. The reflected boxes obey the same estimate. Original and reflected boxes lie in different slabs, which proves joint embedding. Finally the generators at $\theta_{\Pi'}x$ equal $\chi(g_0)$ times those at $\theta_\Pi x$, giving (4.1) on each generator and then on every homogeneous product. □

Let S_+ contain the mass terms at $\mathbb{X}_+$ and the hopping terms with both endpoints in U_+. Put $\Theta = \Theta_\Pi$ and $\Xi = (\Gamma_\parallel, \chi|_{\Gamma_\parallel})$. The finite mode sets used below are $\mathcal{J}_\Pi := \mathrm{idx}_\Xi(\Pi)$ and $\mathcal{J}_{\Pi'} := \mathrm{idx}_\Xi(\Pi')$, with the representatives and orders of Definition 6.11. Thus every sum over I, I' ranges over subsets of these two sets, respectively. For a mode i' at Π' put $\check\zeta_{i'} = \Theta_{\Pi'}\zeta_{i'}$; it belongs to the upper Grassmann algebra. Ordered products are denoted by $\check\zeta_{I'}$. The action and its exponential satisfy

$$S_\mathbb{T} = S_+ + \Theta S_+ + S^\times_\Pi + S^\times_{\Pi'}, \tag{4.2}$$

$$e^{-S_\mathbb{T}} = \sum_{I,I'} \omega_I\omega_{I'}u_I u_{I'}(-\chi(g_0))^{|I'|}\Theta(e^{-S_+})e^{-S_+}\Theta\zeta_I\zeta_I\,\Theta\check\zeta_{I'}\check\zeta_{I'}. \tag{4.3}$$

Indeed every bond is internal to one half or crosses exactly one of the two boundary planes. Reflection maps the lower internal action to the upper one. At Π', the order of the two odd factors is reversed; together with (4.1) this gives the factor $-\chi(g_0)$. On an adapted torus that factor is one. The coefficients of e^{-S_+} are bounded polynomials in upper and boundary link phases.

Lemma 4.4. *For homogeneous upper elements ξ, ξ', ξ'',*

$$\Theta\xi'\,\xi\,\Theta\xi''\,\xi'' = (-1)^{(|\xi|+|\xi'|)|\xi''|}\Theta(\xi'\xi'')\xi\xi'', \tag{4.4}$$

$$\int_{\mathbb{T}} \Theta\xi'\,\xi = \overline{\int_{\mathbb{X}_+} \xi'} \int_{\mathbb{X}_+} \xi. \tag{4.5}$$

Proof. Move ξ past $\Theta\xi''$, and then use reversal of products under Θ; these give the two signs in the first identity. For the second, the top monomial factors into the lower top monomial times the upper one, without a sign since each site contributes eight generators. Reflection takes the upper top monomial to the lower one: the determinant factors are $\det(\gamma_n)^2 = 1$, and reversal of eight generators has sign $(-1)^{28} = 1$. Only the two top coefficients contribute to the integral. This proves the factorization, including the case when an odd element makes both sides zero. □

4.2. **The photon crossing and Gauss' law.** Call a cell upper or lower when its relative interior lies in the corresponding open half, a boundary cell when it lies in $\mathcal{M}$, and a cut cell when its relative interior meets $\mathcal{M}$ and it is not a boundary cell. These notions are periodic modulo Γ. Write $\mathbb{T}_+^{(k)}, \mathbb{T}_-^{(k)}, \mathbb{T}_0^{(k)}, \mathbb{T}_\times^{(k)}$ for the four sets of k-cells.

Lemma 4.5. *Each cell has exactly one of these four kinds and meets at most one boundary plane. Reflection exchanges upper and lower cells, fixes boundary cells with positive orientation, and fixes cut cells with negative orientation. Faces of an upper cell are upper or boundary; faces of a cut cell are upper, lower or cut. A cell with a cut face is cut.*

Let $[\sigma : b]$ and $[Q : \sigma]$ be the incidence numbers. For a cut plaquette and cut cube define d_+ and $d_\times$ by summing their faces of the indicated kind. Then

$$(dA)_\sigma = (d_+A)_\sigma - (d_+\theta^*A)_\sigma + (d_\times A)_\sigma,$$
$$(dz)(Q) = (d_+z)(Q) - (d_+\theta^*z)(Q) + (d_\times z)(Q).$$

Proof. The height variation on a unit cube is at most $|n_\ell|^2$, less than the separation $p/2$ of boundary planes. Connectedness of a cell's relative interior gives the classification. A boundary plane fixes a point in the relative interior of a cut cell. Since distinct lattice cells have disjoint relative interiors and reflection permutes cells, it maps that cell to itself. Its restriction to the affine span is a reflection, so the orientation is negative. On a boundary cell it is the identity. A face of an upper cell that attains boundary height at an interior point is contained in that plane, since an affine function attaining an extremum there is constant. A face of a cut cell cannot be contained in the reflecting plane: reflection in a proper face of a cube translates the cube instead of fixing it. A cut face has points on both sides of the plane, and these are limits of relative-interior points of its coface; hence that coface is cut. Finally $\theta^* d = d\theta^*$ and the negative orientation of a cut cell identify its lower face contributions with the negatives of its reflected upper contributions. Splitting the incidence sum proves both formulas. □

Definition 4.6. Write $A = (A_+, A_-, A_0, a)$ and $z = (z_+, z_-, z_0, z_\times)$ according to this classification. Choose intervals of length $2\pi/q$ for the link variables: $[0, 2\pi/q)$ on upper and boundary links, their reflected signed intervals on lower links, and $[-\pi/q, \pi/q)$ on cut links. These form a measurable fundamental domain. Put

$$\mathcal{W}_+ = \prod_{\sigma\in\mathbb{T}_+^{(2)}} e^{-F_\sigma^2/2} \prod_{Q\in\mathbb{T}_+^{(3)}} 1_{\{dz(Q)=0\}}, \qquad \mathcal{W}_0 = \prod_{\sigma\in\mathbb{T}_0^{(2)}} e^{-F_\sigma^2/2}.$$

Let $\mathcal{Y}_+$ and $\mathcal{Y}_0$ be the product spaces of upper and boundary link variables and integer plaquette variables, with Lebesgue and counting measures.

The half weights depend only on their indicated variables. Closedness and the full Gaussian weight factor as

$$1_{\{dz=0\}} e^{-\|F\|^2/2} = \mathcal{W}_0 \mathcal{W}_+ \Theta \mathcal{W}_+ \prod_{\sigma \in \mathbb{T}_\times^{(2)}} e^{-F_\sigma^2/2} \prod_{Q \in \mathbb{T}_\times^{(3)}} 1_{\{dz(Q)=0\}}. \tag{4.6}$$

Set $C_q = \sup_t \sum_{j\in\mathbb{Z}} e^{-(t+2\pi j/q)^2/2} < \infty$. Dropping cube constraints bounds the integral of a half weight by $(2\pi/q)^{\#\text{links}} C_q^{\#\text{plaquettes}}$, uniformly in its boundary data. This bound will justify all changes of order below.

Lemma 4.7. *For real y, α,*

$$\sum_{j\in\mathbb{Z}} e^{-(y+2\pi j/q)^2/2 + ij\alpha} = \frac{q}{\sqrt{2\pi}} \sum_{k\in\mathbb{Z}} e^{-\beta_k^2/2 - i\beta_k y}, \qquad \beta_k = \frac{q}{2\pi}(\alpha - 2\pi k).$$

For $\alpha \in [0,2\pi)^{\mathbb{T}_\times^{(3)}}$ and $k \in \mathbb{Z}^{\mathbb{T}_\times^{(2)}}$ put

$$\alpha'_\sigma = \sum_Q [Q:\sigma]\alpha_Q, \quad \beta_\sigma = \frac{q}{2\pi}(\alpha'_\sigma - 2\pi k_\sigma), \quad u_{\alpha,k} = \exp\left(i \sum_Q \alpha_Q d_+ z(Q) - i \sum_\sigma \beta_\sigma (d_+ A)_\sigma \right).$$

All sums here are over cut cells. The crossing weight satisfies

$$\sum_{z_\times} \prod_\sigma e^{-F_\sigma^2/2} \prod_Q 1_{\{dz(Q)=0\}} = \int \prod_Q \frac{d\alpha_Q}{2\pi} \sum_k \prod_\sigma \frac{q}{\sqrt{2\pi}} e^{-\beta_\sigma^2/2} u_{\alpha,k} \Theta u_{\alpha,k} e^{iq \sum_{b\in\mathbb{T}_\times^{(1)}} a_b (d^* k)(b)}. \tag{4.7}$$

The positive coefficient measure on (α, k) has mass at most $C_q^{\#\mathbb{T}_\times^{(2)}}$.

Proof. Poisson summation for $s \mapsto e^{-(y+2\pi s/q)^2/2 + i\alpha s}$ gives the first formula by the Fourier transform of a Gaussian. To enforce closedness, insert $1_{\{dz(Q)=0\}} = \int_0^{2\pi} e^{i\alpha_Q dz(Q)} d\alpha_Q/(2\pi)$. The cut-cube formula separates the upper and lower terms; the remaining coefficient of z_σ is α'_σ. Apply the first formula to each cut plaquette. The upper and lower factors give $u_{\alpha,k}\Theta u_{\alpha,k}$. The remaining factor is $e^{-i\sum_\sigma \beta_\sigma (d_\times A)_\sigma}$. For a cut link b,

$$\sum_\sigma [\sigma : b]\alpha'_\sigma = \sum_Q \alpha_Q \sum_\sigma [Q:\sigma][\sigma:b] = 0.$$

The restriction to cut plaquettes does not change the sum, because a coface of a cut link is cut. The last identity is $d^2 = 0$. Thus the remaining factor is the one displayed in (4.7). Absolute convergence follows from C_q before Poisson summation. Afterwards apply the first formula at $y = 0$ and take the modulus of the integer-flux sum to bound each positive k_σ sum by C_q. This proves both the identity and the mass bound, with justified Fubini interchanges. □

Lemma 4.8. *A cut link is perpendicular to a coordinate link plane. For a mode of charge $\mathfrak{q}(i) = 1$ in groups $1, 2$ and -1 in groups $3, 4$, its phase is $e^{iq\mathfrak{q}(i)a_b}$ if its bond lies over that cut link, and is one otherwise. Consequently*

$$\int da\, e^{iq\langle a, d^* k\rangle} u_I u_{I'} = (2\pi/q)^{\#\mathbb{T}_\times^{(1)}} 1_{\{d^* k + \mathfrak{q}(I \cup I') = 0\}}. \tag{4.8}$$

Here the charge is summed separately at each cut link; the constraint is imposed only on those links.

Proof. Reflection fixes a cut link setwise and exchanges its endpoints, so its normal is a coordinate vector and the plane bisects the link. Lemma 2.5 then identifies its photon link phase with the phase of each crossing fermion bond over it. The four groups give the two stated charges. All other crossing bonds have equal nearest photon sites. Each coefficient in $d^*k + \mathfrak{q}(I \cup I')$ is an integer. Integration over one complete period of each a_b therefore vanishes unless that coefficient is zero, when it contributes $2\pi/q$. □

4.3. **A positive representation of the reflected form.**

Definition 4.9. Let $\mathcal{B}^+$ be the span of bounded Borel functions of upper and boundary field strengths and link phases times elements of $\mathrm{Gr}(\mathbb{X}_+)$. Let $\mathcal{G}^+$ allow coefficients Φ with

$$\mathcal{W}_+|\Phi| \le C \prod_{\sigma \in \mathbb{T}^{(2)}_+} e^{-\varepsilon F_\sigma^2}$$

for some $C < \infty$, $\varepsilon > 0$, uniformly in boundary data.

An embedded local product $G = \Phi_0 G_1 \cdots G_r$ in U_+ belongs to $\mathcal{G}^+$. Here is the estimate needed for that assertion. Every link, plaquette and site read by a box $B_j \subset U_+$ belongs to the upper half. The local-lift lemma expresses each coefficient as a Borel function of the upper field strengths and link phases. Where $\mathcal{W}_+ \neq 0$, all cubes of B_j are closed, and the growth bound on that coefficient is

$$C_j \exp\left((\tfrac{1}{2} - \varepsilon_j) \sum_{\sigma \in B_j^{(2)}} F_\sigma^2 \right).$$

Lemma 4.3, with the reflected factor equal to 1, shows that the boxes embed jointly on the torus. Their plaquette classes are therefore distinct. Reading the Grassmann generators on the torus introduces only the prescribed character and sorting signs. Each coefficient of the product is a finite sum of products of local coefficients. Taking $\varepsilon = \min(1/2, \varepsilon_1, \dots, \varepsilon_r)$, multiplying by $\mathcal{W}_+$, and using boundedness of Φ_0 gives exactly the defining upper growth estimate. Outside the set $\mathcal{W}_+ \neq 0$ it holds because its left side is zero. The case of no local factors uses $\varepsilon = 1/2$. Finite sums are covered by a common minimum decay margin and the sum of their constants.

The functions in this definition are initially functions on the configuration space of closed integer fluxes. Their use on the larger half-variable space does not introduce a choice of extension. To see this, fix upper and boundary data for which $\mathcal{W}_+ \neq 0$. Set $z_\times = 0$ and choose the lower fluxes so that $(\theta^* z)_+ = z_+$. Keep the prescribed z_0. All upper cubes are closed by hypothesis, and their reflected lower cubes are closed too. For a cut cube the formula in Lemma 4.5 reads $d_+ z - d_+ \theta^* z + d_\times z = 0$. Thus the completed flux is closed on every cube. Boundary plaquettes are fixed with positive orientation, so the completion respects the prescribed boundary data. The lower link variables can be completed arbitrarily in their intervals. Consequently every half datum with nonzero half weight comes from an actual configuration. Two representations of the same upper function agree there; their products with $\mathcal{W}_+$ therefore agree everywhere. In particular the transform below is a well-defined linear map on $\mathcal{G}^+$, rather than on a choice of formula for its elements.

Theorem 4.10. *On every torus adapted to ℓ, for $G, G' \in \mathcal{G}^+$ the reflected functional has a positive Gram representation*

$$Z_{\mathbb{T}}[\Theta_\Pi G'\, G] = \int \overline{\widehat{G'}(s)} \widehat{G}(s)\, d\mathfrak{m}(s).$$

In particular $Z_{\mathbb{T}}[\Theta_\Pi G\, G] \ge 0$. The finite functional also satisfies $Z_{\mathbb{T}}[\Theta_\Pi G] = \overline{Z_{\mathbb{T}}[G]}$ whenever the integral is absolutely convergent.

Proof. The positive measure is on $s = (A_0, z_0, \alpha, k, I, I')$, with counting measure on the integer variables and mode subsets, and density

$$d\mathfrak{m} = \mathcal{W}_0 \, dA_0 \prod_Q \frac{d\alpha_Q}{2\pi} \prod_\sigma \frac{q}{\sqrt{2\pi}} e^{-\beta_\sigma^2/2} \omega_I \omega_{I'} (2\pi/q)^{\#\mathbb{T}_\times^{(1)}} 1_{\{d^*k + \mathfrak{q}(I \cup I') = 0\}}. \tag{4.9}$$

Define

$$\widehat{G}(s) = \int_{\mathcal{Y}_+} dy_+ \, \mathcal{W}_+ u_{\alpha,k} \int_{\mathbb{X}_+} G e^{-S_+} \zeta_I \check{\zeta}_{I'}. \tag{4.10}$$

Here are explicit bounds on the two expressions. For $a > 0$ set

$$C_{q,a} := \sup_{t \in \mathbb{R}} \sum_{j \in \mathbb{Z}} e^{-a(t + 2\pi j/q)^2} < \infty, \qquad C_q = C_{q,1/2}.$$

Finiteness follows by reducing t to one compact period and using a Gaussian tail, uniformly on that period. Write $N_{\nu,k} = \#\mathbb{T}_\nu^{(k)}$, where $\nu \in \{+, 0, \times\}$. Since S_+ is a finite Grassmann polynomial with uniformly bounded coefficients, its finite exponential has uniformly bounded coefficients as well. There are finitely many mode subsets and their monomials have fixed coefficients. Therefore the defining growth bounds for G give a constant $B_G < \infty$ and a common $\varepsilon_G > 0$ such that, for every I, I',

$$\mathcal{W}_+ \left| \int_{\mathbb{X}_+} G e^{-S_+} \zeta_I \check{\zeta}_{I'} \right| \le B_G \prod_{\sigma \in \mathbb{T}_+^{(2)}} e^{-\varepsilon_G F_\sigma^2}.$$

This estimate is used on data with $\mathcal{W}_+ \neq 0$; on its complement the left side is zero. Since $|u_{\alpha,k}| = 1$, summing each integer plaquette value first and then integrating the compact link intervals gives the uniform bound

$$|\widehat{G}(s)| \le B_G (2\pi/q)^{N_{+,1}} C_{q,\varepsilon_G}^{N_{+,2}}. \tag{4.11}$$

The positive measure in (4.9) has total mass at most

$$(2\pi/q)^{N_{0,1} + N_{\times,1}} C_q^{N_{0,2} + N_{\times,2}} \prod_{i \in \mathcal{J}_\Pi} (1 + \omega_i) \prod_{i' \in \mathcal{J}_{\Pi'}} (1 + \omega_{i'}). \tag{4.12}$$

Indeed the Gauss-law indicator is at most one, the (α, k) coefficient mass is bounded by Lemma 4.7, and the boundary Gaussian sums give $C_q^{N_{0,2}}$. The subset sums equal the two displayed products. Measurability follows from the measurable integrands and absolute convergence. Equations (4.11)–(4.12) now prove $\widehat{G} \in L^2(d\mathfrak{m})$.

Insert (4.3) into the reflected integral. Since $\chi(g_0) = -1$, all its crossing coefficients have positive signs. By (4.4), each term reorganizes as $\Theta(G' e^{-S_+} \zeta_I \check{\zeta}_{I'})(G e^{-S_+} \zeta_I \check{\zeta}_{I'})$. For precision, if G, G' have parities $r, r' \in \{0, 1\}$, the three applications of the product rule produce the total sign $(-1)^{(r+r')(|I|+|I'|)}$. The action exponential is even; the two upper factors have parities $r + |I| + |I'|$ and $r' + |I| + |I'|$. If $r \neq r'$, one is odd and its Berezin integral is zero, because the upper top monomial has even degree $8\#\mathbb{X}_+$. If $r = r'$, the displayed sign is one. Thus the claimed factorization holds for homogeneous elements, and then for all G, G' by linearity. Next insert (4.7) into (4.6). Integration over the cut links gives (4.8), exactly the constraint in (4.9). The remaining upper and lower integrals factor by (4.5). Reflection is a measure-preserving bijection from the chosen lower fundamental domain to $\mathcal{Y}_+$, at fixed boundary data. The two factors are therefore $\widehat{G}$ and $\overline{\widehat{G'}}$, proving the identity.

Before Poisson summation, each coefficient of the original reflected integral has its modulus bounded by a constant times

$$\mathcal{W}_0 \prod_{\sigma \in \mathbb{T}_+^{(2)}} e^{-\varepsilon_G F_\sigma^2} \prod_{\sigma \in \mathbb{T}_-^{(2)}} e^{-\varepsilon_{G'} F_\sigma^2} \prod_{\sigma \in \mathbb{T}_\times^{(2)}} e^{-F_\sigma^2/2}.$$

The cut-cube indicators can be dropped. Summing cut fluxes first gives $C_q^{N_{\times,2}}$, uniformly in the face and link variables. The half and boundary Gaussian sums then prove absolute convergence of the original functional.

To justify the exchanges of order after Poisson summation and before imposing Gauss' law, take absolute values in each of the finitely many Berezin coefficients. The phases $u_{\alpha,k}, u_I, u_{I'}$ and the remaining cut-link exponential all have modulus one. Applying the growth estimates separately to the upper and lower factors gives the integrable majorant with total integral at most

$$B_G B_{G'} (2\pi/q)^{2N_{+,1}+N_{0,1}+N_{\times,1}} C_{q,\varepsilon_G}^{N_{+,2}} C_{q,\varepsilon_{G'}}^{N_{+,2}} C_q^{N_{0,2}+N_{\times,2}} \cdot \prod_{i\in\mathcal{J}_\Pi} (1+\omega_i) \prod_{i'\in\mathcal{J}_{\Pi'}} (1+\omega_{i'}).$$

The equality of upper and lower cell counts follows from reflection. The lower change of variables has unit Jacobian and preserves counting measure. Tonelli's theorem bounds the absolute majorant, and Fubini's theorem permits precisely the reordering used above: the cut links may be integrated first, followed by the two half-variable integrals. This argument also covers the stated Gaussian growth class, without assuming bounded coefficients there. Finally reflection preserves the full photon measure and maps the action to itself, while its Berezin integral conjugates the top coefficient. Changing variables in the absolutely convergent integral proves the covariance identity. □

5. Lattice field strength laws

Definition 5.1. Let $\Omega = \{F \in C^2 : dF = 0\}$, with the topology of coordinatewise convergence on the countable set of plaquettes. It is a closed subspace of a countable product of real lines and is therefore Polish. A local function depends on a finite set of those coordinates. For $\beta \in C_c^2$ set $\varphi_\beta(F) = e^{i\langle\beta,F\rangle}$. A trigonometric polynomial is a finite sum $O = \sum_b a_b \varphi_{\beta_b}$. The real-valued such sums form $\mathcal{T}$. For a chosen expression put $|O|_1 = \sum_b |a_b|$, let $\operatorname{supp} O$ be the union of its cochain supports, and let r_O be an integer at least one such that this union fits in a cube of radius r_O. A chosen expression is retained whenever these bounds are used; no uniqueness of a character representation on the closed-cochain space is needed.

A law means a Borel probability measure on Ω. Gaussian domination means

$$\mu(e^{\langle\beta,F\rangle}) \le e^{\|\beta\|^2/2} \quad (\beta \in C_c^2 \text{ real}). \tag{G_{lat}}$$

For a torus $\mathbb{T}$ define its field strength law by

$$\mu_{\mathbb{T}}(G) = Z_{\mathbb{T}}[G]/Z_{\mathbb{T}}[1]$$

for bosonic functions of the periodic extension of $F = dA + (2\pi/q)z$. The nonnegative determinant weight and the positive partition function make this a probability law on Ω. The same normalised expression on a fermionic local integrand is a linear functional, not an additional probability law. We use the same notation for it only when its integral is absolutely convergent.

Definition 5.2 (Reflection positivity of a lattice law)**.** For a mirror (ℓ, c) let $\Sigma^+(\ell, c)$ be the plaquettes contained in its open positive half-space. For a bounded cylinder function G depending on finitely many such plaquettes set $\Theta G(F) = \overline{G(\theta_{\ell,c}^* F)}$. A law is reflection positive at that mirror if

$$\mu((\Theta G)G) \ge 0$$

for every such bounded Borel G. Invariance at the mirror means invariance under the ordinary tensor pullback $\theta_{\ell,c}^*$.

Lemma 5.3 (Compactness and separation)**.** *The class $\mathcal{K}$ of laws satisfying (G_{lat}) is weakly compact and convex. The class $\mathcal{R}(\ell, c)$ of its members invariant and reflection positive at a fixed mirror is weakly compact and convex. If a law ν has the property that for every real $O \in \mathcal{T}$ there is $\nu_O \in \mathcal{R}(\ell, c)$ with $\nu(O) \le \nu_O(O)$, then $\nu \in \mathcal{R}(\ell, c)$.*

Proof. Compactness of the dominated laws. Let 1_σ have value one at the plaquette σ and zero elsewhere. For $s, R > 0$, Markov's inequality gives

$$\mu(F_\sigma > R) \le e^{-sR}\mu(e^{sF_\sigma}) \le e^{-sR+s^2/2}, \qquad \mu(F_\sigma < -R) \le e^{-sR+s^2/2}.$$

Taking $s = R$ gives $\mu(|F_\sigma| > R) \le 2e^{-R^2/2}$. The bound is uniform in the plaquette as well as the law. Enumerate the plaquettes as $\sigma_1, \sigma_2, \ldots$. Given $\varepsilon > 0$, choose R_j with $2e^{-R_j^2/2} \le \varepsilon 2^{-j}$. The set

$$K_\varepsilon = \Omega \cap \prod_{j\ge 1}[-R_j, R_j]$$

is compact. Indeed, the product is compact, and $dF = 0$ is a collection of closed finite-coordinate conditions. The union bound gives $\mu(K_\varepsilon) \ge 1 - \varepsilon$ for every $\mu \in \mathcal{K}$. Thus $\mathcal{K}$ is tight.

Suppose that $\mu_n \in \mathcal{K}$ converges weakly to μ. For each real compact cochain β and $L > 0$, the function $\min(e^{\langle \beta, F\rangle}, L)$ is bounded and continuous. Consequently

$$\mu\Big(\min(e^{\langle \beta, F\rangle}, L)\Big) = \lim_n \mu_n\Big(\min(e^{\langle \beta, F\rangle}, L)\Big) \le e^{\|\beta\|^2/2}.$$

Monotone convergence as $L \to \infty$ proves domination for μ. Thus $\mathcal{K}$ is closed. Prokhorov's theorem on the Polish space Ω proves its weak compactness. Its defining inequalities are affine in the law, so it is convex.

Closedness of reflection positivity. The map $F \mapsto \theta^* F$ is continuous: it only permutes coordinates and changes their orientation signs. Thus invariance passes to a weak limit on bounded continuous cylinder functions. Such functions determine the finite marginals and hence the law, so the limit is invariant. For a bounded continuous positive-side cylinder function G, the function $(\Theta G)G$ is also bounded and continuous. Its nonnegative expectation therefore passes to the limit. We explain the extension to Borel functions, including the role of invariance.

For any finite collection of positive-side plaquettes, bounded continuous functions are dense in the L^2 space of its finite-dimensional marginal. To see this for an indicator, regularity of that marginal gives, for a Borel set E and $\delta > 0$, a compact $K \subset E$ and open $U \supset E$ with marginal measure of $U \setminus K$ less than δ. A continuous function between zero and one, equal to one on K and zero outside U, differs from 1_E in squared L^2 norm by at most δ. One can construct it from the distances to K and U^c; the cases $K = \varnothing$ or $U^c = \varnothing$ follow by truncation. Simple functions then prove density for every bounded Borel function.

Finite positive-side cylinders themselves are dense in the L^2 space of the sigma-algebra generated by all positive-side plaquettes. For completeness, if a vector is orthogonal to all their indicators, its integral over every cylinder set is zero. The finite signed measure defined by integrating that vector is then zero on the generated sigma-algebra, by the monotone class theorem. The vector is zero almost everywhere. This proves density without an assumption on independence of the coordinates.

For G, H in that L^2 space, invariance gives $\|\Theta G\|_2 = \|G\|_2$. Cauchy–Schwarz consequently gives

$$|\mu((\Theta G)G) - \mu((\Theta H)H)| \le \|G - H\|_2(\|G\|_2 + \|H\|_2).$$

Approximation by the continuous cylinders proves positivity for every positive-side L^2 function, in particular for the bounded Borel cylinders of the definition. This proves closedness of $\mathcal{R}(\ell, c)$ in $\mathcal{K}$. Both invariance and the reflected quadratic inequalities are affine conditions on the law, proving convexity.

Characters distinguish laws. This familiar fact can be proved using only the Gaussian Fourier integral. If finite measures m_1, m_2 on $\mathbb{R}^d$ have equal characteristic functions, let $\gamma_a(x) = (2\pi a)^{-d/2}e^{-|x|^2/(2a)}$, $a > 0$. Fourier inversion of the Gaussian and Fubini's theorem give

$$\int \gamma_a(x - y)\, dm_l(y) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-a|\xi|^2/2} e^{i\langle \xi, x\rangle} \left(\int e^{-i\langle \xi, y\rangle}\, dm_l(y) \right) d\xi.$$

Absolute integrability follows from the total mass of m_l and the integrable Gaussian in ξ. The two convolution densities are equal. For bounded continuous f their integrals against f are therefore

equal. On letting $a \downarrow 0$, Gaussian concentration and continuity give $f * \gamma_a(y) \to f(y)$; bounded convergence gives $m_1(f) = m_2(f)$. Regularity, with the same continuous approximation of indicators used above, gives $m_1 = m_2$. Apply this result to every finite plaquette marginal. Its characteristic functions are exactly the characters with cochain support in that finite set. Equality of all character integrals therefore gives equality on cylinder sets, and the monotone class theorem gives equality of laws on Ω.

Finite-dimensional separation. Map a law to the real and imaginary parts of all its character integrals, in the real product space $\mathbb{R}^I$ indexed by those parts. The map J is continuous and affine, and the preceding argument proves it injective. The set $K = J(\mathcal{R}(\ell, c))$ is compact and convex. The domination hypothesis makes $\mathcal{R}(\ell, c)$ nonempty, by applying it to $O = 0$.

If $J(\nu) \notin K$, compactness makes K closed. There is a basic product neighbourhood of $J(\nu)$, restricting finitely many coordinates $I_0 \subset I$, which misses K. In particular, the projected point $y = \pi_{I_0} J(\nu)$ does not belong to the compact convex set $K_0 = \pi_{I_0} K \subset \mathbb{R}^{I_0}$. Choose a closest point $x_0 \in K_0$ to y. For $x \in K_0$, the segment from x_0 to x stays in K_0; the right derivative at zero of its squared distance to y is nonnegative. Hence

$$\langle y - x_0, x - x_0 \rangle \le 0, \qquad \langle y - x_0, y \rangle > \sup_{x \in K_0} \langle y - x_0, x \rangle.$$

The displayed linear functional uses only finitely many real and imaginary character integrals. Each real or imaginary part is a real trigonometric polynomial, since $\overline{\varphi_\beta} = \varphi_{-\beta}$. It is consequently expectation of some $O \in \mathcal{T}$. The strict inequality says $\nu(O) > \sup_{\mu \in \mathcal{R}(\ell,c)} \mu(O)$, contrary to the hypothesis. Thus $J(\nu) \in K$; injectivity proves $\nu \in \mathcal{R}(\ell, c)$. □

6. Geometry of the transfer operators

The finite comparison repeatedly cuts a torus into slabs, reflects local perturbations into neighboring cells, and closes the slabs to recover a trace. We introduce the geometry and the operator spaces before constructing those traces. A cross-section records the two transverse periods and the fermion boundary signs. A cell records the region occupied by one perturbation, and a fold specifies how its reflected copies are arranged. These data will make it possible to compare tori adapted to different reflection planes.

There are three groups of definitions. The first describes cells, folds and the probability states obtained by repeated reflection. The second assigns Fock spaces to slices and specifies their reflection maps. The third describes operator insertions at a cut, including insertions supported in two neighboring layers. The following section realizes these objects as transfer operators with the full integer-flux sum.

6.1. Cells, folds and comparison states.

Write $W_{\mathrm{c}} = \mathbb{Z}^3 \rtimes \{\mathrm{diag}(s_1, s_2, s_3) : s_\mu = \pm 1\}$ for the subgroup generated by coordinate mirrors. It is normal in W, since signed coordinate permutations preserve its diagonal linear parts. The action on an even local integrand is the action of Definition 2.16; its composition convention is $(G^g)^{g'} = G^{g'g}$. On a bosonic function it is ordinary pullback for an even element and conjugated pullback for an odd element.

Definition 6.1 (Slabs and adapted tori)**.** Fix a family $\ell \in \mathcal{L}$. A *cross-section* for ℓ is a pair $\Xi = (\Gamma_\parallel, \chi_\parallel)$ in which $\Gamma_\parallel$ is a subgroup of rank two of $\mathbb{Z}^3 \cap \{\ell = 0\}$ with $\mathrm{sys}(\Gamma_\parallel) \ge 4$ and $\chi_\parallel \colon \Gamma_\parallel \to \{1, -1\}$ is a character. For $c < c'$ in $\mathrm{Adm}(\ell)$ the open set

$$\{c < \ell < c'\} := \{x \in \mathbb{R}^3 : \ c < \ell(x) < c'\}$$

is the *open slab* between the mirrors (ℓ, c) and (ℓ, c'), and $\mathcal{A}_\Xi[c, c']$ denotes the linear span of those products of $\mathcal{A}(\{c < \ell < c'\})$ that embed in $\Gamma_\parallel$ in the sense of Definition 2.13. The boxes of these products and the plaquettes on which their bounded factors depend lie in the open slab, so that nothing that meets one of the planes $\{\ell = c\}$, $\{\ell = c'\}$ is read.

Let $\mathbb{T} = (\Gamma, \chi)$ be a torus. Its cross-section for ℓ is

$$\Xi(\mathbb{T}, \ell) := \left(\Gamma \cap \{\ell = 0\},\ \chi|_{\Gamma\cap\{\ell=0\}}\right);$$

by Lemma 8.2 it is indeed a cross-section for ℓ. A *closing vector* of $\mathbb{T}$ for ℓ is an element $g \in \Gamma$ with $\Gamma = (\Gamma \cap \{\ell = 0\}) + \mathbb{Z}g$ and $\ell(g) > 0$; such vectors exist (Lemma 8.2). For a closing vector g one has $\ell(\Gamma) = \ell(g)\mathbb{Z}$, so that the number $p(\mathbb{T}, \ell) := \ell(g)$ does not depend on g; it is the *height period* of $\mathbb{T}$ for ℓ.

Write $p := p(\mathbb{T}, \ell)$ and

$$v := \frac{p}{|n_\ell|^2} \cdot n_\ell \in \mathbb{R}^3.$$

Suppose that $v \in \mathbb{Z}^3$ and that the group

$$\Gamma^0 := (\Gamma \cap \{\ell = 0\}) \oplus \mathbb{Z}v$$

satisfies $\mathrm{sys}(\Gamma^0) \geq 4$; the sum is direct, since ℓ vanishes on the first summand and $\ell(v) = p \neq 0$. Then Γ^0 is a period lattice, and the *straightened torus* is

$$\mathbb{T}^0 = \mathbb{T}^0(\ell) := (\Gamma^0, \chi^0),$$

where χ^0 is the character of Γ^0 that agrees with χ on $\Gamma \cap \{\ell = 0\}$ and takes the value -1 at v. In this case $\mathbb{T}^0(\ell)$ is said to be defined, and $|\mathbb{T}^0| = |\mathbb{T}|$ (Lemma 8.2). If one of the two conditions fails, Γ^0 is not a period lattice, and $\mathbb{T}^0(\ell)$ is not defined.

The torus $\mathbb{T}$ is *adapted to* ℓ if $\mathbb{T}^0(\ell)$ is defined and equal to $\mathbb{T}$. Let $\mathbb{T}$ be adapted to ℓ, and let $c \in \mathrm{Adm}(\ell)$ with $c + \frac{p}{2} \in \mathrm{Adm}(\ell)$. Then $\Gamma = \Gamma^0 = (\Gamma \cap \{\ell = 0\}) \oplus \mathbb{Z}v$, in particular $v \in \Gamma$; the linear part of $\theta_{\ell,c}$ fixes $\{\ell = 0\}$ pointwise and reverses n_ℓ, hence maps Γ onto itself, and $\theta_{\ell,c}$ acts on $\mathbb{R}^3/\Gamma$. The explicit reflection formula gives

$$\theta_{\ell,c+p/2} = \tau_v\, \theta_{\ell,c},$$

so that $\theta_{\ell,c}$ and $\theta_{\ell,c+p/2}$ induce the same map of $\mathbb{R}^3/\Gamma$. The two mirrors (ℓ, c) and $(\ell, c + \frac{p}{2})$ cut the torus into two halves, the images in $\mathbb{R}^3/\Gamma$ of the open slabs $\{c < \ell < c + \frac{p}{2}\}$ and $\{c + \frac{p}{2} < \ell < c + p\}$: as $\ell(\Gamma) = p\mathbb{Z}$, these images are disjoint, and their union is the complement of the images of the two planes. The same formula shows that $\theta_{\ell,c}$ maps the first slab onto $\{c - \frac{p}{2} < \ell < c\}$, the translate of the second one by $-v$; thus the map of $\mathbb{R}^3/\Gamma$ induced by $\theta_{\ell,c}$ exchanges the two halves.

Definition 6.2 (Frames and cell structures)**.** A *frame* is a triple $\mathfrak{F} = (\ell^1, \ell^2, \ell^3)$ of families whose normals $n_a := n_{\ell^a}$ are pairwise orthogonal. The *coordinate frame* $\mathfrak{F}_\mathrm{c} := (x_1, x_2, x_3)$ and the *diagonal frame* $\mathfrak{F}_\mathrm{d} := (\ell_-, \ell_+, x_3)$ are frames, and up to the order and the signs of its families every frame is the image of one of these two under an element of O_cub.

Let $\mathfrak{F}$ be a frame, and let p_1, p_2, p_3 be positive integers with

$$\frac{p_a}{|n_a|^2} \cdot n_a \in \mathbb{Z}^3 \qquad \text{and} \qquad p_a \geq 4\,|n_a|^2 \qquad (a = 1, 2, 3).$$

The *rectangular torus* $\mathbb{T}(\mathfrak{F}; p_1, p_2, p_3)$ has the lattice

$$\Gamma = \mathbb{Z}\,\frac{p_1}{|n_1|^2} \cdot n_1 \oplus \mathbb{Z}\,\frac{p_2}{|n_2|^2} \cdot n_2 \oplus \mathbb{Z}\,\frac{p_3}{|n_3|^2} \cdot n_3$$

and the character χ that takes the value -1 at each of the three generators. This lattice has systole at least four (Lemma 8.5), so that $\mathbb{T}(\mathfrak{F}; p_1, p_2, p_3)$ is a torus. Since $\ell^a(n_b) = 0$ for $b \neq a$, the normals being orthogonal, one has $\ell^a(\Gamma) = p_a\mathbb{Z}$: the integer p_a is the height period of $\mathbb{T}(\mathfrak{F}; p_1, p_2, p_3)$ for ℓ^a. The coordinate tori and the adapted tori of Definition 2.9 are rectangular:

$$\mathbb{T}(\mathfrak{F}_\mathrm{c}; p_1, p_2, p_3) = C(p_1, p_2, p_3), \qquad \mathbb{T}(\mathfrak{F}_\mathrm{d}; p_1, p_2, p_3) = D(p_1/2, p_2/2, p_3);$$

for the diagonal frame one has $|n_1|^2 = |n_2|^2 = 2$ and $|n_3|^2 = 1$, and the integrality condition says that p_1 and p_2 are even. For these two frames the conditions $p_a \geq 4\,|n_a|^2$ read $\min(p_1, p_2, p_3) \geq 4$ and $\min(p_1/2, p_2/2, p_3) \geq 4$, respectively, which are the conditions on the sides in Definition 2.9.

A rectangular torus is adapted to each of its three families: $\frac{p_a}{|n_a|^2}\cdot n_a$ is a closing vector for ℓ^a, $\Gamma\cap\{\ell^a=0\}$ is spanned by the two other generators, and the straightened torus $\mathbb{T}^0(\ell^a)$ of Definition 6.1 has the lattice Γ and the character χ.

A *cell structure* $\mathcal{Q}=(\mathfrak{F};o_1,o_2,o_3;h)$ consists of a frame $\mathfrak{F}=(\ell^1,\ell^2,\ell^3)$, positions $o_a\in\mathrm{Adm}(\ell^a)$ and a positive integer h, the *width*. Its *walls* are the mirrors $(\ell^a,o_a+j\cdot h)$, $a=1,2,3$, $j\in\mathbb{Z}$ (note that $o_a+j\cdot h\in\mathrm{Adm}(\ell^a)$, h being an integer), and its *open cells* are the sets

$$Q_\kappa:=\Big\{x\in\mathbb{R}^3:\ o_a+\kappa_a\cdot h<\ell^a(x)<o_a+(\kappa_a+1)\cdot h,\ a=1,2,3\Big\},\qquad \kappa\in\mathbb{Z}^3.$$

The number

$$|Q|:=\frac{h^3}{|n_1|\cdot|n_2|\cdot|n_3|}$$

is the volume of an open cell; it equals h^3 for $\mathfrak{F}_\mathrm{c}$ and $h^3/2$ for $\mathfrak{F}_\mathrm{d}$. The group $W(\mathcal{Q})$ is the group generated by the reflections in the walls. Reflections in walls of different families commute, because $\ell^b\circ\theta_{\ell^a,c}=\ell^b$ for $b\neq a$; so $W(\mathcal{Q})$ is the direct product of three infinite dihedral groups, one for each family. It acts simply transitively on the open cells. Indeed, in each height coordinate its elements are $u\mapsto o+\epsilon(u-o)+2hz$, $\epsilon=\pm1$, $z\in\mathbb{Z}$; they carry the base interval to the interval indexed by $2z$ or $2z-1$, respectively. Each interval index has a unique such expression, and the three height coordinates are independent.

Let $\mathbb{T}=(\Gamma,\chi)$ be a torus. The structure $\mathcal{Q}$ *cuts* $\mathbb{T}$ if $\ell^a(g)\in h\mathbb{Z}$ for all $g\in\Gamma$ and $a=1,2,3$. In that case every translation τ_g, $g\in\Gamma$, permutes the open cells: it maps Q_κ onto $Q_{\kappa'}$ with $\kappa'_a=\kappa_a+\ell^a(g)/h$. The *cells of* $\mathbb{T}$ are the Γ-orbits of open cells. Their number is $|\mathbb{T}|/|Q|$: the map $g\mapsto(\ell^1(g)/h,\ell^2(g)/h,\ell^3(g)/h)$ is an injective homomorphism of Γ into $\mathbb{Z}^3$, the orbits of cells correspond to the cosets of its image, and this image has the index $[\mathbb{Z}^3:\Gamma]\cdot|n_1|\cdot|n_2|\cdot|n_3|\cdot h^{-3}=|\mathbb{T}|/|Q|$ in $\mathbb{Z}^3$. In this sense $|Q|$ is the number of photon sites per cell.

The torus $\mathbb{T}$ *carries* $\mathcal{Q}$ if $\mathbb{T}=\mathbb{T}(\mathfrak{F};p_1,p_2,p_3)$ with $p_a\in2h\mathbb{Z}$ for $a=1,2,3$, that is, if in each of the three directions one period of $\mathbb{T}$ consists of an even number of slabs between consecutive walls. Then $\mathcal{Q}$ cuts $\mathbb{T}$, and the translations τ_g, $g\in\Gamma$, belong to $W(\mathcal{Q})$: by the explicit reflection formula, $\theta_{\ell^a,o_a+h}\,\theta_{\ell^a,o_a}$ is the translation by $\frac{2h}{|n_a|^2}\cdot n_a$, and the generators $\frac{p_a}{|n_a|^2}\cdot n_a$ of Γ are integer multiples of these vectors. The quotient $W(\mathcal{Q})/\Gamma$ acts simply transitively on the cells of $\mathbb{T}$ by the interval action just proved.

Let $\mathcal{Q}'=(\mathfrak{F};o'_1,o'_2,o'_3;h')$ be another cell structure with the same frame $\mathfrak{F}$. When its width h' and the three differences o'_a-o_a of positions all lie in $h\mathbb{Z}$, one calls $\mathcal{Q}'$ *coarser* than $\mathcal{Q}$. For such a structure each wall of $\mathcal{Q}'$ is a wall of $\mathcal{Q}$, and each open cell of $\mathcal{Q}'$ contains exactly $\left(\frac{h'}{h}\right)^3$ open cells of $\mathcal{Q}$, whose closures cover it.

Definition 6.3 (Mirror copies and folds)**.** Let $\mathcal{Q}$ be a cell structure, $\kappa\in\mathbb{Z}^3$, and $G\in\mathcal{A}(Q_\kappa)$ (Definition 2.13); like every element of $\mathcal{A}(U)$, the integrand G is even. For $\kappa'\in\mathbb{Z}^3$ let $g_{\kappa'}$ be the element of $W(\mathcal{Q})$ with $g_{\kappa'}Q_\kappa=Q_{\kappa'}$. The *mirror copy* of G in the cell $Q_{\kappa'}$ is

$$G^{[\kappa']}:=G^{g_{\kappa'}},$$

with the action of W on even integrands given in Definition 2.16. Elements of $\mathcal{A}(Q_\kappa)$ and of $\mathcal{A}(Q_{\kappa'})$ commute, both being even.

Let now $\mathbb{T}=(\Gamma,\chi)$ be a torus that carries $\mathcal{Q}$. The *fold* of G on $\mathbb{T}$ is

$$\mathrm{Fold}_\mathbb{T}\,G:=\prod_{\kappa'}G^{[\kappa']},$$

where κ' ranges over a set of representatives of the cells of $\mathbb{T}$. Let $g\in\Gamma$ and $\tau_gQ_{\kappa'}=Q_{\kappa''}$. Since τ_g lies in $W(\mathcal{Q})$ (Definition 6.2), the element $\tau_g\,g_{\kappa'}$ of $W(\mathcal{Q})$ maps Q_κ onto $Q_{\kappa''}$; hence $\tau_g\,g_{\kappa'}=g_{\kappa''}$, and $(G^{[\kappa']})^{\tau_g}=G^{\tau_g\,g_{\kappa'}}=G^{[\kappa'']}$ by the composition convention. Each local box embeds individually, as follows. Let B be one product box contained in an open cell, and write d_i for its widths in the

three coordinate directions. If its unit enlargement meets its translate by $\gamma \in \Gamma$, then $|\gamma_i| \le d_i + 2$ for each i. Consequently every height change has magnitude strictly less than $h + 2\|n_a\|_1$, since the height range of B is strictly less than h. When $h \ge 4$, the coordinate and diagonal normals satisfy $h + 2\|n_a\|_1 \le 2h$. The carrying condition gives $\ell^a(\gamma) \in 2h\mathbb{Z}$, so all these height changes must vanish, and the frame is a basis, forcing $\gamma = 0$. For $h = 1, 2$, every coordinate width d_i is less than h: this is direct for a coordinate frame, and in a paired diagonal frame the two diagonal height widths are $d_1 + d_2$, while the third is d_3, up to a signed permutation of coordinates. Thus $|\gamma_i| < h + 2 \le 4$ for every i, contrary to $\mathrm{sys}(\Gamma) \ge 4$ for a nonzero γ. Finally, for $h = 3$ the carrying condition gives $\gamma_i \in 6\mathbb{Z}$ in a coordinate frame. In a paired diagonal frame it gives $\gamma_1 - \gamma_2, \gamma_1 + \gamma_2, \gamma_3 \in 6\mathbb{Z}$, so each coordinate lies in $3\mathbb{Z}$. The systole then forces some $|\gamma_i| \ge 6$, whereas intersection would require $|\gamma_i| \le d_i + 2 < 5$ in every coordinate. These alternatives cover all frames and widths. The enlarged box therefore meets no nonzero period translate. Distinct images and their period translates lie in different open cells and hence are disjoint. Thus the product $\mathrm{Fold}_{\mathbb{T}} G$ embeds in $\mathbb{T}$ and the function on $\mathrm{Conf}(\Gamma)$ which it defines does not depend on the choice of the representatives. The complete-fold nonnegativity conclusion of Theorem 8.7 gives $Z_{\mathbb{T}}[\mathrm{Fold}_{\mathbb{T}} G] \ge 0$, and the *chessboard seminorm* of G on $\mathbb{T}$ is

$$\|G\|_{\mathbb{T}} := Z_{\mathbb{T}}\left[\mathrm{Fold}_{\mathbb{T}} G\right]^{\frac{|Q|}{|\mathbb{T}|}},$$

the exponent being the inverse of the number of cells of $\mathbb{T}$.

Definition 6.4 (Arrays)**.** Consider a cell structure $\mathcal{Q} = (\mathfrak{F}_{\mathrm{d}}; c_0, c_+, c_3; h)$ of the diagonal frame with an even width h; thus $c_0, c_+ \in \mathbb{Z}$ and $c_3 \in \frac{1}{2}\mathbb{Z}$. Let Q_0 be its open cell with $\kappa = 0$, and let $O \in \mathcal{T}$ be a real trigonometric polynomial with $\operatorname{supp} O \subset Q_0$ (Definition 5.1). Put

$$\Lambda_h := \frac{h}{2} \cdot (\mathbb{Z}e_- + \mathbb{Z}e_+) + h\mathbb{Z}e_3.$$

The linear bijection $v \mapsto (\ell_-(v), \ell_+(v), v_3)$ of $\mathbb{R}^3$ takes e_-, e_+, e_3 to $(2,0,0)$, $(0,2,0)$, $(0,0,1)$; so a vector $v \in \mathbb{R}^3$ has $\ell_-(v), \ell_+(v), v_3 \in h\mathbb{Z}$ if and only if $v \in \Lambda_h$, and $\Lambda_h \subset \mathbb{Z}^3$ because h is even. Hence

$$\Lambda_h = \left\{ v \in \mathbb{Z}^3 : \ \ell_-(v),\ \ell_+(v),\ v_3 \in h\mathbb{Z} \right\},$$

and $v \mapsto \kappa_v := \frac{1}{h} \cdot (\ell_-(v), \ell_+(v), v_3)$ is a bijection of Λ_h onto $\mathbb{Z}^3$. For $v, \kappa \in \mathbb{Z}^3$ the set $\tau_v Q_\kappa$ is described by the inequalities that define Q_κ, with the bounds for ℓ_-, ℓ_+ and x_3 increased by $\ell_-(v)$, $\ell_+(v)$ and v_3; it is an open cell if and only if these three numbers lie in $h\mathbb{Z}$, that is, if $v \in \Lambda_h$, and then $\tau_v Q_\kappa = Q_{\kappa+\kappa_v}$. Consequently Λ_h consists of the vectors $v \in \mathbb{Z}^3$ for which the translation τ_v maps open cells onto open cells, and it acts simply transitively on the open cells. The polynomial O is a bounded Borel function of finitely many F_σ, $\sigma \subset Q_0$; hence $O \in \mathcal{A}(Q_0)$, its mirror copies $O^{[\kappa]}$ are defined (Definition 6.3), and, O being real, $O^{[\kappa]}(F) = O(g_\kappa^* F)$ for the element $g_\kappa \in W(\mathcal{Q})$ with $g_\kappa Q_0 = Q_\kappa$. The *unfolded array* and the *folded array* of O are the formal sums

$$\widetilde{O} := \sum_{v \in \Lambda_h} \tau_v O \qquad \text{and} \qquad \sum_{\kappa \in \mathbb{Z}^3} O^{[\kappa]}.$$

Each of them has one term in each open cell: $\tau_v O$ reads the plaquettes of $\operatorname{supp} O + v$, which lie in $Q_0 + v$, and $O^{[\kappa]}$ reads plaquettes contained in Q_κ.

Let $\mathbb{T} = (\Gamma, \chi)$ be a torus. If $\mathcal{Q}$ cuts $\mathbb{T}$, then $\Gamma \subset \Lambda_h$, and on $\mathbb{T}$ the unfolded array is the function $\sum_v (\tau_v O)(\widetilde{F})$ on $\mathrm{Conf}(\Gamma)$, where v ranges over a set of representatives of Λ_h/Γ. If $\mathbb{T}$ carries $\mathcal{Q}$, then on $\mathbb{T}$ the folded array is the function $\sum_\kappa O^{[\kappa]}(\widetilde{F})$, where κ ranges over a set of representatives of the cells of $\mathbb{T}$. These values do not depend on the representatives, since $\widetilde{F}$ is Γ-periodic and, in the second case, $\tau_g \in W(\mathcal{Q})$ for $g \in \Gamma$. In both cases the sum has $|\mathbb{T}|/|Q|$ terms, one for each cell of $\mathbb{T}$, and each term has modulus at most $|O|_1$; so both arrays are then bounded functions on $\mathrm{Conf}(\Gamma)$ of modulus at most $\frac{|\mathbb{T}|}{|Q|} \cdot |O|_1$.

Definition 6.5 (Sequences of sides)**.** Let $\mathfrak{n} = (n_i)_{i\geq 1}$ be a sequence of positive integers with $n_i \to \infty$, and let

$$\mathcal{D}(\mathfrak{n}) := \{k \in \mathbb{N} : \ k \text{ divides } n_i \text{ for all but finitely many } i\}.$$

This set contains every divisor of each of its elements and the least common multiple of any two of them. The sequence $\mathfrak{n}$ satisfies the *divisibility condition* if

$$8 \in \mathcal{D}(\mathfrak{n}), \qquad |\mathcal{D}(\mathfrak{n})| = \infty. \tag{Div}$$

For an even integer $n \geq 8$ put

$$D(n) := D(n/2, n, n), \qquad D'(n) := D'(n/2, n, n);$$

the cubic torus is $C(n) = C(n, n, n)$, as in Definition 2.9. For every index i with n_i even and $n_i \geq 8$ the four tori

$$C(n_i) \text{ (cubic)}, \qquad D'(n_i), \qquad D(n_i) \text{ (adapted)}, \qquad C(n_i/2, 2n_i, n_i) \text{ (elongated)}$$

are defined, their lattices having systole at least four (Lemma 8.5), and each of them has n_i^3 photon sites. For integers $a_1, a_2, a_3 \geq 4$ the *pressure* of the coordinate torus $C(a_1, a_2, a_3)$ is

$$f(a_1, a_2, a_3) := \frac{1}{a_1 \cdot a_2 \cdot a_3} \cdot \log Z_{C(a_1,a_2,a_3)}[1],$$

the partition function being positive by Theorem 3.4. For the indices i above put

$$\Delta_i := f(n_i/2, 2n_i, n_i) - f(n_i, n_i, n_i), \qquad \mu_i := \mu_{C(n_i)};$$

thus μ_i is the law of the cubic torus $C(n_i)$ (Definition 2.11).

Remark 6.6. If every positive integer divides n_i for all large i, then $\mathcal{D}(\mathfrak{n}) = \mathbb{N}$, and (Div) holds. It holds as well for $n_i = 8 \cdot L^{\lfloor \log_L \bar{m}^{-1} \rfloor} \cdot L^{k_i}$ with an odd integer $L \geq 3$, a number $0 < \bar{m} \leq 1$ and integers $k_i \to \infty$, which is the form of the sides in Definition 15.6: here $8L^j \in \mathcal{D}(\mathfrak{n})$ for every $j \geq 0$. For $n_i = 2p_i$ with primes p_i one has $\mathcal{D}(\mathfrak{n}) = \{1, 2\}$, and (Div) fails. Condition (Div) is imposed on a single sequence of sides, from which the four tori above, the data of Definition 6.7 and the comparison states of Definition 6.8 are all built. We retain this divisibility condition in the infinite-volume theorem; the cutoff sequences used in this manuscript satisfy it.

Definition 6.7 (Data and perturbed functionals)**.** Suppose that $\mathfrak{n}$ satisfies (Div). *Data* are $\mathfrak{d} = (c_0, c_+, c_3; h; O)$, where $c_0, c_+ \in \mathbb{Z}$, $c_3 \in \mathbb{Z} + \frac{1}{2}$, h is an even positive integer with $4h \in \mathcal{D}(\mathfrak{n})$, and $O \in \mathcal{T}$ satisfies $\operatorname{supp} O \subset Q_0$ for the open cell Q_0 with $\kappa = 0$ of the cell structure

$$\mathcal{Q}_\mathfrak{d} := (\mathfrak{F}_\mathrm{d}; c_0, c_+, c_3; h),$$

and where moreover

$$h > 8r_O + 8.$$

Thus the walls of $\mathcal{Q}_\mathfrak{d}$ are the mirrors $(\ell_-, c_0 + j \cdot h)$, $(\ell_+, c_+ + j \cdot h)$ and $(x_3, c_3 + j \cdot h)$, $j \in \mathbb{Z}$, and $|Q_0| = h^3/2$. The *folded array of the data* is the folded array of O with respect to $\mathcal{Q}_\mathfrak{d}$ (Definition 6.4),

$$O_\mathfrak{d} := \sum_{\kappa \in \mathbb{Z}^3} O^{[\kappa]},$$

and $\widetilde{O}$ denotes the unfolded array of O with respect to $\mathcal{Q}_\mathfrak{d}$.

For every index i with $n_i \in 2h\mathbb{Z}$ and $n_i \geq 8$, the structure $\mathcal{Q}_\mathfrak{d}$ cuts $C(n_i)$ and $D'(n_i)$, and $D(n_i)$ carries it. Indeed the three diagonal-frame heights of every period generator are multiples of $2h$. For $D(n_i)$ the three height periods are $n_i, 2n_i, n_i$, and its periods are along the frame normals, which is exactly the carrying condition of Definition 6.2; the torus $D(n_i)$ then has $n_i^3/|Q_0|$ cells. Every $i \geq i_0(\mathfrak{d})$ is such an index, because then $4h$ divides n_i and $n_i \geq 72$ (Definition 6.20); so these

assertions hold for all large i. Let i be an index for which $D(n_i)$ carries $\mathcal{Q}_\mathfrak{d}$, write $\mathbb{T} = (\Gamma, \chi) := D(n_i)$, and let $t \geq 0$. For $G \in \mathcal{A}(\mathbb{R}^3)$ that embeds in $\mathbb{T}$ the *perturbed functional* is

$$\mu_i^t(G) := \frac{Z_\mathbb{T}\left[\exp(t \cdot O_\mathfrak{d}) \cdot G\right]}{Z_\mathbb{T}\left[\exp(t \cdot O_\mathfrak{d})\right]}.$$

Its denominator is positive: on $\mathbb{T}$ one has $|O_\mathfrak{d}| \leq \frac{n_i^3}{|Q_0|} \cdot |O|_1$ (Definition 6.4), and Lemma 3.5 together with Theorem 3.4 gives

$$Z_\mathbb{T}\left[\exp(t \cdot O_\mathfrak{d})\right] \geq \exp\left(-t \cdot |O|_1 \cdot \frac{n_i^3}{|Q_0|}\right) \cdot Z_\mathbb{T}[1] > 0.$$

On bosonic G the functional μ_i^t is an expectation. Indeed, the determinant $\det D_A^\mathbb{T}$ is non-negative (Lemma 2.10), so that

$$\frac{\exp\left(t \cdot O_\mathfrak{d}(\widetilde{F})\right) \cdot \exp\left(-\frac{1}{2}\|F\|^2\right) \cdot \det D_A^\mathbb{T}}{Z_\mathbb{T}\left[\exp(t \cdot O_\mathfrak{d})\right]}$$

is the density of a probability measure on $\mathrm{Conf}(\Gamma)$ with respect to the measure of Definition 2.6; the law of $\widetilde{F}$ under this probability measure is a Borel probability measure on Ω, again denoted μ_i^t, and $\mu_i^t(G) = \int \mathrm{d}\mu_i^t(F)\, G(F)$ for every bosonic G as above.

Only these data in the specified diagonal frame are used below.

Definition 6.8 (Comparison states)**.** Let $\mathfrak{n}$ satisfy (Div), and let $\mathfrak{d}$ be data. Consider a subsequence $(i_m)_{m\geq 1}$ of the indices and a sequence $(t_k)_{k\geq 1}$ of positive numbers tending to 0; for every k the law $\mu_{i_m}^{t_k}$ of Definition 6.7 is defined once m is large. If Borel probability measures ν_{t_k}, $k \geq 1$, and ν_0 on Ω satisfy

$$\mu_{i_m}^{t_k} \to \nu_{t_k} \text{ weakly as } m \to \infty \text{ for every } k, \qquad \nu_{t_k} \to \nu_0 \text{ weakly as } k \to \infty,$$

then ν_0 is called a *comparison state* for $\mathfrak{d}$. Given a further subsequence $(i'_m)_{m\geq 1}$ of the indices, a comparison state ν_0 for $\mathfrak{d}$ is *subordinate* to (i'_m) if it is obtained in this way from some subsequence (i_m) of (i'_m), with suitable numbers t_k and measures ν_{t_k}. In every limit of this kind the volume limit $i \to \infty$ is taken first and the limit $t \to 0+$ in the strength second; where a regularisation parameter ε occurs as well, the limit $\varepsilon \to 0$ is taken last.

Definition 6.9 (Amplification factor)**.** For a sequence $\mathfrak{n}$ with (Div) and data $\mathfrak{d} = (c_0, c_+, c_3; h; O)$, for $k \in \mathbb{N}$, $t \geq 0$ and every index i with n_i even and $n_i \geq 8$, so that Δ_i is defined, put

$$\gamma_{t,i}(k) := \exp\left(2t \cdot |O|_1 \cdot k^3 + |Q_0| \cdot k^3 \cdot \Delta_i\right).$$

By (Div), 8 divides n_i for all large i, and $n_i \to \infty$; so these indices include all large i, and $\Delta_i \to 0$ as $i \to \infty$ (Theorem 8.10). Hence, for every fixed k,

$$\lim_{t\to 0+} \limsup_{i\to\infty} \gamma_{t,i}(k) = 1.$$

6.2. **Slices and Fock spaces.** In this subsection the coordinate x_1 is singled out. A point of $\mathbb{Z}^3$ is written (j, y) with $j \in \mathbb{Z}$ and $y \in \mathbb{Z}^2$, where $y = (y_2, y_3)$ collects the coordinates x_2, x_3, and e_2, e_3 also denote the unit vectors of $\mathbb{Z}^2$.

Definition 6.10 (Slices)**.** *The slice lattice.* The slice lattice is $\mathbb{Z}^2$, with the coordinates (x_2, x_3). Its links are the segments $b = (y, \nu) = [y, y + e_\nu]$ with $y \in \mathbb{Z}^2$ and $\nu \in \{2, 3\}$, and its plaquettes are the closed unit squares $\sigma' = y + [0, 1]^2$. A function on the plaquettes is written as a function of the lower corner y of σ', and a one-cochain of the slice is also read as an odd function of oriented pairs of neighbours, as in Definition 2.1. The coboundary d_s of the slice lattice acts on a function a on the sites and on a one-cochain A by

$$(d_\mathrm{s} a)(y, \nu) = a(y + e_\nu) - a(y), \qquad (d_\mathrm{s} A)(y) = A(y, 2) + A(y + e_2, 3) - A(y + e_3, 2) - A(y, 3).$$

Slice variables. Let A be a real one-cochain on $\mathbb{Z}^3$ and $j \in \mathbb{Z}$. The links of the plane $\{x_1 = j\}$ carry the one-cochain A_j of the slice, and the links that join the plane $\{x_1 = j\}$ to the plane $\{x_1 = j+1\}$ carry the function a_j on $\mathbb{Z}^2$:

$$A_j(y,\nu) := A_\nu(j,y) \quad (\nu \in \{2,3\}), \qquad a_j(y) := A_1(j,y).$$

For a real two-cochain F on $\mathbb{Z}^3$, the electric part F_j^{el}, a one-cochain of the slice, and the magnetic part F_j^{mag}, a function on the plaquettes of the slice, are

$$F_j^{\mathrm{el}}(y,\nu) := F_{1\nu}(j,y) \quad (\nu \in \{2,3\}), \qquad F_j^{\mathrm{mag}}(y) := F_{23}(j,y);$$

where the two-cochain has to be named, they are written $F_j^{\mathrm{el}}(F)$ and $F_j^{\mathrm{mag}}(F)$. The formulas for d in Definition 2.1 show that $F = dA$ has the parts

$$F_j^{\mathrm{el}} = A_{j+1} - A_j - d_{\mathrm{s}} a_j, \qquad F_j^{\mathrm{mag}} = d_{\mathrm{s}} A_j,$$

and that every two-cochain F satisfies $(dF)(j,y) = F_{j+1}^{\mathrm{mag}}(y) - F_j^{\mathrm{mag}}(y) - (d_{\mathrm{s}} F_j^{\mathrm{el}})(y)$. Thus $dF = 0$ if and only if $F_{j+1}^{\mathrm{mag}} - F_j^{\mathrm{mag}} = d_{\mathrm{s}} F_j^{\mathrm{el}}$ for every $j \in \mathbb{Z}$.

Reflections of the slice variables. Let $j_0 \in \mathbb{Z}$ and $c := j_0 + \frac{1}{2}$, so that (x_1, c) is a link plane. Let further $\nu \in \{2,3\}$, $c' \in \frac{1}{2}\mathbb{Z}$, and let θ_{s} be the reflection of $\mathbb{R}^2$ in the line $\{x_\nu = c'\}$; it permutes the sites, the links and the plaquettes of $\mathbb{Z}^2$. For a one-cochain u of the slice, $\theta_{\mathrm{s}}^* u$ is the pull-back, $(\theta_{\mathrm{s}}^* u)(y,y') = u(\theta_{\mathrm{s}} y, \theta_{\mathrm{s}} y')$, and a function on the sites or on the plaquettes is composed with θ_{s} through the action of θ_{s} on sites or plaquettes. For an isometry θ of $\mathbb{R}^3$ with $\theta\mathbb{Z}^3 = \mathbb{Z}^3$, $(\theta^* A)_j$ and $(\theta^* a)_j$ denote the slice variables of the one-cochain $\theta^* A$, and $F_j^{\mathrm{el}}(\theta^* F)$, $F_j^{\mathrm{mag}}(\theta^* F)$ the parts of the two-cochain $\theta^* F$ (pull-backs as in Definition 2.1). The reflection $\theta_{x_1,c}$ maps (j,y) to $(2j_0 + 1 - j, y)$; it reverses the orientation of the links parallel to e_1 and of the plaquettes $(x; 1\mu)$, $\mu \in \{2,3\}$, and preserves that of all other links and plaquettes. The reflection $\theta_{x_\nu,c'}$ maps (j,y) to $(j, \theta_{\mathrm{s}} y)$; it reverses the orientation of the links parallel to e_ν and of the plaquettes $(x; 1\nu)$ and $(x; 23)$, and preserves that of all other links and plaquettes. It follows that, for every $j \in \mathbb{Z}$,

$$\begin{aligned}
(\theta_{x_1,c}^* A)_j &= A_{2j_0+1-j}, & (\theta_{x_1,c}^* a)_j &= -a_{2j_0-j},\\
F_j^{\mathrm{el}}(\theta_{x_1,c}^* F) &= -F_{2j_0-j}^{\mathrm{el}}(F), & F_j^{\mathrm{mag}}(\theta_{x_1,c}^* F) &= F_{2j_0+1-j}^{\mathrm{mag}}(F),\\
(\theta_{x_\nu,c'}^* A)_j &= \theta_{\mathrm{s}}^* A_j, & (\theta_{x_\nu,c'}^* a)_j &= a_j \circ \theta_{\mathrm{s}},\\
F_j^{\mathrm{el}}(\theta_{x_\nu,c'}^* F) &= \theta_{\mathrm{s}}^* F_j^{\mathrm{el}}(F), & F_j^{\mathrm{mag}}(\theta_{x_\nu,c'}^* F) &= -F_j^{\mathrm{mag}}(F) \circ \theta_{\mathrm{s}}.
\end{aligned} \tag{6.1}$$

Rectangular cross-sections. Let $n_2, n_3 \geq 4$ be integers. The rectangular cross-section with the sides n_2, n_3 is the cross-section $\Xi = (n_2\mathbb{Z}e_2 + n_3\mathbb{Z}e_3; \chi_\parallel)$ for the family x_1 (Definition 6.1) with $\chi_\parallel(n_2 e_2) = \chi_\parallel(n_3 e_3) = -1$. Its slice torus is $\Lambda_{\mathrm{s}} := \mathbb{Z}^2/(n_2\mathbb{Z} \times n_3\mathbb{Z})$. It has $|\Lambda_{\mathrm{s}}| = n_2 n_3$ sites, the set $\Lambda_{\mathrm{s}}^{(1)}$ of its $2|\Lambda_{\mathrm{s}}|$ links and the set $\Lambda_{\mathrm{s}}^{(2)}$ of its $|\Lambda_{\mathrm{s}}|$ plaquettes, and d_{s} acts on its cochains by the formulas above. The subgroup

$$\Gamma_{\mathrm{s}} := \left\{ (2\pi/q) \cdot k : \ k \in \mathbb{Z}^{\Lambda_{\mathrm{s}}^{(1)}}, \ d_{\mathrm{s}} k = 0 \right\}$$

of $\mathbb{R}^{\Lambda_{\mathrm{s}}^{(1)}}$ consists of $2\pi/q$ times the closed integer one-cochains of Λ_{s}, and the quotient $\mathbb{R}^{\Lambda_{\mathrm{s}}^{(1)}}/\Gamma_{\mathrm{s}}$ carries the measure induced by Lebesgue measure. A plaquette $\sigma_0 \in \Lambda_{\mathrm{s}}^{(2)}$ is fixed; it is the position of the flux string: in the slice operator of Definition 6.12 the flux $(2\pi/q) \cdot k$ of the sector k is placed on σ_0. Finally, for $a > 0$ and $s \in \mathbb{R}$,

$$g_a(s) := (2\pi a)^{-\frac{1}{2}} \cdot \exp\left(-\frac{s^2}{2a}\right).$$

Hulls. For a non-empty finite set Λ of sites of $\mathbb{Z}^2$ let $y_\nu^- := \min_{y\in\Lambda} y_\nu$ and $y_\nu^+ := \max_{y\in\Lambda} y_\nu$ for $\nu = 2, 3$. The hull

$$\mathrm{hull}(\Lambda) := [y_2^-, y_2^+] \times [y_3^-, y_3^+]$$

is the smallest product of two closed intervals with integer end points that contains Λ. For a finite set S of links of $\mathbb{Z}^2$ whose end points lie in Λ, $\mathrm{hull}(S, \Lambda) := \mathrm{hull}(\Lambda)$. The hull *embeds* in the rectangular cross-section Ξ if the sets $\mathrm{hull}(\Lambda) + [-1, 1]^2 + g$, $g \in n_2\mathbb{Z} \times n_3\mathbb{Z}$, are pairwise disjoint, and it *contains* the plaquette σ_0 of Λ_s if a representative of σ_0 in $\mathbb{Z}^2$, which is a closed unit square, is a subset of $\mathrm{hull}(\Lambda)$.

Definition 6.11 (Modes and Fock spaces)**.** Let $\rho \in \mathbb{R}$ and $w_0, \dots, w_3 > 0$ be the Wilson parameter and the hopping weights of the chamber operator (Definition 2.8). Bonds and their types are those of Definition 2.4, and A is a real one-cochain on $\mathbb{Z}^3$.

Modes at a mirror. Let $\Pi = (\ell, c)$ be a mirror, $n := \hat{n}_\ell$ and $\theta := \theta_\Pi$. Just as θ_Π is also written $\theta_{\ell,c}$, the reflection Θ_Π of Definition 2.16 is also written $\Theta_{\ell,c}$. A bond *crosses* Π if its end points lie on different sides of Π; by Lemma 2.5 it is then of the form $\{x, \theta x\}$, and the labels are chosen so that $\ell(x) < c$: the site x is the lower end of the bond and θx its upper end. Let $\{x, \theta x\}$ be a bond that crosses Π, let $j \in \{0, 1, 2, 3\}$ be its type and $k \in \{1, 2\}$. The bond carries four groups of modes. For each group and each k, the following list gives an element ζ of the Grassmann algebra $\mathrm{Gr}(\mathbb{X})$ (Definition 2.7), a coefficient ω and a phase u; the vectors $e^\pm_{n,k}$ are those of Definition 2.7:

$$\begin{aligned}
&\text{group 1:} && \zeta = (e^-_{n,k})^*\psi(\theta x), && \omega = \tfrac{w_j}{2}(1+\rho), && u = U_A(x, \theta x),\\
&\text{group 2:} && \zeta = (e^+_{n,k})^*\psi(\theta x), && \omega = \tfrac{w_j}{2}(1-\rho), && u = U_A(x, \theta x),\\
&\text{group 3:} && \zeta = \bar\psi(\theta x)\, e^+_{n,k}, && \omega = \tfrac{w_j}{2}(1+\rho), && u = \overline{U_A(x, \theta x)},\\
&\text{group 4:} && \zeta = \bar\psi(\theta x)\, e^-_{n,k}, && \omega = \tfrac{w_j}{2}(1-\rho), && u = \overline{U_A(x, \theta x)}.
\end{aligned}$$

The set $\mathrm{idx}(\Pi)$ of the modes at Π consists of the triples formed by a bond that crosses Π, a group and an index $k \in \{1, 2\}$; there are eight modes for each such bond. For a mode $i \in \mathrm{idx}(\Pi)$, its element, its coefficient and its phase, as given by the list, are written ζ_i, ω_i and u_i. The weights being positive, $\omega_i \geq 0$ for every $i \in \mathrm{idx}(\Pi)$ if $-1 \leq \rho \leq 1$, and $\omega_i > 0$ for every $i \in \mathrm{idx}(\Pi)$ if $-1 < \rho < 1$, in particular if $0 < \rho < 1$. The four vectors $e^\pm_{n,k}$ form an orthonormal basis of $\mathbb{C}^4$, and a site is an end of at most one bond that crosses Π. Therefore the ζ_i, $i \in \mathrm{idx}(\Pi)$, form a basis of the linear span of the generators $\psi_\alpha(x')$, $\bar\psi_\alpha(x')$ at the upper ends x' of the bonds that cross Π; since Θ_Π maps the span of the generators at θx antilinearly and bijectively onto the span of those at x (Definition 2.16), the $\Theta_\Pi(\zeta_i)$ form a basis of the span of the generators at the lower ends.

Translations and orders. For $v \in \mathbb{Z}^3$ and a mode i at Π, the triple $i + v$ formed by the translated bond, the same group and the same k is a mode at $(\ell, c + \ell(v))$. The type of a bond and the vectors $e^\pm_{n,k}$ do not change under translations, and $\theta_{\ell,c+\ell(v)}(x + v) = \theta_\Pi(x) + v$ by Definition 2.2; hence ζ_{i+v} arises from ζ_i by the translation of the fields by v, and $\omega_{i+v} = \omega_i$. Each set $\mathrm{idx}(\Pi)$ carries a total order, fixed once and for all, and the orders are compatible with the normal translations: $i \mapsto i + n_\ell$ is an order preserving bijection of $\mathrm{idx}((\ell, c))$ onto $\mathrm{idx}((\ell, c + |n_\ell|^2))$. (It suffices to choose the orders for the positions c in a set of representatives of $\mathrm{Adm}(\ell)$ modulo $|n_\ell|^2\mathbb{Z}$ and to transport them.)

Cross-sections. Let $\Xi = (\Gamma_\parallel, \chi_\parallel)$ be a cross-section for ℓ (Definition 6.1). The Grassmann algebra of Ξ is generated by the fields at the sites taken modulo $\Gamma_\parallel$, subject to $\psi(x + g) = \chi_\parallel(g)\psi(x)$ and $\bar\psi(x + g) = \chi_\parallel(g)\bar\psi(x)$ for $g \in \Gamma_\parallel$, as for tori in Definition 2.8; in it, $\zeta_{i+g} = \chi_\parallel(g) \cdot \zeta_i$ for $i \in \mathrm{idx}(\Pi)$ and $g \in \Gamma_\parallel$. Since $\theta_\Pi(x + g) = \theta_\Pi(x) + g$ for $x \in \mathbb{R}^3$ and $g \in \Gamma_\parallel$ (Definition 2.2, as $\ell(g) = 0$) and $\chi_\parallel$ is real, the formulas (2.3) show that Θ_Π maps these relations to linear combinations of them; so Θ_Π acts on the Grassmann algebra of Ξ, again antilinearly and reversing the order of products. The group $\Gamma_\parallel$, which lies in $\{\ell = 0\}$, acts on $\mathrm{idx}(\Pi)$ by $i \mapsto i + g$; let

$$\mathrm{idx}_\Xi(\Pi) := \mathrm{idx}(\Pi)/\Gamma_\parallel,$$

a finite set: the bonds that cross Π fall into finitely many classes under the translations by $\mathbb{Z}^3 \cap \{\ell = 0\}$, and $\Gamma_\parallel$ has finite index in this group of rank two. A set of representatives of its elements is fixed, and $\mathrm{idx}_\Xi(\Pi)$ is ordered as the representatives are ordered in $\mathrm{idx}(\Pi)$. The element and the coefficient of a class are those of its representative, and so is its phase, which for $\Gamma_\parallel$-periodic A does not depend on the choice of the representative. The representatives for $(\ell, c + |n_\ell|^2)$ are the translates by n_ℓ of those for (ℓ, c); so the orders of the sets $\mathrm{idx}_\Xi(\Pi)$ are again compatible with τ_{n_ℓ}. Translations along the mirror behave differently: for $v \in \mathbb{Z}^3 \cap \{\ell = 0\}$ with $v \notin \Gamma_\parallel$, the map $i \mapsto i + v$ permutes the finite set $\mathrm{idx}_\Xi(\Pi)$ without fixed points, and therefore it preserves no total order of $\mathrm{idx}_\Xi(\Pi)$. Accordingly, let $v \in \mathbb{Z}^3$ with $\ell(v) = 0$. The unitary V_v of Corollary 7.19 maps the space $\mathcal{H}(c)$ of that corollary onto itself; it acts on the bosonic mirror variables through the pull-back τ_v^*, and it moves the value at a set $I = \{i_1 < \cdots < i_d\} \subset \mathrm{idx}_\Xi(\Pi)$ of modes to the set $I + v = \{i_1 + v, \dots, i_d + v\}$, multiplied by the sign of the permutation that brings $i_1 + v, \dots, i_d + v$ into increasing order and by the product of the numbers $\chi_\parallel(g)$, where $g \in \Gamma_\parallel$ ranges over the d vectors by which the translates by v of the representatives of $i_1, \dots, i_d$ differ from the representatives of $i_1 + v, \dots, i_d + v$.

Fock spaces. For a finite set $\mathcal{J}$, the Fock space $\mathcal{F}(\mathcal{J})$ is the Hilbert space with the orthonormal basis $(e_I)_{I \subset \mathcal{J}}$. The fermion parity is the unitary involution U_f with $U_\mathrm{f} e_I := (-1)^{\#|I|} e_I$; an operator $\Upsilon \in \operatorname{End} \mathcal{F}(\mathcal{J})$ is *even* if it commutes with U_f, and $\Upsilon_{IJ} := \langle e_I, \Upsilon e_J \rangle$ are its matrix entries.

Ordered monomials; matrices and elements. Let $\mathcal{J}$ be a finite subset of $\mathrm{idx}(\Pi)$ or of $\mathrm{idx}_\Xi(\Pi)$, with the induced order, and let $I = \{i_1 < \cdots < i_d\} \subset \mathcal{J}$. Put $\omega_I := \prod_{i \in I} \omega_i$ and $u_I := \prod_{i \in I} u_i$, and let $\omega := \mathrm{diag}(\omega_I)_{I \subset \mathcal{J}}$ be the diagonal operator of the coefficients on $\mathcal{F}(\mathcal{J})$. For $\Upsilon \in \operatorname{End} \mathcal{F}(\mathcal{J})$,

$$\begin{aligned} \zeta_I &:= \zeta_{i_1} \zeta_{i_2} \cdots \zeta_{i_d}, \qquad \Theta_\Pi(\zeta_I) = \Theta_\Pi(\zeta_{i_d}) \cdots \Theta_\Pi(\zeta_{i_2})\, \Theta_\Pi(\zeta_{i_1}), \\ \mathrm{Ins}_\Pi[\Upsilon] &:= \sum_{I,J \subset \mathcal{J}} \Upsilon_{IJ} \cdot \Theta_\Pi(\zeta_J)\, \zeta_I, \end{aligned} \tag{6.2}$$

with $\zeta_\emptyset = \omega_\emptyset = u_\emptyset = 1$; the second formula holds because Θ_Π reverses products. In $\mathrm{Ins}_\Pi[\Upsilon]$ the row index I belongs to the upper side of Π and the column index J to the lower side. The elements $\Theta_\Pi(\zeta_i)$ are odd and anticommute; reversing the order of the factors gives $\Theta_\Pi(\zeta_{i_1}) \cdots \Theta_\Pi(\zeta_{i_d}) = (-1)^{\binom{\#|I|}{2}} \cdot \Theta_\Pi(\zeta_I)$, so that $(-1)^{\binom{\#|I|}{2}}$ is the sign of the reversal of $\#|I|$ odd factors.

Localisation. Let $\mathcal{J}$ again be a finite subset of $\mathrm{idx}(\Pi)$ or of $\mathrm{idx}_\Xi(\Pi)$ with the induced order, and let $\mathcal{J}' \subset \mathcal{J}$, ordered as a subset of $\mathcal{J}$. Every subset of $\mathcal{J}$ is, in exactly one way, of the form $I \cup R$ with $I \subset \mathcal{J}'$ and $R \subset \mathcal{J} \setminus \mathcal{J}'$. For $\Upsilon \in \operatorname{End} \mathcal{F}(\mathcal{J}')$, the operator $\iota(\Upsilon) \in \operatorname{End} \mathcal{F}(\mathcal{J})$ is defined by its matrix entries

$$\iota(\Upsilon)_{I \cup R,\, J \cup R'} := \delta_{RR'} \cdot (-1)^{\#\{(i,r) \in I \times R\colon\, i > r\}} \cdot (-1)^{\#\{(i,r) \in J \times R\colon\, i > r\}} \cdot \Upsilon_{IJ}$$

for $I, J \subset \mathcal{J}'$ and $R, R' \subset \mathcal{J} \setminus \mathcal{J}'$. By Proposition 7.22(b), $\iota : \operatorname{End} \mathcal{F}(\mathcal{J}') \to \operatorname{End} \mathcal{F}(\mathcal{J})$ is a unital injective $*$-homomorphism, hence isometric.

Coordinate planes. Let $j \in \mathbb{Z}$. The bonds that cross the link plane $(x_1, j + \frac{1}{2})$ are of type 3 (Lemma 2.5), and there are eight of them over each $y \in \mathbb{Z}^2$, that is, over the link from (j, y) to $(j+1, y)$. The lower end x of such a bond has $\pi(x) = (j, y)$ and its upper end has $\pi(\theta_{x_1, j+1/2} x) = (j + 1, y)$, so that $U_A(x, \theta_{x_1, j+1/2} x) = \exp(iq\, a_j(y))$. For a finite set $\Lambda \subset \mathbb{Z}^2$, $\mathcal{J}^1(\Lambda)$ is the set of the $64 \cdot \#|\Lambda|$ modes of the bonds over the sites of Λ, ordered as a subset of $\mathrm{idx}((x_1, j + \frac{1}{2}))$. A mode i of a bond over $y \in \Lambda$ has the charge $\mathfrak{q}(i) := \delta_y$ if it belongs to the group 1 or 2, and $\mathfrak{q}(i) := -\delta_y$ if it belongs to the group 3 or 4, where $\delta_y \in \mathbb{Z}^\Lambda$ is the indicator of y; for $I \subset \mathcal{J}^1(\Lambda)$ let $\mathfrak{q}(I) := \sum_{i \in I} \mathfrak{q}(i) \in \mathbb{Z}^\Lambda$. Then $u_I = \exp(iq \langle a_j, \mathfrak{q}(I) \rangle)$, and for $a \in \mathbb{R}^\Lambda$ the diagonal unitary

$$u(a) := \mathrm{diag}\left(\exp(iq \langle a, \mathfrak{q}(I) \rangle)\right)_{I \subset \mathcal{J}^1(\Lambda)} \quad \text{on } \mathcal{F}(\mathcal{J}^1(\Lambda))$$

satisfies $\mathrm{diag}(u_I)_{I \subset \mathcal{J}^1(\Lambda)} = u(a_j)$, with a_j restricted to Λ. The bonds that cross the site plane (x_1, j) are of type 0 (Lemma 2.5), and they carry no phase, since $\pi(x) = \pi(\theta_{x_1, j} x)$ for such a bond

$\{x, \theta_{x_1,j}x\}$, $x_1 < j$. This photon site is (j, y) for some $y \in \mathbb{Z}^2$, because x_1 and $2j - x_1$ both differ from the integer $\pi(x)_1$ by less than $\frac{1}{2}$ (Definition 2.4); the bond is then said to lie over y. There are eight such bonds over each $y \in \mathbb{Z}^2$: by Lemma 2.5, the bond of type 0 at the site $(j, y) + \vec{g}\, v_0$, $\vec{g} \in O_{\mathrm{cub}}$, crosses (x_1, j) if and only if the wall $(j, y) + \vec{g}\,\Pi_0$ of the chamber of this site is the plane $\{x_1 = j\}$, that is, $\vec{g}\, e_3 = \pm e_1$; this holds for 16 of the 48 matrices $\vec{g}$, and these 16 sites are the end points of eight bonds. For a finite set $\Lambda \subset \mathbb{Z}^2$, $\mathcal{J}^{\mathrm{s}}(\Lambda)$ is the set of the $64 \cdot \#|\Lambda|$ modes over Λ at (x_1, j), ordered as a subset of $\mathrm{idx}((x_1, j))$.

For a rectangular cross-section Ξ (Definition 6.10) the sets of the modes over Λ_{s} are

$$\mathcal{J}^{\mathrm{l}}(\Lambda_{\mathrm{s}}) := \mathrm{idx}_{\Xi}((x_1, j + \tfrac{1}{2})), \qquad \mathcal{J}^{\mathrm{s}}(\Lambda_{\mathrm{s}}) := \mathrm{idx}_{\Xi}((x_1, j)),$$

the charges $\mathfrak{q}(i)$ and $\mathfrak{q}(I)$ are defined as above, with values in $\mathbb{Z}^{\Lambda_{\mathrm{s}}}$, and

$$\mathcal{F} := \mathcal{F}(\mathcal{J}^{\mathrm{l}}(\Lambda_{\mathrm{s}})), \qquad \mathcal{F}' := \mathcal{F}(\mathcal{J}^{\mathrm{s}}(\Lambda_{\mathrm{s}})).$$

The diagonal operators of the coefficients on $\mathcal{F}$ and on $\mathcal{F}'$ are ω and ω', and for $a \in \mathbb{R}^{\Lambda_{\mathrm{s}}}$ the diagonal unitary $u(a)$ on $\mathcal{F}$ is given by the same formula as above, with Λ_{s} in the place of Λ. If A is $\Gamma_{\parallel}$-periodic, then a_j is a function on Λ_{s}, and, the phase of a class being that of its representative, $u_I = \exp\left(iq\langle a_j, \mathfrak{q}(I)\rangle\right)$ for $I \subset \mathcal{J}^{\mathrm{l}}(\Lambda_{\mathrm{s}})$, that is, $\mathrm{diag}(u_I)_{I \subset \mathcal{J}^{\mathrm{l}}(\Lambda_{\mathrm{s}})} = u(a_j)$.

Signed relabelling. Let Ξ be a rectangular cross-section and let $\Lambda \subset \mathbb{Z}^2$ be a finite set on which the reduction map $\mathbb{Z}^2 \to \Lambda_{\mathrm{s}}$ is injective; this holds if $\mathrm{hull}(\Lambda)$ embeds in Ξ. Every mode $i \in \mathcal{J}^{\mathrm{l}}(\Lambda)$ is $i = r(i) + g_i$ with a unique representative $r(i)$ of a class of $\mathcal{J}^{\mathrm{l}}(\Lambda_{\mathrm{s}})$ and a unique $g_i \in \Gamma_{\parallel}$, and r is injective on $\mathcal{J}^{\mathrm{l}}(\Lambda)$. For $I = \{i_1 < \cdots < i_d\} \subset \mathcal{J}^{\mathrm{l}}(\Lambda)$ let

$$s(I) := \chi_{\parallel}(g_{i_1}) \cdots \chi_{\parallel}(g_{i_d}) \cdot \mathrm{sgn}\left(\text{the permutation that sorts } (r(i_1), \ldots, r(i_d))\right),$$

the sorting being done in the order of $\mathcal{J}^{\mathrm{l}}(\Lambda_{\mathrm{s}})$, and let $U_\Lambda : \mathcal{F}(\mathcal{J}^{\mathrm{l}}(\Lambda)) \to \mathcal{F}(r(\mathcal{J}^{\mathrm{l}}(\Lambda)))$ be the unitary with $U_\Lambda e_I := s(I) \cdot e_{r(I)}$. In the Grassmann algebra of Ξ one has $\zeta_i = \chi_{\parallel}(g_i) \cdot \zeta_{r(i)}$, hence $\zeta_I = s(I) \cdot \zeta_{r(I)}$; thus U_Λ is the unitary of Proposition 7.22(f) for the relabelling r. On Ξ, the localisation of $\Upsilon \in \mathrm{End}\,\mathcal{F}(\mathcal{J}^{\mathrm{l}}(\Lambda))$ is the operator

$$\iota(\Upsilon) := \iota\left(U_\Lambda \Upsilon U_\Lambda^*\right) \in \mathrm{End}\,\mathcal{F},$$

where on the right ι is formed with the subset $r(\mathcal{J}^{\mathrm{l}}(\Lambda))$ of $\mathcal{J}^{\mathrm{l}}(\Lambda_{\mathrm{s}})$. If $\mathcal{J}^{\mathrm{l}}(\Lambda)$ consists of representatives, then U_Λ is the identity. Conjugation by U_Λ preserves norms, adjoints and products. Moreover $\omega_{r(i)} = \omega_i$, and $\mathfrak{q}(r(i))$ is the image of $\mathfrak{q}(i)$ under the reduction map. Write $[y] \in \Lambda_{\mathrm{s}}$ for the class of $y \in \mathbb{Z}^2$; then, for every $a \in \mathbb{R}^{\Lambda_{\mathrm{s}}}$,

$$U_\Lambda\, \omega\, U_\Lambda^* = \omega, \qquad U_\Lambda\, u\left(a([\,\cdot\,])\right) U_\Lambda^* = u(a) \quad \text{on } \mathcal{F}(r(\mathcal{J}^{\mathrm{l}}(\Lambda))),$$

where on the left ω and $u(a([\,\cdot\,]))$ act on $\mathcal{F}(\mathcal{J}^{\mathrm{l}}(\Lambda))$, $a([\,\cdot\,])$ being the function $y \mapsto a([y])$ on Λ, and on the right ω and $u(a)$ are restricted to the span of the e_I, $I \subset r(\mathcal{J}^{\mathrm{l}}(\Lambda))$.

Half slab. Let Ξ be a rectangular cross-section, let $j \in \mathbb{Z}$, and let A be $\Gamma_{\parallel}$-periodic, so that A_j is a one-cochain of Λ_{s}. The half slab $\mathbb{X}_{\mathrm{h}}$ is the set of the fermion sites x with $j < x_1 < j + \frac{1}{2}$, taken modulo $\Gamma_{\parallel} = n_2\mathbb{Z}e_2 + n_3\mathbb{Z}e_3$; its fields satisfy the relations of the Grassmann algebra of Ξ, and $\int_{\mathbb{X}_{\mathrm{h}}}$ is the Berezin integral of Definition 2.7 over one representative of each class. Let S_{h} be the sum of the terms of $\bar{\psi} D_A \psi$ (Definition 2.8) whose sites all lie in $\mathbb{X}_{\mathrm{h}}$. A bond with both ends in $\mathbb{X}_{\mathrm{h}}$ that carries a phase is of type 3 and lies over a link of the plane $\{x_1 = j\}$ (Lemma 2.5); so S_{h} depends on A only through the phases $\exp(iqA_j(b))$, $b \in \Lambda_{\mathrm{s}}^{(1)}$, and it is written $S_{\mathrm{h}}(A_j)$, or $S_{\mathrm{h}}(A)$ with an arbitrary $A \in \mathbb{R}^{\Lambda_{\mathrm{s}}^{(1)}}$ in the place of A_j. Let $\zeta'_{i'}$, $i' \in \mathcal{J}^{\mathrm{s}}(\Lambda_{\mathrm{s}})$, be the modes of the site plane (x_1, j), $\zeta'_{I'}$ their ordered products as in (6.2), and ζ_i, $i \in \mathcal{J}^{\mathrm{l}}(\Lambda_{\mathrm{s}})$, the modes of the link plane $(x_1, j + \frac{1}{2})$; the elements $\zeta'_{i'}$ and $\Theta_{x_1,j+1/2}(\zeta_i)$ belong to the Grassmann algebra of $\mathbb{X}_{\mathrm{h}}$. For $A \in \mathbb{R}^{\Lambda_{\mathrm{s}}^{(1)}}$, the half-slab

matrix $\mathfrak{N}(A) : \mathcal{F}' \to \mathcal{F}$ has the matrix entries

$$\mathfrak{N}(A)_{I,I'} := \int_{\mathbb{X}_{\rm h}} \zeta'_{I'} \exp\left(-S_{\rm h}(A)\right) \Theta_{x_1,j+1/2}(\zeta_I), \qquad I \subset \mathcal{J}^{\rm l}(\Lambda_{\rm s}),\ I' \subset \mathcal{J}^{\rm s}(\Lambda_{\rm s}).$$

If $-1 \le \rho \le 1$, the diagonal operators ω and ω' have non-negative entries; let $\omega^{\frac12}$ and $\omega'^{\frac12}$ be the diagonal operators with the non-negative square roots of these entries, and put

$$\widehat{\mathfrak{N}}(A) := \omega^{\frac12}\, \mathfrak{N}(A)\, \omega'^{\frac12} :\ \mathcal{F}' \to \mathcal{F}.$$

Independence of j. For every $y \in \mathbb{Z}^2$, the translation τ_{e_1} carries the bonds over y that cross (x_1, j) and $(x_1, j+\frac12)$ onto those over y that cross $(x_1, j+1)$ and $(x_1, j+\frac32)$. So $i \mapsto i + e_1$ maps $\mathcal{J}^{\rm s}(\Lambda)$ and $\mathcal{J}^{\rm l}(\Lambda)$, formed at j, onto the same sets formed at $j+1$; it preserves the orders (they are compatible with τ_{n_ℓ}, and $n_{x_1} = e_1$), the groups, the coefficients and the charges. As the representatives for $(\ell, c + |n_\ell|^2)$ are the translates by n_ℓ of those for (ℓ, c), the representatives of the classes at $(x_1, j+1)$ and at $(x_1, j+\frac32)$ are the translates by e_1 of those at (x_1, j) and at $(x_1, j+\frac12)$. Hence $i \mapsto i + e_1$ also maps $\mathcal{J}^{\rm s}(\Lambda_{\rm s})$ and $\mathcal{J}^{\rm l}(\Lambda_{\rm s})$ onto the same sets formed at $j+1$, with the same properties, and in the signed relabelling $r(i+e_1) = r(i) + e_1$, $g_{i+e_1} = g_i$ and $s(I + e_1) = s(I)$. The translation τ_{e_1} carries the half slab of j onto that of $j+1$, and $S_{\rm h}(A)$, formed at j, to $S_{\rm h}(A)$, formed at $j+1$, for every $A \in \mathbb{R}^{\Lambda_{\rm s}^{(1)}}$, because the hopping matrices $w(x, x')$ of Definition 2.8 do not change under translations. It carries $\zeta'_{I'}$ to $\zeta'_{I'+e_1}$ and, by Definition 2.2, $\Theta_{x_1,j+1/2}(\zeta_I)$ to $\Theta_{x_1,j+3/2}(\zeta_{I+e_1})$, and the Berezin integrals over the two half slabs correspond to each other under it. Identifying e_I with e_{I+e_1}, therefore, the Fock spaces $\mathcal{F}(\mathcal{J}^{\rm l}(\Lambda))$, $\mathcal{F}(\mathcal{J}^{\rm s}(\Lambda))$, $\mathcal{F}$ and $\mathcal{F}'$, the operators ω, ω' and $u(a)$ on them, the charges, the unitary U_Λ, the localisation ι on Ξ, and the operators $\mathfrak{N}(A)$ and $\widehat{\mathfrak{N}}(A)$ do not depend on j.

Definition 6.12 (Operators of a cross-section)**.** Let Ξ be a rectangular cross-section, with $\Lambda_{\rm s}$, $\Gamma_{\rm s}$, σ_0 and g_a as in Definition 6.10 and $\mathcal{F}$, $\mathcal{F}'$ and $u(a)$ as in Definition 6.11. The Hilbert spaces of the link plane and of the site plane are

$$\mathcal{H}^{\rm l} := \bigoplus_{k\in\mathbb{Z}} L^2\left(\mathbb{R}^{\Lambda_{\rm s}^{(1)}}/\Gamma_{\rm s}; \mathcal{F}\right), \qquad \mathcal{H}^{\rm s} := \bigoplus_{k\in\mathbb{Z}} L^2\left(\mathbb{R}^{\Lambda_{\rm s}^{(1)}}/\Gamma_{\rm s}; \mathcal{F}'\right),$$

the L^2 spaces being formed with the measure of Definition 6.10. The summation index k is the flux index, and a vector of a summand is written as a $\Gamma_{\rm s}$-periodic function $\Phi(A)$ of $A \in \mathbb{R}^{\Lambda_{\rm s}^{(1)}}$. All operators below are diagonal in k. An operator on $L^2(\mathbb{R}^{\Lambda_{\rm s}^{(1)}}/\Gamma_{\rm s})$ acts on $L^2(\mathbb{R}^{\Lambda_{\rm s}^{(1)}}/\Gamma_{\rm s}; \mathcal{F}) = L^2(\mathbb{R}^{\Lambda_{\rm s}^{(1)}}/\Gamma_{\rm s}) \otimes \mathcal{F}$ through the first factor, and in the same way on every summand.

The electric field. For $b \in \Lambda_{\rm s}^{(1)}$ let $E_b := -i\,\partial/\partial A(b)$, the self-adjoint generator of the translations of the coordinate $A(b)$ on $L^2(\mathbb{R}^{\Lambda_{\rm s}^{(1)}}/\Gamma_{\rm s})$. These operators commute. Let $E := (E_b)_{b\in\Lambda_{\rm s}^{(1)}}$; for a bounded Borel function g on $\mathbb{R}^{\Lambda_{\rm s}^{(1)}}$ the operator $g(E)$ is defined by the joint spectral calculus of the commuting self-adjoint operators E_b. With $|x|^2 := \sum_b x(b)^2$, one has for $x \in \mathbb{R}^{\Lambda_{\rm s}^{(1)}}$ and $a > 0$

$$\left(\exp(-i\langle x, E\rangle)\Phi\right)(A) = \Phi(A - x),$$
$$\left(\exp\left(-\tfrac{a}{2}|E|^2\right)\Phi\right)(A) = \int_{\mathbb{R}^{\Lambda_{\rm s}^{(1)}}/\Gamma_{\rm s}} \mathrm{d}A' \sum_{\gamma\in\Gamma_{\rm s}} \prod_{b\in\Lambda_{\rm s}^{(1)}} g_a\left(A(b) - A'(b) + \gamma(b)\right) \cdot \Phi(A') :$$

$\exp\left(-\frac{a}{2}|E|^2\right)$ is the convolution with the function $\sum_{\gamma\in\Gamma_{\rm s}} \prod_b g_a(A(b) + \gamma(b))$ (Lemma 7.24(b)).

The gauge group. For $a \in \mathbb{R}^{\Lambda_{\rm s}}$ let $U_{\rm G}(a)$ be the operator on $\mathcal{H}^{\rm l}$ given by

$$\left(U_{\rm G}(a)\Phi\right)(A) := u(a)\, \Phi(A - d_{\rm s}a).$$

For $a \in (2\pi/q)\cdot\mathbb{Z}^{\Lambda_{\rm s}}$ one has $u(a) = 1$, the charges being integers, and $d_{\rm s}a \in \Gamma_{\rm s}$; hence $U_{\rm G}$ is a unitary representation of the compact group $(\mathbb{R}/(2\pi/q)\mathbb{Z})^{\Lambda_{\rm s}}$. Let $P_{\rm G}$ be the orthogonal projection

onto its invariant vectors. Then

$$\int_{[0,2\pi/q)^{\Lambda_{\rm s}}} \mathrm{d}a\, U_{\rm G}(a) = (2\pi/q)^{|\Lambda_{\rm s}|}\cdot P_{\rm G},$$

and $P_{\rm G}$ commutes with every $g(E)$ (Lemma 7.24(c)).

Crossing operators. For $\beta\in\mathbb{R}^{\Lambda_{\rm s}^{(1)}}$ and a subset S of $\Lambda_{\rm s}^{(1)}$ let $|x|_S^2:=\sum_{b\in S}x(b)^2$, and

$$K:=(2\pi/q)^{|\Lambda_{\rm s}|}\cdot\exp\left(-\tfrac12|E|^2\right)P_{\rm G},\qquad K^{1/2}:=(2\pi/q)^{\frac{|\Lambda_{\rm s}|}{2}}\cdot\exp\left(-\tfrac14|E|^2\right)P_{\rm G},$$
$$K_\beta:=(2\pi/q)^{|\Lambda_{\rm s}|}\cdot\exp\left(-\tfrac12|E-\beta|^2\right)P_{\rm G},\qquad K_{\beta,S}:=K_\beta\exp\left(\tfrac14|E-\beta|_S^2\right).$$

Since $P_{\rm G}$ commutes with the functions of E, the operator $K^{1/2}$ is non-negative and $(K^{1/2})^2=K$. Likewise $K_{\beta,S}=(2\pi/q)^{|\Lambda_{\rm s}|}\cdot\exp\left(-\frac12|E-\beta|^2+\frac14|E-\beta|_S^2\right)P_{\rm G}$, a bounded operator, the exponent being at most $-\frac14|E-\beta|^2$.

The slice operator. Let $-1\le\rho\le1$, so that the operators $\widehat{\mathfrak{N}}(A)$ of Definition 6.11 are defined. For $k\in\mathbb{Z}$ and $A\in\mathbb{R}^{\Lambda_{\rm s}^{(1)}}$ let

$$\mathfrak{M}_k(A):=\exp\left(-\tfrac14\left\|d_{\rm s}A+(2\pi/q)\cdot k\cdot 1_{\sigma_0}\right\|^2\right)\cdot\widehat{\mathfrak{N}}(A):\ \mathcal{F}'\to\mathcal{F},$$

where 1_{σ_0} is the indicator of σ_0 and $\|\cdot\|$ the Euclidean norm of $\mathbb{R}^{\Lambda_{\rm s}^{(2)}}$. The function $A\mapsto\mathfrak{M}_k(A)$ is $\Gamma_{\rm s}$-periodic, because $d_{\rm s}\gamma=0$ for $\gamma\in\Gamma_{\rm s}$ and $\widehat{\mathfrak{N}}(A)$ depends on A only through the phases $\exp(iqA(b))$; it is bounded uniformly in k, since the entries of $\mathfrak{N}(A)$ are polynomials in the phases $\exp(\pm iqA(b))$. The slice operator

$$\mathfrak{M}:=\bigoplus_{k\in\mathbb{Z}}\mathfrak{M}_k:\ \mathcal{H}^{\rm s}\to\mathcal{H}^{\rm l}$$

acts on the summand k as the operator of multiplication by $\mathfrak{M}_k(A)$; it is bounded, and it depends on the choice of σ_0. The operators

$$\mathfrak{M}\mathfrak{M}^*\quad\text{and}\quad K^{1/2}\mathfrak{M}\mathfrak{M}^*K^{1/2}=\left(K^{1/2}\mathfrak{M}\right)\left(K^{1/2}\mathfrak{M}\right)^*\qquad\text{on }\mathcal{H}^{\rm l},$$
$$\mathfrak{M}^*K\mathfrak{M}=\left(K^{1/2}\mathfrak{M}\right)^*\left(K^{1/2}\mathfrak{M}\right)\qquad\text{on }\mathcal{H}^{\rm s}$$

are non-negative; through $\mathfrak{M}$ they depend on σ_0 as well.

A commutation relation. Let $\delta\in\mathbb{R}^{\Lambda_{\rm s}^{(1)}}$ satisfy $\langle\delta,\gamma\rangle\in2\pi\mathbb{Z}$ for every $\gamma\in\Gamma_{\rm s}$. Then the function $A\mapsto\exp(i\langle\delta,A\rangle)$ is $\Gamma_{\rm s}$-periodic, and the operator of multiplication by it, written $\exp(i\langle\delta,A\rangle)$, is unitary. Since $\exp(-i\langle\delta,A\rangle)\,E_b\exp(i\langle\delta,A\rangle)=E_b+\delta(b)$ for every link b, every bounded Borel function g on $\mathbb{R}^{\Lambda_{\rm s}^{(1)}}$ satisfies

$$g(E)\exp(i\langle\delta,A\rangle)=\exp(i\langle\delta,A\rangle)\,g(E+\delta).\tag{6.3}$$

For a finite set S of links of $\mathbb{Z}^2$ the same formula $E_b=-i\,\partial/\partial A(b)$, $b\in S$, defines commuting self-adjoint operators on $L^2(\mathbb{R}^S)$, and $E_S:=(E_b)_{b\in S}$; on $L^2(\mathbb{R}^S)$ the identity (6.3) holds with E_S in the place of E, for every $\delta\in\mathbb{R}^S$ and every bounded Borel function g on $\mathbb{R}^S$.

6.3. **Insertions and two-layer elements.**

Definition 6.13 (The mid-plane algebra)**.** Let S be a finite set of links of the slice lattice $\mathbb{Z}^2$ (Definition 6.10), and let Λ be a non-empty finite set of sites of $\mathbb{Z}^2$ that contains the end points of the links of S. Put

$$\mathcal{H}_S:=L^2(\mathbb{R}^S)\otimes\mathcal{F}\left(\mathcal{J}^{\rm l}(\Lambda)\right),$$

where $\mathcal{F}(\mathcal{J}^{\rm l}(\Lambda))$ is the Fock space of the modes over Λ at a link plane (Definition 6.11). The variable of $L^2(\mathbb{R}^S)$ is $A\in\mathbb{R}^S$, and $E_S=(E_b)_{b\in S}$ is the family of commuting self-adjoint operators on $L^2(\mathbb{R}^S)$ of Definition 6.12; for $S'\subseteq S$ and a bounded Borel function g on $\mathbb{R}^{S'}$, the operator $g(E_{S'})$ is formed

with the joint spectral calculus of $E_{S'} = (E_b)_{b\in S'}$. Functions of A and of E_S act on the first factor of $\mathcal{H}_S$, operators on the Fock space on the second. For $a \in \mathbb{R}^\Lambda$, $u(a)$ is the diagonal unitary on $\mathcal{F}(\mathcal{J}^1(\Lambda))$ of Definition 6.11. An operator Υ on this Fock space is *homogeneous of charge* $\mathfrak{q}(\Upsilon) \in \mathbb{Z}^\Lambda$ if

$$u(a)\,\Upsilon\,u(a)^{-1} = \exp\left(iq\langle a, \mathfrak{q}(\Upsilon)\rangle\right)\cdot\Upsilon \qquad \text{for all } a \in \mathbb{R}^\Lambda.$$

The *mid-plane algebra* $\mathfrak{W}(S,\Lambda)$ is the unital $*$-algebra of operators on $\mathcal{H}_S$ generated by

(a) the operators $g(E_{S'})$, $S' \subseteq S$, where g lies in the linear span of the constant function 1 and of the functions $x \mapsto \exp\left(-\mathfrak{Q}(x) + \langle\lambda, x\rangle\right)$ on $\mathbb{R}^{S'}$, with $\mathfrak{Q}$ a positive definite real quadratic form on $\mathbb{R}^{S'}$ and $\lambda \in \mathbb{C}^{S'}$;

(b) the operators $\exp(i\langle\delta, A\rangle)\cdot\Upsilon$, where $\delta \in \mathbb{R}^S$, $\Upsilon \in \operatorname{End}\mathcal{F}(\mathcal{J}^1(\Lambda))$ is homogeneous of a charge $\mathfrak{q}(\Upsilon)$, and $d_{\mathrm{s}}^*\delta = q\cdot\mathfrak{q}(\Upsilon)$ on $\mathbb{Z}^2$; here d_{s}^* is the adjoint of d_{s} on finitely supported cochains, and δ and $\mathfrak{q}(\Upsilon)$ are extended by zero to all links and all sites of $\mathbb{Z}^2$.

Every element of $\mathfrak{W}(S,\Lambda)$ is even, that is, it commutes with the parity U_{f} of the Fock space, and it is gauge invariant: it commutes with the operators $\Phi \mapsto u(a)\,\Phi\,(\,\cdot - (d_{\mathrm{s}}a)|_S)$, $a \in \mathbb{R}^\Lambda$ (Lemma 11.2).

For $\beta \in \mathbb{R}^S$, α_β denotes the $*$-automorphism of $\mathfrak{W}(S,\Lambda)$ that maps $g(E_{S'})$ to $g(E_{S'} - \beta|_{S'})$ and fixes the generators (b); it is the conjugation by the unitary operator $\exp(i\langle\beta, A\rangle)$ (Lemma 11.2).

Let Ξ be a rectangular cross-section such that $\mathrm{hull}(S,\Lambda)$ embeds in Ξ and does not contain the plaquette σ_0 (Definition 6.10). The realisation $X \mapsto X_\Xi$ of $\mathfrak{W}(S,\Lambda)$ on the Hilbert space $\mathcal{H}^{\mathrm{l}}$ of Definition 6.12 acts in the same way on every summand of $\mathcal{H}^{\mathrm{l}}$: the links of S are read as links of Λ_{s}, the operators $g(E_{S'})$ and $\exp(i\langle\delta, A\rangle)$ are given by the same formulas on $L^2\left(\mathbb{R}^{\Lambda_{\mathrm{s}}^{(1)}}/\Gamma_{\mathrm{s}}\right)$, and Υ is replaced by $\iota(\Upsilon)$, the localisation with the signed relabelling of Definition 6.11. By Lemma 11.2 this is a well defined unital $*$-representation. For $X \in \mathfrak{W}(S,\Lambda)$ put

$$\|X\|_{\#} := \sup_{\Xi,\,\beta} \|(\alpha_\beta X)_\Xi\|\,,$$

the supremum over all rectangular cross-sections Ξ as above and all $\beta \in \mathbb{R}^S$.

Let $S \subseteq S_1$ and $\Lambda \subseteq \Lambda_1$, where S_1 is a finite set of links and Λ_1 a finite set of sites of $\mathbb{Z}^2$ that contains the end points of the links of S_1. Then $\mathfrak{W}(S,\Lambda)$ is a subalgebra of $\mathfrak{W}(S_1,\Lambda_1)$: an element of $\mathfrak{W}(S,\Lambda)$ is identified with its tensor product with the identity of $L^2(\mathbb{R}^{S_1\setminus S})$, its Fock part being carried to $\mathcal{F}(\mathcal{J}^1(\Lambda_1))$ by the localisation ι. The union of all the algebras $\mathfrak{W}(S,\Lambda)$ is a $*$-algebra, denoted by $\mathfrak{W}$. By Lemma 11.2, for each single cross-section Ξ as above the number $\sup_{\beta\in\mathbb{R}^S}\|(\alpha_\beta X)_\Xi\|$ is the operator norm of X on $\mathcal{H}_S$. Hence $\|\cdot\|_{\#}$ is the operator norm on $\mathcal{H}_S$: a C*-norm on $\mathfrak{W}(S,\Lambda)$, invariant under every α_β, with $\|1\|_{\#} = 1$; it does not change under the inclusions just described, and so it is a C*-norm on $\mathfrak{W}$. One and the same algebra $\mathfrak{W}(S,\Lambda)$ is associated with each of the coordinate link planes (x_μ, c), $\mu \in \{1,2,3\}$, $c \in \mathbb{Z} + \frac{1}{2}$.

Definition 6.14 (Insertions)**.** For a link b and a site y of $\mathbb{Z}^2$ and for $j' \in \mathbb{Z}$, let (j', b) be the link of $\mathbb{Z}^3$ above b in the plane $\{x_1 = j'\}$ and (j', y) the site above y in this plane. Let $c = j + \frac{1}{2}$ with $j \in \mathbb{Z}$, let S and Λ be as in Definition 6.13, and let $X \in \mathfrak{W}(S,\Lambda)$. Let $k_X(A', A) \in \operatorname{End}\mathcal{F}(\mathcal{J}^1(\Lambda))$, $A', A \in \mathbb{R}^S$, be the integral kernel of the operator $\exp\left(-\frac{1}{4}|E_S|^2\right) X \exp\left(-\frac{1}{4}|E_S|^2\right)$ on $\mathcal{H}_S$. Let $S_c^\times(\Lambda)$ be the sum of the terms of $\bar\psi D_A\psi$ that belong to the bonds over Λ crossing the plane $\{x_1 = c\}$, that is,

$$S_c^\times(\Lambda) := -\sum_{(x,x')} \bar\psi(x)\,w(x,x')\,U_A(x,x')\,\psi(x'),$$

with the hopping matrices $w(x,x')$ and the parallel transports $U_A(x,x')$ of Definition 2.8, the sum being taken over the ordered pairs (x,x') of fermion sites such that $\{x,x'\}$ is a bond that crosses $\{x_1 = c\}$ and $\{\pi(x),\pi(x')\} = \{(j,y),(j+1,y)\}$ for some $y \in \Lambda$. The *insertion* of X at c is the

integrand

$$G_X = G_X^{(c)} := \mathrm{Ins}_c\left[\omega^{\frac12}\, k_X\Big(\hat A_{j+1}|_S,\, (\hat A_j + d_{\mathrm s}\hat a_j)|_S\Big)\, u(\hat a_j)\, \omega^{\frac12}\right] \cdot \prod_{b\in S} g_1\Big(F_j^{\mathrm{el}}(b)\Big)^{-1} \cdot \exp\left(S_c^{\times}(\Lambda)\right).$$

Here $\hat A$ is a local lift and $F = d\hat A$ (Definition 2.13); F_j^{el} is the electric part of F, and $\hat A_j$, $\hat A_{j+1}$ and $\hat a_j$ are the slice variables of $\hat A$, that is, those of Definition 6.10 with $\hat A$ in the place of A; Ins_c is formed at the mirror (x_1, c) with the modes of $\mathcal J^1(\Lambda)$, so that its argument is a matrix indexed by the subsets of $\mathcal J^1(\Lambda)$; ω is the diagonal operator of the crossing coefficients on $\mathcal F(\mathcal J^1(\Lambda))$, and $u(\hat a_j)$ is formed with the restriction of $\hat a_j$ to Λ; g_1 is the Gaussian density of Definition 6.10. In G_X the crossing terms $S_c^{\times}(\Lambda)$ are formed with $\hat A$ in the place of A: the term of an ordered pair (x, x') with $\pi(x) = (j, y)$ and $\pi(x') = (j+1, y)$ carries the phase $U_{\hat A}(x, x') = \exp\left(iq\hat a_j(y)\right)$, and the term of the pair (x', x) the phase $U_{\hat A}(x', x) = \exp\left(-iq\hat a_j(y)\right)$. The last factor, $\exp(S_c^{\times}(\Lambda))$, is the inverse of $\exp(-S_c^{\times}(\Lambda))$; both exponentials are even, since $S_c^{\times}(\Lambda)$ is.

Let $\mathcal K(c, S, \Lambda)$ be the union of the links (j, b) and $(j+1, b)$, $b \in S$, of the links from (j, y) to $(j+1, y)$, $y \in \Lambda$, and of the cubes $(j, y) + \left[-\frac12, \frac12\right]^3$ and $(j+1, y) + \left[-\frac12, \frac12\right]^3$, $y \in \Lambda$. The midpoint of a product box is the vector of its interval midpoints, and its supremum radius is the largest of its half-widths. A cubic box has the form $x + [-r, r]^3$ and is determined by its centre x and radius r. The canonical insertion box $B(c, S, \Lambda)$ is specifically the cubic box of least radius containing $\mathcal K(c, S, \Lambda)$ and, among those cubic boxes, has the lexicographically least centre. The centre condition selects one of the possibly several least-radius cubes; it does not restrict the general product boxes of Definition 2.13. If $S = \Lambda = \varnothing$, the insertion is constant and can be read in any chosen box. The following canonical box formulas are used only for nonempty $\mathcal K$; one may first enlarge Λ to a nonempty finite set when applying them to a constant insertion. Let $\prod_\mu [x_\mu^-, x_\mu^+]$ be the smallest product of intervals that contains $\mathcal K(c, S, \Lambda)$, and put

$$x^0 := \Big(\tfrac12(x_\mu^- + x_\mu^+)\Big)_{\mu=1,2,3}, \qquad r_0 := \max_\mu \tfrac12(x_\mu^+ - x_\mu^-), \qquad B^{\#}(c, S, \Lambda) := x^0 + [-2r_0, 2r_0]^3;$$

the number r_0 is the radius of $B(c, S, \Lambda)$. An identity between integrands read in different boxes means equality as functions of a cochain and its fields in a common containing box. Their evaluations agree on every common local lift, and hence on every torus where that box embeds.

Remark 6.15. Over (S, Λ), the insertion G_X puts the kernel of $K^{\frac12} X K^{\frac12}$ in the place of the crossing weight $\prod_{b\in S} g_1(F_j^{\mathrm{el}}(b)) \cdot \exp(-S_c^{\times}(\Lambda))$. The box $B(c, S, \Lambda)$ is neither covariant under reflections nor monotone in (S, Λ). Indeed, the reflection θ_c in the plane $\{x_1 = c\}$ leaves c, S, Λ and $\mathcal K(c, S, \Lambda)$ unchanged, but, since $x_1^- = j - \frac12$ and $x_1^+ = j + \frac32$, the first coordinate of the centre of $B(c, S, \Lambda)$ is $j + \frac32 - r_0$, which differs from c as soon as $r_0 > 1$; then $\theta_c B(c, S, \Lambda) \neq B(c, S, \Lambda)$. For $c = \frac12$, the box $B(c, \emptyset, \{(0,0)\}) = \left[-\frac12, \frac32\right] \times \left[-\frac32, \frac12\right]^2$ is not contained in $B(c, \emptyset, \{(0,0), (1,0)\}) = \left[-\frac12, \frac32\right]^2 \times \left[-\frac32, \frac12\right]$. The box $B^{\#}(c, S, \Lambda)$ is covariant under $W_{\mathrm c}$ because signed coordinate maps carry its coordinate extrema and their midpoints to the corresponding extrema and midpoint. It contains every radius-r_0 box containing $\mathcal K$: the centre of such a box differs from x^0 by at most r_0 in each coordinate, and its points consequently differ by at most $2r_0$. In particular it contains $B(c, S, \Lambda)$ and every transported least-radius box.

Definition 6.16 (Bounded elements)**.** Let B be a box and $\mathbb T = (\Gamma, \chi) = C(n_1, n_2, n_3)$ a coordinate torus (Definition 2.9). A *placement* of B in $\mathbb T$ is a finite family $g_1, \dots, g_r$ of elements of $W_{\mathrm c}$ (Definition 2.2), $r \geq 1$, with the following two properties.

(a) The boxes $g_1 B, \dots, g_r B$ embed disjointly in Γ (Definition 2.13).

(b) (Free columns.) For each $\mu \in \{1, 2, 3\}$ there is a plaquette σ orthogonal to e_μ such that none of the sets $g_j B + v$, $1 \leq j \leq r$, $v \in \Gamma$, contains one of the plaquettes $\sigma + k\, e_\mu$, $k \in \mathbb Z$.

Let $G \in \mathcal{G}^{\mathrm{ev}}(B)$ and $C \geq 0$. The integrand G is *bounded by C with the box B* if for every coordinate torus $\mathbb{T} = (\Gamma, \chi)$, every placement $g_1, \dots, g_r$ of B in $\mathbb{T}$ and every $\beta \in C^2(\Gamma)$

$$\left| Z_{\mathbb{T}} \left[\varphi_\beta \prod_{j=1}^{r} G^{g_j} \right] \right| \leq C^r \cdot Z_{\mathbb{T}}[1],$$

where $\varphi_\beta(F) := \exp(i\langle \beta, F \rangle)$ with the pairing $\langle \cdot, \cdot \rangle$ of $C^2(\Gamma)$, G^{g_j} is the image of G under g_j Definition 2.16, and $Z_{\mathbb{T}}$ is the functional of Definition 2.11.

Lemma 6.17 (Placements)**.** *Let B be a box and $\mathbb{T} = (\Gamma, \chi) = C(n_1, n_2, n_3)$ a coordinate torus.*

- *(a) If $g_1, \dots, g_r$ is a placement of B in $\mathbb{T}$ and $g \in W$, then $g\mathbb{T}$ is a coordinate torus and $gg_1g^{-1}, \dots, gg_rg^{-1}$ is a placement of the box gB in $g\mathbb{T}$.*
- *(b) Let $\mathcal{Q} = (\mathfrak{F}_{\mathrm{c}}; o_1, o_2, o_3; h)$ be a coordinate cell structure with $o_\mu \in \mathbb{Z} + \frac{1}{2}$, $\mu = 1, 2, 3$, that cuts $\mathbb{T}$ (Definition 6.2), and let $g_1, \dots, g_r \in W_{\mathrm{c}}$ be such that each of the boxes $g_1B, \dots, g_rB$ lies in an open cell of $\mathcal{Q}$. Then $g_1, \dots, g_r$ have the property (b) of Definition 6.16.*

Proof. Consider first (a). Let $\vec{g} \in O_{\mathrm{cub}}$ be the linear part of g, so that $g(x + v) = g(x) + \vec{g}\, v$ for $x, v \in \mathbb{R}^3$. The matrix $\vec{g}$ is a signed permutation matrix: $\vec{g}\, e_\mu = \pm e_{\mu'}$, where $\mu \mapsto \mu'$ is a permutation of $\{1, 2, 3\}$. Hence $\vec{g}\,\Gamma = \sum_\mu n_\mu \mathbb{Z} e_{\mu'}$, and the character $\chi \circ \vec{g}^{-1}$ of $g\mathbb{T} = (\vec{g}\,\Gamma, \chi \circ \vec{g}^{-1})$ takes the value -1 on the generators $n_\mu e_{\mu'}$, because $\chi(\pm n_\mu e_\mu) = -1$; so $g\mathbb{T}$ is a coordinate torus. The subgroup W_{c} is normal in W, so the elements gg_jg^{-1} lie in W_{c}; moreover $(gg_jg^{-1})(gB) = g(g_jB)$. The set gB is again a product box. Its side lengths are those of B in the order permuted by $\vec{g}$; its midpoint is the image under g of the midpoint of B. Thus its supremum radius equals that of B. These statements follow by applying the signed permutation to the three coordinate intervals. For $v \in \Gamma$ one has

$$g\left(g_jB + [-1, 1]^3 + v\right) = g(g_jB) + [-1, 1]^3 + \vec{g}\, v, \qquad g\left(g_jB + v\right) = g(g_jB) + \vec{g}\, v.$$

As g is a bijection of $\mathbb{R}^3$ and $v \mapsto \vec{g}\, v$ a bijection of Γ onto $\vec{g}\,\Gamma$, the sets on the left are pairwise disjoint if and only if the corresponding sets on the right are; hence the boxes $g(g_jB)$ embed disjointly in $\vec{g}\,\Gamma$. Let $\mu \in \{1, 2, 3\}$ and let σ be a plaquette orthogonal to e_μ as in property (b). Since g maps $\mathbb{Z}^3$ onto itself, $g\sigma$ is a plaquette orthogonal to $e_{\mu'}$, and $g(\sigma + k\, e_\mu) = g\sigma \pm k\, e_{\mu'}$. A set $g(g_jB) + \vec{g}\, v = g(g_jB + v)$ contains $g(\sigma + k\, e_\mu)$ only if $g_jB + v$ contains $\sigma + k\, e_\mu$, which is excluded. So the plaquette $g\sigma$ has the property required for the direction μ', and since $\mu \mapsto \mu'$ is a bijection, every direction is provided for.

For (b), note that the coordinates of every $v \in \Gamma$ lie in $h\mathbb{Z}$, because $\mathcal{Q}$ cuts $\mathbb{T}$; the translation by v therefore permutes the open cells of $\mathcal{Q}$, and every set $g_jB + v$ lies in an open cell. Fix μ and choose $\nu \neq \mu$. As $o_\nu - \frac{1}{2}$ is an integer, there is a plaquette σ orthogonal to e_μ whose projection to the x_ν-axis is the interval $\left[o_\nu - \frac{1}{2}, o_\nu + \frac{1}{2}\right]$; it meets the wall $\{x_\nu = o_\nu\}$ of $\mathcal{Q}$. The translation by $k\, e_\mu$ does not change the coordinate x_ν, so every plaquette $\sigma + k\, e_\mu$, $k \in \mathbb{Z}$, meets the same wall. An open cell of $\mathcal{Q}$ lies strictly between two consecutive walls orthogonal to e_ν and does not meet $\{x_\nu = o_\nu\}$. Hence no plaquette $\sigma + k\, e_\mu$ is contained in an open cell, and a fortiori none is contained in one of the sets $g_jB + v$. □

Definition 6.18 (Two-layer elements)**.** For $j \in \mathbb{Z}$ the *thick layer* j is the open slab $\left\{|x_1 - j| < \frac{1}{2}\right\}$. The photon links that it contains are those of the plane $\{x_1 = j\}$, which carry the variables A_j, and its fermion sites are the $x \in \mathbb{X}$ with $x_1 - j \in \{-v_1, -v_2, -v_3, v_3, v_2, v_1\}$ (Definition 2.4). The sets $\mathbb{X}_{j,1}, \dots, \mathbb{X}_{j,6}$ of the fermion sites with $x_1 - j = -v_1, -v_2, -v_3, v_3, v_2, v_1$, respectively, are the *thin layers* of the thick layer j; they are ordered by x_1. The successor $\mathbb{X}^+_{j,k}$ of $\mathbb{X}_{j,k}$ is $\mathbb{X}_{j,k+1}$ if $k \leq 5$ and $\mathbb{X}_{j+1,1}$ if $k = 6$; $\mathbb{X}^{++}_{j,k}$ is the successor of $\mathbb{X}^+_{j,k}$, and the predecessor $\mathbb{X}^-_{j,k}$ is the thin layer whose successor is $\mathbb{X}_{j,k}$.

Every $P \in \mathcal{P}$ (Definition 2.20) has a unique representation

$$P = \sum_{\alpha} a_\alpha \cdot \exp\left(i\langle \delta_\alpha, \hat{A}\rangle\right) \cdot \Psi_\alpha$$

with finitely many terms, coefficients $a_\alpha \neq 0$ and pairwise different pairs $(\delta_\alpha, \Psi_\alpha)$, where each Ψ_α is a monomial, written as a product of components of the fields in a fixed total order (the monomials are linearly independent, and so are the functions $\hat{A} \mapsto \exp(i\langle \delta, \hat{A}\rangle)$ for pairwise different δ). The element P *reads* the link b if $\delta_\alpha(b) \neq 0$ for some α, and it reads the site x if some Ψ_α contains a field at x. As an element of $\mathcal{P}$, a character φ_β, $\beta \in C^2_{\mathrm{c}}$, equals $\exp(i\langle d^*\beta, \hat{A}\rangle)$, and the links that it reads are those of $\operatorname{supp} d^*\beta$. The least (largest) integer j for which P reads a site of the thick layer j or a link meeting the plane $\{x_1 = j\}$ is the *lowest* (*highest*) *plane* of P.

Let $c = j_0 + \frac{1}{2}$ with $j_0 \in \mathbb{Z}$. A *two-layer monomial* at c is a gauge invariant element $\exp(i\langle \delta, \hat{A}\rangle)\Psi$ of $\mathcal{P}$ in which δ is supported on the links of the planes $\{x_1 = j_0\}$, $\{x_1 = j_0 + 1\}$ and on the links between these two planes, and Ψ is a monomial in the fields of $\mathbb{X}_{j_0,6} \cup \mathbb{X}_{j_0+1,1}$, the two ends of the bonds that cross the plane $\{x_1 = c\}$. The linear span $\mathcal{P}_1(c)$ of the two-layer monomials at c is the space of *two-layer elements* at c. For such a δ and for a link b and a site y of $\mathbb{Z}^2$, let $\delta_0(b)$ and $\delta_1(b)$ be the values of δ on the links (j_0, b) and $(j_0 + 1, b)$, and $\delta_\perp(y)$ its value on the link from (j_0, y) to $(j_0 + 1, y)$; the triple $(\delta_0, \delta_1, \delta_\perp)$ is the *exponent* of the monomial. Two-layer monomials with $\delta \in q\, C^1(\mathbb{Z})$ are called *integer*.

Let $Y = \exp(i\langle \delta, \hat{A}\rangle)\Psi$ be a two-layer monomial at c with $Y \neq 1$. Let S be the set of the links b of $\mathbb{Z}^2$ with $\delta_0(b) \neq 0$ or $\delta_1(b) \neq 0$, and let Λ_Y, the set of the *sites in play*, be a finite set of sites of $\mathbb{Z}^2$ that contains the end points of the links of S, the sites y with $\delta_\perp(y) \neq 0$, and every y for which Ψ contains a field at a site x with $\pi(x) \in \{(j_0, y), (j_0 + 1, y)\}$. Since $Y \neq 1$, the set Λ_Y is not empty. As $|\rho| < 1$, the crossing coefficients are positive and ω is invertible; $\Upsilon_Y \in \operatorname{End} \mathcal{F}(\mathcal{J}^{\mathrm{l}}(\Lambda_Y))$ is the operator with

$$\left(\Psi \cdot \exp\left(-S^{\times}_c(\Lambda_Y)\right)\right)\big|_{\hat{a}_{j_0}=0} = \mathrm{Ins}_c\left[\omega^{\frac{1}{2}}\, \Upsilon_Y\, \omega^{\frac{1}{2}}\right],$$

here $S^{\times}_c(\Lambda_Y)$ is formed with $\hat{A}$ as in Definition 6.14, and the evaluation at $\hat{a}_{j_0} = 0$ makes all phases $U_{\hat{A}}(x, x') = \exp\left(\pm iq\hat{a}_{j_0}(y)\right)$, $y \in \Lambda_Y$, of the crossing bonds equal to 1. The operator Υ_Y exists and is unique by Proposition 11.7.

Let Ξ be a rectangular cross-section as in Definition 6.13, with Λ_Y in the place of Λ. On $\mathcal{H}_S$ and on $\mathcal{H}^{\mathrm{l}}$ put

$$k^0_Y := \exp\left(i\langle \delta_0 + \delta_1, A\rangle\right) \cdot \exp\left(-\tfrac{1}{2}|E + \delta_0|^2\right) \cdot \Upsilon_Y,$$

where $E = E_S$ on $\mathcal{H}_S$, while on $\mathcal{H}^{\mathrm{l}}$ the vector $E = (E_b)_{b \in \Lambda^{(1)}_{\mathrm{s}}}$ is the one of Definition 6.12, δ_0 and δ_1 are read as cochains of Λ_{s}, and Υ_Y acts through ι; on $\mathcal{H}^{\mathrm{l}}$ put moreover

$$k_Y := (2\pi/q)^{|\Lambda_{\mathrm{s}}|} \cdot k^0_Y \cdot P_{\mathrm{G}}.$$

On $\mathcal{H}^{\mathrm{l}}$ the factor $\exp(i\langle \delta_0 + \delta_1, A\rangle)$ is a function on $\mathbb{R}^{\Lambda^{(1)}_{\mathrm{s}}}/\Gamma_{\mathrm{s}}$, because $d^*_{\mathrm{s}}(\delta_0 + \delta_1) \in q\mathbb{Z}^{\mathbb{Z}^2}$ (Proposition 11.7 and Lemma 11.2). On $\mathcal{H}_S$ the relation (6.3), which holds on $L^2(\mathbb{R}^S)$ as well, gives $k^0_Y = \exp(i\langle \delta_1, A\rangle) \cdot \exp\left(-\frac{1}{2}|E_S|^2\right) \cdot \Upsilon_Y \cdot \exp(i\langle \delta_0, A\rangle)$; on $\mathcal{H}^{\mathrm{l}}$ the two factors $\exp(i\langle \delta_0, A\rangle)$ and $\exp(i\langle \delta_1, A\rangle)$ are in general not defined, since in general only $\delta_0 + \delta_1$ has its divergence in $q\mathbb{Z}^{\mathbb{Z}^2}$.

With $\widetilde{E} := E_S + \delta_0$ and $\varepsilon > 0$, let

$$X'_Y := \exp\left(i\langle \delta_0 + \delta_1, A\rangle\right) \cdot \exp\left(-\tfrac{1}{4}|\widetilde{E}|^2 + \tfrac{1}{4}\langle \widetilde{E}, \delta_1 - \delta_0\rangle + \tfrac{1}{8}\left(|\delta_0|^2 + |\delta_1|^2\right)\right) \cdot \Upsilon_Y,$$

$$X_{Y,\varepsilon} := \exp\left(i\langle \delta_0 + \delta_1, A\rangle\right) \cdot \exp\left(-\varepsilon|\widetilde{E} + \delta_1|^2 - \varepsilon|\widetilde{E} - \delta_0|^2 + \tfrac{1}{2}\langle \widetilde{E}, \delta_1 - \delta_0\rangle + \tfrac{1}{4}\left(|\delta_0|^2 + |\delta_1|^2\right)\right) \cdot \Upsilon_Y,$$

$$P_\varepsilon := \exp\left(-\varepsilon|E_S|^2\right),$$

operators on $\mathcal{H}_S$. They are bounded and lie in $\mathfrak{W}(S,\Lambda_Y)$, and on $\mathcal{H}^{\mathrm{l}}$

$$k_Y = K_{0,S}^{\frac12}\,(X'_Y)_\Xi\,K_{0,S}^{\frac12}, \qquad K^{\frac12}\,(X_{Y,\varepsilon})_\Xi\,K^{\frac12} = (P_\varepsilon)_\Xi\,k_Y\,(P_\varepsilon)_\Xi \tag{6.4}$$

(Proposition 11.7). Thus the bounded operators X'_Y and $X_{Y,\varepsilon}$ take the places of the formal expressions $K_{0,S}^{-\frac12}k_YK_{0,S}^{-\frac12}$ and $P_\varepsilon K^{-\frac12}k_YK^{-\frac12}P_\varepsilon$. Put

$$\xi_Y(\varepsilon) := \left\|X'_Y - P_\varepsilon X'_Y P_\varepsilon\right\|_{\#}.$$

The monomial 1 is given no set of sites, and hence no box $B^{\#}(c,S,\Lambda_Y)$; for it, $\delta=0$ and $S=\emptyset$. One puts $\Upsilon_1 := 1$, the identity of the Fock spaces $\mathcal{F}(\mathcal{J}^{\mathrm{l}}(\Lambda))$, and $X'_1 := X_{1,\varepsilon} := 1$, the unit of $\mathfrak{W}$. On $\mathcal{H}_S$ one puts $k_1^0 := 1$ as well, whereas on $\mathcal{H}^{\mathrm{l}}$

$$k_1^0 := \exp\left(-\tfrac12|E|^2\right), \qquad k_1 := (2\pi/q)^{|\Lambda_{\mathrm{s}}|}\cdot k_1^0\cdot P_{\mathrm{G}} = K.$$

These are the values of the formulas above for $\delta=0$ and $\Upsilon_1=1$: for $S=\emptyset$ the vector E_S has no components, while on $\mathcal{H}^{\mathrm{l}}$ the vector E has one component for each link of Λ_{s}. Moreover, for $S=\emptyset$ one has $P_\varepsilon=1$ and $K_{0,\emptyset}=K$; as the realisation is unital, the two relations (6.4) hold for $Y=1$ as well. The maps $Y\mapsto\Upsilon_Y$, $Y\mapsto k_Y^0$, $Y\mapsto k_Y$, $Y\mapsto X'_Y$ and $Y\mapsto X_{Y,\varepsilon}$ are extended linearly to $\mathcal{P}_1(c)$, operators attached to different monomials being compared through the inclusions of Definition 6.13. After this extension the relations (6.4) hold monomial by monomial: for each two-layer monomial, the operators $K_{0,S}$ and P_ε in (6.4) are built from the set S of that monomial. The map $Y\mapsto\xi_Y(\varepsilon)$ is not linear; its formula is applied, unchanged, to the finite combinations Y of two-layer monomials that all have the same exponent, since such monomials share one set S and one operator P_ε. Let $\Theta_c := \Theta_{(x_1,c)}$ be the reflection of Definition 2.16 at the mirror (x_1,c), acting on $\mathcal{P}$ by Definition 2.16. If Y is a two-layer monomial with the exponent $(\delta_0,\delta_1,\delta_\perp)$, then Θ_cY is a finite combination of two-layer monomials with the exponent $(-\delta_1,-\delta_0,\delta_\perp)$, hence with the same set S (Proposition 11.7). Indeed, $\Theta_cY = \exp(-i\langle\delta,\theta_c^*\hat A\rangle)\cdot\Theta_c(\Psi)$ by Definition 2.16, where $\langle\delta,\theta_c^*\hat A\rangle = \langle\delta_0,\hat A_{j_0+1}\rangle + \langle\delta_1,\hat A_{j_0}\rangle - \langle\delta_\perp,\hat a_{j_0}\rangle$ by (6.1), and $\Theta_c(\Psi)$ is a combination of monomials in the fields of the two thin layers $\mathbb{X}_{j_0,6}$ and $\mathbb{X}_{j_0+1,1}$, which θ_c exchanges.

Definition 6.19 (Other coordinate directions)**.** The objects of Subsections 6.2 and 6.3 are attached to the direction e_1, the time direction. For $\mu\in\{1,2,3\}$ let $g_\mu\in O_{\mathrm{cub}}$ be the cyclic permutation matrix with $g_\mu e_1 = e_\mu$; thus $g_1=1$, g_2 maps $e_1\mapsto e_2\mapsto e_3\mapsto e_1$, and $g_3=g_2^{-1}$. Each g_μ is an even element of W, so that $G\mapsto G^{g_\mu}$ is linear Definition 2.16. The objects attached to a coordinate link plane (x_μ,c), $c\in\mathbb{Z}+\frac12$, are the images under g_μ of the corresponding objects at (x_1,c); for instance, the insertion of $X\in\mathfrak{W}(S,\Lambda)$ at (x_μ,c) is $(G_X^{(c)})^{g_\mu}$. Moreover,

$$\Theta_{\mu,c} := \Theta_{(x_\mu,c)}, \qquad \mathcal{P}_1(\mu,c) := \{Y^{g_\mu} : Y\in\mathcal{P}_1(c)\}.$$

This cyclic choice suffices for all coordinate-direction statements.

Definition 6.20 (Threshold)**.** Let the sequence $\mathfrak{n}=(n_i)$ satisfy (Div), and let $\mathfrak{d}=(c_0,c_+,c_3;h;O)$ be data (Definition 6.7). The *threshold* $i_0(\mathfrak{d})$ is the smallest index j for which all the sides n_i with $i\ge j$ are multiples of $4h$. It exists, because $4h\in\mathcal{D}(\mathfrak{n})$, that is, $4h$ divides all but finitely many of the n_i. Since $r_O\ge1$, the inequality $h>8r_O+8$ gives $h>16$, and as h is even, $h\ge18$. Consequently, for every $i\ge i_0(\mathfrak{d})$ the side n_i is even and $n_i\ge4h\ge72$, and the four objects $C(n_i)$, $D'(n_i)$, $D(n_i)$ and $C(n_i/2,2n_i,n_i)$ of Definition 6.5 are tori in the sense of Definition 2.8: all their sides are integers not smaller than $n_i/2\ge36$, so that the systoles of their period lattices are at least four.

7. Transfer operators with the full flux sum

We construct transfer operators without choosing a periodic potential for the field strength. The integer plaquette variables are retained until a Fourier transformation at each cut. Their closedness

becomes a boundary Gauss constraint. The construction works at every real value of the mass; no inverse fermion matrix occurs in it.

7.1. **Cells and configurations of a cylinder.** Fix $\ell \in \mathcal{L}$, write $n = n_\ell$, and fix a cross-section $\Xi = (\Gamma_\parallel, \chi_\parallel)$ as in Definition 6.1. All cells in this subsection are cells of the cylinder $\mathbb{Z}^3/\Gamma_\parallel$. For $c \in \mathrm{Adm}(\ell)$ put $c^\pm = c \pm |n|^2/2$. We suppress ℓ in θ_c and Θ_c.

Definition 7.1. The incidence $[\sigma : \sigma']$ is the coefficient of the oriented face σ' in $\partial\sigma$. Thus $(du)(\sigma) = \sum_{\sigma'}[\sigma : \sigma']u(\sigma')$. Define the following sets of k-cells:

$$\begin{aligned}\mathcal{M}^k(c) &= \{\sigma : \ell \equiv c \text{ on } \sigma\},\\ \mathcal{X}^k(c) &= \{\sigma : \min_\sigma \ell < c < \max_\sigma \ell\},\\ \mathcal{B}^k(c,c') &= \{\sigma : \sigma \subset \{c \le \ell \le c'\},\ \sigma \notin \mathcal{M}^k(c) \cup \mathcal{M}^k(c')\}.\end{aligned}$$

The cells of the closed slab are $[c,c']^{(k)} = \mathcal{M}^k(c) \sqcup \mathcal{B}^k(c,c') \sqcup \mathcal{M}^k(c')$. A slab configuration is a pair $\varpi = (A, z)$ on its links and plaquettes, with real A, integer z, and $dz = 0$ on $\mathcal{B}^3(c,c')$. Write this set as $\mathfrak{C}[c,c']$. Its field strength is $F = dA + 2\pi z/q$. For an integer one-cochain a on the closed slab put

$$\varpi + [a] = (A + 2\pi a/q, z - da).$$

This operation preserves F, the link phases and dz.

The mirror data are $\mathfrak{m} = (A|_{\mathcal{M}^1(c)}, z|_{\mathcal{M}^2(c)})$, and similarly $\mathfrak{m}'$ at c'. The remaining variables form $\mathring{\varpi}$; thus $\varpi = \mathfrak{m} \vee \mathring{\varpi} \vee \mathfrak{m}'$. The interior measure and weight are

$$\begin{aligned}\int d\mathring{\varpi}\, \Phi &= \sum_{z \in \mathbb{Z}^{\mathcal{B}^2(c,c')}} \int_{[0,2\pi/q)^{\mathcal{B}^1(c,c')}} dA\, \Phi,\\ \mathcal{W}_{c,c'}(\varpi) &= \prod_{\sigma \in \mathcal{B}^2(c,c')} e^{-F_\sigma^2/2} \prod_{Q \in \mathcal{B}^3(c,c')} \mathbf{1}[(dz)(Q) = 0].\end{aligned}$$

An expression containing an integrand defined only on $\mathfrak{C}[c,c']$ is assigned the value zero where $\mathcal{W}_{c,c'} = 0$.

The cylinder incidence is unambiguous: two distinct faces of a cell cannot be identified by $\Gamma_\parallel$, since a cell has diameter at most two in the maximum norm and the systole is at least four. Oriented pullback commutes with d and $d^2 = 0$, by cancellation of the two occurrences of every codimension-two face.

Lemma 7.2. *A cell is cut by at most one admissible mirror. If it is cut by c, its minimum and maximum heights are c^- and c^+; reflection in c fixes it setwise and reverses its orientation. A cell contained in the mirror is fixed with its orientation. Reflection in c' carries such a cell at c to its translate by*

$$v_{c,c'} = 2(c' - c)n/|n|^2 \in \mathbb{Z}n.$$

Every face of a cut cell is either cut, above the mirror, or below it; none is contained in the mirror. Reflection exchanges the upper and lower faces. If $d_c^+ u$ denotes the incidence sum over upper faces and $d_c u$ the incidence sum over cut faces, then on cut cells

$$du = d_c^+ u - d_c^+ \theta_c^* u + d_c u, \qquad d_c^2 = 0. \tag{7.1}$$

For $c < c' < c''$,

$$\mathcal{B}(c,c'') = \mathcal{B}(c,c') \sqcup \mathcal{M}(c') \sqcup \mathcal{X}(c') \sqcup \mathcal{B}(c',c''). \tag{7.2}$$

If an integer one-cochain a vanishes on cut links and on the closed lower half, then on cut cubes

$$d_c^+(da) + d_c(d_c^+ a) = 0. \tag{7.3}$$

All the sets of cells on a bounded height interval are finite.

Proof. An axis height has span zero or one on a cell. A diagonal height has span zero, one or two, and its endpoints are integers. Consequently an axis cell can only be cut at the midpoint of a unit interval; a diagonal cell can only be cut when its span is two, at its integer midpoint. In the axis case reflection replaces its normal coordinate a by $1-a$. For a diagonal with the two varying coordinates, it either interchanges them or replaces the pair by $(1-a_\nu, 1-a_\mu)$. Its determinant on the direction space of the cut cell is -1. On a cell in the mirror the reflection is pointwise the identity. These observations prove the orientation claims and the assertions about faces. For an upper face σ' of a cut cell σ, incidence covariance gives

$$[\sigma : \theta_c\sigma']u(\theta_c\sigma') = -[\sigma : \sigma'](\theta_c^* u)(\sigma').$$

Adding the upper, lower and cut faces gives (7.1). The cut faces alone have the same codimension-two cancellation as an ordinary boundary, proving $d_c^2 = 0$. Every cell of a larger slab is on one of the two sides, in the intervening mirror, or cut by it, which proves (7.2).

For (7.3), compute $d(da) = 0$ on a cut cube. Its lower faces contribute zero. The contributions from upper faces are $d_c^+(da)$, and those from cut faces are $d_c(d_c^+ a)$, because a vanishes on their cut and lower edges. Finally a rank-two lattice in $\ker \ell$ has a bounded fundamental parallelogram there; its product with a bounded interval in the normal direction is bounded. It contains representatives for the finitely many cells under consideration. □

Lemma 7.3. *Let Φ be invariant under integer shifts supported on the interior links of a slab, and suppose its interior absolute integral is finite. The integral does not change if any link interval $[0, 2\pi/q)$ is replaced by any other interval of length $2\pi/q$.*

Proof. After summing over z, the resulting function of each link variable is $2\pi/q$ periodic. Indeed a shift of that link is compensated by the bijection $z \mapsto z - da$. An integrable periodic function has the same integral over any interval of a period: split the translated interval at the endpoints of the original period and translate the pieces. Fubini permits this operation successively at every link. □

7.2. **Slab integrands and their evaluation.**

Definition 7.4. For a height interval I, let $\mathrm{Gr}_\Xi(I)$ be the Grassmann algebra of the sites with $\ell(x) \in I$, modulo $\psi(x+g) = \chi_\parallel(g)\psi(x)$ and the same relation for $\bar\psi$, $g \in \Gamma_\parallel$. When I is bounded it is a finite-dimensional algebra. Its Berezin integral $\int_I$ is the coefficient of the top monomial, with the convention of Definition 2.7.

The class $\mathcal{I}_\Xi[c, c']$ consists of Borel maps $G : \mathfrak{C}[c, c'] \to \mathrm{Gr}_\Xi((c, c'))$ invariant under every integer shift, whose coefficients in a monomial basis obey

$$|a_\Psi(\varpi)| \le C \exp\left((\tfrac{1}{2} - \varepsilon_G) \sum_{\sigma \in \mathcal{B}^2(c,c')} F_\sigma^2 \right), \qquad 0 < \varepsilon_G \le \tfrac{1}{2}. \tag{7.4}$$

The analogous torus class $\mathcal{I}(\mathbb{T})$ uses $\|F\|^2$ on the right. A local product is *read on* $\mathbb{T}$ when all its period-translated boxes $B_j + g$, $g \in \Gamma$, are pairwise disjoint. Their linear span is $\mathcal{A}(\mathbb{T})$. This is weaker than embedding: embedding also requires disjoint translates of each enlarged box $B_j + [-1, 1]^3$. Products on adjacent slabs are read in increasing height order. Translation acts on coefficients by pullback and on fields by $\tau_v\psi(x) = \psi(x+v)$. Reflection acts on coefficients by conjugation and pullback and on fields by Θ_c. Both operations preserve these classes.

Changing the representatives or the ordering of the generators changes monomial coefficients only by signs. The bound therefore has an intrinsic meaning. Adjacent products satisfy the bound with $\min \varepsilon_G$, since their interior plaquette sets are disjoint. Even and odd parts, and finite sums, remain in the class. The identities

$$\tau_v \Theta_c \tau_v^{-1} = \Theta_{c+\ell(v)}, \qquad \Theta_{c'}\Theta_c = \tau_{v_{c,c'}}, \qquad \Theta_{2c'-c}\tau_{v_{c,c'}} = \Theta_{c'}$$

follow first on a generator from the definition, and then on products. In particular two parallel reflections produce the stated translation, including on odd fields.

Lemma 7.5. *Every closed integer two-cochain on the full cubical lattice, or on the cell complex of a closed axis or diagonal slab, has an integer one-cochain primitive. Admissible slab products in $\mathcal{A}_\Xi[c, c']$, and torus products read on $\mathbb{T}$, therefore have a well-defined evaluation in $\mathcal{I}_\Xi[c, c']$ and $\mathcal{I}(\mathbb{T})$, respectively. Evaluation is linear on the space of maps, commutes with adjacent products, translations and reflections, and agrees with the box evaluation of Lemma 2.14. No periodic primitive is asserted.*

Proof. Here is an integer construction of the required slab primitive. Choose a coordinate direction μ_0 absent from ℓ and use it as the vertical direction. The complex is $K_2 \times \mathbb{Z}$, where K_2 is the full cell complex of a plane or an axis or diagonal strip, including its edges when it contains no squares. Write the two planar coordinates as (a, r) and let I_r be the integer positions of the edges $[(a, r), (a, r + 1)]$ contained in the strip. It is an interval of integers. The square $Q(a, r)$ between two such edges is present exactly when $a, a + 1 \in I_r$. Set the component of the primitive in the a direction equal to zero. For each nonempty I_r choose $a_r \in I_r$ and set its transverse component at a to the oriented sum of $u(Q(s, r))$ from $s = a_r$ to $s = a - 1$, reversing the sign when $a < a_r$. A singleton interval therefore gives its edge the value zero. The difference at $a + 1$ and a is exactly $u(Q(a, r))$, so the coboundary is the desired one on $K_2 \times \{0\}$, with integer values. Only empty edge intervals require no assignment.

Set the vertical component of the primitive equal to zero. For a horizontal link b of direction ν, extend its value to level j by the signed sum of the mixed plaquette values between levels 0 and j. The sign is -1 if $\nu < \mu_0$ and $+1$ if $\nu > \mu_0$, with oriented sums for negative j. The coboundary then has the desired values on every mixed plaquette. The residual two-cochain is closed, vanishes on mixed plaquettes, and vanishes on the horizontal plaquettes at level zero. Its cube identity says that each horizontal plaquette value is constant between successive levels, so the residual vanishes everywhere. This proves the claim using integer additions only.

Lift a slab configuration to its ordinary nonperiodic slab complex, write $z = da$, and put $\widehat{A} = A + 2\pi a/q$. This is one common potential for all its local boxes. If two potentials have the same field strength and phases on a box, their difference is $2\pi/q$ times a closed integer one-cochain. Integrating that cochain along coordinate paths gives an integer zero-cochain primitive; the plaquette relation makes the path integral independent of order. Gauge covariance of a local integrand has multiplier one for this integer gauge transformation. Hence its value is independent of the lift, as already established in Lemma 2.14. Using one common lift for all factors proves that evaluation respects an equality of maps, rather than merely an equality of formal expressions. It also proves linearity. Disjoint boxes have disjoint sets of field generators and interior plaquettes; their product coefficients are signed products and satisfy (7.4). The fixed integer construction, restricted to finitely many boxes, is Borel. Translating or reflecting a lift gives a lift of the transformed configuration. Independence of the lift proves the claimed commutation properties. On a torus one first extends the cochains to the full lattice; a common lift there gives the identical conclusion, including configurations of nonzero flux. □

7.3. **Berezin matrices and the closing sign.**

Lemma 7.6. *Berezin integration factorizes over disjoint sets of sites without a sign, even for inhomogeneous factors. Translation preserves the integral, and reflection conjugates it:*

$$\int_{I+\ell(v)} \tau_v G = \int_I G, \qquad \int_{2c-I} \Theta_c G = \overline{\int_I G}.$$

Proof. A top monomial at one site contains eight generators, an even number. Thus top monomials on disjoint sites commute. For factorization, only the top coefficient of each factor can contribute to

the top coefficient of their product. If one factor is odd, its integral and the integral of the product both vanish. This also proves the inhomogeneous statement by linearity. A change of representative multiplies the top monomial by $\chi_\parallel(g)^8 = 1$. Translation just renames these pairs. Under reflection the interchange of the four ψ and four $\bar\psi$ generators, together with reversal of their order, preserves the top monomial. The two spin transformations contribute $(\det \gamma_{\hat n})^2 = 1$: its eigenvalues are $1, 1, -1, -1$. The coefficients are conjugated, which proves the reflection formula. □

Let $\mathcal{J}(c)$ be ordered representatives of the crossing modes modulo $\Gamma_\parallel$, as in Definition 6.11. Choose them compatibly with translation by n: $\mathcal{J}(c + |n|^2) = \mathcal{J}(c) + n$ with the transported order. This can be done independently for the two mirror classes. For $I \subset \mathcal{J}(c)$ write $\zeta_I^{(c)}$ for the ordered monomial, $\omega_I = \prod_{i\in I} \omega_i$, and let $\mathcal{F}(c)$ have orthonormal basis e_I. Its parity is $U_{\mathrm f} e_I = (-1)^{|I|} e_I$.

Definition 7.7. The mode $i + v$ has the translated bond and the same group and spin index as i. Let $r_v(i) \in \mathcal{J}(c + \ell(v))$ represent its class and write $i + v = r_v(i) + g_v(i)$ with $g_v(i) \in \Gamma_\parallel$. Let $s_v(I)$ be the sign for sorting the representatives $r_v(i)$ into their chosen order, multiplied by $\prod_{i\in I} \chi_\parallel(g_v(i))$. Define

$$U_v e_I = s_v(I) e_{r_v(I)}.$$

Then

$$\tau_v \zeta_I^{(c)} = s_v(I) \zeta_{r_v(I)}^{(c+\ell(v))}, \qquad U_v U_{v'} = U_{v+v'}, \qquad U_v \omega_c = \omega_{c+\ell(v)} U_v, \tag{7.5}$$

where $\omega_c = \mathrm{diag}(\omega_I)$. Every U_v is a real unitary, preserves parity, and is $\chi_\parallel(v)^{|I|}$ on e_I when $v \in \Gamma_\parallel$. For $v \in \mathbb{Z}n$ the sign is one and $r_v(I) = I + v$. In particular

$$\Theta_{2c'-c} \zeta_{I+v_{c,c'}}^{(2c'-c)} = \Theta_{c'} \zeta_I^{(c)}. \tag{7.6}$$

These assertions follow by sorting the translated degree-one factors. Translation bijects their classes and preserves their coefficients. The group law follows from its action on the monomials, which are linearly independent: at the upper endpoints they are an exterior basis of the relevant field generators. The last identity is $\Theta_{2c'-c} \tau_{v_{c,c'}} = \Theta_{c'}$ applied to that basis.

Lemma 7.8. *Every crossing bond at c has its endpoints in (c^-, c^+) and meets no other admissible mirror. Let $S_{(c,c')}$ be the action of bonds and mass terms whose sites lie in the open slab. Then*

$$S_{(c,c'')} = S_{(c,c')} + S_{(c',c'')} + S_{c'}^\times.$$

The phases of $S_{(c,c')}$ use only links of the closed slab. At a cut the crossing action depends only on the cut links, and only for a coordinate link plane. With $u_I(A) = \prod_{i\in I} u_i(A)$,

$$e^{-S_c^\times(A)} = \sum_{I \subset \mathcal{J}(c)} \omega_I u_I(A) \Theta_c(\zeta_I^{(c)}) \zeta_I^{(c)}. \tag{7.7}$$

Proof. The chamber geometry in Lemma 2.5 places every crossing bond in the two chambers adjacent to its wall. Its nearest photon link, if nonconstant, is contained in the closed slab and is cut precisely at the corresponding coordinate link plane. Thus every action term belongs either to one open slab or to its intervening crossing action. The mode expansion of Proposition 4.1 writes $-S_c^\times$ as a sum of $\omega_i u_i \Theta_c(\zeta_i)\zeta_i$. These terms are commuting even elements of square zero. Their exponential is the product of $1 + \omega_i u_i \Theta_c(\zeta_i)\zeta_i$. In each term of the expanded product, reversing the lower factors in Θ_c cancels the sign from bringing each lower-upper pair together. This gives (7.7). □

Definition 7.9. For an operator Υ on $\mathcal{F}(c)$ define

$$\mathrm{Ins}_c[\Upsilon] = \sum_{I,J} \Upsilon_{IJ} \Theta_c(\zeta_J^{(c)}) \zeta_I^{(c)}.$$

For $G \in \mathrm{Gr}_\Xi((c,c'))$ its slab matrix $\mathfrak{N}_{c,c'}[G] : \mathcal{F}(c) \to \mathcal{F}(c')$ is

$$\mathfrak{N}_{c,c'}[G]_{I',I} = \int_{(c,c')} \zeta_I^{(c)} G \Theta_{c'}(\zeta_{I'}^{(c')}).$$

The row index is on the upper side, and the column index on the lower side. In particular (7.7) is $\mathrm{Ins}_c[\omega_c \operatorname{diag}(u_I)]$.

Lemma 7.10. *For $c < c' < c''$ the matrices satisfy*

$$\mathfrak{N}_{c,c''}[G \,\mathrm{Ins}_{c'}[\Upsilon] G'] = \mathfrak{N}_{c',c''}[G'] \Upsilon \mathfrak{N}_{c,c'}[G], \tag{7.8}$$

$$\mathfrak{N}_{c',2c'-c}[\Theta_{c'} G] = U_{v_{c,c'}} \mathfrak{N}_{c,c'}[G]^*, \tag{7.9}$$

$$\mathfrak{N}_{c+\ell(v),c'+\ell(v)}[\tau_v G] = U_v \mathfrak{N}_{c,c'}[G] U_v^{-1}. \tag{7.10}$$

For homogeneous G, $U_{\mathrm{f}} \mathfrak{N}[G] = (-1)^{\deg G} \mathfrak{N}[G] U_{\mathrm{f}}$. If $\mathbb{T} = (\Gamma, \chi)$ has closing vector g and an admissible chain $c_0 < \cdots < c_r = c_0 + \ell(g)$, then for even Υ_j and homogeneous $G_j \in \mathrm{Gr}_\Xi((c_{j-1}, c_j))$,

$$\int_{\mathbb{T}} \mathrm{Ins}_{c_0}[\Upsilon_0] G_1 \,\mathrm{Ins}_{c_1}[\Upsilon_1] \cdots \mathrm{Ins}_{c_{r-1}}[\Upsilon_{r-1}] G_r \\ = \operatorname{tr}\left(U_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} \mathfrak{N}_r \Upsilon_{r-1} \cdots \mathfrak{N}_1 \Upsilon_0 \right). \tag{7.11}$$

Both sides are zero when the total degree of the G_j is odd.

Proof. Expand the insertion at c' and group the integrand as

$$[\zeta_I^{(c)} G \Theta_{c'}(\zeta_J^{(c')})][\zeta_K^{(c')} G' \Theta_{c''}(\zeta_{I''}^{(c'')})].$$

There is no permutation here. The two brackets live on disjoint site sets, since no fermion site lies in the mirror. Berezin factorization gives matrix multiplication, with coefficients Υ_{KJ}. Apply reflection to the entire first bracket, reverse its factors and use (7.6); the conjugated integral is exactly the matrix entry in (7.9). Translation instead contributes the two sorting signs in (7.5), proving (7.10). A matrix entry has total parity $|I| + |I'| + \deg G$ and its integral vanishes if this is odd. This proves the parity assertion.

For closure, every site class of $\mathbb{T}$ has exactly one representative in (c_0, c_r), since its heights are one coset of $\ell(g)\mathbb{Z}$ and no endpoint height occurs. Fields just below c_0 read as $\chi(g)$ times their translates by g. In particular the seam monomial reads as

$$\Theta_{c_0}(\zeta_J^{(c_0)}) = \chi(g)^{|J|} \sum_{J'} (U_g)_{J'J} \Theta_{c_r}(\zeta_{J'}^{(c_r)}) \quad \text{in } \mathrm{Gr}_{\mathbb{T}}.$$

Expand Υ_0. In a contributing term $|I|$ and $|J|$ have the same parity. If the total degree of the G_j is even, moving this seam monomial from the beginning to the end costs $(-1)^{|J|}$. Its complete seam factor is therefore $(-\chi(g))^{|J|}$, the diagonal of $U_{\mathrm{f}}^{(1+\chi(g))/2}$. Repeated composition gives $\mathfrak{N}_r \Upsilon_{r-1} \cdots \mathfrak{N}_1$. Because U_g is a real unitary, $(U_g)_{J'J} = (U_g^{-1})_{JJ'}$; summing the remaining indices and cyclically moving Υ_0 gives (7.11). If the total degree is odd, the Berezin integrand is odd and the matrix product anticommutes with parity, so its trace is zero. Thus an antiperiodic closing character gives an ordinary trace, and a periodic one inserts parity. □

7.4. **The Fourier boundary measure.**

Lemma 7.11. *For real y, s and $q > 0$,*

$$\sum_{j \in \mathbb{Z}} e^{-(y+2\pi j/q)^2/2 + ijs} = \frac{q}{\sqrt{2\pi}} \sum_{k \in \mathbb{Z}} e^{-\varkappa_k^2/2 - i\varkappa_k y}, \qquad \varkappa_k = \frac{q}{2\pi}(s - 2\pi k). \tag{7.12}$$

Also

$$C_{q,\varepsilon} := \sup_y \sum_{j \in \mathbb{Z}} e^{-\varepsilon(y+2\pi j/q)^2} \le 2 + \frac{q}{2\pi}\sqrt{\pi/\varepsilon} < \infty.$$

An L^2 scalar or finite-matrix kernel defines a Hilbert–Schmidt operator, with norm equal to its kernel L^2 norm. For two such operators, their product is trace class, its kernel is the integral of the two kernels, and

$$\operatorname{Tr}(XY) = \int\!\!\int \operatorname{tr}(k_X(u,v)k_Y(v,u))\, d\mu(u)d\nu(v),$$

with absolute convergence.

Proof. The Fourier transform of $e^{-x^2/2}$ is $\sqrt{2\pi}e^{-t^2/2}$. Applying Fourier series to its $2\pi/q$ periodization, with the character e^{ijs}, gives (7.12); both series are absolutely convergent. For the Gaussian sum remove the at most two terms nearest its maximum and bound each remaining monotone tail by the Gaussian integral divided by the spacing $2\pi/q$. This proves the bound. For the kernel assertions choose orthonormal bases of the two L^2 spaces and apply Parseval entry by entry to the kernel. The sum of squared matrix coefficients is its squared L^2 norm. First compute composition and trace for finite-rank kernels; L^2 approximation and Cauchy–Schwarz give both formulas and absolute convergence in the limit. Indeed $\|XY\|_1 \le \|X\|_2\|Y\|_2$, and the integral of the modulus of the product kernels has this same bound. □

Definition 7.12. At a mirror c use variables $(\vartheta, k) \in [0,2\pi)^{\mathcal{X}^3(c)} \times \mathbb{Z}^{\mathcal{X}^2(c)}$ and set

$$\varkappa = \frac{q}{2\pi}(d_c^*\vartheta - 2\pi k),$$
$$d\mu_c^\times(\vartheta,k) = \prod_{Q\in\mathcal{X}^3(c)} \frac{d\vartheta_Q}{2\pi} \prod_{\sigma\in\mathcal{X}^2(c)} \frac{q}{\sqrt{2\pi}} e^{-\varkappa_\sigma^2/2},$$
$$\mathfrak{e}_c((\vartheta,k);\varpi) = \exp\{i\langle\vartheta, d_c^+ z\rangle - i\langle\varkappa, d_c^+ A\rangle\}.$$

The integer variables carry counting measure. For $I \subset \mathcal{J}(c)$ let $\mathfrak{q}(I)(b)$ count selected modes over a cut link b in groups 1 and 2 positively and those in groups 3 and 4 negatively. Define

$$\mathfrak{D}_c(k,I) = (2\pi/q)^{|\mathcal{X}^1(c)|} \prod_{b\in\mathcal{X}^1(c)} \mathbf{1}[(d_c^*k)(b) + \mathfrak{q}(I)(b) = 0].$$

When no link is cut this factor is one. Only coordinate link planes cut links. There $u_I(A) = e^{iq\langle\mathfrak{q}(I),A\rangle}$; at all other mirrors $u_I = 1$.

The phase has modulus one. The Fourier measure is finite, since (7.12) with $y = 0$ gives uniformly in ϑ

$$\frac{q}{\sqrt{2\pi}} \sum_{k_\sigma\in\mathbb{Z}} e^{-\varkappa_\sigma^2/2} \le \sum_{j\in\mathbb{Z}} e^{-2\pi^2 j^2/q^2}. \tag{7.13}$$

At a coordinate site plane all cut sets are empty, so this measure is a unit mass and the phase is one.

Lemma 7.13. *Fix the configuration on the upper and lower faces of the cells cut by c. Then, with the field strength and closedness computed from these faces and the variables $A^\times, z^\times$ on the cut cells,*

$$\sum_{z^\times} \int_{[0,2\pi/q)^{\mathcal{X}^1(c)}} dA^\times \prod_{\sigma\in\mathcal{X}^2(c)} e^{-F_\sigma^2/2} \prod_{Q\in\mathcal{X}^3(c)} \mathbf{1}[(dz)(Q) = 0] u_I(A^\times)$$
$$= \int d\mu_c^\times(\vartheta,k)\, \mathfrak{D}_c(k,I) \mathfrak{e}_c((\vartheta,k);\varpi)\overline{\mathfrak{e}_c((\vartheta,k);\theta_c^*\varpi)}. \tag{7.14}$$

Both sides converge absolutely, uniformly in the face data.

Proof. By (7.1), write

$$F = A^\partial + d_c A^\times + 2\pi z^\times/q, \qquad dz = z^\partial + d_c z^\times,$$

where $A^\partial = d_c^+ A - d_c^+\theta_c^* A$ and $z^\partial = d_c^+ z - d_c^+\theta_c^* z$. Use $\mathbf{1}[j=0] = \int_0^{2\pi} e^{i\vartheta j} d\vartheta/(2\pi)$ at every cut cube. The integer sum at each cut plaquette is now (7.12) with $s = (d_c^*\vartheta)_\sigma$ and $y = A_\sigma^\partial + (d_c A^\times)_\sigma$. After taking these sums, the dependence on the cut links is

$$e^{-i\langle d_c^*\varkappa, A^\times\rangle} e^{iq\langle \mathfrak{q}(I), A^\times\rangle} = e^{iq\langle d_c^* k + \mathfrak{q}(I), A^\times\rangle},$$

since $d_c^* d_c^* = 0$. Its compact integral is precisely $\mathfrak{D}_c(k, I)$, including its factor $2\pi/q$ per link. The remaining face factor is $e^{i\langle\vartheta, z^\partial\rangle - i\langle\varkappa, A^\partial\rangle}$, the product of the two phases in (7.14). Before the transformation, the absolute sum is bounded by $(2\pi/q)^{|\mathcal{X}^1|} C_{q,1/2}^{|\mathcal{X}^2|}$. After it, the phases have modulus one, the Gauss factor is bounded, and (7.13) bounds the measure. These bounds justify all rearrangements and are uniform in the face data. □

Lemma 7.14. *An interior integer shift does not change either boundary phase of a slab. Translations rename the Fourier variables and preserve $\mu_c^\times$, $\varkappa$ and the Gauss factor, with the mode index changed by r_v.*

Proof. At the lower boundary suppose a vanishes on the cut links and the closed lower half. The change in the exponent of $\mathfrak{e}_c$ under $[a]$ is, modulo $2\pi i\mathbb{Z}$,

$$-i\langle\vartheta, d_c^+(da) + d_c(d_c^+ a)\rangle = 0$$

by (7.3). Reflecting proves the assertion at the upper boundary. Translation preserves oriented incidence and upper faces; hence it commutes with d_c^+ and d_c^* when the mirror is translated as well. The formulas in Definition 7.12 then prove the measure and phase claims. A translated mode preserves its group and lies over the translated link, so the charge and Gauss factor transform with r_v. □

7.5. **Mirror Hilbert spaces and transfer kernels.**

Definition 7.15. The mirror variable is $\xi = (\mathfrak{m}, \vartheta, k, I)$, where A in $\mathfrak{m}$ ranges over $[0, 2\pi/q)^{\mathcal{M}^1(c)}$, its z ranges over $\mathbb{Z}^{\mathcal{M}^2(c)}$, and $I \subset \mathcal{J}(c)$. On their space $\mathcal{Y}_c$ use the positive measure

$$d\mu_c(\xi) = dA \prod_{\sigma\in\mathcal{M}^2(c)} e^{-F_\sigma^2/2} d\mu_c^\times(\vartheta, k)\mathfrak{D}_c(k, I), \tag{7.15}$$

with counting measure for z and I. Put $\mathcal{H}(c) = L^2(\mathcal{Y}_c, \mu_c)$ and $U_{\mathrm{f}} f(\xi) = (-1)^{|I|} f(\xi)$. If

$$\xi^v = (\tau_{-v}^*\mathfrak{m}, \tau_{-v}^*\vartheta, \tau_{-v}^* k, r_v(I)),$$

define $(V_v f)(\xi^v) = s_v(I) f(\xi)$.

The measure is finite, with the explicit bound

$$\mu_c(\mathcal{Y}_c) \le (2\pi/q)^{|\mathcal{M}^1|+|\mathcal{X}^1|} C_{q,1/2}^{|\mathcal{M}^2|} \left(\sum_{j\in\mathbb{Z}} e^{-2\pi^2 j^2/q^2}\right)^{|\mathcal{X}^2|} 2^{|\mathcal{J}(c)|}. \tag{7.16}$$

Indeed sum the mirror plaquette variables at fixed links using the Gaussian-sum bound, then use (7.13) and the bound on the Gauss factor. It is a Borel measure on a countable union of finite boxes, so its L^2 space is separable. Translation carries μ_c to $\mu_{c+\ell(v)}$. Thus V_v is a unitary; (7.5) gives its group law and parity commutation. For $v \in \Gamma_\parallel$, it is the identity on even functions and multiplication by $\chi_\parallel(v)$ on odd functions. Zero Gauss weights are harmless: they remove those mode components from the L^2 space.

Definition 7.16. For $G \in \mathcal{I}_\Xi[c, c']$ define the kernel, with $\varpi = \mathfrak{m} \vee \mathring{\varpi} \vee \mathfrak{m}'$,

$$\mathfrak{S}[G](\xi', \xi) = \sqrt{\omega_{I'}\omega_I} \int d\mathring{\varpi}\, \mathcal{W}_{c,c'}(\varpi)\mathfrak{e}_c((\vartheta, k); \varpi)\overline{\mathfrak{e}_{c'}((\vartheta', k'); \theta_{c'}^*\varpi)} \cdot \mathfrak{N}_{c,c'}[G(\varpi)e^{-S_{(c,c')}(A)}]_{I',I}. \tag{7.17}$$

It acts from $\mathcal{H}(c)$ to $\mathcal{H}(c')$ by integration against μ_c. The square root is the nonnegative one. For an admissible product use its evaluation from Lemma 7.5.

Theorem 7.17. *The kernel is bounded and Hilbert–Schmidt. Its dependence on G is linear. For homogeneous G it has the corresponding parity. For adjacent slabs, and for every integer translation,*

$$\mathfrak{S}[G']\mathfrak{S}[G] = \mathfrak{S}[GG'], \tag{7.18}$$

$$V_v \mathfrak{S}[G] V_v^{-1} = \mathfrak{S}[\tau_v G], \tag{7.19}$$

$$\mathfrak{S}[\Theta_{c'} G] = V_{v_{c,c'}} \mathfrak{S}[G]^*. \tag{7.20}$$

If g closes a torus and $c_0 < \cdots < c_r = c_0 + \ell(g)$, $r \geq 2$, then

$$Z_{\mathbb{T}}[G_1 \cdots G_r] = \mathrm{Tr}_{\mathcal{H}(c_0)} \left(V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} \mathfrak{S}[G_r] \cdots \mathfrak{S}[G_1] \right). \tag{7.21}$$

The product on the left is read on the torus. These identities hold for arbitrary entries, by decomposition into their even and odd parts. They hold for all real M, and also when some crossing coefficients vanish.

Proof. The action of a finite open slab is quadratic in its finitely many Grassmann generators. Its coefficients are polynomials in link phases of modulus one. Its exponential is a finite polynomial and all its coefficients are uniformly bounded in A. Multiplying by the fixed endpoint monomials and extracting a Berezin coefficient therefore bounds the matrix entry in the kernel by

$$C_0 C \exp\left((\tfrac{1}{2} - \varepsilon_G) \sum_{\mathcal{B}^2} F_\sigma^2 \right).$$

The phases have modulus one. Drop the closure indicators and sum the resulting majorant over the independent plaquette integers. Each plaquette contributes at most C_{q,ε_G}, and the compact link integrals contribute $(2\pi/q)^{|\mathcal{B}^1|}$. The finite endpoint weights can be absorbed into $C_{c,c'}$, giving

$$|\mathfrak{S}[G](\xi', \xi)| \leq C_{c,c'} C C_{q,\varepsilon_G}^{|\mathcal{B}^2(c,c')|}. \tag{7.22}$$

The bound is uniform in both boundaries and is also a uniform bound on the absolute interior integral. Borel measurability follows by integration of the Borel coefficients. The finite boundary measures and Lemma 7.11 now give the Hilbert–Schmidt property, with norm at most the right side of (7.22) times $\sqrt{\mu_c(\mathcal{Y}_c)\mu_{c'}(\mathcal{Y}_{c'})}$. Linearity follows from absolute convergence. The matrix parity rule passes directly to the kernel.

For composition, expand both kernels and integrate their common boundary ξ'. At fixed I' the Fourier variables occur only in

$$\int d\mu_{c'}^{\times}\, \mathfrak{D}_{c'}(k', I') \mathfrak{e}_{c'}(\cdot; \varpi_2) \overline{\mathfrak{e}_{c'}(\cdot; \theta_{c'}^* \varpi_1)}.$$

Replace this by the left side of (7.14). The two endpoint square roots multiply to $\sqrt{\omega_{I''}\omega_I}\,\omega_{I'}$. The sum over I' is consequently matrix composition with $\omega_{c'}\,\mathrm{diag}(u_{I'})$, namely insertion of $e^{-S_{c'}^{\times}}$. Equation (7.8) and the split of the action give the slab matrix of $GG' e^{-S_{(c,c'')}}$, in precisely this order. The weights of the two interiors, the mirror plaquettes, and the newly restored cut cells combine to $\mathcal{W}_{c,c''}$ by (7.2). Every interior link and plaquette appears once, and every cube constraint appears once, since no cube lies in a mirror. The two outer phases read the same faces in the assembled slab. This is the kernel of $\mathfrak{S}[GG']$. The uniform absolute interior bound and the finite common boundary measure justify Fubini at each step; the restored crossing integral has its own uniform absolute bound in Lemma 7.13.

For translation use the substitution $\varpi \mapsto \tau_{-v}^* \varpi$. It just renames links, plaquettes and cubes, preserving their ranges and weights. The Fourier phases transform by Lemma 7.14; the matrix

acquires $s_v(I')s_v(I)$ by (7.10). The kernel identity is

$$\mathfrak{S}[\tau_v G](\xi'^v, \xi^v) = s_v(I')s_v(I)\mathfrak{S}[G](\xi', \xi),$$

which is (7.19).

For reflection use $\check{\varpi} = \theta_{c'}^* \varpi$ on $[c', 2c' - c]$. The mirror data at c' are fixed with their orientations; those at c are translated by $v_{c,c'}$. The interior pullback is a signed permutation of coordinates. It preserves the weights, exchanges and conjugates the two phases, and conjugates the matrix by (7.9). A reflected interior link interval can be $(-2\pi/q, 0]$ rather than $[0, 2\pi/q)$. The complete interior integrand is invariant under an integer shift of that link: the weight depends on F, dz, the action on phases, G is shift invariant, and the two boundary phases are invariant by Lemma 7.14. Its absolute integral is bounded by the preceding estimate. Lemma 7.3 therefore restores the original compact intervals. We obtain the exact kernel identity

$$\mathfrak{S}[\Theta_{c'} G](\xi^{v_{c,c'}}, \xi') = \overline{\mathfrak{S}[G](\xi', \xi)},$$

which proves (7.20).

It remains to close the cylinder. Composition reduces the claim to two slabs. A product of two Hilbert–Schmidt operators is trace class, so its trace is the absolutely convergent double kernel integral. Set ξ_0^g to be the translated mirror variable at $c_2 = c_0 + \ell(g)$. The twisted first factor has kernel

$$(V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} \mathfrak{S}[G_2])(\xi_0, \xi_1) = (-\chi(g))^{|I_0|} s_g(I_0) \mathfrak{S}[G_2](\xi_0^g, \xi_1).$$

This identifies both the seam translation and its parity sign. Apply (7.14) at c_1 and at c_0; at the latter the lower face data are $\tau_g^* \varpi_2$. Indeed $\tau_g^* \theta_{c_2}^* = \theta_{c_0}^* \tau_g^*$, so the phases there are exactly the paired phases of that crossing identity. The new cut-link factors are u_{I_0} and u_{I_1}. The mode sum is

$$\operatorname{tr}\left(U_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} \mathfrak{N}_2 \omega_{c_1} \operatorname{diag}(u_I) \mathfrak{N}_1 \omega_{c_0} \operatorname{diag}(u_I) \right),$$

which by (7.11) is the Berezin integral of $G_1 G_2$ times the exponential of the full torus action. If its degree is odd both expressions vanish; the same computation therefore applies without a parity restriction.

Finally the cylinder cells in

$$\bigcup_{j=0}^{1} (\mathcal{M}(c_j) \cup \mathcal{X}(c_j)) \ \cup \ \mathcal{B}(c_0, c_1) \cup \mathcal{B}(c_1, c_2)$$

are one set of representatives of all torus cells: translation by g advances the corresponding sets by one full period. Thus each link has one compact integral and each plaquette one integer sum. The weights are exactly $e^{-\|F\|^2/2}$ and the indicator $dz = 0$ on every torus cube. This proves (7.21), including all flux sectors. The product satisfies the torus growth bound by disjointness of the interior plaquette sets, so its left side is well defined. Expanding each entry into even and odd parts proves the inhomogeneous statement. □

Definition 7.18. For $G \in \mathcal{I}_\Xi[c, c']$, its reflection in the lower mirror is the member of $\mathcal{I}_\Xi[2c - c', c]$ defined by

$$(\Theta_c G)(\varpi') = \Theta_c(G(\theta_c^* \varpi')).$$

Pullback bijects the two configuration spaces, preserves integer shifts and the interior field strength norm, so the reflected coefficients have the same positive decay margin in (7.4). The map is an antilinear involution and preserves parity. Its transfer operator satisfies

$$\mathfrak{S}[\Theta_c G] = \mathfrak{S}[G]^* V_v, \qquad v = 2(c' - c)n/|n|^2. \tag{7.23}$$

Indeed apply (7.20) to the lower reflected slab: $\mathfrak{S}[G] = V_v \mathfrak{S}[\Theta_c G]^*$. Taking adjoints and using unitarity proves the displayed formula. Here V_v maps $\mathcal{H}(2c - c')$ to $\mathcal{H}(c')$; the composite thus has exactly the required boundary spaces.

Corollary 7.19. *All composition, translation, reflection and closing-trace identities of Theorem 7.17 hold for the admissible local products $\mathcal{A}_{\Xi}[c, c']$ and their finite sums. In a closed chain their product is read on the torus, belongs to $\mathcal{A}(\mathbb{T})$, and is evaluated in $\mathcal{I}(\mathbb{T})$. It is not required to embed with its enlarged boxes.*

Proof. Evaluation commutes with these operations by Lemma 7.5. Boxes in different open slabs cannot meet after a period translation except when they belong to the same slab and differ by an element of $\Gamma_{\|}$; in that case the stipulated embedding condition excludes their intersection. Each box thus retains its prescribed evaluation on the torus. Apply the theorem to these evaluations and use multilinearity for finite sums. □

7.6. **Coordinate slice factorisation.** Throughout this subsection and the following slice arguments fix $q > 0$, $w_t > 0$, $M \in \mathbb{R}$ and $|\rho| \le 1$. Fix a rectangular torus $C(n_1, n_2, n_3)$, with $n_\mu \ge 4$, and take e_1 as time. Write $s = n_2 n_3$ and $V = n_1 s$. The transverse torus is Λ_{s}; its $2s$ links carry A_j, and the s time links from slice j to slice $j+1$ carry a_j. The conventions of Definitions 6.10–6.12 give

$$F_j^{\mathrm{el}} = A_{j+1} - A_j - d_{\mathrm{s}} a_j, \qquad F_j^{\mathrm{mag}} = d_{\mathrm{s}} A_j$$

for an exact field strength. The site-plane and link-plane Fock spaces are $\mathcal{F}'$ and $\mathcal{F}$, respectively. In $\mathrm{Ins}[\Upsilon] = \sum_{I,J} \Upsilon_{IJ} \Theta(\zeta_J)\zeta_I$, the row I is on the upper side and the column J on the lower side. In particular the reflected ordered monomial has its factors in reverse order. This convention fixes the signs in the following identities.

Let $S_j^{\times}$ and $S_{j+1/2}^{\times}$ denote the crossing actions at the indicated planes. The crossing expansion gives

$$e^{-S_j^{\times}} = \mathrm{Ins}_j[\omega'], \qquad e^{-S_{j+1/2}^{\times}} = \mathrm{Ins}_{j+1/2}[\omega^{1/2} u(a_j) \omega^{1/2}]. \tag{7.24}$$

The first formula has no phase: both ends of a site-plane crossing bond have the same nearest photon site. The coefficients $\omega_i = w_t(1 \pm \rho)/2$ are nonnegative for $|\rho| \le 1$, so all square roots here are the nonnegative diagonal square roots.

Proposition 7.20 (Slice factorisation)**.** *Let $q > 0$, $w_t > 0$, $M \in \mathbb{R}$ and $|\rho| \le 1$. For even $\Upsilon_j \in \mathrm{End}\,\mathcal{F}$ and a real torus one-cochain A, one has*

$$\begin{aligned} &\int_{C(n)} e^{-S_{C(n)}(A)} \prod_{j=0}^{n_1-1} \left(e^{S_{j+1/2}^{\times}(A)} \,\mathrm{Ins}_{j+1/2}[\omega^{1/2} \Upsilon_j \omega^{1/2}] \right) \\ &\qquad = \mathrm{tr}_{\mathcal{F}'} \prod_{j=n_1-1,\dots,0} \left(\widehat{\mathfrak{N}}(A_{j+1})^* \Upsilon_j \widehat{\mathfrak{N}}(A_j) \right), \qquad A_{n_1} = A_0. \end{aligned} \tag{7.25}$$

Later times stand to the left. For $\Upsilon_j = u(a_j)$ the left member is $\det D_A^{C(n)}$.

Proof. Every action term belongs either to one of the open half slabs $(j, j+1/2)$, $(j+1/2, j+1)$ or to their intervening crossing planes: no fermion site lies on a plane and each bond crosses at most one plane. Thus the action is the sum of these pieces. All pieces are even, and the factors $e^{S_{j+1/2}^{\times}}$ cancel the link-plane crossing exponentials. The remaining site-plane factors are $\mathrm{Ins}_j[\omega']$. The lower half-slab matrix is $\mathfrak{N}(A_j)$. Reflection in $j+1/2$ carries the lower half-slab action with slice A_{j+1} to the upper half-slab action; the adjoint rule of Lemma 7.10 therefore makes the upper matrix $\mathfrak{N}(A_{j+1})^*$, after translation identifies the two site-plane Fock spaces.

Apply the composition and closure rules of the same lemma to the chain $0, 1/2, 1, \dots, n_1$. The boundary character satisfies $\chi(n_1 e_1) = -1$, so its closing parity factor is the identity. The resulting trace is

$$\mathrm{tr}_{\mathcal{F}'} \prod_{j=n_1-1,\dots,0} \left(\mathfrak{N}(A_{j+1})^* \omega^{1/2} \Upsilon_j \omega^{1/2} \mathfrak{N}(A_j) \omega' \right).$$

Split every ω' into two square roots and cyclically move the final square root to the front. Since $\widehat{\mathfrak{N}} = \omega^{1/2}\mathfrak{N}\omega'^{1/2}$, this is the stated trace. No odd factor was interchanged. For $\Upsilon_j = u(a_j)$, (7.24) restores the canceled crossing exponentials and the Berezin determinant identity proves the last assertion. □

Lemma 7.21 (Half-slab matrices). *(a) The entries of $\widehat{\mathfrak{N}}(A)$ are polynomials in $e^{\pm iqA(b)}$. Consequently this matrix is continuous, bounded and invariant under $A \mapsto A + 2\pi k/q$, with integer k.*

(b) A site-plane mode over y has charge $+\delta_y$ in groups $1,2$ and $-\delta_y$ in groups $3,4$. Add these charges over I' to define $u'(\lambda)e_{I'} = e^{iq\langle\lambda,\mathfrak{q}'(I')\rangle}e_{I'}$. Then

$$\widehat{\mathfrak{N}}(A - d_{\mathrm{s}}\lambda) = u(\lambda)^{-1}\widehat{\mathfrak{N}}(A)u'(\lambda). \tag{7.26}$$

In particular $\mathfrak{M}\mathfrak{M}^*$ commutes with every $U_{\mathrm{G}}(\lambda)$ and with P_{G}.

Proof. The half-slab action is quadratic in finitely many Grassmann generators and polynomial in the link phases. Its exponential is a finite polynomial, and taking its Berezin coefficient preserves polynomial dependence. On real A every phase has modulus one, which proves boundedness. The change of variables $\psi(x) \mapsto e^{iq\lambda(\pi(x))}\psi(x)$, $\bar\psi(x) \mapsto e^{-iq\lambda(\pi(x))}\bar\psi(x)$ preserves the Berezin top monomial and the action. In the defining entry

$$\mathfrak{N}(A)_{I,I'} = \int_{\mathbb{X}_{\mathrm{h}}} \zeta'_{I'} e^{-S_{\mathrm{h}}(A)}\Theta(\zeta_I)$$

the boundary monomials acquire respectively $e^{iq\langle\lambda,\mathfrak{q}'(I')\rangle}$ and $e^{-iq\langle\lambda,\mathfrak{q}(I)\rangle}$. This proves (7.26); its diagonal weights commute with the charge representations. The magnetic factor is unchanged by $A \mapsto A - d_{\mathrm{s}}\lambda$. Thus $\mathfrak{M}_k(A - d_{\mathrm{s}}\lambda)\mathfrak{M}_k(A - d_{\mathrm{s}}\lambda)^* = u(\lambda)^{-1}\mathfrak{M}_k(A)\mathfrak{M}_k(A)^*u(\lambda)$, which proves the commutation statement by the definition of U_{G} and Haar averaging. □

7.7. **Localisation of even Fock operators.** The signs in localisation are useful both for the word formula and for transporting a local insertion through a periodic seam. For disjoint ordered sets I, R put $\epsilon(I,R) = (-1)^{\#\{(i,r)\in I\times R : i>r\}}$. Then $\zeta_{I\cup R} = \epsilon(I,R)\zeta_I\zeta_R$ and $\epsilon(I,R)\epsilon(R,I) = (-1)^{|I||R|}$.

Proposition 7.22 (Localisation). *(a) For disjoint I, R and J, R with $|I| + |J|$ even,*

$$\Theta(\zeta_{J\cup R})\zeta_{I\cup R} = \epsilon(I,R)\epsilon(J,R)\Theta(\zeta_J)\zeta_I\Theta(\zeta_R)\zeta_R.$$

(b) For an ordered subset $\mathcal{J}' \subset \mathcal{J}$, the map

$$\iota(T)_{I\cup R, J\cup R'} = \delta_{RR'}\epsilon(I,R)\epsilon(J,R)T_{IJ}$$

is a unital isometric $*$-homomorphism. It is transitive under inclusion.

(c) Localisations on disjoint subsets commute whenever at least one of the two operators is even.

(d) For an even T, multiplicative diagonal weights ω_I, u_I, and $\mathcal{J}'' = \mathcal{J} \setminus \mathcal{J}'$, one has

$$\begin{aligned}&\mathrm{Ins}^{\mathcal{J}}[\omega^{1/2}\iota(T)\,\mathrm{diag}(u_I)\omega^{1/2}]\\ &\quad= \mathrm{Ins}^{\mathcal{J}'}[\omega^{1/2}T\,\mathrm{diag}(u_I)\omega^{1/2}]\prod_{i\in\mathcal{J}''}(1 + \omega_i u_i\Theta(\zeta_i)\zeta_i).\end{aligned} \tag{7.27}$$

(e) For several disjoint subsets and even T_l, the first factor on the right is replaced by the product of their insertions and the last product runs over the unused modes.

(f) A signed relabelling $\zeta_I = \sigma(I)\widetilde{\zeta}_{r(I)}$ is implemented by the unitary $Ue_I = \sigma(I)e_{r(I)}$; it carries T to UTU^* and preserves all these identities.

Proof. The unitary $U_\otimes(e_I\otimes e_R) = \epsilon(I,R)e_{I\cup R}$ implements $\iota(T) = U_\otimes(T\otimes 1)U_\otimes^*$. Under the canonical map $e_I \mapsto \zeta_I$, it implements multiplication $\zeta_I\zeta_R$. This proves the homomorphism, norm and

transitivity assertions. The commutator on disjoint sets has precisely the sign $(-1)^{(|I_1|+|J_1|)(|I_2|+|J_2|)}$; a nonzero entry of an even operator has its indicated sum even, so the sign is one.

For a nonzero entry T_{IJ}, $|I|+|J|$ is even. Since Θ reverses products, sorting gives

$$\Theta(\zeta_{J\cup R})\zeta_{I\cup R} = \epsilon(I,R)\epsilon(J,R)\Theta(\zeta_J)\zeta_I\Theta(\zeta_R)\zeta_R.$$

The two sorting signs cancel those in the matrix entry. Sum over R and use $\Theta(\zeta_R)\zeta_R = \prod_{i\in R}\Theta(\zeta_i)\zeta_i$ to obtain (7.27). Applying the same identity successively proves the assertion for several subsets; all inserted elements are even. Finally $\Theta(\zeta_J)\zeta_I = \sigma(I)\sigma(J)\Theta(\widetilde{\zeta}_{r(J)})\widetilde{\zeta}_{r(I)}$. These are exactly the signs in UTU^*, proving relabelling. □

7.8. **Unfolding the integer fluxes.** Put $X = \mathbb{R}^{\Lambda_{\mathrm{s}}^{(1)}}$ and $\Gamma_{\mathrm{s}} = (2\pi/q)(\mathbb{Z}^{\Lambda_{\mathrm{s}}^{(1)}} \cap \ker d_{\mathrm{s}})$, and set $Q = X/\Gamma_{\mathrm{s}}$. This quotient is noncompact in the transverse field strength directions. All its integrals use the measure induced by Lebesgue measure, without a normalization factor.

Lemma 7.23 (Unfolding)**.** *A path is $(A_0,\dots,A_{n_1};a_0,\dots,a_{n_1-1})$ with $A_j\in X$, $A_{n_1}=A_0+g$, $g\in\Gamma_{\mathrm{s}}$, and $a_j\in[0,2\pi/q)^{\Lambda_{\mathrm{s}}}$. Define its measure by*

$$\int d\mathsf{p} = \int_Q d[A_0]\sum_{g\in\Gamma_{\mathrm{s}}}\int_{X^{n_1-1}}\prod_{j=1}^{n_1-1}dA_j\int\prod_{j=0}^{n_1-1}da_j. \tag{7.28}$$

For each $k\in\mathbb{Z}$, the path represents a configuration with

$$F_j^{\mathrm{el}} = A_{j+1}-A_j-d_{\mathrm{s}}a_j, \qquad B_j = F_j^{\mathrm{mag}} = d_{\mathrm{s}}A_j + (2\pi/q)k1_{\sigma_0}. \tag{7.29}$$

Every nonnegative invariant Borel function Φ on the configuration representatives satisfies

$$\int_{\mathrm{Conf}(\Gamma)}\Phi = \sum_{k\in\mathbb{Z}}\int d\mathsf{p}\,\Phi(\varpi(\mathsf{p},k)). \tag{7.30}$$

The same identity holds with absolute convergence for every integrable complex Φ.

Proof. For a closed integer two-cochain write its slice components as z_j^{mag} and z_j^{el}. Closedness is $z_{j+1}^{\mathrm{mag}} - z_j^{\mathrm{mag}} = d_{\mathrm{s}}z_j^{\mathrm{el}}$. Set $h_0=0$ and $h_{j+1}=h_j+z_j^{\mathrm{el}}$. It follows that $z_j^{\mathrm{mag}} = z_0^{\mathrm{mag}} + d_{\mathrm{s}}h_j$ and $d_{\mathrm{s}}h_{n_1}=0$. Conversely these relations recover every closed z uniquely from z_0^{mag} and $h_1,\dots,h_{n_1}$.

The integer shift with spatial slice h_j adds $2\pi h_j/q$ to A_j and replaces z by a cochain whose magnetic slices are all z_0^{mag} and whose electric slices are zero except at the last step, where they equal h_{n_1}. The sums over h_j, $1\le j<n_1$, unfold their angular boxes to X, and $g=2\pi h_{n_1}/q$ is the path's closing vector.

The integer range of d_{s} on the two-dimensional torus consists exactly of the integer plaquette functions with sum zero. Here is an elementary integral proof. Orient the dual graph so that the column of a primal link is the incidence column of its dual edge. Choose a spanning tree and a root. Starting at the leaves, transport each prescribed integer plaquette value to its parent along the unique tree edge. The remaining value at the root is the total sum and is zero. The resulting integer edge flow solves the incidence equation. Thus $z_0^{\mathrm{mag}} = k1_{\sigma_0} + d_{\mathrm{s}}h$, where $k=\sum_\sigma z_0^{\mathrm{mag}}(\sigma)$ and h is unique modulo closed integer one-cochains. Shift every spatial slice by $2\pi h/q$. The angular box for A_0, together with the sum over these cosets, becomes a fundamental domain for Γ_{s}. Indeed the boxes $[0,2\pi/q)^{2s}+2\pi h/q$ tile X, and grouping them by closed integer shifts produces precisely that quotient domain. This proves (7.30). Translation of all A_j by Γ_{s} leaves its integrated integrand unchanged, so $[A_0]$ is legitimate.

The representative $\varpi(\mathsf{p},k)$ has magnetic integers $k1_{\sigma_0}$ and electric integers zero except for $qg/(2\pi)$ at the closing step. Since $d_{\mathrm{s}}g=0$, it is closed and has exactly (7.29). All transformations preserve Lebesgue or counting measure. Tonelli proves the nonnegative case; applying it first to $|\Phi|$ then using Fubini proves the complex case. □

7.9. **Gaussian kernels and trace class estimates.** For a vector $x \in X$ abbreviate the product of the scalar Gaussian densities by $g_a(x) = \prod_b g_a(x_b)$, and $f^{\mathrm{per}}([x]) = \sum_{g\in\Gamma_{\mathrm{s}}} f(x+g)$. The coordinate electric generators are $E_b = -i\partial_{A(b)}$, so $e^{-i\langle x,E\rangle}\Phi(A) = \Phi(A-x)$. The Haar probability measure of the slice gauge group is

$$d\mu_{\mathrm{G}}(a) = (q/2\pi)^s\, da, \qquad a \in [0, 2\pi/q)^{\Lambda_{\mathrm{s}}}.$$

Thus $P_{\mathrm{G}} = \int U_{\mathrm{G}}(a) d\mu_{\mathrm{G}}(a)$, where $U_{\mathrm{G}}(a)\Phi(A) = u(a)\Phi(A - d_{\mathrm{s}}a)$. Raw angular integration contributes the factor $c_{\mathrm{G}} = (2\pi/q)^s$.

Lemma 7.24 (Convolutions and Gaussian kernels)**.** *(a) For $f \in L^1(X)$, convolution on Q with kernel $f^{\mathrm{per}}([A'-A])$ equals $\widehat{f}(E)$, where $\widehat{f}(\xi) = \int_X f(x)e^{-i\langle\xi,x\rangle}dx$.*

(b) In particular

$$(e^{i\langle\beta,x\rangle}g_a(x))^{\mathrm{per}} \longleftrightarrow e^{-a|E-\beta|^2/2}, \tag{7.31}$$

$$(g_a(x+\beta))^{\mathrm{per}} \longleftrightarrow e^{-a|E|^2/2}e^{i\langle\beta,E\rangle}. \tag{7.32}$$

(c) Functions of E commute with U_{G} and P_{G}. For $S \subset \Lambda_{\mathrm{s}}^{(1)}$,

$$K_{\beta,S}^{1/2} = c_{\mathrm{G}}^{1/2} e^{-|E-\beta|^2/4 + |E-\beta|_S^2/8} P_{\mathrm{G}}. \tag{7.33}$$

(d) For $\varepsilon > 0$ and a real magnetic slice η,

$$\sum_{k\in\mathbb{Z}} \int_Q e^{-\varepsilon\|d_{\mathrm{s}}A + (2\pi/q)k1_{\sigma_0} + \eta\|^2}\, d[A] < \infty. \tag{7.34}$$

(e) If a flux-diagonal kernel is bounded in norm by $h^{\mathrm{per}}([A'-A])$, where $0 \le h(x) \le Ce^{-\varepsilon|x|^2}$, and a multiplication operator T into $\mathcal{H}^1$ obeys $\sum_k \int_Q \|T_k(A)\|^2 d[A] < \infty$, then $K'T$ and $K'P_{\mathrm{G}}T$ are Hilbert–Schmidt. Their norms are at most

$$\sqrt{\dim \mathcal{F}}\, \|h^{\mathrm{per}}\|_{L^2(Q)} \left(\sum_k \int_Q \|T_k(A)\|^2 d[A] \right)^{1/2}.$$

Proof. Integrating $f(x)e^{-i\langle x,E\rangle}$ and applying the joint spectral theorem gives $\widehat{f}(E)$. Partitioning X into translates of a quotient domain gives its stated kernel. The Fourier transform of the normalized Gaussian is $e^{-a|\xi|^2/2}$; modulation replaces ξ by $\xi - \beta$, whereas translating the Gaussian by $+\beta$ gives the phase $e^{+i\langle\beta,\xi\rangle}$. This proves both signs. The equality $U_{\mathrm{G}}(a) = u(a)e^{-i\langle d_{\mathrm{s}}a,E\rangle}$ proves commutation. The charges of $u(a)$ are integer valued, and $d_{\mathrm{s}}m$ is an integer closed one-cochain for an integer zero-cochain m. Hence $U_{\mathrm{G}}(a + 2\pi m/q) = U_{\mathrm{G}}(a)$ on the quotient, and $U_{\mathrm{G}}(a+b) = U_{\mathrm{G}}(a)U_{\mathrm{G}}(b)$. It is therefore a unitary representation of the compact group $(\mathbb{R}/(2\pi/q)\mathbb{Z})^{\Lambda_{\mathrm{s}}}$. Normalized Haar averaging is a self-adjoint projection: replacing a by $-a$ in the average proves $P_{\mathrm{G}}^* = P_{\mathrm{G}}$, and replacing (a,b) by $(a+b,b)$ in its square proves $P_{\mathrm{G}}^2 = P_{\mathrm{G}}$. The same change of variables shows that its range consists of invariant vectors; it fixes every such vector. Consequently all factors in (7.33) commute and are nonnegative, and squaring proves that formula.

The adjoint d_{s}^* has kernel the constant plaquette functions, so $\operatorname{ran} d_{\mathrm{s}} = 1^{\perp}$. Orthogonal decomposition and translation of A therefore give

$$\int_Q e^{-\varepsilon\|d_{\mathrm{s}}A + (2\pi/q)k1_{\sigma_0} + \eta\|^2} d[A] = C_\varepsilon \exp\left\{ -\frac{\varepsilon}{s}\left(\frac{2\pi k}{q} + \sum_\sigma \eta_\sigma \right)^2 \right\},$$

where $C_\varepsilon = \int_Q e^{-\varepsilon\|d_{\mathrm{s}}A\|^2} d[A]$ is finite. Indeed $\ker d_{\mathrm{s}}$ is a rational subspace, its integer vectors form a full lattice there, and $Q = (\ker d_{\mathrm{s}}/\Gamma_{\mathrm{s}}) \times (\ker d_{\mathrm{s}})^{\perp}$. The first factor is compact and the second has a nondegenerate Gaussian. The displayed Gaussian series in k converges, proving (7.34).

For completeness, write $A = A_0 + A_\perp$ in $\ker d_{\mathrm{s}} \oplus (\ker d_{\mathrm{s}})^\perp$ and take A_0 in a compact lattice parallelepiped of radius R. For a Gaussian $h(x) = Ce^{-a|x|^2}$,

$$\sum_{g\in\Gamma_{\mathrm{s}}} h(A+g) \le C \sum_{g\in\Gamma_{\mathrm{s}}} e^{-a(|g|-R)_+^2}.$$

The number of lattice points in a shell $m \le |g| < m+1$ is at most a fixed multiple of $(m+1)^{\dim\ker d_{\mathrm{s}}}$; disjoint small lattice balls inside a larger ball give this elementary bound. The Gaussian series therefore converges uniformly on the parallelepiped. The perpendicular factor can only decrease the bound. Unfolding gives $\|h^{\mathrm{per}}\|_1 = \|h\|_1$, and $\|h^{\mathrm{per}}\|_2^2 \le \|h^{\mathrm{per}}\|_\infty \|h^{\mathrm{per}}\|_1 < \infty$. The kernel of $K' P_{\mathrm{G}}$ is

$$c_{\mathrm{G}}^{-1} \int da\, K'([A'],[A+d_{\mathrm{s}}a])u(a). \tag{7.35}$$

Its majorant is the average of translates of h^{per} and has no larger L^2 norm, by Minkowski. Multiplying the kernels on the right by $T_k(A)$ and integrating their squared Hilbert–Schmidt norms proves the bound; the matrix rank is at most $\dim \mathcal{F}$. □

Corollary 7.25 (Trace class)**.** *$K_{\beta,S}^{1/2}\mathfrak{M}$ is Hilbert–Schmidt. In particular*

$$T_{\mathrm{l}} = K^{1/2}\mathfrak{M}\mathfrak{M}^* K^{1/2}, \qquad T_{\mathrm{s}} = \mathfrak{M}^* K \mathfrak{M}$$

are nonnegative trace-class operators on the link and site Hilbert spaces, respectively.

Proof. The Gaussian convolution in (7.33) has a Gaussian kernel majorant: use variance $1/2$ on unused links and $1/4$ on S. Moreover

$$\|\mathfrak{M}_k(A)\|^2 \le \sup_A \|\widehat{\mathfrak{N}}(A)\|^2 e^{-\|d_{\mathrm{s}}A + (2\pi/q)k1_{\sigma_0}\|^2/2}.$$

Lemma 7.24 proves the assertion. If $X_0 = K^{1/2}\mathfrak{M}$, the two trace-class operators are $X_0X_0^*$ and $X_0^*X_0$. □

7.10. **Paths and the word formula.**

Lemma 7.26 (A trace over paths)**.** *Let $n_1 \ge 2$. Suppose magnetic weights W_j and matrix kernels $\widetilde{K}_j$ satisfy, with fixed $C, \varepsilon > 0$,*

$$|W_j(B)| \le Ce^{-\varepsilon\|B\|^2}, \quad \|\widetilde{K}_j(A',A)\| \le Ce^{-\varepsilon|A'-A|^2}, \quad \widetilde{K}_j(A'+g, A+g) = \widetilde{K}_j(A',A).$$

The last relation holds for $g \in \Gamma_{\mathrm{s}}$. Let K_j have kernel $\sum_g \widetilde{K}_j(A'+g,A)$ on Q, and let $T_j(A,k) = W_j(B(A,k))\widehat{\mathfrak{N}}(A)\widehat{\mathfrak{N}}(A)^*$, where $B(A,k) = d_{\mathrm{s}}A + (2\pi/q)k1_{\sigma_0}$. Then $c_{\mathrm{G}}K_jP_{\mathrm{G}}T_j$ is Hilbert–Schmidt, and the absolutely convergent identity

$$\begin{aligned}\sum_k \int d\mathsf{p} \prod_j W_j(B(A_j,k))\,\mathrm{tr}_{\mathcal{F}'} \prod_{j=n_1-1,\dots,0} \Big(\widehat{\mathfrak{N}}(A_{j+1})^* \widetilde{K}_j(A_{j+1}, A_j + d_{\mathrm{s}}a_j)u(a_j)\widehat{\mathfrak{N}}(A_j)\Big) \\ = \mathrm{Tr}_{\mathcal{H}^{\mathrm{l}}} \prod_{j=n_1-1,\dots,0} (c_{\mathrm{G}}K_jP_{\mathrm{G}}T_j)\end{aligned} \tag{7.36}$$

holds.

Proof. Lemma 7.24 admits each factor and gives its kernel

$$\int da \sum_g \widetilde{K}_j(A'+g, A+d_{\mathrm{s}}a)u(a)W_j(B(A,k))\widehat{\mathfrak{N}}(A)\widehat{\mathfrak{N}}(A)^*.$$

Products of at least two Hilbert–Schmidt factors have a trace given by the integral of their cyclic kernel product. To see absolute convergence here, replace every matrix kernel by its Gaussian norm bound and every magnetic weight by $Ce^{-\varepsilon\|B\|^2}$, together with the constant bound for $\widehat{\mathfrak{N}}\widehat{\mathfrak{N}}^*$. The resulting nonnegative scalar operators H_j are also Hilbert–Schmidt. Their cyclic integral is

at most $\prod_j \|H_j\|_2$. Indeed the two-factor integral is their L^2 kernel pairing, bounded by Cauchy–Schwarz. For more factors, the convolution of the first $r-1$ kernels represents $H_{r-1}\cdots H_1$; its Hilbert–Schmidt norm is at most $\prod_{j<r}\|H_j\|_2$, since an operator norm is at most its Hilbert–Schmidt norm. Pairing this kernel with that of H_r gives the stated bound. For nonnegative kernels Tonelli justifies convolution before finiteness has been established; the bound then proves that the cyclic integral is finite. The original matrix integrand is at most $\dim\mathcal{F}$ times this majorant. Finite-rank approximation proves the two-factor trace formula in the L^2 kernel norm; kernel composition and the integrable majorant extend it to the finite product. Thus all subsequent exchanges of sums and integrals, including the flux sum, are justified by absolute convergence.

Choose a quotient domain and write $A_j = A_j^\flat + g_j$, with $g_0 = 0$ and $g_{n_1} = A_{n_1} - A_0$. Simultaneous periodicity turns a step kernel into $\widetilde{K}_j(A_{j+1}^\flat + g_{j+1} - g_j, A_j^\flat + d_{\mathrm{s}}a_j)$. The map $(g_1,\dots,g_{n_1}) \mapsto (g_1 - g_0,\dots,g_{n_1} - g_{n_1-1})$ is a bijection, so the periodised cyclic integral unfolds to the path integral. Finally move the last $\widehat{\mathfrak{N}}(A_0)^*$ across the finite-dimensional trace. Since $\widehat{\mathfrak{N}}(A_{n_1}) = \widehat{\mathfrak{N}}(A_0)$, this changes the trace on $\mathcal{F}$ into the trace on $\mathcal{F}'$ shown in (7.36). □

A local step datum consists of a finite link set S, a finite site set Λ containing its endpoints, and an even matrix kernel $k(A', A)$ on the Fock modes over Λ, such that

$$\begin{aligned} u(a)^{-1}k(A' + d_{\mathrm{s}}a, A + d_{\mathrm{s}}a)u(a) &= k(A', A), \\ \|k(A', A)\| &\le Ce^{-\varepsilon_k|A'-A|_S^2}. \end{aligned} \tag{7.37}$$

At time $j + 1/2$ its local integrand is

$$G[k] = \mathrm{Ins}^{\mathcal{J}^1(\Lambda)}[\omega^{1/2}k(A_{j+1}, A_j + d_{\mathrm{s}}a_j)u(a_j)\omega^{1/2}]e^{S^\times_{j+1/2}(\Lambda)}\prod_{b\in S} g_1(F_j^{\mathrm{el}}(b))^{-1}. \tag{7.38}$$

The arguments of k are restricted to S.

Proposition 7.27 (The word formula)**.** *(a) Each $G[k]$ is an even gauge-invariant local integrand in its box $B(j+1/2, S, \Lambda)$, with Gaussian growth constant $\min(\varepsilon_k, 1/2)$.*

(b) For each step j, let u_j be a bounded magnetic function, v_j an electric function satisfying $|v_j(x)| \le Ce^{(1/2-\varepsilon_v)|x|^2}$, and $(S_{j,l}, \Lambda_{j,l}, k_{j,l})$ finitely many local step data. Assume that their boxes embed disjointly in $C(n)$, that no periodic translate of a local box contains a magnetic plaquette representing σ_0, and that v_j is independent of the coordinates in $\bigcup_l S_{j,l}$. Then

$$\begin{aligned} &Z_{C(n)}\left[\prod_j u_j(F_j^{\mathrm{mag}})v_j(F_j^{\mathrm{el}})\prod_l G[k_{j,l}]\right] \\ &\quad = (2\pi)^V \operatorname{Tr}_{\mathcal{H}^1}\prod_{j=n_1-1,\dots,0}(c_{\mathrm{G}}K_jP_{\mathrm{G}}\mathfrak{M}u_j\mathfrak{M}^*)\,. \end{aligned} \tag{7.39}$$

Here K_j is the periodisation of the matrix kernel

$$\widetilde{K}_j(A', A) = v_j(A' - A)\prod_{b\notin\cup_l S_{j,l}} g_1(A'_b - A_b)\prod_l \iota(k_{j,l}(A', A)), \tag{7.40}$$

and u_j acts by $u_j(B(A,k))$. Every step under the trace is Hilbert–Schmidt.

(c) With no local data, $v_j(x) = e^{i\langle\beta_j, x\rangle}$ yields $c_{\mathrm{G}}K_jP_{\mathrm{G}} = K_{\beta_j}$; including the factor $\prod_{b\in S}\sqrt{2}e^{-x_b^2/2}$ yields $K_{\beta_j,S}$. In particular

$$Z_{C(n)}[1] = (2\pi)^V \operatorname{Tr} T_{\mathrm{l}}^{n_1} = (2\pi)^V \operatorname{Tr} T_{\mathrm{s}}^{n_1}. \tag{7.41}$$

Proof. In (7.38), all matrix entries and Grassmann elements are even. Apart from the inverse Gaussian, its coefficients are bounded by $Ce^{-\varepsilon_k|F_j^{\mathrm{el}}|_S^2}$. The inverse Gaussian multiplies this by $(2\pi)^{|S|/2}e^{|F_j^{\mathrm{el}}|_S^2/2}$, proving the claimed growth. Under gauge functions λ_j, λ_{j+1}, the two kernel

arguments both shift by $-d_{\mathrm{s}}\lambda_{j+1}$, while a_j shifts by $-\lambda_{j+1}+\lambda_j$. Equation (7.37) therefore changes the inserted matrix to $u(\lambda_{j+1})^{-1}(\cdots)u(\lambda_j)$. The monomial $\Theta(\zeta_J)\zeta_I$ acquires the inverse entry phase, so each term is invariant.

Unfold the torus integral by Lemma 7.23. On a local box avoiding the string, the path values are a local lift of the configuration: their phases and field strengths agree. Two such lifts on the rectangular hull differ by an integer gradient times $2\pi/q$, because their difference is a closed integer one-cochain on a contractible rectangle. This follows by path integration from a fixed vertex, with plaquette closedness ensuring independence of the path. Such a gradient has trivial phases on the fermions, and gauge invariance makes $G[k]$ independent of its choice.

At each time the localisation identity (7.27) replaces the product of local crossing insertions by the full-plane insertion with $\Upsilon_j=\prod_l \iota(k_{j,l}(A_{j+1},A_j+d_{\mathrm{s}}a_j))u(a_j)$. The unused crossing factors are restored by the exponential in (7.38). Apply Proposition 7.20. The inverse Gaussians remove precisely the electric Gaussian factors on $\bigcup_l S_{j,l}$. The remaining electric weight is (7.40); its magnetic weight is $W_j(B)=e^{-\|B\|^2/2}u_j(B)$. There are $2V$ electric plaquettes and $e^{-t^2/2}=\sqrt{2\pi}g_1(t)$, so the common constant is $(2\pi)^V$.

The independence assumption on v_j prevents double counting of its permitted Gaussian growth on inserted links. It and the disjointness of the $S_{j,l}$ give $\|\widetilde{K}_j(A',A)\|\le Ce^{-\varepsilon|A'-A|^2}$, where ε is the minimum of the finitely many positive decay constants and $1/2$. Simultaneous translation by $g\in\Gamma_{\mathrm{s}}$ preserves this kernel: on each local hull, $qg/(2\pi)$ is an integer gradient, so kernel covariance and integer charges prove the assertion. The same growth estimates show that the original integrand, multiplied by $e^{-\|F\|^2/2}$, is integrable. Lemma 7.26 now proves (7.39) with absolute convergence. Gaussian Fourier transformation proves the two stated special kernels, using $\sqrt{2}e^{-t^2/2}g_1(t)=g_{1/2}(t)$. For the partition function take all insertions equal to one and cyclically split K into its two square roots. Corollary 7.25 justifies cyclicity; the nonzero spectra of $X_0X_0^*$ and $X_0^*X_0$ coincide, proving (7.41). □

7.11. **The real external-field identity.** For a real magnetic slice η, define

$$\mathfrak{M}_{(\eta),k}(A)=e^{-\|B(A,k)+\eta\|^2/4}\widehat{\mathfrak{N}}(A),\qquad X_j=K^{1/2}\mathfrak{M}_{(\beta_j^{\mathrm{mag}})}.$$

The operator $\mathfrak{M}_{(\eta)}$ is multiplication from $\mathcal{H}^{\mathrm{s}}$ to $\mathcal{H}^{\mathrm{l}}$. Set $U_j=e^{i\langle\beta_j^{\mathrm{el}},E\rangle}$.

Proposition 7.28 (External field)**.** *For every real torus two-cochain β,*

$$\begin{aligned}&Z_{C(n)}[e^{-\langle\beta,F\rangle-\|\beta\|^2/2}]\\&\quad=(2\pi)^V\operatorname{Tr}_{\mathcal{H}^{\mathrm{s}}}\prod_{j=n_1-1,\dots,0}(X_{j+1}^*U_jX_j),\qquad X_{n_1}=X_0.\end{aligned}\tag{7.42}$$

Every X_j is Hilbert–Schmidt. Equivalently the jth factor is $\mathfrak{M}^*_{(\beta_{j+1}^{\mathrm{mag}})}K^{1/2}e^{i\langle\beta_j^{\mathrm{el}},E\rangle}K^{1/2}\mathfrak{M}_{(\beta_j^{\mathrm{mag}})}$.

Proof. Completion of the square replaces the photon weight by $e^{-\|F+\beta\|^2/2}$. This is integrable: a fixed real shift leaves Gaussian decay after the estimate $\|F+\beta\|^2\ge\|F\|^2/2-\|\beta\|^2$. Unfold the configurations and use the determinant case of Proposition 7.20. The magnetic weight at slice j is $e^{-\|B(A_j,k)+\beta_j^{\mathrm{mag}}\|^2/2}$; the electric kernel is $g_1(A_{j+1}-A_j-d_{\mathrm{s}}a_j+\beta_j^{\mathrm{el}})$. Its Fourier transform is $e^{-|E|^2/2}e^{+i\langle\beta_j^{\mathrm{el}},E\rangle}$, by (7.32). Lemma 7.26 therefore gives the trace on $\mathcal{H}^{\mathrm{l}}$ of

$$\prod_{j=n_1-1,\dots,0}\left(K^{1/2}U_jK^{1/2}\mathfrak{M}_{(\beta_j^{\mathrm{mag}})}\mathfrak{M}^*_{(\beta_j^{\mathrm{mag}})}\right),$$

with the unchanged normalization $(2\pi)^V$. Lemma 7.24, its shifted flux sum (7.34), and boundedness of $\widehat{\mathfrak{N}}$ show that each X_j is Hilbert–Schmidt. Move the last $\mathfrak{M}^*_{(\beta_0^{\mathrm{mag}})}$ to the front and group the neighboring factors. The cyclic move is legitimate: the remaining product is trace class since $n_1\ge 2$

and it contains a product of two Hilbert–Schmidt operators with bounded operators between and around them. This gives exactly (7.42), including the $j+1$ index on its adjoint magnetic factor. □

7.12. **The other coordinate time directions.**

Proposition 7.29 (Cyclic permutation of the axes)**.** *All coordinate slice, word and external-field identities hold with either e_2 or e_3 as time. Under the cyclic permutation g with $ge_1 = e_\mu$, they have the same matrices, constants and operator norms, with the slice variables relabelled by g.*

Proof. For $\mu = 2$ take the ordered pair of diagonal mirrors $(x_1 - x_2, 0), (x_2 - x_3, 0)$; for $\mu = 3$ reverse that order. Their reflections compose to g. If their upward unit normals are n', n'', put $U = \gamma_{n''}\gamma_{n'}$. The Clifford identity gives $U^{-1}\gamma_v U = \gamma_{gv}$. Composing the two antilinear Grassmann reflections yields the linear map

$$\psi(x) \mapsto U\psi(gx), \qquad \bar\psi(x) \mapsto \bar\psi(gx)U^{-1}.$$

It preserves the Berezin integral, either by composing the two reflected-integral identities or directly because the two Jacobian determinants cancel. It carries each hopping term to the corresponding term on $gC(n)$, preserves the mass term, and conjugates the reflection at $x_1 = c$ to that at $x_\mu = c$.

Choose the eigenbases at e_μ to be $e^{\pm}_{e_\mu,k} = U^{-1}e^{\pm}_{e_1,k}$, and transport the mode orders and representatives by $i \mapsto gi$. Then every mode, reflected mode, insertion and half-slab action is the image of its e_1 counterpart. Its coefficient and charge are unchanged. Thus the half-slab matrices and the operators $K, P_{\mathrm{G}}, \mathfrak{M}$ are literally the same in the transported coordinates. Bosonic oriented pullback is orthogonal and preserves the integer quotient measure. Applying this change of variables to both sides of each proved identity establishes the assertion. In the word formula its embedding and string-avoidance hypotheses are transported with the boxes and σ_0. The passage uses a fixed two-reflection word on odd modes, so no unmentioned sign choice is made there. □

8. Trace inequalities, straightening and pressure

The comparison argument needs two finite-volume estimates. The first removes the glide and fermion parity in a closing trace. The second bounds the pressure lost when a cubic torus is replaced by a torus adapted to a diagonal reflection. Both follow from the slab operators constructed above. Throughout this section fix $0 < \rho < 1$, $w_0, \dots, w_3 > 0$, $M \in \mathbb{R}$ and $q > 0$. In particular the preceding normalization theorem gives $0 < Z_{\mathbb{T}}[1] < \infty$. We use the Hilbert–Schmidt, reflection, translation and closing trace rules of Theorem 7.17.

8.1. **The trace inequality.** For a compact map Y between Hilbert spaces let $s_j(Y)$ be its singular values in decreasing order and put $\|Y\|_p = (\sum_j s_j(Y)^p)^{1/p}$, $1 \le p < \infty$. Thus, for a Hilbert–Schmidt map $Y : H \to H'$,

$$\|Y\|_{2k}^{2k} = \operatorname{Tr}_H (Y^*Y)^k.$$

Lemma 8.1 (Schatten trace inequality)**.** *Let $Y_j : H_{j-1} \to H_j$ be compact, let $U : H_r \to H_0$ be unitary, and let $p_j \ge 1$ satisfy $\sum_{j=1}^r p_j^{-1} = 1$. If $\|Y_j\|_{p_j} < \infty$, then the closing product is trace class and*

$$|\operatorname{Tr}_{H_0}(UY_r \cdots Y_1)| \le \prod_{j=1}^r \|Y_j\|_{p_j}. \tag{8.1}$$

Singular-value norms are unchanged by unitary factors and adjoints, and $\|Y\|_p$ decreases with p. If $T \ne 0$ is positive and trace class, then $s \mapsto \log \operatorname{Tr} T^s$ is convex for $s \ge 1$.

Proof. Here is a finite-dimensional proof of the inequality, also explaining why the spaces may differ. For square matrices $A_1, \dots, A_r$, the operator norm on the kth exterior power gives

$$\prod_{j=1}^k s_j(A_r \cdots A_1) \le \prod_{i=1}^r \prod_{j=1}^k s_j(A_i).$$

Indeed, the singular values of $\bigwedge^k A$ are the products of k singular values of A, and exterior powers preserve products. For invertible matrices take logarithms. The decreasing vectors $x_j = \log s_j(A_r \cdots A_1)$ and $y_j = \sum_i \log s_j(A_i)$ obey $\sum_{j\le k} x_j \le \sum_{j\le k} y_j$ for every k. Consequently

$$\sum_j (x_j - t)_+ = \max_{0\le k\le d} \left(\sum_{j\le k} x_j - kt \right) \le \sum_j (y_j - t)_+ .$$

Integrating against $e^t\,dt$ on $\mathbb{R}$, using $e^x = \int_{-\infty}^{\infty} (x-t)_+ e^t\,dt$, proves

$$\|A_r \cdots A_1\|_1 \le \sum_j \prod_i s_j(A_i) \le \prod_i \left(\sum_j s_j(A_i)^{p_i} \right)^{1/p_i} .$$

The last step is scalar Hölder. Approximation by invertible matrices removes the invertibility restriction, and $|\operatorname{Tr} B| \le \|B\|_1$ follows from the polar decomposition.

To treat different spaces, embed $H_0, \dots, H_{r-1}$ in their orthogonal direct sum. Extend each Y_j, $j < r$, by zero from the $(j-1)$st to the jth summand, and extend UY_r from the last summand to the first. These extensions have the same singular values as the original maps. Their product is the closing product on H_0 and zero on its orthogonal complement. Finite-rank approximation, with the displayed nuclear-norm bound applied also to differences, now gives trace-class convergence and (8.1). Unitary invariance and monotonicity are properties of singular values and scalar ℓ^p norms. Finally, if λ_j are the eigenvalues of T, scalar Hölder gives

$$\sum_j \lambda_j^{(1-t)a+tb} \le \left(\sum_j \lambda_j^a \right)^{1-t} \left(\sum_j \lambda_j^b \right)^t , \qquad 0 \le t \le 1,$$

which proves the convexity assertion. □

8.2. **A fold removes the closing twist.** Let $\ell(x) = n_\ell \cdot x$ be a reflection family. A closing vector g of $\mathbb{T} = (\Gamma, \chi)$ satisfies $\Gamma = \Gamma_\parallel \oplus \mathbb{Z}g$, where $\Gamma_\parallel = \Gamma \cap \{\ell = 0\}$ and $p := \ell(g) > 0$ generates $\ell(\Gamma)$. The straightened torus replaces g by

$$g^0 = \frac{p}{|n_\ell|^2} n_\ell, \qquad \Gamma^0 = \Gamma_\parallel \oplus \mathbb{Z}g^0, \qquad \chi^0(g^0) = -1, \qquad \chi^0|_{\Gamma_\parallel} = \chi|_{\Gamma_\parallel}.$$

It is used only when $g^0 \in \mathbb{Z}^3$ and Γ^0 has systole at least four.

Lemma 8.2 (Closing vectors)**.** *The cross-section has rank two and systole at least four. One has $\ell(\Gamma) = p\mathbb{Z}$ for a positive integer p; closing vectors are precisely its elements of height p. When straightening is defined, it preserves the cross-section, height period and number of photon sites, and gives a torus adapted to ℓ.*

Proof. The integral form ℓ is nonzero on the rank-three lattice, so its image is $p\mathbb{Z}$ with $p > 0$. An element g of height p exists, and every $\gamma \in \Gamma$ decomposes uniquely as $\gamma_\parallel + (\ell(\gamma)/p)g$. This proves the closing description and the rank of the kernel. Its systole is at least that of Γ. For the site count choose $u \in \mathbb{Z}^3$ with $\ell(u) = 1$. The surjection $\mathbb{Z}^3/\Gamma \to \mathbb{Z}/p\mathbb{Z}$ has kernel $(\mathbb{Z}^3 \cap \{\ell = 0\})/\Gamma_\parallel$, whence

$$[\mathbb{Z}^3 : \Gamma] = p[\mathbb{Z}^3 \cap \{\ell = 0\} : \Gamma_\parallel] = [\mathbb{Z}^3 : \Gamma^0].$$

Since $\ell(g^0) = p$, the kernel of ℓ on Γ^0 is the same $\Gamma_\parallel$, and its image is the same $p\mathbb{Z}$. Repeating the straightening therefore gives Γ^0 and χ^0 again. This is precisely adaptation to ℓ. □

Choose equally spaced admissible mirrors $c_j = c_0 + jh'$, with $p = 2kh'$, $k \ge 1$, and put $v = 2h'n_\ell/|n_\ell|^2$. The product of two consecutive mirror reflections is translation by v. A content G in slab j has copies in every slab: translations by integer multiples of v give the copies of the same parity, and reflection in c_j gives the adjacent copy. The product over $2k$ consecutive slabs is $\operatorname{fold}_\ell G$

on $\mathbb{T}^0$. It closes because $kv = g^0$. All contents here are even local integrands of the class admitted by the slab theorem, supported strictly inside their open slabs; their boxes embed in the common transverse cross-section.

Proposition 8.3 (Hölder with a closing twist)**.** *For contents G_j in the $2k$ successive slabs, one has*

$$Z_{\mathbb{T}^0}[\mathrm{fold}_\ell\, G_j] = \|\mathfrak{S}[G_j]\|_{2k}^{2k} \geq 0, \tag{8.2}$$

$$|Z_{\mathbb{T}}[G_1 \cdots G_{2k}]| \leq \prod_{j=1}^{2k} Z_{\mathbb{T}^0}[\mathrm{fold}_\ell\, G_j]^{1/(2k)}. \tag{8.3}$$

Proof. Write $Y = \mathfrak{S}[G_j] : \mathcal{H}(c_{j-1}) \to \mathcal{H}(c_j)$. The reflection and translation rules of the slab theorem give, for the copies beginning with G_j,

$$\mathfrak{S}[G^{(j+2s)}] = V_{sv} Y V_{sv}^{-1}, \qquad \mathfrak{S}[G^{(j+2s+1)}] = V_{(s+1)v} Y^* V_{sv}^{-1}.$$

Multiplication in trace order therefore telescopes to $V_{kv}(Y^*Y)^k$. The closing trace on $\mathbb{T}^0$ is untwisted in fermion parity because $\chi^0(g^0) = -1$, and its translation factor $V_{g^0}^{-1}$ cancels V_{kv}. This proves (8.2), including the normalization. On the original torus the exact closing trace is

$$Z_{\mathbb{T}}[G_1 \cdots G_{2k}] = \mathrm{Tr}_{\mathcal{H}(c_0)} \left(V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} \mathfrak{S}[G_{2k}] \cdots \mathfrak{S}[G_1] \right).$$

The factor $V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2}$ is unitary. Apply Lemma 8.1 with all exponents $2k$ and use the fold identity. This proves (8.3). The glide and parity are retained until this application of Hölder; no positivity of the twisted trace is assumed. □

Corollary 8.4 (Adapted reflection form)**.** *If $\mathbb{T}$ is adapted to ℓ, and G, G' are admitted contents in one open half of it, then*

$$Z_{\mathbb{T}}[(\Theta_{\ell,c} G')G] = \mathrm{Tr}(\mathfrak{S}[G']^* \mathfrak{S}[G]). \tag{8.4}$$

The form is Hermitian and nonnegative on the diagonal. The same holds with a bounded bosonic weight $(\Theta_{\ell,c}\Phi_+)\Phi_+$, where Φ_+ depends on positive-half field strength only. For a coordinate cross-section with fixed transverse sides, there is a positive trace-class operator T, independent of the longitudinal integer side $n \geq 4$, such that

$$Z_{C(n,n_2,n_3)}[1] = \mathrm{Tr}\, T^n. \tag{8.5}$$

Proof. In (8.4), the reflected lower-half operator is $\mathfrak{S}[G']^* V_{g^0}$. Insert it in the closing trace, cancel V_{g^0} and cycle the two Hilbert–Schmidt factors. Replacing G, G' by $\Phi_+ G, \Phi_+ G'$ proves the weighted statement; the scalar weight commutes with the Grassmann factors. For (8.5), take a coordinate half-step $Y = \mathfrak{S}[1]_{[c,c+1/2]}$ and set $T = Y^*Y$. Reflection gives the next half-step as $V_{e_1} Y^*$, so the unit-step operator is $V_{e_1} T$. Translation covariance makes the product of n unit steps equal $V_{ne_1} T^n$. The closing trace cancels V_{ne_1}, with no parity factor because the longitudinal character is -1. The half-step is Hilbert–Schmidt, so T is positive trace class. The cross-section and half-step construction contain no longitudinal side. □

8.3. **The three straightening operations.**

Lemma 8.5 (Systoles)**.** *For positive integer sides,*

$$\mathrm{sys}(n_1\mathbb{Z}e_1 + n_2\mathbb{Z}e_2 + n_3\mathbb{Z}e_3) = \min_j n_j, \qquad \mathrm{sys}(a\mathbb{Z}e_- + b\mathbb{Z}e_+ + c\mathbb{Z}e_3) = \min(a,b,c).$$

In any orthogonal frame with integer normal closing generators $p_a n_a/|n_a|^2$, the systole is $\min_a p_a/|n_a|^2$.

Proof. A diagonal vector $iae_- + jbe_+ + kce_3$ has maximum norm $\max(|i|a + |j|b, |k|c)$. Its smallest nonzero value is $\min(a, b, c)$. The coordinate statement follows directly from the three coordinates. A family orthogonal to $x_1 - x_2$ is either $x_1 + x_2$ or x_3; a diagonal family involving just one of indices one and two has nonzero scalar product with it. Thus a frame containing a diagonal family is a signed coordinate permutation of the diagonal frame, and a frame without one is a coordinate frame. These permutations preserve the maximum norm. □

Proposition 8.6. *Put $e_- = e_1 - e_2$, $e_+ = e_1 + e_2$. The following table gives the cross-section, height period, original closing character, and straightened torus. In the first two rows $n \geq 8$ is even, and in the last row $a, c \geq 4$.*

$\mathbb{T}$	ℓ	$\Gamma_\parallel$	g	$p,\ \chi(g)$	$\mathbb{T}^0$
$C(n,n,n)$	ℓ_-	$n\mathbb{Z}e_+ + n\mathbb{Z}e_3$	ne_1	$n,\ -1$	$D'(n/2,n,n)$
$D'(n/2,n,n)$	ℓ_+	$(n/2)\mathbb{Z}e_- + n\mathbb{Z}e_3$	ne_+	$2n,\ +1$	$D(n/2,n,n)$
$D(a,2a,c)$	x_1	$4a\mathbb{Z}e_2 + c\mathbb{Z}e_3$	ae_-	$a,\ -1$	$C(a,4a,c)$

Here D' has character $+1$ along its e_+ generator and -1 along its other two generators, whereas D has character -1 along all three generators.

Proof. On $n\mathbb{Z}^3$, the kernel of ℓ_- is $n\mathbb{Z}e_+ + n\mathbb{Z}e_3$, and $\ell_-(ne_1) = n$. The character at ne_+ is $(-1)^2 = +1$; replacing ne_1 by $(n/2)e_-$ proves the first row. On the new lattice, ℓ_+ vanishes on e_-, e_3 and takes the value $2n$ at ne_+. Straightening leaves the lattice unchanged and changes this closing character to -1. Finally a vector $iae_- + j(2a)e_+ + kce_3$ has first two coordinates $a(i+2j), a(2j-i)$. Its first coordinate vanishes exactly when $i = -2j$, giving $4aje_2 + kce_3$ and character $(-1)^{j+k}$. The height period is a, and straightening replaces ae_- by ae_1. These calculations give the last row. Lemma 8.5 gives systole at least four for every lattice in the table, including the new straightened lattices. □

8.4. **Cell-wise comparison.** For an orthogonal frame (ℓ^1, ℓ^2, ℓ^3) and admissible origins o_a, choose an integer width $h > 0$ and open cells

$$Q_\kappa = \{o_a + \kappa_a h < \ell^a < o_a + (\kappa_a + 1)h, a = 1, 2, 3\}.$$

Their volume is $|Q| = h^3/(|n_1||n_2||n_3|)$. The reflections in their walls commute in different frame directions. In each direction they generate an infinite dihedral group acting simply transitively on the layers; their product therefore acts simply transitively on the cells. For a content G in one cell, write $G^{[\kappa]}$ for its image in Q_κ under this unique wall-group element. Reflections act antilinearly on coefficients, as prescribed by Θ.

A torus is cut evenly when $\ell^a(\Gamma) \subset 2h\mathbb{Z}$ for every a. A rectangular torus of this frame carries the cells when its height periods p_a lie in $2h\mathbb{Z}$. On such a torus let

$$\mathrm{Fold}_{\mathbb{T}}\, G = \prod_{\kappa \bmod \Gamma} G^{[\kappa]}, \qquad \|G\|_{\mathbb{T}} = Z_{\mathbb{T}}[\mathrm{Fold}_{\mathbb{T}}\, G]^{|Q|/|\mathbb{T}|}.$$

Period translations belong to the wall group, so the product does not depend on the cell representatives. Even contents commute, which removes any ordering ambiguity. The sign and the inequalities below also show that the displayed root is defined.

Theorem 8.7 (Chessboard comparison)**.** *Let an evenly cut torus $\mathbb{T}_0$ be successively straightened in directions ℓ^1, ℓ^2, ℓ^3, with all three straightenings defined. Write p_a for the height period at the step in direction a. The final torus is the rectangular torus $\mathbb{T}_3$ with height periods (p_1, p_2, p_3) and negative characters along all generators. For a periodic family of admitted even contents G_κ in its cells,*

$$|Z_{\mathbb{T}_0}[\prod_\kappa G_\kappa]| \leq \prod_\kappa \|G_\kappa\|_{\mathbb{T}_3}. \tag{8.6}$$

There are $p_1p_2p_3/h^3$ cells on both sides, and every complete fold on $\mathbb{T}_3$ has nonnegative functional. In particular this gives ordinary chessboard comparison on a rectangular torus, comparison from

$C(n,n,n)$ and $D'(n/2,n,n)$ to $D(n/2,n,n)$ in the diagonal frame when $n \in 2h\mathbb{Z}$, and comparison from $D(a,2a,c)$ to $C(a,4a,c)$ in the coordinate frame when $a,c \in 2h\mathbb{Z}$.

Proof. Embedding, including the enlarged boxes. Consider one of the local boxes $B = y+\prod_{i=1}^{3}[-r_i,r_i]$ making up a content in Q_κ, and put $d_i = 2r_i$. Its width in frame direction a is

$$H_a = \sum_i |(n_a)_i| d_i < h,$$

because this compact box lies in the open cell. Each coordinate occurs in at least one frame normal, so $d_i < h$ for every i. Suppose that its enlargement meets a nontrivial period translate. For the corresponding $\gamma \in \Gamma \setminus \{0\}$, intersection of the coordinate intervals gives

$$|\gamma_i| \le d_i + 2 \quad (i = 1,2,3). \tag{8.7}$$

For $h \ge 4$ at least one height $\ell^a(\gamma)$ is nonzero, since the frame is a basis. Even cutting and (8.7) then give

$$2h \le |\ell^a(\gamma)| \le \sum_i |(n_a)_i|(d_i+2) = H_a + 2|n_a|^2 < h + 4 \le 2h,$$

a contradiction. Here $|n_a|^2 \le 2$ for every admitted normal.

For $h = 1,2$ the same intersection condition gives $|\gamma|_\infty < h+2 \le 4$, contrary to the systole. For $h = 3$ use the frame classification. Up to a signed coordinate permutation the frame is either (x_1,x_2,x_3) or (x_1-x_2,x_1+x_2,x_3). In the first case even cutting puts every γ_i in $6\mathbb{Z}$. In the second it puts $\gamma_1-\gamma_2$, $\gamma_1+\gamma_2$ and γ_3 in $6\mathbb{Z}$, so all three coordinates belong to $3\mathbb{Z}$. Thus $|\gamma|_\infty \ge 4$ forces $|\gamma|_\infty \ge 6$ in either case, whereas (8.7) gives $|\gamma|_\infty < h+2 = 5$. This contradiction covers the last integer width. Therefore every admitted rectangular local box embeds with its one-unit enlargement. The proof applies to each intermediate evenly cut torus, and embedding in its full period lattice implies embedding in its transverse subgroup.

If boxes from two chosen cell representatives have intersecting period translates, the corresponding open cells intersect. They must be the same open cell, because the cells partition the complement of the walls. The two representatives therefore have the same period orbit. Once that orbit is fixed, equality of the translated cell indices gives equality of the period vectors: the three height forms are independent. Within one content the boxes are disjoint by admissibility. This proves the joint embedding required for a layer product; the associated local plaquette supports are disjoint as well. The use of open cells is essential for these strict inequalities.

One-direction copying. For direction a, let $g^a_{t\leftarrow j}$ denote the unique element of its wall group taking layer j to layer t, and write $\kappa[a{:}j]$ for the index obtained by replacing the a-th entry of κ by j. Define

$$(\mathfrak{R}^j_a\mathfrak{G})_\kappa := (G_{\kappa[a:j]})^{g^a_{\kappa_a\leftarrow j}}.$$

The map carries the indicated open cell to Q_κ, so each entry is admitted there. Put $\Gamma_\parallel = \Gamma \cap \ker \ell^a$ and $p = p_a$ at this step. Even cutting implies $p/h = 2k$ for an integer $k > 0$. The straightened lattice is $\Gamma^0 = \Gamma_\parallel \oplus \mathbb{Z}g^0$, where $g^0 = pn_a/|n_a|^2$; it too is evenly cut, because $\ell^b(g^0) = 0$ for $b \ne a$.

Here is the periodicity check for the copied family. A tangential translation τ_γ, $\gamma \in \Gamma_\parallel$, commutes with every direction-a wall reflection, since its displacement has zero ℓ^a height. Also $(\kappa_\gamma)_a = 0$. The periodicity of the original family therefore gives

$$(\mathfrak{R}^j_a\mathfrak{G})_{\kappa+\kappa_\gamma} = \big((\mathfrak{R}^j_a\mathfrak{G})_\kappa\big)^{\tau_\gamma}.$$

The new normal translation τ_{g^0} belongs to this wall group and moves a layer by $2k$. Simple transitivity implies $g^a_{t+2k\leftarrow j} = \tau_{g^0} g^a_{t\leftarrow j}$. This proves the same identity for $\gamma = g^0$ and hence for all of Γ^0. No old, skew closing translation is used in this argument.

Choose representatives $\mathcal{K}_\parallel$ of the transverse cell indices, and let

$$\mathcal{K} = \{\kappa : 0 \le \kappa_a < p/h,\ \kappa[a{:}0] \in \mathcal{K}_\parallel\}.$$

This set represents the cells of both tori. Indeed if g is an original closing vector, write a period uniquely as $\gamma_{\|} + mg$. Its a-th cell displacement is mp/h. First reduce that coordinate into the displayed window, then reduce the transverse index modulo $\Gamma_{\|}$. These two reductions are unique. For the straightened torus the identical argument uses g^0 in place of g.

Group each original layer into its admitted slab product. Proposition 8.3 applies to the even number p/h of layers and gives

$$\left| Z_{\mathbb{T}} \left[\prod_{\kappa \in \mathcal{K}} G_\kappa \right] \right| \leq \prod_{0 \leq j < p/h} Z_{\mathbb{T}^0} \left[\prod_{\kappa \in \mathcal{K}} (\mathfrak{R}_a^j \mathfrak{G})_\kappa \right]^{h/p}. \tag{8.8}$$

Each factor on the right is nonnegative: it is the repeated two-sided reflection fold of that layer, whose transfer operator is an alternating product of an operator and its adjoint. This is precisely the positive trace in (8.2).

The three-direction iteration. Let Γ_0 be the initial lattice. Choose a closing vector g_1 for ℓ^1 in Γ_0, then a closing vector g_2 for ℓ^2 in $\Gamma_0 \cap \ker \ell^1$, and a positive generator g_3 of the common kernel. Successive direct-sum decomposition gives

$$\Gamma_0 = (\Gamma_0 \cap \ker \ell^1) \oplus \mathbb{Z} g_1, \qquad \Gamma_0 \cap \ker \ell^1 = \mathbb{Z} g_2 \oplus \mathbb{Z} g_3.$$

After the first step the new normal generator is orthogonal to the other two height forms. Hence the second height period is that of $\Gamma_0 \cap \ker \ell^1$, namely p_2; after the second step the third period is that of $\mathbb{Z} g_3$, namely p_3. Reducing cell coordinates in the order $1, 2, 3$ shows that the common representative set can be taken to be the brick $0 \leq \kappa_a < p_a/h$ for all three coordinates. Each step replaces its triangular generator by $\gamma_a = p_a n_a / |n_a|^2$. Consequently $\Gamma_3 = \bigoplus_a \mathbb{Z} \gamma_a$. Later steps keep the earlier γ_a in their transverse subgroup, so their characters remain negative; the last step makes its own character negative too.

Apply (8.8) first in direction 1, then to each resulting family in direction 2, and then in direction 3. The exponents in the three nested geometric means are respectively h/p_1, h/p_2, h/p_3. For a chosen triple (j_1, j_2, j_3) the last family is $\mathfrak{R}_3^{j_3} \mathfrak{R}_2^{j_2} \mathfrak{R}_1^{j_1} \mathfrak{G}$. The wall-group actions in distinct frame directions commute on even contents. Its entry in cell κ is therefore the unique wall-group image of the single original content $G_{(j_1,j_2,j_3)}$. Its product is exactly that content's complete fold on $\mathbb{T}_3$. It follows that

$$|Z_{\mathbb{T}_0}[\prod_{\kappa} G_\kappa]| \leq \prod_{\substack{0 \leq j_a < p_a/h \\ a=1,2,3}} Z_{\mathbb{T}_3}[\mathrm{Fold}_{\mathbb{T}_3} G_{(j_1,j_2,j_3)}]^{h^3/(p_1 p_2 p_3)}.$$

There are $p_1 p_2 p_3 / h^3$ factors. Since the rectangular volume is $p_1 p_2 p_3 / (|n_1||n_2||n_3|)$, the final exponent equals $|Q|/|\mathbb{T}_3|$, proving (8.6). A complete mirror family is unchanged by one-direction copying. Applying (8.2) to it already proves its nonnegativity; no limiting argument is needed. The stated special cases follow from Proposition 8.6, with height triples $(n, 2n, n)$ and $(a, 4a, c)$, respectively. □

8.5. **Pressure limits from separate convexity.**

Lemma 8.8. *Let u be real on a product of unbounded positive subsets of $\mathbb{R}$. Suppose in each variable, with the others fixed, u is convex and $u(s)/s$ is non-increasing. Then $u(n)/(n_1 n_2 n_3)$ is bounded below and converges to its finite infimum as all three coordinates tend to infinity in any manner.*

Proof. Choose $0 < a_j < b_j$ in each coordinate set. For $n_j \geq b_j$ put $\alpha_j = (n_j - a_j)/(b_j - a_j)$. Convexity gives the extrapolation bound $u(s) \geq \alpha u(b_j) - (\alpha - 1) u(a_j)$, while ratio monotonicity gives $u(s) \leq (s/a_j) u(a_j)$ for $s \geq a_j$. Apply the former successively in variables one, two and three, using the latter on the terms with negative coefficients. This gives

$$\begin{aligned} u(n) \geq{}& \alpha_1 \alpha_2 \alpha_3 u(b_1, b_2, b_3) - \alpha_1 \alpha_2 (\alpha_3 - 1) u(b_1, b_2, a_3) \\ & - \alpha_1 (\alpha_2 - 1)(n_3/a_3) u(b_1, a_2, a_3) - (\alpha_1 - 1)(n_2 n_3/(a_2 a_3)) u(a_1, a_2, a_3). \end{aligned}$$

After division by $n_1n_2n_3$, each coefficient is bounded, since $0 \le \alpha_j - 1 \le \alpha_j \le n_j/(b_j - a_j)$. The four fixed finite values therefore give a uniform lower bound. For arbitrary n, enlarge its coordinates to at least b_j; ratio monotonicity bounds the original quotient below by the enlarged one. The quotient is separately non-increasing. If f_* is its infimum, choose n^0 where it is below $f_* + \varepsilon$. It then lies in $[f_*, f_* + \varepsilon)$ whenever $n_j \ge n_j^0$ for all j, proving the asserted unrestricted convergence. □

Proposition 8.9 (Coordinate pressure)**.** *For integer sides $n_j \ge 4$ define*

$$f(n_1, n_2, n_3) = \frac{\log Z_{C(n_1,n_2,n_3)}[1]}{n_1n_2n_3}.$$

It is non-increasing in each side, is bounded below, and tends to a finite number $f_\infty = \inf f$ as $\min_j n_j \to \infty$. With two sides fixed, the unnormalized log partition function is the restriction of a convex function of the third side.

Proof. By (8.5), in direction one $Z_C[1] = \operatorname{Tr} T^{n_1}$, with T fixed by the two transverse sides. Positivity of $Z_C[1]$ implies $T \ne 0$. Consequently $f = (n_2n_3)^{-1} \log \|T\|_{n_1}$, and Lemma 8.1 proves both the ratio monotonicity and convexity in this direction. The other two directions give the same properties. Apply Lemma 8.8 to $u(n) = \log Z_C[1]$ on $\{4, 5, \dots\}^3$. This proves the lower bound and limit. The infimum is also bounded above by the finite value $f(4, 4, 4)$. □

Theorem 8.10 (Four tori and the pressure error)**.** *For even $n \ge 8$ all four tori below have n^3 photon sites, and*

$$Z_{C(n,n,n)}[1] \le Z_{D'(n/2,n,n)}[1] \le Z_{D(n/2,n,n)}[1] \le Z_{C(n/2,2n,n)}[1]. \tag{8.9}$$

Put $\Delta(n) = f(n/2, 2n, n) - f(n, n, n)$. Then

$$0 \le \Delta(n) \le f(n/2, n/2, n/2) - f_\infty \longrightarrow 0. \tag{8.10}$$

The logarithm of any ratio between these four partition functions has modulus at most $n^3\Delta(n) = o(n^3)$.

Proof. Apply (8.3) to constant contents 1 in each of the three rows of Proposition 8.6. Every fold remains 1, giving $Z_{\mathbb{T}}[1] \le Z_{\mathbb{T}^0}[1]$. Two equal slabs suffice: the corresponding half periods are $n/2, n, n/4$, respectively. These are admissible mirror spacings, including coordinate link planes when $n \equiv 2 \pmod 4$. This proves (8.9). The index of a diagonal lattice with sides a, b, c is $2abc$, so all four indices equal n^3. Thus their extreme logarithms differ by $n^3\Delta(n)$. Monotonicity of f gives $f(n/2, 2n, n) \le f(n/2, n/2, n/2)$, and $f(n, n, n) \ge f_\infty$. The coordinate-pressure limit now proves (8.10) and the bound on every intermediate ratio. □

Proposition 8.11 (Mirror-periodic pressure)**.** *Fix a cell structure and an admitted content G of its base cell. On rectangular carrying tori with even numbers m_a of cells in direction a, put $Z_G(m) = Z_{\mathbb{T}(m)}[\operatorname{Fold}_{\mathbb{T}(m)} G]$. If Z_G is positive for one shape, it is positive for every admissible shape, and*

$$\frac{\log Z_G(m)}{m_1m_2m_3}$$

decreases in each variable and tends to a finite infimum as $\min_a m_a \to \infty$. For fixed transverse shape,

$$Z_G(m) = \operatorname{Tr}(Y^*Y)^{m_a/2}, \tag{8.11}$$

where Y is Hilbert–Schmidt and independent of m_a.

Proof. Multiply the mirror copies in one layer to form its slab content S. Its transverse cross-section and S depend on the two transverse cell counts only. Take $Y = \mathfrak{S}[S]$. The complete fold is the one-direction fold of this layer, so (8.2) proves (8.11). If $Y = 0$ the trace vanishes for all longitudinal shapes; otherwise it is positive for all of them. Starting from one positive shape, changing the three coordinates successively proves positivity at every shape. The admissible shape set is a product: $hm_a \ge 4$ for a coordinate normal, and $hm_a \ge 8$ for a diagonal normal, with each m_a even. The

logarithm is separately convex and its ratio by m_a is non-increasing, by the singular-value lemma. Lemma 8.8 therefore proves the conclusion. □

9. Gaussian domination with every flux sector

The pressure comparison controls errors in the volume limit. Tightness needs a different estimate: a Maxwell shift cannot increase the partition function on a coordinate torus. Its proof uses the coordinate trace identity and retains the full sum over integer magnetic fluxes.

For a real two-cochain β on $\mathbb{T}$ define

$$Z_{\mathbb{T}}(\beta) = \int_{\mathrm{Conf}(\Gamma)} e^{-\|F+\beta\|^2/2} \det D_A^{\mathbb{T}}\, d(A,z) = Z_{\mathbb{T}}[e^{-\langle\beta,F\rangle - \|\beta\|^2/2}]. \tag{9.1}$$

With direction e_1 as time, write $\beta_j^{\mathrm{el}} = (\beta_{12}(j,\cdot), \beta_{13}(j,\cdot))$ and $\beta_j^{\mathrm{mag}} = \beta_{23}(j,\cdot)$.

Lemma 9.1 (Shifted Hilbert–Schmidt maps)**.** *On a coordinate slice Λ_{s} with transverse sides at least four, let $K^{1/2}$ be the gauge-projected heat operator of the coordinate construction. For every real magnetic shift b, the map*

$$X_b = K^{1/2}\mathfrak{M}_{(b)}, \qquad \mathfrak{M}_{k,(b)}(A) = e^{-\|d_{\mathrm{s}}A + (2\pi/q)k1_{\sigma_0} + b\|^2/4}\widehat{\mathfrak{N}}(A)$$

is Hilbert–Schmidt on the direct sum of all flux sectors $k \in \mathbb{Z}$.

Proof. The quotient kernel estimate of Lemma 7.24 applies to

$$K^{1/2} = (2\pi/q)^{|\Lambda_{\mathrm{s}}|/2} e^{-|E|^2/4} P_{\mathrm{G}} :$$

the scalar heat kernel is the periodization of $g_{1/2}$. Using the normalized Haar measure, an explicit scalar majorant for its gauge-averaged matrix kernel is

$$\widehat{g}([x]) = c_{\mathrm{G}}^{1/2} \int g_{1/2}^{\mathrm{per}}([x - d_{\mathrm{s}}a])\, d\mu_{\mathrm{G}}(a).$$

The unitary fermion factor $u(a)$ has norm one. The proved periodisation estimate gives $g_{1/2}^{\mathrm{per}} \in L^1 \cap L^\infty$, and Minkowski's inequality gives $\|\widehat{g}\|_2 \le c_{\mathrm{G}}^{1/2}\|g_{1/2}^{\mathrm{per}}\|_2 < \infty$. These bounds control the noncompact directions of the quotient X/Γ_{s}. The finite-dimensional matrix $\widehat{\mathfrak{N}}(A)$ is uniformly bounded by a constant C. Integrating first in A' bounds the squared kernel norm of X_b by

$$\dim \mathcal{F}'\, C^2 \|\widehat{g}\|_2^2 \sum_{k\in\mathbb{Z}} \int dA\, e^{-\|d_{\mathrm{s}}A + (2\pi/q)k1_{\sigma_0} + b\|^2/2}.$$

The link quotient is $(\ker d_{\mathrm{s}}/\Gamma_{\mathrm{s}}) \times (\ker d_{\mathrm{s}})^\perp$, where $\Gamma_{\mathrm{s}} = (2\pi/q)(\mathbb{Z}^{\Lambda_{\mathrm{s}}^{(1)}} \cap \ker d_{\mathrm{s}})$ is a full lattice in the kernel. Full rank follows by rational row reduction of the integral coboundary matrix. The first factor is compact, and d_{s} is injective on the second. Let $N_{\mathrm{s}} = |\Lambda_{\mathrm{s}}|$ and $B = \sum_\sigma b_\sigma$. Every link occurs with opposite signs in its two plaquette boundaries, so $\sum_\sigma (d_{\mathrm{s}}A)_\sigma = 0$. Projecting onto the constant plaquette vector and onto $\operatorname{ran} d_{\mathrm{s}}$, and translating the latter component of A, therefore gives

$$\sum_{k\in\mathbb{Z}} \int dA\, e^{-\|d_{\mathrm{s}}A + (2\pi/q)k1_{\sigma_0} + b\|^2/2} \le C_{\mathrm{s}} \sum_{k\in\mathbb{Z}} e^{-((2\pi/q)k + B)^2/(2N_{\mathrm{s}})} < \infty. \tag{9.2}$$

Here C_{s} is the compact kernel volume times the finite Gaussian integral on $(\ker d_{\mathrm{s}})^\perp$, independent of k, b. Any additional orthogonal component of b decreases the integral. Thus the full direct-sum kernel is square integrable, proving the claim. □

Theorem 9.2 (Coordinate Gaussian domination)**.** *On every coordinate torus with sides at least four, every real two-cochain β satisfies*

$$0 < Z_{\mathbb{T}}(\beta) \le Z_{\mathbb{T}}(0), \qquad \mathbb{E}_{\mathbb{T}} e^{\langle\beta,F\rangle} \le e^{\|\beta\|^2/2}. \tag{9.3}$$

For real two-cochains β_i and complex z_i,

$$\left|\mathbb{E}_{\mathbb{T}} e^{\sum_i z_i \langle \beta_i, F\rangle}\right| \le \exp\left(\tfrac{1}{2}\left\|\sum_i (\operatorname{Re} z_i)\beta_i\right\|^2\right).$$

The estimates hold for the normalized determinant-weighted law and with the complete flux sum in its definition.

Proof. Let $V = n_1 n_2 n_3$, $X_j = X_{\beta_j^{\mathrm{mag}}}$ and $U_j = e^{i\langle \beta_j^{\mathrm{el}}, E\rangle}$. The latter is the unitary translation $(U_j\Phi)(A) = \Phi(A + \beta_j^{\mathrm{el}})$ in every flux sector. The coordinate identity of Proposition 7.28 reads exactly

$$(2\pi)^{-V} Z_{\mathbb{T}}(\beta) = \operatorname{Tr}_{\mathcal{H}^{\mathrm{s}}}\left(X_0^* U_{n_1-1} X_{n_1-1} X_{n_1-1}^* U_{n_1-2} X_{n_1-2} \cdots X_1^* U_0 X_0\right).$$

Lemma 9.1 permits cyclically moving X_0^* to the right and changing the trace space:

$$(2\pi)^{-V} Z_{\mathbb{T}}(\beta) = \operatorname{Tr}_{\mathcal{H}^{\mathrm{l}}}(U_{n_1-1} X_{n_1-1} X_{n_1-1}^* \cdots U_0 X_0 X_0^*).$$

For two Hilbert–Schmidt maps, this cyclic identity follows by Cauchy–Schwarz on their matrix elements; the double series is absolutely convergent. Trace Hölder and unitary invariance now give

$$(2\pi)^{-V} Z_{\mathbb{T}}(\beta) \le \prod_{j=0}^{n_1-1} \left(\operatorname{Tr}(X_j^* X_j)^{n_1}\right)^{1/n_1}.$$

We used the nonnegativity of the shifted determinant integral, so its value equals the modulus of the trace. The nonzero eigenvalues of $X_j X_j^*$ and $X_j^* X_j$ agree.

Let $\beta^{(j)}$ have zero electric part and the same magnetic part β_j^{mag} at every time. The same coordinate identity, with all U equal to 1 and all X equal to X_j, identifies its partition function as $(2\pi)^V \operatorname{Tr}(X_j^* X_j)^{n_1}$. The factors $(2\pi)^V$ cancel since the Hölder exponents sum to one. Thus

$$Z_{\mathbb{T}}(\beta) \le \prod_{j=0}^{n_1-1} Z_{\mathbb{T}}(\beta^{(j)})^{1/n_1}. \tag{9.4}$$

In particular a purely electric shift has $Z_{\mathbb{T}}(\beta) \le Z_{\mathbb{T}}(0)$.

Each $\beta^{(j)}$ is supported on 23 plaquettes only. Choose direction e_2 as time; this same field is now purely electric, so the preceding bound gives $Z_{\mathbb{T}}(\beta^{(j)}) \le Z_{\mathbb{T}}(0)$. Equivalently, interchange coordinates one and two. The determinant and configuration measure are covariant under that isometry, the torus sides are interchanged, and oriented pullback preserves the two-cochain norm. No assumption of equal side lengths is needed. Substitution in (9.4) proves the upper bound. For strict positivity, $Z_{\mathbb{T}}(0) > 0$ and $\det D_A^{\mathbb{T}} \ge 0$ imply a positive-measure set on which the determinant is positive. The shifted Maxwell factor is strictly positive on that same set. Finally (9.1) with $-\beta$ gives the normalized exponential bound. Taking the modulus of a complex exponential inside the positive probability integral proves the complex version. □

Corollary 9.3 (Moments and simultaneous tails)**.** *For $X = \langle \beta, F\rangle$ one has*

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 \le \|\beta\|^2, \quad \mathbb{E}X^{2r} \le (2re\|\beta\|^2)^r, \quad \mathbb{E}X^4 \le \tfrac{3}{2}e^2\|\beta\|^4.$$

The fourth joint cumulant obeys $|\kappa_4(\langle\beta_1, F\rangle, \dots, \langle\beta_4, F\rangle)| \le 15 \prod_i \|\beta_i\|$. For nonzero β_i of disjoint supports and $\xi_i \ge 0$,

$$\mathbb{P}\{|\langle\beta_i, F\rangle| \ge \xi_i\|\beta_i\|,\ i = 1, \dots, s\} \le \prod_{i=1}^{s} 2e^{-\xi_i^2/2}.$$

The Gaussian domination bound also holds for a finitely supported infinite-lattice cochain whose supporting box embeds in the torus, paired with the periodically extended field strength.

Proof. The two exponential bounds imply $\mathbb{E}e^{t|X|} < \infty$ for every $t > 0$. Differentiating the bound at zero from both sides gives $\mathbb{E}X = 0$ and $\mathbb{E}X^2 \le \|\beta\|^2$. Alternatively the second bound follows from $\mathbb{E}\cosh(tX) \le e^{t^2\|\beta\|^2/2}$ and $\cosh(tX) \ge 1+t^2X^2/2$. For $\beta \ne 0$, the inequality $t^{2r}X^{2r} \le (2r)!\cosh(tX)$ with $t^2 = 2r/\|\beta\|^2$ yields

$$\mathbb{E}X^{2r} \le (2r)!\left(\frac{\|\beta\|^2}{2r}\right)^r e^r.$$

Use $(2r)! \le (2r)^{2r}$, or $4! = 24$ for the fourth bound. The cumulant formula has one joint fourth moment and three products of covariances. Scalar Hölder and the preceding bounds give the constant $\frac{3}{2}e^2 + 3 < 15$. For a fixed choice of signs, take $\beta = \sum_i \pm\xi_i\beta_i/\|\beta_i\|$. Its squared norm is $\sum_i \xi_i^2$. On the corresponding one-sided event $\langle\beta, F\rangle \ge \sum_i \xi_i^2$, so exponential Markov gives probability at most $e^{-\sum_i \xi_i^2/2}$. Sum over all 2^s sign choices. Finally periodization preserves both the pairing and the cochain norm when the supporting box embeds, because no two supported plaquettes are identified. □

10. Folded perturbations and comparison states

Fix $0 < \rho < 1$, $w_i > 0$, $M \in \mathbb{R}$ and $q > 0$ throughout the finite-volume argument. The torus parameters remain fixed while its side tends to infinity. Let $n_i \to \infty$ satisfy (Div), and write

$$C_i = C(n_i), \quad D_i = D(n_i/2, n_i, n_i), \quad E_i = C(n_i/2, 2n_i, n_i), \qquad V_i = n_i^3.$$

All three tori have V_i photon sites. The pressure comparison gives

$$Z_{C_i}[1] \le Z_{D_i}[1] \le Z_{E_i}[1], \qquad 0 \le V_i^{-1}\log\frac{Z_{E_i}[1]}{Z_{C_i}[1]} = \Delta_i \longrightarrow 0.$$

We will use this error at fixed perturbation strength. There is no volume-independent bound on the partition-function ratio.

10.1. **Cell structures and perturbations.** In the diagonal frame put $\ell_- = x_1 - x_2$, $\ell_+ = x_1 + x_2$ and $e_\pm = e_1 \pm e_2$. A cell structure has walls $\ell_\pm = c_\pm + jh$ and $x_3 = c_3 + jh$, with h even, $c_-, c_+ \in \mathbb{Z}$ and $c_3 \in \mathbb{Z} + \frac{1}{2}$. Its cells are

$$Q_\kappa = \{c_a + \kappa_a h < \ell^a < c_a + (\kappa_a + 1)h,\ a = 1, 2, 3\}, \qquad |Q| = h^3/2.$$

The wall reflections generate a direct product of three infinite dihedral groups, acting simply transitively on the open cells. For a cell integrand G, let $G^{[\kappa]}$ be its image under the unique wall-group element carrying the original cell to Q_κ. Images of even integrands commute. On a torus carrying this cell structure, products and sums of these images range over one representative of each cell modulo the period lattice. They do not depend on the representatives, since period translations belong to the wall group.

Let $O = \sum_j a_j e^{i\langle\beta_j, F\rangle}$ be a real trigonometric polynomial supported in Q_0, and put $|O|_1 = \sum_j |a_j|$. Choose an even $h > 8r_O + 8$ with $4h \in \mathcal{D}(\mathfrak{n})$. The datum $\mathfrak{d} = (c_-, c_+, c_3; h; O)$ defines the folded array

$$O_\mathfrak{d} = \sum_\kappa O^{[\kappa]}$$

on D_i. The unfolded array places translated copies of O in the cells of C_i, using the translation lattice

$$\Lambda_h = \tfrac{h}{2}(\mathbb{Z}e_- + \mathbb{Z}e_+) + h\mathbb{Z}e_3.$$

Both arrays have $V_i/|Q|$ terms and modulus at most $|O|_1 V_i/|Q|$. All these constructions are defined for sufficiently large i; deleting the finitely many earlier indices has no effect on any limit below.

For $t \ge 0$ define

$$\mu_i^t(G) = \frac{Z_{D_i}[e^{tO_\mathfrak{d}}G]}{Z_{D_i}[e^{tO_\mathfrak{d}}]}.$$

The denominator is at least $e^{-t|O|_1 V_i/|Q|} Z_{D_i}[1] > 0$. On bosonic observables this is expectation under a probability measure. It is also a linear normalized functional on the local fermionic integrands used below.

Proposition 10.1 (Normalized comparison)**.** *For all sufficiently large i and every $t > 0$,*

$$\mu_{C_i}(O) \le \mu_i^t(O) + \frac{|Q|\Delta_i}{t}.$$

Proof. Set $A_i = V_i/|Q|$. Translation invariance of the cubic law and Jensen's inequality give

$$tA_i \mu_{C_i}(O) \le \log \frac{Z_{C_i}[e^{t\widetilde{O}}]}{Z_{C_i}[1]}.$$

Apply Theorem 8.7 to the purely translated cell contents $G_\kappa = G_0^{\tau_{v_\kappa}}$, where $G_0 = e^{tO}$ and $v_\kappa \in \Lambda_h$ carries Q_0 to Q_κ. These contents have the same adapted-torus chessboard seminorm. Indeed, τ_{v_κ} conjugates the wall group to itself, since every height $\ell^a(v_\kappa)$ is a multiple of h. Conjugation permutes the cells modulo the period lattice and therefore, directly from the product definition,

$$\mathrm{Fold}_{D_i}(G_0^{\tau_{v_\kappa}}) = (\mathrm{Fold}_{D_i} G_0)^{\tau_{v_\kappa}}.$$

Translation covariance of the finite functional identifies the two partition functions. Thus all A_i seminorms in the chessboard product equal $Z_{D_i}[\mathrm{Fold}_{D_i} G_0]^{1/A_i}$, and the comparison gives

$$Z_{C_i}[e^{t\widetilde{O}}] \le Z_{D_i}[e^{tO_\mathfrak{d}}].$$

Here $\mathrm{Fold}_{D_i} G_0 = e^{tO_\mathfrak{d}}$: antilinear reflection commutes with the real exponential and multiplication of its bosonic coefficients. All contents are bounded and admitted, and both tori have the same number A_i of cells.

The function $b_i(t) = \log Z_{D_i}[e^{tO_\mathfrak{d}}]$ is differentiable and convex, since $O_\mathfrak{d}$ is bounded and the bosonic law is positive. Thus $b_i(t) - b_i(0) \le t b_i'(t)$. Wall invariance of the perturbed law gives $b_i'(t) = A_i \mu_i^t(O)$. Consequently

$$tA_i \mu_{C_i}(O) \le tA_i \mu_i^t(O) + \log \frac{Z_{D_i}[1]}{Z_{C_i}[1]} \le tA_i \mu_i^t(O) + V_i \Delta_i.$$

Division by tA_i proves the estimate. □

Proposition 10.2 (Positivity at perturbed walls)**.** *For every $t \ge 0$ the law μ_i^t is invariant and reflection positive at each wall of its cell structure.*

Proof. Fix a wall $\ell = c$ and let p be the torus height period in that direction. The carrying condition gives $p \in 2h\mathbb{Z}$, so $\ell = c + p/2$ is another wall. These two walls bound the open positive torus half; the other half is its reflection. There are p/h layers of fine cells around the period. Relative to the chosen wall, select indices $r = 0, \dots, p/(2h) - 1$ for the positive half and $r = p/(2h), \dots, p/h - 1$ for the negative half. Reflection sends r to $-r - 1$ modulo p/h and therefore pairs the two lists without fixed cells. Transverse cell representatives are paired at the same time. The wall group action on even integrands is an actual group action, so reflection sends each source copy to the copy in its paired cell, including its scalar conjugation.

Let O^+ be the sum of copies in the positive list. Every such copy is supported strictly inside its open cell, so it reads only field strength plaquettes in that half. No source plaquette lies in either boundary wall. Its negative-half partner is ΘO^+ and hence $O_\mathfrak{d} = O^+ + \Theta O^+$ on the torus. The original observable is real and each copy is real on real closed fields, so O^+ is real. In particular the complete source is fixed by Θ. For real t,

$$e^{tO_\mathfrak{d}} = (\Theta e^{tO^+}) e^{tO^+}.$$

For a positive-side bounded cylinder observable G, finite adapted reflection positivity applied to $e^{tO^+}G$ gives

$$Z_{D_i}[e^{tO_{\mathfrak{d}}}(\Theta G)G] \geq 0.$$

Indeed e^{tO^+} is a bounded bosonic function on this finite torus and reads only the positive half. Multiplication by it therefore preserves the positive-half class in the finite positivity theorem. For coupled local integrands it changes only the coefficient bound, not its positive Gaussian decay margin. Its reflected factor multiplies ΘG, and the two commute as scalar functions, giving the displayed form. Divide by the positive denominator established before the proposition.

The ordinary tensor pullback by the reflection preserves the configuration measure and determinant. It fixes the real source, as just shown. Changing variables thus gives $\mu_i^t(f \circ \theta^*) = \mu_i^t(f)$ for every bounded bosonic f. This proves the invariance used in the reflection-positive law definition. The same argument works at every fine wall and consequently at every aligned coarse wall used later. □

10.2. **Local amplification.** We use bounded elements in the sense of Definition 6.16. Besides disjoint embedding, a placement requires a free column in each coordinate direction. This requirement allows the flux string of a slice to be chosen outside every insertion. It cannot be replaced by disjointness alone.

Lemma 10.3 (Compatible coarse cells)**.** *For a box $B = y + [-r, r]^3$ there are even widths $h'' \in h\mathbb{Z}$ and $kh \in h''\mathbb{Z}$, independent of y and of the perturbation strength, such that*

$$h'' \geq 2r + 6h, \quad kh \geq 16h'', \quad 4h'', 2kh \in \mathcal{D}(\mathfrak{n}).$$

There are a diagonal cell structure $\mathcal{Q}'$ of width kh coarser than $\mathcal{Q}_{\mathfrak{d}}$, and a coordinate cell structure $\mathcal{Q}''$ of width h'' with half-integer walls, with these properties. The box B lies in an open cell of $\mathcal{Q}'$. Every image gB, $g \in W(\mathcal{Q}')$, lies in an open coordinate cell, and different images lie in different coordinate cells. No coordinate wall meets any wall-group image of $\operatorname{supp} O$*. For all sufficiently large i, the diagonal structures are carried by D_i, $\mathcal{Q}''$ cuts D_i and is carried by E_i, and the coordinate folds of each gB form placements in E_i. The numbers of coarse diagonal and coordinate cells are respectively*

$$m_i = V_i/(|Q|k^3), \qquad s_i = V_i/h''^3.$$

The threshold depends only on $\mathfrak{n}, h, r$.

Proof. The eventual divisor set is closed under divisors and least common multiples. Its infinitude therefore allows us to choose $4h''$ as a sufficiently large common multiple of $4h$, and then $2kh$ as a sufficiently large common multiple of $4h''$. These choices give the displayed inequalities; they involve no position or strength. Choose aligned diagonal origins c'_-, c'_+, c'_3 so that, writing $u_a = \ell^a(y) - c'_a$,

$$u_- = 2h'' + s_-, \quad u_+ = 4h'' + s_+, \quad u_3 = h'' + s_3, \qquad 0 \leq s_a < h.$$

Such choices exist because each coset of $h\mathbb{Z}$ meets an interval of length h. More precisely $c'_a \in c_a + h\mathbb{Z}$, and reduce $\ell^a(y) - c_a - u_a^0$ modulo h, where $(u_-^0, u_+^0, u_3^0) = (2h'', 4h'', h'')$. The remainder is $s_a \in [0, h)$ and fixes c'_a. Every coarse wall is therefore a fine wall: its origin is aligned and kh is a multiple of h. The ranges of the diagonal forms on B have radii $2r, 2r, r$, so B lies strictly between the origins and the next walls at distance kh.

Every element of the diagonal wall group has a unique expression

$$g = \tau_{kh(z_-e_- + z_+e_+ + 2z_3e_3)}\theta_-^{a_-}\theta_+^{a_+}\theta_3^{a_3}, \quad z \in \mathbb{Z}^3,\ a \in \{0,1\}^3,$$

where the three reflections are at the chosen origins. Its image centre is

$$gy = y + h''\kappa(g) + s(a), \quad |s(a)|_\infty < 2h,$$

where

$$\kappa(g) = (-2a_- - 4a_+,\ 2a_- - 4a_+,\ -2a_3) + \frac{kh}{h''}(z_- + z_+,\ z_+ - z_-,\ 2z_3).$$

To verify the normal form, in one height coordinate the two neighbouring wall reflections generate translation by $2kh$. Every word is either such a translation or that translation followed by reflection at the origin; its sign and translation integer are unique. The three normal directions are orthogonal, so their groups commute and the three normal forms combine. Reflection in $\ell_- = c'_-$ displaces the centre by $-u_- e_-$, reflection in $\ell_+ = c'_+$ by $-u_+ e_+$, and reflection in $x_3 = c'_3$ by $-2u_3 e_3$. Substitution of the chosen u_a gives the displayed vector, with remainder

$$s(a) = (-a_- s_- - a_+ s_+,\ a_- s_- - a_+ s_+,\ -2a_3 s_3).$$

This also verifies $|s(a)|_\infty < 2h$. The eight first vectors are distinct and their differences have supremum norm at most six. Since $kh/h'' \geq 16$, κ is injective: equality first forces the second integer vectors to agree and then forces a and z to agree. In detail a nonzero difference of the translation vectors has an integer coordinate of absolute value at least one, hence after multiplication by kh/h'' has size at least sixteen. The reflection-vector difference has size at most six and cannot cancel it. The translation map on z is injective, and in the reflection vector its first two coordinates determine a_+, a_- by their sum and difference while its third determines a_3.

For each coordinate choose a half-integer origin o_μ in

$$(y_\mu + r + 2h - h'',\ y_\mu - r - 2h).$$

This interval has length at least $2h$. A wall-group image of the centre of $\operatorname{supp} O$ has at most four possible residues modulo h in either of the first two coordinates, and at most two in the third. Its support has coordinate radius at most r_O. At most $4(2r_O + 1) = 8r_O + 4 < h$ half-integer residue classes are therefore forbidden. Choose o_μ in a remaining class; the interval above meets every class. The walls $o_\mu + h''\mathbb{Z}$ then avoid all the supports, since $h'' \in h\mathbb{Z}$. For clarity the residue count follows from the actual fine reflections. In the first two coordinates their images of (x_1, x_2) are

$$(x_1, x_2), \quad (x_2 + c_-, x_1 - c_-), \quad (c_+ - x_2, c_+ - x_1), \quad (c_- + c_+ - x_1, c_+ - c_- - x_2).$$

All translations in that wall group have coordinate values in $h\mathbb{Z}$, so these four possibilities give all residues. The third coordinate has only x_3 and $2c_3 - x_3$. The centre of every copy of the radius-r_O source box has one of these residues. For each residue at most $2r_O + 1$ half-integer wall positions can meet its interval of radius r_O. The forbidden union consequently has the stated size. The permissible origin interval has length at least $2h$, and therefore contains representatives of all h half-integer classes modulo h; a permissible class can be chosen in it. The origin inequalities also give

$$[y_\mu - r - 2h, y_\mu + r + 2h] \subset (o_\mu, o_\mu + h''),$$

so gB lies in the coordinate cell indexed by $\kappa(g)$. Injectivity proves distinctness.

For the threshold choose $4h, 4h'', 2kh$ to divide every n_i. The period lattices are

$$\Gamma_i = \mathbb{Z}\frac{n_i}{2} e_- + \mathbb{Z} n_i e_+ + \mathbb{Z} n_i e_3, \quad \Gamma'_i = \mathbb{Z}\frac{n_i}{2} e_1 + \mathbb{Z}2n_i e_2 + \mathbb{Z} n_i e_3.$$

Their diagonal heights are multiples of $2kh$, and all their coordinates are multiples of $2h''$. This proves carrying and cutting. A nonzero period has supremum norm at least $n_i/2 \geq kh > 2r + 2$, so every radius-r box embeds. If two image cells differ by $v \in \Gamma_i$, compare $g'B$ with $\tau_v gB$; both wall-group elements lie in the same coordinate cell and injectivity forces $g' = \tau_v g$. Thus distinct diagonal cells modulo periods give distinct coordinate cells modulo periods.

Finally, the coordinate wall group modulo Γ'_i acts simply transitively on the cells of E_i. Its images of gB embed disjointly because they lie strictly in those cells. They have free columns: for a direction e_μ choose another coordinate ν and a plaquette orthogonal to e_μ meeting the wall $x_\nu = o_\nu$. Every translate along e_μ meets that same wall, hence lies in none of the open cells. This proves the required placement condition. The cell counts follow by dividing V_i by the respective cell volumes. In the free-column construction the chosen plaquette has a unit edge straddling the half-integer wall o_ν. Its whole column parallel to e_μ continues to meet that wall. Every period

translate of a placement box remains in an open coordinate cell, since the periods change its layer indices by even integers. Such a box cannot contain any plaquette of the wall-meeting column. This checks the condition against every period translate, rather than only against one collection of cell representatives. □

Lemma 10.4 (Chessboard comparison with the folded source)**.** *Use the coarse diagonal structure of Lemma 10.3 on D_i, and let m_i be its number of cells. For an admitted even content G in one of those open cells, put $H = \prod_{\kappa'} G^{[\kappa']}$, its coarse fold. For $t \geq 0$ and all sufficiently large i,*

$$|\mu_i^t(G)|^{m_i} \leq \frac{Z_{D_i}[e^{tO_\mathfrak{d}} H]}{Z_{D_i}[e^{tO_\mathfrak{d}}]}. \tag{10.1}$$

The numerator is nonnegative. The assertion includes admitted unbounded contents with a strict Gaussian growth margin.

Proof. Every coarse wall is a fine wall: the origins are aligned modulo h and the coarse width is kh. Thus its wall group is a subgroup of the fine wall group. Every fine open cell lies in one coarse open cell, and each coarse cell contains exactly k^3 fine cells. For a coarse cell $Q'_{\kappa'}$ define

$$O'_{\kappa'} = \sum_{\kappa:\, Q_\kappa \subset Q'_{\kappa'}} O^{[\kappa]}, \qquad S_{\kappa'} = e^{tO'_{\kappa'}}.$$

Each sum has k^3 terms. Its supporting plaquettes lie strictly inside the corresponding coarse cell, since the original source lies strictly inside every fine cell. It is a bounded bosonic content, with modulus at most $e^{t|O|_1 k^3}$. Multiplying G by it enlarges the bosonic support within that same coarse cell and leaves the positive Gaussian growth margin unchanged. More precisely, in each defining product of $\mathcal{A}(Q'_{\kappa'})$ from Definition 2.13, absorb $S_{\kappa'}$ into its bounded bosonic factor. The local coupled factors and their disjoint boxes are unchanged. Their coefficient bounds acquire only the finite scalar constant $e^{t|O|_1 k^3}$. Thus the product remains admitted in that coarse cell, and all the contents satisfy the hypotheses of the ordinary chessboard theorem.

The coarse wall group maps the fine source copies into one another and maps the contained fine cells bijectively onto those in the image coarse cell. It follows that $S_{\kappa'}$ is the wall-group image of a single coarse source content. Scalar conjugation under an odd image commutes with the real exponential. On D_i the product of these contents is $e^{tO_\mathfrak{d}}$: the carrying conditions let the fine and coarse cells be represented compatibly, and each fine cell occurs once. Consequently the complete coarse fold of any one $S_{\kappa'}$ is $e^{tO_\mathfrak{d}}$.

Let κ'_0 be the cell reading G. Assign the content $S_{\kappa'_0} G$ to that cell and $S_{\kappa'}$ to every other coarse cell. The complete fold of the first content is $e^{tO_\mathfrak{d}} H$, because the source factors are scalar and all contents are even. Theorem 8.7 on the rectangular torus D_i therefore gives

$$|Z_{D_i}[e^{tO_\mathfrak{d}} G]| \leq Z_{D_i}[e^{tO_\mathfrak{d}} H]^{1/m_i} Z_{D_i}[e^{tO_\mathfrak{d}}]^{(m_i-1)/m_i}.$$

The complete-fold conclusion makes the first factor nonnegative, and the source denominator is strictly positive. Divide by it and raise to the m_ith power. This proves (10.1). The proof used the admitted growth class throughout; it did not require pointwise boundedness of G. □

Lemma 10.5 (Amplification)**.** *Let G be bounded by C in a box of radius r. There are an integer $k = k(\mathfrak{d}, r)$ and a threshold i_0 such that*

$$|\mu_i^t(G)| \leq C\gamma_{t,i}(k), \qquad \gamma_{t,i}(k) = e^{2t|O|_1 k^3 + |Q| k^3 \Delta_i}$$

for $t \geq 0$ and $i \geq i_0$.

Proof. Use the two structures of Lemma 10.3. The weighted comparison of Lemma 10.4 gives

$$|\mu_i^t(G)|^{m_i} \leq \frac{Z_{D_i}[e^{tO_\mathfrak{d}} H]}{Z_{D_i}[e^{tO_\mathfrak{d}}]}, \qquad H = \prod_{\kappa'} G^{[\kappa']}.$$

The numerator is nonnegative. Expand the source in characters: $e^{tO_{\mathfrak{d}}} = \sum_\alpha a_\alpha \varphi_\alpha$, with

$$\sum_\alpha |a_\alpha| \le e^{t|O|_1 V_i/|Q|}.$$

The series converges uniformly, and absolute convergence of the integral with H allows integration term by term. Each plaquette in a character lies in an open coordinate cell, by the avoidance property of the lemma. Each copy of G also lies in one such cell, and different copies occupy different cells modulo periods.

Collect the factors of $\varphi_\alpha H$ by coordinate cells and apply the cross-frame chessboard estimate from D_i to E_i. There are s_i cell contents. A content without G is a character; its fold has modulus one, so its cell norm is at most $Z_{E_i}[1]^{1/s_i}$. A content with G^g is a character times G^g. Its coordinate fold is a character times a placement of G^g with s_i members. The placement bound gives cell norm at most $CZ_{E_i}[1]^{1/s_i}$. Exactly m_i contents are of the second kind. Consequently, uniformly in α,

$$|Z_{D_i}[\varphi_\alpha H]| \le C^{m_i} Z_{E_i}[1], \quad |Z_{D_i}[e^{tO_{\mathfrak{d}}} H]| \le e^{t|O|_1 V_i/|Q|} C^{m_i} Z_{E_i}[1].$$

Use also

$$Z_{D_i}[e^{tO_{\mathfrak{d}}}] \ge e^{-t|O|_1 V_i/|Q|} Z_{D_i}[1], \qquad \log \frac{Z_{E_i}[1]}{Z_{D_i}[1]} \le V_i \Delta_i.$$

These estimates bound $|\mu_i^t(G)|^{m_i}$ by $C^{m_i} \exp(2t|O|_1 V_i/|Q| + V_i\Delta_i)$. Taking the m_i-th root proves the stated formula. □

Lemma 10.6 (Domination of local exponentials)**.** *Let a real cochain β be supported in a box of radius r. The geometric integer $k = k(\mathfrak{d}, r)$ and the threshold of Lemma 10.3 satisfy, for every $t \ge 0$,*

$$\mu_i^t(e^{\langle \beta, F\rangle}) \le \exp\left(\frac{\|\beta\|^2}{2}\right) \gamma_{t,i}(k) \qquad (i \ge i_0).$$

Neither k nor i_0 depends on the coefficients of β or on the position of its supporting box.

Proof. The exponential is unbounded, so the bounded insertion estimate alone would not prove this assertion. We use positivity of the bosonic measure and the coordinate Gaussian estimate after a second chessboard comparison. Set $G = e^{\langle \beta, F\rangle}$. It is an admitted bosonic integrand with positive Gaussian growth margin, since

$$0 < G \le \exp\left(\|\beta\|^2 + \frac{1}{4} \sum_{\sigma \in \operatorname{supp} \beta} F_\sigma^2\right).$$

Here $|\beta_\sigma F_\sigma| \le \beta_\sigma^2 + F_\sigma^2/4$. Disjoint copies retain this margin, so all the following chessboard integrals are absolutely convergent.

Use the coarse structures of Lemma 10.3. There are $m_i = V_i/(|Q|k^3)$ coarse diagonal cells. The first chessboard estimate gives the following bound, by Lemma 10.4:

$$\mu_i^t(G)^{m_i} \le \frac{Z_{D_i}[e^{tO_{\mathfrak{d}}} H]}{Z_{D_i}[e^{tO_{\mathfrak{d}}}]}, \qquad H = \prod_{\kappa'} G^{[\kappa']}.$$

The left side is positive. Since G is real, its copy under an odd wall-group element has no additional scalar conjugation effect. Orthogonality of tensor pullback gives

$$G^{[\kappa']} = e^{\langle \beta^{\kappa'}, F\rangle}, \qquad \beta^{\kappa'} = (g_{\kappa'}^{-1})^* \beta, \qquad \|\beta^{\kappa'}\| = \|\beta\|.$$

In particular $H > 0$. The source bound and positivity of the bosonic functional give

$$Z_{D_i}[e^{tO_{\mathfrak{d}}} H] \le e^{t|O|_1 V_i/|Q|} Z_{D_i}[H], \qquad Z_{D_i}[e^{tO_{\mathfrak{d}}}] \ge e^{-t|O|_1 V_i/|Q|} Z_{D_i}[1].$$

Place the m_i copies into their distinct coordinate cells. There are $s_i = V_i/h''^3$ such cells. A nonempty content is $e^{\langle \beta', F\rangle}$, where β' is a period translate of some $\beta^{\kappa'}$ into the chosen cell. Its

support lies strictly inside that cell and its norm is $\|\beta\|$. The other contents are one. The coordinate fold of a nonempty content in E_i has the form $e^{\langle\beta'',F\rangle}$, where β'' is the sum of its s_i wall-group images. These images occupy distinct open cells, so their plaquette supports are disjoint even after periodization. Thus

$$\|\beta''\|^2 = s_i\|\beta\|^2, \qquad Z_{E_i}[e^{\langle\beta'',F\rangle}] \le e^{s_i\|\beta\|^2/2} Z_{E_i}[1]$$

by Theorem 9.2. The cross-frame chessboard product consequently gives

$$Z_{D_i}[H] \le e^{m_i\|\beta\|^2/2} Z_{E_i}[1].$$

There are exactly m_i nonempty contents, each contributing $e^{\|\beta\|^2/2} Z_{E_i}[1]^{1/s_i}$; every empty content contributes $Z_{E_i}[1]^{1/s_i}$.

Finally $Z_{E_i}[1]/Z_{D_i}[1] \le e^{V_i\Delta_i}$. Combining the three estimates and taking the m_ith root gives

$$\mu_i^t(G) \le e^{\|\beta\|^2/2 + 2t|O|_1 V_i/(|Q|m_i) + V_i\Delta_i/m_i} = e^{\|\beta\|^2/2}\gamma_{t,i}(k).$$

All geometric choices and side thresholds came from the support radius in Lemma 10.3; neither the amplitude nor the location enters them. □

Proposition 10.7 (Existence of comparison states)**.** *Every subsequence of the cubic indices has a further subsequence along which the perturbed laws converge at each of a sequence of strengths $t_k \downarrow 0$. A further limit of those volume-limit laws is a comparison state ν_0. It satisfies Gaussian domination, is invariant and reflection positive at all cell walls, and, if the cubic laws converge to ν along the initial subsequence, satisfies*

$$\nu(O) \le \nu_0(O).$$

Proof. Tightness at a fixed strength. Apply Lemma 10.6 to $\beta = \pm s1_\sigma$. Every plaquette fits in a radius-one box, so one integer $k_1 = k(\mathfrak{d}, 1)$ and one threshold work simultaneously for every σ and $s > 0$. For $R, s > 0$, Markov's inequality gives

$$\mu_i^t(|F_\sigma| > R) \le 2e^{-sR+s^2/2}\gamma_{t,i}(k_1).$$

Take $s = R$. Since $\Delta_i \to 0$, the numbers $\gamma_{t,i}(k_1)$ are bounded in i after the common threshold; denote an upper bound by $B_t < \infty$. Thus

$$\sup_{i\ge i_0}\sup_{\sigma} \mu_i^t(|F_\sigma| > R) \le 2B_t e^{-R^2/2}.$$

Enumerate the plaquettes and choose radii R_j with $2B_t e^{-R_j^2/2} \le \varepsilon 2^{-j}$. The compact set $\Omega \cap \prod_j[-R_j, R_j]$ has probability at least $1-\varepsilon$ for every $i \ge i_0$. The common threshold is essential here: a threshold depending on j would not suffice for the infinite union bound. Each of the finitely many preceding laws is itself tight. Taking the union of this compact set with finitely many compact sets for those laws proves tightness of the whole defined family.

Volume limits and then strength limits. Starting from the given subsequence, apply Prokhorov successively at $t_1 = 1, t_2 = 1/2, \ldots$, each time retaining a subsequence of the preceding one. The diagonal indices i_m tend to infinity and satisfy $\mu_{i_m}^{t_k} \to \nu_{t_k}$ for every fixed k. For a fixed compact real β, truncation of its nonnegative exponential as in Lemma 5.3 and the preceding estimate give

$$\nu_{t_k}(e^{\langle\beta,F\rangle}) \le e^{\|\beta\|^2/2 + 2t_k|O|_1 k_\beta^3}.$$

The term $|Q|k_\beta^3\Delta_{i_m}$ vanishes in this volume limit; no simultaneous choice of i_m and a varying t_k has been made.

For $\beta = \pm s1_\sigma$, the last bound is uniform in k and σ, since $t_k \le 1$ and $k_\beta = k_1$. It gives the common tail bound $2e^{2|O|_1 k_1^3} e^{-R^2/2}$. The same compact-product construction therefore proves

tightness of $(\nu_{t_k})_k$. Take a subsequence with $\nu_{t_{k_j}} \to \nu_0$. For each fixed compact β, truncation once more gives

$$\nu_0(e^{\langle \beta, F\rangle}) \le \liminf_j e^{\|\beta\|^2/2 + 2t_{k_j}|O|_1 k_\beta^3} = e^{\|\beta\|^2/2}.$$

This proves Gaussian domination of the comparison state.

Positivity and comparison. Fix one of the cell walls and a bounded continuous cylinder G in its open positive half-space. Its finitely many coordinates and their reflections fit inside the corresponding positive and negative torus halves for all sufficiently large i. Proposition 10.2 applies there. Its invariance identities and the expectation of $(\Theta G)G$ pass through the volume limit and then through the strength limit, because these are bounded continuous functions. The density and isometry argument in Lemma 5.3 extends positivity to the required Borel functions. This holds for every cell wall separately.

If the initial cubic laws converge to ν, their retained subsequence still does so. At a fixed $t_k > 0$, Proposition 10.1 gives

$$\mu_{C_{i_m}}(O) \le \mu_{i_m}^{t_k}(O) + |Q|\Delta_{i_m}/t_k.$$

The error tends to zero because this strength is fixed. The bounded continuous observable O therefore gives $\nu(O) \le \nu_{t_k}(O)$. Passing along the strength subsequence gives $\nu(O) \le \nu_0(O)$, as asserted. □

11. Operators inserted at a link plane

The comparison of partition functions controls bosonic sources. To restore translations of the limiting law we must also control local fermionic integrands. We do this by representing them, after a finite algebraic reduction, by bounded operators at a coordinate link plane. The operator norm is independent of the cross-section and of an electric character multiplying the integrand.

11.1. **Signs and covariance.** At $c = j + \frac{1}{2}$ write $\Theta_c = \Theta_{(x_1,c)}$. The modes and ordered products are those of Definition 6.11. The matrix coefficient convention is

$$\mathrm{Ins}_c[T] = \sum_{I,J} T_{IJ}\Theta_c(\zeta_J)\zeta_I.$$

Thus the row belongs to the upper side and the column to the lower side. This convention fixes all the adjoints below.

Lemma 11.1 (Covariance of inserted matrices)**.** *For a matrix-valued coefficient $T(\hat{A})$,*

$$\Theta_c \,\mathrm{Ins}_c[T](\hat{A}) = \mathrm{Ins}_c[T(\theta_c^*\hat{A})^*].$$

For a translation v, let $U_v e_I$ be e_{I+v} times the sign of the permutation sorting the translated indices. Then

$$(\mathrm{Ins}_c[T])^{\tau_v}(\hat{A}) = \mathrm{Ins}_{c+v_1}[U_v T(\tau_v^*\hat{A})U_v^*].$$

For a spatial mirror θ_w orthogonal to e_ν, $\nu = 2, 3$, let u_w be the matrix on the one-mode space defined by $\Theta_w\zeta_i = \sum_{i'}(u_w)_{i'i}\zeta_{i'}$. Let $re_I = (-1)^{\binom{|I|}{2}}e_I$ and $U_w = (\bigwedge u_w)r$. For even T,

$$\Theta_w \,\mathrm{Ins}_c[T](\hat{A}) = \mathrm{Ins}_c[U_w\overline{T(\theta_w^*\hat{A})}U_w^*].$$

All these finite-dimensional maps are unitary, preserve the crossing weights ω, and transport the lower-end phase $u(a)$ with the corresponding pullback. In the spatial case, $U_w\overline{u(a\circ\theta_{\mathrm{s}})}U_w^* = u(a)$.

Proof. The reflection reverses products and squares to the identity, hence $\Theta_c(\Theta_c\zeta_J\,\zeta_I) = \Theta_c\zeta_I\,\zeta_J$. Conjugating the scalar coefficient produces exactly the adjoint matrix. Translation replaces each mode by its translated mode; sorting the upper and lower ordered products gives the two signs in the matrix of $U_vTU_v^*$. A mode and its translate have the same crossing coefficient and their charges are translated.

Here are the signs in the spatial case. For any spinor e,

$$\Theta_w(e^*\psi(x)) = -\bar{\psi}(\theta_w x)\gamma_\nu e, \qquad \Theta_w(\bar{\psi}(x)e) = -(\gamma_\nu e)^*\psi(\theta_w x).$$

Since $\gamma_\nu\gamma_1 = -\gamma_1\gamma_\nu$, γ_ν maps the two eigenspaces of γ_1 isometrically onto one another. The four mode groups are sent respectively to groups $3, 4, 1, 2$ at the reflected bond, with coefficients the negative matrix of this isometry, or its conjugate. Thus u_w is unitary, preserves ω_i, and changes the charge to its negative reflected value. Expanding products of distinct modes gives the minors of u_w; Cauchy–Binet shows that $\bigwedge u_w$ is unitary.

Reversing $|I|$ odd factors gives the sign $(-1)^{\binom{|I|}{2}}$. On a homogeneous element, $\Theta_w\Theta_c = (-1)^{\deg}\Theta_c\Theta_w$ because the normal Clifford matrices anticommute. Applying these two rules to $\Theta_c\zeta_J\,\zeta_I$, then returning the lower factor to the left, gives

$$\Theta_w(\Theta_c\zeta_J\,\zeta_I) = (-1)^{\binom{|I|}{2}+\binom{|J|}{2}}\Theta_c(\zeta_J^w)\zeta_I^w$$

when $|I|$ and $|J|$ have the same parity. This is precisely the displayed matrix formula with U_w. Evenness of T ensures the parity condition on every nonzero entry. Its charge rule gives the identity for $u(a)$, entry by entry. The remaining weight identities follow by multiplying the one-mode identities. □

11.2. **The norm independent of the cross-section.**

Lemma 11.2 (Normal form and norm)**.** *An element of $\mathfrak{W}(S,\Lambda)$ has a unique finite normal form*

$$X = \sum_{\delta} e^{i\langle\delta,A\rangle}\Upsilon_\delta(E),$$

where each Υ_δ is a finite sum of bounded Gaussian matrix symbols and has charge $q^{-1}d_{\mathrm{s}}^*\delta$. Realisation on an admitted rectangular cross-section is a unital $*$-homomorphism. The norm defined in Definition 6.13 is a C^*-norm and equals the operator norm on $L^2(\mathbb{R}^S)\otimes\mathcal{F}_\Lambda$. For every one admitted cross-section Ξ,

$$\|X\|_{\#} = \sup_{\beta\in\mathbb{R}^S}\|(\alpha_\beta X)_\Xi\|.$$

Enlarging S or Λ gives an isometric inclusion.

Proof. The Weyl identity $f(E)e^{i\langle\delta,A\rangle} = e^{i\langle\delta,A\rangle}f(E+\delta)$ puts any word in normal form. Products of the Gaussian symbols remain bounded Gaussian symbols, because their positive quadratic parts add. Distinct characters are linearly independent: conjugate an allegedly vanishing finite normal form by translations of A, multiply by a selected inverse character of the translation vector, and average over larger cubes. Every other frequency averages to zero, while the selected operator is fixed. Thus each coefficient operator vanishes. Fourier transformation in A then forces its continuous symbol to vanish. The charge condition is preserved by the Weyl rule. It gives gauge covariance because conjugating the matrix coefficient by $u(a)$ multiplies it by the character with exponent $d_{\mathrm{s}}^*\delta$, which cancels the change of $e^{i\langle\delta,A\rangle}$ under $A\mapsto A - d_{\mathrm{s}}a$. To check periodicity, take $\gamma = (2\pi/q)k \in \Gamma_{\mathrm{s}}$. The rectangular hull of the support embeds in the slice. On this hull the closed integer cochain k has an integer primitive λ: fix its value at one vertex, integrate k along a grid path, and use the zero plaquette sums to interchange two successive coordinate steps. All paths then give the same value. Thus, summing by parts only over links supporting δ,

$$\langle\delta,\gamma\rangle = (2\pi/q)\langle d_{\mathrm{s}}^*\delta,\lambda\rangle \in 2\pi\mathbb{Z}.$$

The last inclusion uses the integer charge of the coefficient. Consequently the character is well defined on the quotient. The realised word therefore satisfies the same relations and adjoint rules.

We give the norm argument because a bound depending on the slice area would be insufficient. Signed localisation of the Fock space is an injective matrix $*$-homomorphism, hence isometric; it suffices to retain the finite factor $\mathcal{F}_\Lambda$. On the enlarged space

$$L^2(\mathbb{R}^S, dz) \otimes L^2(\mathbb{R}^{\Lambda_{\mathrm{s}}^{(1)}}/\Gamma_{\mathrm{s}}, d[A]) \otimes \mathcal{F}_\Lambda$$

take a measurable section $A^\flat$ of the quotient and the two unitaries

$$(U_1\Phi)(z,[A]) = \Phi(z + A^\flat|_S, [A]), \qquad (U_2\Phi)(z,[A]) = \Phi(z, [A - z]).$$

Here z is extended by zero off S; translations preserve the quotient measure, so these maps are unitary. If $E' = -i\partial_z$, direct substitution gives

$$U_2U_1(e^{i\langle\delta,z\rangle}\Upsilon_\delta(E'))U_1^*U_2^* = e^{i\langle\delta,A\rangle}\Upsilon_\delta(E_S + E').$$

The character of A is well defined on the quotient by the periodicity already proved. After Fourier transformation in z, the last operator is the direct integral of $(\alpha_{-p}X)^0_\Xi$. Uniform continuity of the finite Gaussian symbols makes its fibre norm continuous in p. Its essential supremum is therefore its supremum. Unitary conjugation identifies this number with the original local operator norm. This proves the stated equality for every Ξ and, consequently, the C^*-identity. Enlargement is tensoring with an identity, with the signed Fock localisation on the finite factor; both operations preserve the norm. □

11.3. **Kernels and words.** For $0 < s \le \frac14$ let $k_X^{(s)}$ be the integral kernel of $e^{-s|E_S|^2}Xe^{-s|E_S|^2}$. In the normal form above it is

$$k_X^{(s)}(A', A) = \sum_\delta e^{i\langle\delta,A'\rangle}\left(e^{-s|\xi+\delta|^2}\Upsilon_\delta(\xi)e^{-s|\xi|^2}\right)^\vee(A' - A),$$

with inverse Fourier transform $(2\pi)^{-|S|}\int e^{i\langle\xi,x\rangle}\cdot d\xi$. In particular, these kernels are continuous and have Gaussian decay in $A' - A$. We give the calculation, since the decay must survive division by the Maxwell factor in the definition of an insertion.

Consider one normal-form term $e^{i\langle\delta,A\rangle}\Upsilon e^{-\langle E,QE\rangle+\langle\lambda,E\rangle}$, where $Q \ge 0$ is extended by zero from its original link set. Set $m = |S|$, $B = Q + 2sI$ and $L = \lambda - 2s\delta$. Its regularized Fourier multiplier is

$$e^{-s|\xi+\delta|^2-\langle\xi,Q\xi\rangle+\langle\lambda,\xi\rangle-s|\xi|^2} = e^{-s|\delta|^2}e^{-\langle\xi,B\xi\rangle+\langle L,\xi\rangle}.$$

All eigenvalues of B are at least $2s > 0$. Diagonalize B orthogonally and integrate each coordinate. The real Gaussian integral first gives the formula for real L; both sides are entire in L and the same formula follows for complex L:

$$k_X^{(s)}(A', A) = (4\pi)^{-m/2}\det(B)^{-1/2}e^{i\langle\delta,A'\rangle-s|\delta|^2}\exp\Big(\tfrac14\langle L + ix, B^{-1}(L + ix)\rangle\Big)\,\Upsilon, \qquad x = A' - A. \tag{11.1}$$

The scalar pairing here is bilinear. Since B^{-1} is real and symmetric, the scalar factor multiplying Υ has modulus

$$C_s\exp\Big(-\tfrac14\langle x + \operatorname{Im}\lambda, B^{-1}(x + \operatorname{Im}\lambda)\rangle\Big), \quad C_s = (4\pi)^{-m/2}\det(B)^{-1/2}e^{-s|\delta|^2+\langle\operatorname{Re}L,B^{-1}\operatorname{Re}L\rangle/4}.$$

Thus $\|k_X^{(s)}(A', A)\|$ is bounded by $Ce^{-c|x|^2}$ with $c > 0$. For example, $B^{-1} \ge \|B\|^{-1}I$ and $|x+\operatorname{Im}\lambda|^2 \ge |x|^2/2 - |\operatorname{Im}\lambda|^2$ permit $c = (8\|B\|)^{-1}$. This proves the claimed continuity, integrability and decay, also for a finite sum of terms. The empty link set has the evident interpretation, with determinant and scalar Gaussian integral equal to one.

There are two useful identities. Multiplying the kernel by $e^{i\langle\beta,A'-A\rangle}$ conjugates its operator by $e^{i\langle\beta,A\rangle}$ on the local real space. It therefore replaces E by $E - \beta$, including in the two heat factors, and replaces X by $\alpha_\beta X$. Taking the adjoint of the heat sandwich gives $k_{X^*}^{(s)}(A', A) = k_X^{(s)}(A, A')^*$.

Finally simultaneous gauge translation of the two arguments leaves x unchanged. The charge equation for Υ gives

$$u(a)^{-1}k_X^{(s)}(A'+d_{\mathrm{s}}a, A+d_{\mathrm{s}}a)u(a) = e^{i\langle\delta,d_{\mathrm{s}}a\rangle - iq\langle z(\delta),a\rangle}k_X^{(s)}(A',A) = k_X^{(s)}(A',A).$$

The kernel is even because the coefficient has total charge zero. Thus it satisfies every local-step hypothesis, rather than only a scalar integrability estimate.

Definition 11.3. Using the local-step construction $G[k]$, set

$$G_X = G[k_X^{(1/4)}], \qquad G'_X = G[k_X^{(1/8)}].$$

Thus G_X agrees with Definition 6.14; in G'_X only the two heat factors in the kernel are changed. Both are read in the same boxes B and $B^{\#}$.

Lemma 11.4 (The mixed word formula)**.** *Suppose local insertion boxes embed disjointly in $C(n_1,n_2,n_3)$ and avoid a common flux column parallel to e_1. Choose its plaquette as σ_0. At time j let S_j be the union of the link sets of the G' insertions, and let β_j and β_j^{mag} be the electric and magnetic parts of a real two-cochain β. Write $u_j(F) = e^{i\langle\beta_j^{\mathrm{mag}},F\rangle}$ and $B_j = \prod_{a:j_a=j}(\alpha_{\beta_j}X_a)_\Xi$. The factors in this last product commute. Then*

$$Z_C\left[\varphi_\beta\prod_a G_a^\circ\right] = (2\pi)^{n_1n_2n_3}\,\mathrm{Tr}\prod_{j=n_1-1,\dots,0} K^{1/2}_{\beta_j,S_j}B_jK^{1/2}_{\beta_j,S_j}\mathfrak{M}u_j\mathfrak{M}^*.$$

The choice $G_a^\circ = G_{X_a}$ contributes no links to S_j; the choice $G_a^\circ = G'_{X_a}$ contributes its set S_a.

Proof. The general local-step word formula applies because the boxes are disjoint and avoid the flux string. Their slice hulls embed and avoid σ_0: a representative of that plaquette in a hull would give a plaquette in the corresponding box on the chosen column. Hence every local realisation is defined.

Its scalar electric weight must be independent of the inserted links. To meet that hypothesis, let T_j be the union of the link sets of all G and G' insertions at step j, and split β_j into its restrictions to these disjoint local sets and to T_j^c. For the local kernel k_a, use

$$k_{a,\beta}(A',A) = e^{i\langle\beta_j|_{S_a},A'-A\rangle}k_a(A',A).$$

The added factor has modulus one and is unchanged by a common gauge shift of A' and A, so it preserves both local-step hypotheses. Evaluation in $G[k_{a,\beta}]$ multiplies the local integrand by $e^{i\langle\beta_j|_{S_a},F_j^{\mathrm{el}}\rangle}$. Use only $e^{i\langle\beta_j|_{T_j^c},F_j^{\mathrm{el}}\rangle}$ as v_j. It is bounded and independent of every inserted-link coordinate. The magnetic character remains in u_j.

For clarity, the operator calculation identifying its step is as follows. On the links belonging to a G insertion the electric kernel is that of $e^{-|E|^2/4}Xe^{-|E|^2/4}$; for a G' it is that of $e^{-|E|^2/8}Xe^{-|E|^2/8}$. Off all inserted links it is the kernel of $e^{-|E|^2/2}$. The electric character shifts every E to $E-\beta_j$. Thus the common factors on either side of the inserted product are $e^{-|E-\beta_j|^2/4+|E-\beta_j|^2_{S_j}/8}$. Gauge integration contributes $(2\pi/q)^{|\Lambda_{\mathrm{s}}|}P_G$; its square root can be put on each side since all inserted operators commute with the gauge action. These are exactly the two factors $K^{1/2}_{\beta_j,S_j}$.

Local sets in the same step have disjoint link and site supports. Their bosonic operators act on distinct variables, and the signed Fock localisations of even operators on disjoint sets commute. This justifies their product. Inserting the identified steps in the general word formula proves the result, including its order and its normalisation. □

Proposition 11.5 (A trace bound with insertions)**.** *For arbitrary real electric sources β_j, link sets S_j, bounded B_j and Borel magnetic factors $|u_j|\le 1$,*

$$\left|\mathrm{Tr}\prod_{j=n_1-1,\dots,0} K^{1/2}_{\beta_j,S_j}B_jK^{1/2}_{\beta_j,S_j}\mathfrak{M}u_j\mathfrak{M}^*\right| \le (2\pi)^{-n_1n_2n_3}Z_C[1]\prod_j 2^{|S_j|/2}\|B_j\|.$$

The word is trace class.

Proof. Set $T_j = K^{1/2}_{\beta_j,S_j}\mathfrak{M}$. These maps from $\mathcal{H}^{\mathrm{s}}$ to $\mathcal{H}^{\mathrm{l}}$ are Hilbert–Schmidt by the transfer kernel estimate. Cyclicity of trace for a trace-class factor and a bounded factor, and then the word formula without local insertions, give

$$\operatorname{Tr}(T_j^*T_j)^{n_1} = (2\pi)^{-n_1n_2n_3}Z_C\left[\prod_{a=0}^{n_1-1}\left(e^{i\langle\beta_j,F_a^{\mathrm{el}}\rangle}\prod_{b\in S_j}\sqrt{2}\,e^{-F_a^{\mathrm{el}}(b)^2/2}\right)\right] \le 2^{|S_j|n_1/2}(2\pi)^{-n_1n_2n_3}Z_C[1].$$

Indeed the Fourier multiplier $e^{-|E-\beta_j|^2/2+|E-\beta_j|^2_{S_j}/4}$ has electric kernel obtained by replacing g_1 on those links by $g_{1/2} = \sqrt{2}\,e^{-F^2/2}g_1$ and multiplying by the electric character. The bosonic integrand in the display has modulus at most $2^{|S_j|n_1/2}$, so positivity of the bosonic weight proves the inequality.

Move the leftmost heat factor to the right end of the word. Its trace is now that of $\prod_j B_jT_ju_jT_{j-1}^*$, with indices modulo n_1. Regard the alternating maps as operators on $\mathcal{H}^{\mathrm{l}}\oplus\mathcal{H}^{\mathrm{s}}$ by zero extension; their nonzero singular values are unchanged. Schatten Hölder with $2n_1$ equal exponents $2n_1$ bounds the trace by $\prod_j\|B_j\|\|T_j\|^2_{2n_1}$. The displayed power estimate gives the asserted bound. Products of two Hilbert–Schmidt maps are trace class, while all remaining factors are bounded; this also justifies the cyclic trace used above. □

11.4. **Uniform bounds for local integrands.**

Proposition 11.6 (Insertion bounds)**.** *The insertion maps are linear and $G_1 = 1$. They produce even gauge-invariant local integrands with a strict Gaussian growth margin, and*

$$\Theta_cG_X = G_{X^*}, \qquad \Theta_cG'_X = G'_{X^*}.$$

Coordinate-group images of an insertion are insertions of transported operators with the same norm. Consequently G_X is bounded by $\|X\|_\#$ and G'_X by $2^{|S|/2}\|X\|_\#$ with the box $B^\#$. These assertions hold in any coordinate direction. Boundedness is preserved by the full coordinate symmetry group, by enlargement of the box, and by multiplication by a character whose support is included in that box.

Proof. Linearity is immediate from the kernel formula. The kernel for $X = 1$ is $\prod_{b\in S}g_1(A'(b)-A(b))$. The crossing expansion $\operatorname{Ins}_c[\omega^{1/2}u(a)\omega^{1/2}] = e^{-S_c^\times}$ cancels the compensating exponential in $G[k]$, while the electric Gaussian factors cancel their inverses. Hence $G_1 = 1$. Gauge invariance follows from $d_{\mathrm{s}}^*\delta = q$ charge Υ_δ in normal form and from the lower-end phase: a gauge change by a multiplies the lower fields by $e^{-iq\langle a,\mathfrak{q}_-\rangle}$, exactly the column factor $u(a)$. Evenness follows because the total charge is zero, and Fock parity is the gauge phase at the constant value $a = \pi/q$.

Here is the coefficient growth after the compensating Maxwell factor is divided out. For the single symbol used in (11.1), the coefficient of G_X, apart from its finite Grassmann factor, is

$$\det(I+2Q)^{-1/2}e^{-|\delta|^2/4+|x|^2/2}\exp\Big(\tfrac{1}{2}\langle\lambda-\delta/2+ix,(I+2Q)^{-1}(\lambda-\delta/2+ix)\rangle\Big)\,e^{i\langle\delta,\hat{A}_{j+1}\rangle}.$$

Indeed the factor $(2\pi)^{-m/2}$ in its kernel cancels the same factor in $\prod_{b\in S}g_1(x_b)$. Writing $R = (I+2Q)^{-1}$, the modulus of this coefficient is

$$C\exp\{\langle x,Q'x\rangle - \langle\operatorname{Im}\lambda,Rx\rangle\}, \qquad Q' = \tfrac{1}{2}(I-R) = Q(I+2Q)^{-1}.$$

The eigenvalue corresponding to an eigenvalue $a \ge 0$ of Q is $a/(1+2a) < 1/2$. For G'_X the same computation instead has $B = Q + I/4$ and quadratic growth matrix $Q'_{1/8} = I/2 - B^{-1}/4$. This matrix may have negative eigenvalues, which improve the bound; its largest eigenvalue is still strictly less than $1/2$. Choose $\varepsilon > 0$ smaller than half the gap to $1/2$ for every one of the finitely many terms. The elementary inequality $L|x| \le \varepsilon|x|^2 + L^2/(4\varepsilon)$ absorbs each linear term without losing that gap. The remaining factor is $\operatorname{Ins}_c[\omega^{1/2}\Upsilon u(a)\omega^{1/2}]e^{S_c^\times}$. Its coefficients are finite sums of phases of modulus one, so they are uniformly bounded on the real potentials. Thus each Grassmann

coefficient is at most $Ce^{(1/2-\varepsilon)\|F|_B\|^2}$ for some positive margin. The general local-step proof supplies the gauge invariance and integer-shift invariance on the actual box. These facts also justify absolute convergence on every admitted torus.

We next verify the transport used in a placement. For a translation v, write $v_{\mathrm{s}} = (v_2, v_3)$ and let

$$(\mathfrak{T}_v\Phi)(A) = U_v\Phi(\tau_{v_{\mathrm{s}}}^* A).$$

This is a unitary between the two local real Hilbert spaces. It transports each electric coordinate and preserves its squared Euclidean norm. The kernel of $\mathfrak{T}_v X \mathfrak{T}_v^{-1}$ is the transported kernel $U_v k_X(\tau_{\mathrm{s}}^* A', \tau_{\mathrm{s}}^* A)U_v^*$. The inserted-matrix covariance lemma gives the same signed relabelling on both matrix indices, while a_j and the crossing bonds are translated. Therefore

$$(G_X^{(c)})^{\tau_v} = G^{(c+v_1)}_{\mathfrak{T}_v X \mathfrak{T}_v^{-1}}.$$

For reflection in the insertion plane, the upper and lower potentials are interchanged and the time-link potential changes sign. The matrix is adjointed. Writing the two slice potentials as A', A, simultaneous gauge covariance gives

$$k(A, A' - d_{\mathrm{s}}a) = u(a)^{-1}k(A + d_{\mathrm{s}}a, A')u(a).$$

Consequently the reflected matrix factor is

$$[k(A, A' - d_{\mathrm{s}}a)u(-a)]^* = u(a)k(A, A' - d_{\mathrm{s}}a)^* = k(A + d_{\mathrm{s}}a, A')^* u(a).$$

This is the adjoint kernel evaluated at $(A', A + d_{\mathrm{s}}a)$, with its lower phase restored to the right. The electric difference is unchanged. The adjoint kernel identity then gives $\Theta_c G_X = G_{X^*}$. Reflection in another time plane c' is $\tau_{2(c'-c)e_1}\theta_c$, and hence transports the operator to $\mathfrak{T}_{2(c'-c)e_1} X^* \mathfrak{T}^{-1}_{2(c'-c)e_1}$. These calculations apply to both heat parameters.

For a spatial mirror set

$$(\mathfrak{J}\Phi)(A) = U_w\overline{\Phi(\theta_{\mathrm{s}}^* A)}.$$

Oriented pullback is an orthogonal signed permutation of link variables, so $\mathfrak{J}$ is antiunitary. Its kernel transformation is $k^w(A', A) = U_w\overline{k(\theta_{\mathrm{s}}^* A', \theta_{\mathrm{s}}^* A)}U_w^*$. The squared heat norm is unchanged. In the local-step formula the spatial matrix identity gives exactly this kernel, and $U_w\overline{u(a \circ \theta_{\mathrm{s}})}U_w^* = u(a)$ restores the lower-end phase. The crossing exponential and the inverse electric Gaussian are pulled back to their counterparts in the reflected box. Thus the transported operator is $\mathfrak{J}X\mathfrak{J}^{-1}$ and its insertion is the reflected integrand.

Every element of W_{c} has a normal word consisting of a translation and at most one reflection in each coordinate. Compose the three rules along that word. Each preserves the local operator norm, so the transported operator X^g has $\|X^g\|_\# = \|X\|_\#$. The formulas do not depend on an alternative word. To see this directly, the insertion map is injective in our range: multiplying by $e^{-S_c^\times}$ and the electric Gaussian recovers its matrix kernel; the modes are a basis and every ω_i is positive. A zero kernel means $e^{-s|E|^2}Xe^{-s|E|^2} = 0$. The heat operator has dense range and trivial kernel, so $X = 0$. Two transported operators producing the same even group image therefore coincide. The symmetric box is covariant: $gB^\#(c, S, \Lambda) = B^\#(g(c), gS, g\Lambda)$. Its definition includes both the original kernel box and its time-reflected one, so every displayed comparison is made in one common containing box.

Let $(g_1, \dots, g_r)$ now be an arbitrary placement of $B^\#$ in a coordinate torus, and write $c_a = g_a(c) = j_a + 1/2$, $S_a = g_a S$, $\Lambda_a = g_a\Lambda$. The group W_{c} does not permute the time direction, so all these insertions have that same time direction. The preceding calculation identifies the ath image with $G^\circ_{X_a}$, where $X_a = X^{g_a}$ and $\|X_a\|_\# = \|X\|_\#$. Use the free column parallel to e_1 to select σ_0. Its column is outside every period translate of every placement box. Since each actual kernel box is contained in its symmetric box, its slice hull embeds and avoids that magnetic string.

Reduce j_a to its residue $\bar{j}_a \in \{0, \dots, n_1 - 1\}$. Translation by $(j_a - \bar{j}_a)e_1$ is a torus period, has no slice component, and changes neither local electric variables nor the ordering of modes. A period

acts trivially on the even integrand; the two seam fermion signs therefore cancel. The families of period translates of the boxes are unchanged. The mixed word formula may consequently be applied at these reduced times without changing the integral. Put $r_j = \#\{a : \bar{j}_a = j\}$ and form

$$\mathcal{B}_j = \prod_{a:\bar{j}_a=j} (\alpha_{\beta_j} X_a)_{\Xi}.$$

Every factor has norm at most $\|X\|_{\#}$ and hence $\|\mathcal{B}_j\| \le \|X\|_{\#}^{r_j}$. Disjoint embedding makes the corresponding S_a pairwise disjoint in the slice. For unprimed insertions $S_j = \varnothing$; for primed insertions $|S_j| = r_j|S|$. The trace bound applied to the mixed word identity gives

$$\left| Z_C \left[\varphi_\beta \prod_{a=1}^{r} (G_X^{\circ})^{g_a} \right] \right| \le Z_C[1] \prod_{j=0}^{n_1-1} (\|\mathcal{B}_j\| 2^{|S_j|/2}).$$

The factor $(2\pi)^{|\mathbb{T}|}$ from the word formula cancels its inverse in the trace estimate. Since $\sum_j r_j = r$, this is $\|X\|_{\#}^{r} Z_C[1]$ for G and $(2^{|S|/2}\|X\|_{\#})^{r} Z_C[1]$ for G'. The character, placement and torus were arbitrary, which is the full boundedness assertion. Cyclic coordinate permutations give the other two time directions.

For completeness the general closure properties require no insertion representation. If g is a signed coordinate permutation followed by a translation, conjugation takes a placement of B to a placement of gB in $g\mathbb{T}$. It takes coordinate tori to coordinate tori, because it permutes their period generators and their negative characters. It preserves enlarged-box disjointness and sends a free column in direction e_μ to a free column in the permuted direction. Also W_{c} is normal in W, so the conjugated placement maps still belong to W_{c}. Pulling the character back changes its exponent to $g^*\beta$ for an even g and to $-g^*\beta$ for an odd g. Covariance of the finite functional then identifies the absolute values and the denominators. This proves invariance of boundedness under the full group.

Every placement of a larger box is a placement of its contained smaller box; the free columns of the larger box also serve the smaller one. Enlargement therefore preserves the bound. Finally the image of a local character is another real-exponent character. Its exponent is $(-1)^{\mathrm{parity}(g)}(g^{-1})^*\beta$ and its support lies in the image of the larger box. Periodize these finitely supported exponents and add them to the character already present in the definition of boundedness. The bound for G with that new total exponent is exactly the bound for $\varphi_\beta G$. This proves character absorption, including reflections and arbitrary real coefficients. □

11.5. **Two adjacent layers.**

Proposition 11.7 (Regularised two-layer representation)**.** *Let $Y = e^{i\langle\delta,\hat{A}\rangle}\Psi \in \mathcal{P}_1(c)$, with $\delta = (\delta_0, \delta_1, \delta_\perp)$ and local sets S, Λ_Y as in Definition 6.18. Its lower and upper charges satisfy*

$$d_{\mathrm{s}}^*\delta_0 - \delta_\perp = q\mathfrak{q}_-, \qquad d_{\mathrm{s}}^*\delta_1 + \delta_\perp = q\mathfrak{q}_+.$$

The matrix Υ_Y in that definition exists uniquely and has charge $q^{-1}d_{\mathrm{s}}^*(\delta_0+\delta_1)$. The operators $X'_Y, X_{Y,\varepsilon}$ defined there lie in $\mathfrak{W}$, and

$$Y = G'_{X'_Y}, \qquad Y - G_{X_{Y,\varepsilon}} = G'_{X'_Y - P_\varepsilon X'_Y P_\varepsilon}.$$

For a finite C_Y independent of ε,

$$\xi_Y(\varepsilon) = \|X'_Y - P_\varepsilon X'_Y P_\varepsilon\|_{\#} \le C_Y\varepsilon.$$

All reflected operators are adjoints; in particular $X_{\Theta_c Y,\varepsilon} = X_{Y,\varepsilon}^*$. More explicitly, with $d = \delta_1 - \delta_0$ and $s_\delta = |\delta_0|^2 + |\delta_1|^2$,

$$\|X'_Y\|_{\#} = \|\Upsilon_Y\| e^{|d|^2/16 + s_\delta/8}, \qquad \|X_{Y,\varepsilon}\|_{\#} = \|\Upsilon_Y\| e^{(1/2-2\varepsilon)^2|d|^2/(8\varepsilon) + (1/4-\varepsilon)s_\delta}.$$

Proof. There are three separate points: the matrix representing the two layers, the exact heat-sandwich formulas, and the small error after regularization. Keeping them separate also explains which constants remain uniform when the volume grows.

The Gauss laws and the matrix. Gauge invariance is $d^{*}\delta = q\mathfrak{q}(\Psi)$, where

$$(d^{*}\delta)(x) = \sum_{\mu=1}^{3} \{\delta(x - e_{\mu}, \mu) - \delta(x, \mu)\}.$$

There are no incoming time links below the lower plane and no outgoing ones above the upper plane in this monomial. Thus the time contribution is $-\delta_{\perp}$ at the lower plane and $+\delta_{\perp}$ at the upper plane. The remaining contributions are $d_{\mathrm{s}}^{*}\delta_0$ and $d_{\mathrm{s}}^{*}\delta_1$, proving the displayed two equations.

The upper modes and their lower reflected modes form a basis of all Grassmann generators at the ends of the crossing bonds over Λ_Y. Therefore the products $\Theta_c(\zeta_J)\zeta_I$ form a basis of that Grassmann algebra. Put

$$\Psi\ \mathrm{Ins}_c[\omega] = \mathrm{Ins}_c[T_Y], \qquad \Upsilon_Y = \omega^{-1/2} T_Y \omega^{-1/2}.$$

The basis determines T_Y uniquely. Every crossing coefficient is $w_3(1+\rho)/2$ or $w_3(1-\rho)/2$, so ω is invertible. This is exactly the matrix equation defining Υ_Y. To compute its charge, expand $\mathrm{Ins}_c[\omega] = \sum_R \omega_R \Theta_c(\zeta_R)\zeta_R$. The summand multiplied by Ψ has upper charge $\mathfrak{q}_+ + \mathfrak{q}(R)$ and lower charge $\mathfrak{q}_- - \mathfrak{q}(R)$. A nonzero matrix entry (I, J) consequently satisfies

$$\mathfrak{q}(I) = \mathfrak{q}_+ + \mathfrak{q}(R), \qquad \mathfrak{q}(J) = -\mathfrak{q}_- + \mathfrak{q}(R).$$

Subtracting shows that every entry has charge $z = \mathfrak{q}_+ + \mathfrak{q}_-$: $u(a)\Upsilon_Y u(a)^{-1} = e^{iq\langle a, z\rangle}\Upsilon_Y$. The sum of the Gauss laws is $d_{\mathrm{s}}^{*}(\delta_0 + \delta_1) = qz$. Its pairing with the constant function is zero, so the total charge is zero and the matrix is even.

Recovery of the time-link phase. Define

$$k_Y^0(A', A) = e^{i\langle \delta_1, A'\rangle} \prod_{b \in S} g_1(A'(b) - A(b)) e^{i\langle \delta_0, A\rangle} \Upsilon_Y.$$

Its modulus has Gaussian decay $e^{-|A'-A|^2/2}$. A common gauge shift contributes $e^{i\langle \delta_0 + \delta_1, d_{\mathrm{s}} a\rangle}$, which is canceled by the charge phase of the matrix. Hence it is a local step datum. Let α_a^{-} multiply only the lower Grassmann generators by the inverse of their gauge phases. On an ordered product $\Theta_c(\zeta_J)\zeta_I$ its multiplier is $e^{iq\langle \mathfrak{q}(J), a\rangle}$, so

$$\alpha_a^{-}\ \mathrm{Ins}_c[T] = \mathrm{Ins}_c[Tu(a)], \qquad \alpha_a^{-}\Psi = e^{-iq\langle \mathfrak{q}_-, a\rangle}\Psi.$$

Apply this automorphism to $\Psi\,\mathrm{Ins}_c[\omega] = \mathrm{Ins}_c[\omega^{1/2}\Upsilon_Y\omega^{1/2}]$. The diagonal matrices ω and $u(a)$ commute, and the crossing expansion at nonzero time-link potential gives

$$\mathrm{Ins}_c[\omega^{1/2}\Upsilon_Y u(a)\omega^{1/2}] = e^{-iq\langle \mathfrak{q}_-, a\rangle}\Psi e^{-S_c^{\times}}.$$

In $G[k_Y^0]$ the Gaussian in the kernel cancels the inverse Gaussian at $x = F_{j_0}^{\mathrm{el}}|_S$. Its remaining scalar phase is

$$e^{i\langle \delta_1, \hat{A}_{j_0+1}\rangle + i\langle \delta_0, \hat{A}_{j_0}\rangle + i\langle d_{\mathrm{s}}^{*}\delta_0, \hat{a}_{j_0}\rangle}.$$

Use the matrix identity just obtained, cancel the crossing exponential and subtract $q\mathfrak{q}_-$ from the last exponent. The first Gauss law makes that exponent $\delta_{\perp}$. Thus $G[k_Y^0] = Y$ exactly, including the time-link phase.

The primed sandwich. Write $D = \delta_0 + \delta_1$ and $\widetilde{E} = E_S + \delta_0$. Fourier inversion of $e^{-|\xi+\delta_0|^2/2}$ gives $e^{-i\langle \delta_0, x\rangle}\prod_b g_1(x_b)$. It follows that the operator with the displayed kernel is

$$k_Y^0 = e^{i\langle D, A\rangle} e^{-|\widetilde{E}|^2/2}\Upsilon_Y.$$

The definition of X_Y' is this multiplication character times the Gaussian symbol

$$f'(x) = e^{-|x|^2/4 + \langle x, d\rangle/4 + s_\delta/8}, \qquad x = \widetilde{E}.$$

This is bounded; the charge of its matrix is $q^{-1}d_{\mathrm{s}}^{*}D$, so X'_Y belongs to the local algebra. Move the left heat factor through the character by the Weyl rule. The two heat exponents are $-|x+\delta_1|^2/8$ and $-|x-\delta_0|^2/8$. The identity

$$\frac{|x+\delta_1|^2+|x-\delta_0|^2}{8}=\frac{|x|^2}{4}+\frac{\langle x,d\rangle}{4}+\frac{s_\delta}{8}$$

then proves

$$e^{-|E_S|^2/8}X'_Y e^{-|E_S|^2/8}=k^0_Y.$$

The equality of kernels proves $Y=G'_{X'_Y}$. Periodization and gauge averaging give the asserted slice realization: off S the ordinary heat factors remain, on S the factors are $e^{-|E_S|^2/8}$, and the common gauge factor is $(2\pi/q)^{|\Lambda_{\mathrm{s}}|}P_{\mathrm{G}}$. This is precisely $K_{0,S}^{1/2}(X'_Y)_{\Xi}K_{0,S}^{1/2}$. Multiplication of the kernel by an electric character conjugates the local operator and shifts E to $E-\beta$, so the same identity holds with $K_{\beta,S}$ and $\alpha_\beta X'_Y$. The already proved local norm formula is the scalar supremum times $\|\Upsilon_Y\|$. Since $-|x|^2/4+\langle x,d\rangle/4=-|x-d/2|^2/4+|d|^2/16$, it gives the stated norm of X'_Y.

Regularization with ordinary heat factors. In the same variable $x=\widetilde{E}$, the symbol defining $X_{Y,\varepsilon}$ is

$$f_\varepsilon(x)=e^{-2\varepsilon|x|^2+(1/2-2\varepsilon)\langle x,d\rangle+(1/4-\varepsilon)s_\delta}.$$

It is a bounded Gaussian for every $\varepsilon>0$ with the same charge character. Moving the left ordinary heat factor through that character contributes $-(|x+\delta_1|^2+|x-\delta_0|^2)/4$. Adding its exponent to the exponent of f_ε gives

$$-\frac{|x|^2}{2}-\varepsilon|x+\delta_1|^2-\varepsilon|x-\delta_0|^2.$$

The last two terms are exactly the left and right $P_\varepsilon=e^{-\varepsilon|E_S|^2}$ factors after commutation. Therefore

$$e^{-|E_S|^2/4}X_{Y,\varepsilon}e^{-|E_S|^2/4}=P_\varepsilon k^0_Y P_\varepsilon.$$

The same equality after periodization proves its slice version. Completing the square in f_ε gives the stated norm: its maximum exponent is $(1/2-2\varepsilon)^2|d|^2/(8\varepsilon)+(1/4-\varepsilon)s_\delta$.

The primed heat factors commute with P_ε. Thus $P_\varepsilon k^0_Y P_\varepsilon$ is also the primed sandwich of $P_\varepsilon X'_Y P_\varepsilon$. Equality of kernels and linearity now give the exact difference of integrands in the statement. The error norm is the supremum of

$$\|\Upsilon_Y\|e^{-|x|^2/4+\langle x,\delta_1-\delta_0\rangle/4+(|\delta_0|^2+|\delta_1|^2)/8}(1-e^{-\varepsilon|x+\delta_1|^2-\varepsilon|x-\delta_0|^2}).$$

Set $a(x)=|x+\delta_1|^2+|x-\delta_0|^2$. The inequality $1-e^{-\varepsilon a}\le\varepsilon a$ gives, for every $\varepsilon>0$,

$$\xi_Y(\varepsilon)\le C_Y\varepsilon,\qquad C_Y=\|\Upsilon_Y\|e^{|d|^2/16+s_\delta/8}\sup_x e^{-|x-d/2|^2/4}a(x)<\infty.$$

Finiteness follows by expanding the quadratic polynomial about $d/2$ and using the boundedness of $r^2e^{-r^2/4}$. The placement bound for primed insertions proves the claimed bound for the error integrand.

Reflected monomials. Reflection exchanges the two spatial slice exponents with a minus sign and reverses the time-link orientation twice, once by pullback and once by scalar conjugation. Its exponent is $(-\delta_1,-\delta_0,\delta_\perp)$, with the same supports. Reflect the defining matrix equation. The crossing exponential at zero time link is fixed and the inserted-matrix rule takes the matrix to its adjoint; hence $\Upsilon_{\Theta_c Y}=\Upsilon_Y^*$. The two-layer kernel is correspondingly $k^0_{\Theta_c Y}(A',A)=k^0_Y(A,A')^*$. Taking adjoints of either proved heat-sandwich identity and using the injectivity of the positive heat sandwich proves $X'_{\Theta_c Y}=(X'_Y)^*$ and $X_{\Theta_c Y,\varepsilon}=X^*_{Y,\varepsilon}$. The argument is linear for a finite combination of monomials; one may first enlarge all supports to common finite sets. □

The regularized norm need not remain bounded as $\varepsilon \downarrow 0$. If $\delta_1 \neq \delta_0$, its displayed formula can grow exponentially in $1/\varepsilon$. We will therefore fix ε while taking the volume and strength limits. Only after the strength limit removes the coefficient of that norm will the $O(\varepsilon)$ error be sent to zero.

Theorem 11.8 (Approximate reflection identity)**.** *For $X \in \mathfrak{W}$ read in a fixed insertion box, choose the amplification integer for its symmetric box and write $\gamma = \gamma_{t,i}(k)$. Then*

$$|\mu_i^t(G_X)| \le \gamma \|X\|_{\#}, \qquad |\mu_i^t(\Theta_c G_X) - \overline{\mu_i^t(G_X)}| \le 4\sqrt{\gamma^2 - 1}\,\|X\|_{\#}.$$

For a two-layer monomial Y with link set S,

$$|\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| \le 2\,2^{|S|/2}\xi_Y(\varepsilon)\gamma + 4\sqrt{\gamma^2 - 1}\,\|X_{Y,\varepsilon}\|_{\#}.$$

Hence for every two-layer polynomial Y,

$$\lim_{t \downarrow 0} \limsup_{i \to \infty} |\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| = 0.$$

The same statements hold at every coordinate link plane.

Proof. The functional $o(X) = \mu_i^t(G_X)$ is linear, unital and has norm at most γ by amplification. In particular $\gamma \ge 1$. If $X = X^*$ and $\|X\|_{\#} \le 1$, the C^*-identity gives $\|X + isI\|_{\#}^2 = \|X^2 + s^2 I\|_{\#} \le 1 + s^2$. Writing $o(X) = u + iv$ and using $o(I) = 1$ yields $(v + s)^2 \le |o(X + isI)|^2 \le \gamma^2(1 + s^2)$ for every real s. Equivalently

$$(\gamma^2 - 1)s^2 - 2vs + \gamma^2 - v^2 \ge 0 \qquad (s \in \mathbb{R}).$$

When $\gamma > 1$, its discriminant is nonpositive; it equals $4\gamma^2(v^2 - \gamma^2 + 1)$. Thus $|v| \le \sqrt{\gamma^2 - 1}$. When $\gamma = 1$, the same inequality has no quadratic term and forces $v = 0$ by taking s of either sign with arbitrarily large modulus. Scaling proves $|o(H) - \overline{o(H)}| \le 2\sqrt{\gamma^2 - 1}\,\|H\|_{\#}$ for every self-adjoint H. For a general X, put $H = (X + X^*)/2$ and $K = (X - X^*)/(2i)$. Both are in the local algebra and have norm at most $\|X\|_{\#}$. Since

$$o(X^*) - \overline{o(X)} = o(H) - \overline{o(H)} - i\{o(K) - \overline{o(K)}\},$$

the triangle inequality gives $|o(X^*) - \overline{o(X)}| \le 4\sqrt{\gamma^2 - 1}\,\|X\|_{\#}$. The insertion adjoint identity proves the first assertion.

Apply this bound to $X_{Y,\varepsilon}$ and bound its two error integrands by $2^{|S|/2}\xi_Y(\varepsilon)$ using the previous propositions. Explicitly write $Y = G_{X_{Y,\varepsilon}} + G'_{X'_Y - P_\varepsilon X'_Y P_\varepsilon}$, and the adjoint identity gives the corresponding decomposition of $\Theta_c Y$. Each of the two primed errors has the same placement constant $2^{|S|/2}\xi_Y(\varepsilon)$. Amplification therefore bounds their expectations by that constant times γ; the preceding estimate bounds the remaining ordinary insertion. This proves the displayed two-layer estimate.

At fixed ε, $\Delta_i \to 0$ gives $\gamma_{t,i}(k) \to e^{2t|O|_1 k^3}$ as $i \to \infty$. Consequently the upper volume limit of the defect is at most

$$2\,2^{|S|/2}\xi_Y(\varepsilon)e^{2t|O|_1 k^3} + 4\sqrt{e^{4t|O|_1 k^3} - 1}\,\|X_{Y,\varepsilon}\|_{\#}.$$

Let $t \downarrow 0$ with ε still fixed. The second term vanishes because the norm is finite and independent of i, t. The result is at most $2\,2^{|S|/2}\xi_Y(\varepsilon)$, which tends to zero when $\varepsilon \downarrow 0$. The geometric integer and the threshold are independent of ε, because its regularisation changes no support. Finite sums follow by the triangle inequality. Coordinate covariance gives the other directions. The regularisation is removed last: the norm of $X_{Y,\varepsilon}$ need not stay bounded as $\varepsilon \downarrow 0$. □

12. Eliminating the dependence on outer layers

An insertion at a link plane satisfies an approximate reflection identity, but a general local character depends on more than the two adjacent layers. We now reduce that character, and more generally any element of $\mathcal{P}$, to those two layers. The reduction is finite and exact before the source is imposed. With the folded source its error is $O(t)$, uniformly in volume. We first prove the uniform bound needed to estimate this error; it does not use contour shifts.

12.1. **Fermion recursion.** The six thin layers in the thick layer j have first coordinates $j - v_1, j - v_2, j - v_3, j + v_3, j + v_2, j + v_1$. Denote the two successors of a thin layer $\mathbb{L}$ by $\mathbb{L}^+, \mathbb{L}^{++}$ and its predecessors by $\mathbb{L}^-, \mathbb{L}^{--}$.

Lemma 12.1 (Bonds between layers)**.** *Every site has two neighbours in its own thin layer, one in the next layer and one in the preceding layer. The successive upward bond types are*

$$2, 1, 0, 1, 2, 3$$

as the layer index runs from one to six. The same-layer bond types are $0, 1$ *in layers one and six,* $0, 3$ *in layers two and five, and* $2, 3$ *in layers three and four. Only type* 3 *carries a link phase. Its bonds within a thick layer lie in thin layers two through five and project to links of the slice at* j*; the type* 3 *bonds from layer six to the next layer one project to time links from* j *to* $j + 1$*. Each photon link has eight such bonds, two in each of the four interior thin layers for a spatial link. Along any bond the photon-site displacement is zero or a signed coordinate unit vector.*

Proof. Write $x = \pi(x) + y$, where the coordinates of y are a signed permutation of v_1, v_2, v_3. The type 0 bond reverses the sign of the coordinate of modulus v_3; type 1 exchanges the positions of v_2 and v_3 while retaining their signs; type 2 exchanges v_1 and v_2. These operations do not change $\pi(x)$. Type 3 changes $\pi(x)$ by the signed unit vector in the coordinate of modulus v_1 and reverses that coordinate of y. If the first modulus is v_3, the first two operations lead to the adjacent v_3 and v_2 layers. If it is v_2, types one and two lead to the v_3 and v_1 layers. If it is v_1, type two leads to v_2 and type three to the next or previous thick layer. The other two bonds leave the first coordinate unchanged. This proves the list and displacement assertion.

For a link in direction e_ν, its lower bond ends are the sites with $y_\nu = v_1$. The remaining two moduli have two possible orders and four choices of signs, giving eight ends and eight bonds. For a spatial link the first coordinate is each of $-v_2, -v_3, v_3, v_2$ twice; for a time link it is v_1, with the upper ends having first coordinate $j + 1 - v_1$. The phase assertions now follow from $U_A(x, x') = e^{iqA(\pi(x),\pi(x'))}$. □

Proposition 12.2 (Exact fermion replacements)**.** *All hopping matrices are invertible for* $0 < \rho < 1$*. With the replaced field placed at the far left, the following rules preserve the finite integral with every bounded bosonic factor* $\Phi(F)$*:*

$$\bar\psi(x) \longmapsto \left(M\bar\psi(x') - \sum_{x''\sim x',\, x''\neq x} \bar\psi(x'')w(x'', x')U_A(x'', x') + \frac{\partial}{\partial\psi(x')} \right) w(x, x')^{-1}U_A(x', x),$$

$$\psi(x) \longmapsto U_A(x, x')w(x', x)^{-1} \left(M\psi(x') - \sum_{x''\sim x',\, x''\neq x} w(x', x'')U_A(x', x'')\psi(x'') - \frac{\partial}{\partial\bar\psi(x')} \right).$$

The derivatives are left Grassmann derivatives and act on the factor to the right. Choosing x' *as the next-layer neighbour of* x *defines, by a finite recursion, a linear map* $R^+_{\mathbb{L}} : \mathcal{P} \to \mathcal{P}$ *eliminating every field of* $\mathbb{L}$*. Its new fields lie in* $\mathbb{L}^+$ *or* $\mathbb{L}^{++}$*; its new link phases are those of type* 3 *bonds with an end in* $\mathbb{L}^+$*. If the original integrand is read in* B*, the result is read in* $B^{+1} = B + [-1, 1]^3$*. The integral identity holds on every torus in which this enlarged box embeds. The maps commute with bosonic characters and introduce no fields below* $\mathbb{L}$ *when none were present. The downward maps*

$R_{\mathbb{L}}^{-}$ have the corresponding reversed properties. The identities remain exact after reflecting all their local integrands and boxes in any lattice mirror.

Proof. The replacement is an identity of integrated Grassmann coefficients, not a pointwise equation of fields. We keep the derivative term because it is essential when the factor to its right contains a field at x'. The Berezin integral of a left derivative is zero. Since the fermion action S is even,

$$\partial_{\psi} e^{-S} = (\bar{\psi} D)_{\alpha} e^{-S}, \qquad \partial_{\bar{\psi}} e^{-S} = -(D\psi)_{\alpha} e^{-S}.$$

Apply the product rule to $e^{-S}G$ and integrate. This gives

$$Z[\Phi(\bar{\psi} D)_{\alpha}(x')G] = -Z[\Phi\, \partial_{\psi_{\alpha}(x')} G], \quad Z[\Phi(D\psi)_{\alpha}(x')G] = Z[\Phi\, \partial_{\bar{\psi}_{\alpha}(x')} G].$$

The factor Φ has no Grassmann variables. Solving these two equations for the summand at the neighbour x of x' gives the two displayed replacement rules, including their opposite derivative signs. The matrices $w(x, x')$ have nonzero eigenvalues proportional to $1 + \rho$ and $1 - \rho$, so their inverses exist. Every term preserves the gauge charge: a hopping phase transports one unit of charge between its photon ends, and the derivative removes the corresponding generator. Thus the rules preserve $\mathcal{P}$.

Fix an order of generators in the thin layer. In a monomial, move its first remaining layer field to the far left with its Grassmann sign, and replace it using its next-layer neighbour. The neighbours of x' other than x lie in $\mathbb{L}^{+}$ or $\mathbb{L}^{++}$, by the layer lemma. The derivative may remove fields there but creates none in $\mathbb{L}$. Each replacement therefore decreases the number of fields of $\mathbb{L}$ by one. The recursion terminates after at most their original number in each branch. More explicitly, if a monomial has N fields of $\mathbb{L}$, first reorder it to $\epsilon\, \xi G$ with the selected such field ξ at the left. Insert the appropriate row or column replacement. Expand its finite neighbour sum and its derivative of G into the fixed ordered monomial basis. Every surviving summand has at most $N - 1$ fields of $\mathbb{L}$; a derivative can delete additional fields, never add them. Apply the same operation to those summands. Induction on N proves that the terminal finite sum has no field of $\mathbb{L}$ and has the same integral against Φ. Making the ordering and selected field fixed on every basis monomial defines a linear map on their span; it does not assert that a different choice gives the same pointwise polynomial. The new phases are those appearing at x'; its bond to x is included among them. This gives the asserted phase support.

The geometry in the layer lemma shows that these new sites have photon cubes contained in B^{+1}, and the new photon links lie there as well. Indeed an interior step has unchanged photon site until its possible type 3 spatial hop; a step across a time link has photon displacement e_1, and the remaining bonds keep that new photon site fixed. No step has a sum of two independent photon displacements. Thus the box enlargement is one. The integrands have bounded real phase coefficients, so all the finite integrals are absolutely convergent. Iterating the pointwise Berezin identities proves the integrated identity with every bounded Φ.

Characters commute with all generators and derivatives and are unchanged by this recursion. Choosing the preceding neighbour proves the downward statements. Finally the finite reflection covariance of the functional gives the reflected identities on the reflected torus; this uses no reflection invariance of the original torus or of Φ. □

Proposition 12.3 (An exact normal form for integral estimates)**.** *For $P \in \mathcal{P}$ and a coordinate link plane c, there are finitely many $\lambda_a \in \mathbb{C}$, compactly supported real two-cochains β_a, integer two-layer monomials Y_a at c, and a box $B'(P, c)$, depending only on P, c and the fixed model, such that*

$$Z_{\mathbb{T}}[\Phi P] = \sum_{a} \lambda_a Z_{\mathbb{T}}[\Phi \varphi_{\beta_a} Y_a]$$

for every bounded bosonic factor Φ and every torus in which B' embeds. All terms and all β_a are read in that box. Their reflected identities have coefficients $\overline{\lambda_a}$.

Proof. We first reduce only the Grassmann support; phases far from the two target slices will subsequently be assigned to a bosonic character. Expand P in its finite phase-monomial expression. Beginning with its lowest thin layer, apply the upward map to each layer below $\mathbb{X}_{j,6}$ in increasing order. Stop after $\mathbb{X}_{j,5}$. Previously emptied layers are not reintroduced. A step produces fields only in its next two layers, so the last step produces fields at most in $\mathbb{X}_{j,6}$ and $\mathbb{X}_{j+1,1}$. There are only finitely many layers between the original lowest layer and the target, and each recursion on a layer is finite. Next apply the downward maps in decreasing order to every layer above $\mathbb{X}_{j+1,1}$, stopping after $\mathbb{X}_{j+1,2}$. The last downward step produces fields no lower than $\mathbb{X}_{j,6}$, and no subsequent map acts below it. All surviving fields therefore lie at the two ends of the target link plane.

If N_+ and N_- are the numbers of layers processed in these two stages, repeated use of the support assertion puts every resulting term in $B_1 = B + (N_+ + N_-)[-1,1]^3$. These numbers depend on the original finite support and target, not on the torus. Every intermediate integral identity is valid when B_1 embeds, since each of its smaller boxes then embeds. The finite sum produced by the recursion has coefficients depending on the fixed mass and invertible hopping matrices. It uses no inverse of the full fermion matrix.

For each resulting monomial $e^{i\langle\delta,\hat{A}\rangle}\Psi$ the integer charge $\mathfrak{q}(\Psi)$ is supported on these two photon slices and has total sum zero. The latter follows by summing $d^*\delta = q\mathfrak{q}(\Psi)$; every supported link contributes with opposite signs at its ends. Choose a finite rectangular set $R \subset \mathbb{Z}^2$ containing all charged sites on the two slices. The graph consisting of its links in both slices and the time links joining equal sites is connected. Choose a spanning tree of this graph and a root. Orient its edges from a child towards its parent. On the edge from v to its parent set

$$J(v, \mathrm{parent}(v)) = -\sum_{u \in \mathrm{subtree}(v)} \mathfrak{q}(\Psi)(u).$$

All these values are integers; set the current to zero off the tree and use opposite values on oppositely oriented edges. At a nonroot vertex its incoming child currents minus its outgoing current equal $\mathfrak{q}(\Psi)(v)$. At the root the same equality uses the zero total sum. Thus $d^*J = \mathfrak{q}(\Psi)$ at every site, and its support is entirely within the two target slices and their joining time links. The current $\delta - qJ$ is divergence free. The finite-box filling lemma, Lemma 2.19, gives a real two-cochain β in an enlarged finite box with $d^*\beta = \delta - qJ$. All links supporting $\delta - qJ$ lie in a fixed finite rectangle containing B_1 and the tree. The filling lemma is applied on the infinite lattice to this finitely supported current, not to a periodic current on a torus. Choose a larger box containing the resulting finite two-chain. On each local lift in that larger box, summation by parts gives $\langle \delta - qJ, \hat{A}\rangle = \langle \beta, d\hat{A}\rangle = \langle \beta, F\rangle$. Hence, on every local lift there,

$$e^{i\langle\delta,\hat{A}\rangle}\Psi = \varphi_\beta(F) e^{iq\langle J,\hat{A}\rangle}\Psi.$$

The last factor is gauge invariant, its phase exponent divided by q is integer, and its fields and phase links occupy just the prescribed two layers. It is therefore an integer two-layer monomial. Perform this construction for each surviving term. There are finitely many, so one larger box contains all their tree currents, fillings and recursion boxes. That is $B'(P,c)$. The pointwise phase splitting and the already proved integral recursions give the stated identity against any bounded Φ, with the same terms for every torus admitting this box. For a reflected identity use the original identity on the reflected torus, with the pulled-back bosonic factor, and apply finite functional covariance. Reflection conjugates the scalar coefficients and sends the containing box to its reflected box. All constructions took place on $\mathbb{Z}^3$; no volume, source strength or bosonic factor entered them. □

Proposition 12.4 (A uniform local bound)**.** *For $P \in \mathcal{P}$ choose the preceding normal form at $c = \frac{1}{2}$ and set*

$$C_P = \sum_a |\lambda_a| \cdot 2^{|S_a|/2} \|X'_{Y_a}\|_{\#} < \infty,$$

with the contribution for $Y_a = 1$ equal to $|\lambda_a|$. For every fixed box B_0 there are k_P, i_0 such that

$$\sup_{\operatorname{supp}\beta\subset B_0} |\mu_i^t(\varphi_\beta P)| \le C_P \gamma_{t,i}(k_P) \quad (t \ge 0, i \ge i_0).$$

The same bound, with the same C_P and possibly a different k_P, i_0, holds for $\Theta_\Pi P$ at any fixed mirror.

Proof. Put all the normal-form boxes, the symmetric insertion boxes, and B_0 in one fixed box. Each Y_a is bounded by $2^{|S_a|/2}\|X'_{Y_a}\|_\#$, and multiplying by $\varphi_{\beta+\beta_a}$ changes none of these constants. The normal form applies with the bounded factor $e^{tO_\flat}\varphi_\beta$. Divide by its positive perturbed partition function and apply amplification in the common box. The result is the displayed sum with a single geometric integer and threshold, uniformly over all the real coefficients of β. Reflecting the exact normal form and using covariance of boundedness proves the last assertion. □

12.2. **Imaginary contour shifts.** An entire bosonic factor Φ is bounded on each horizontal strip, as in Definition 2.20. Set

$$\Phi^{[\eta]}(F) = \Phi(F - i\eta), \qquad P^{(\delta)}(\hat{A}) = P(\hat{A} - i\delta),$$

where η is a real two-cochain and δ a real one-cochain. All their compact supports are periodised when used on a torus. The factor reflection is $\Phi^\theta(F) = \overline{\Phi(\theta^*\overline{F})}$. Consequently a reflected shift is $-\theta^*\eta$, rather than $\theta^*\eta$.

Proposition 12.5 (A local contour identity)**.** *Let δ be supported on a finite set S of links in one coordinate site plane. Let B contain P and $b + [-1,1]^3$ for every $b \in S$, and suppose B embeds in $\mathbb{T}$. Define the local even Grassmann polynomial*

$$Q_\delta = S(A) - S(A - i\delta), \qquad V_\delta = e^{Q_\delta}.$$

Only bonds over the support of δ enter Q_δ; each forward and reverse hopping term is multiplied respectively by $e^{q\delta_b} - 1$ and $e^{-q\delta_b} - 1$. In particular $Q_\delta^r = 0$ for $r > 64|S|$, so V_δ is a finite polynomial in $\mathcal{P}$. Then for every entire factor Φ,

$$Z_{\mathbb{T}}[\Phi P] = Z_{\mathbb{T}}\left[\Phi^{[d\delta]} e^{\|d\delta\|^2/2} e^{i\langle d^*d\delta, \hat{A}\rangle} V_\delta P^{(\delta)}\right].$$

Every new local term is read in B.

Proof. The eight bonds over each link have disjoint ends. The $|S|$ links therefore involve at most $16|S|$ sites and $128|S|$ Grassmann generators. Each term of Q_δ has degree two, proving the nilpotence bound. The hopping phase and its two fields are gauge invariant, so $Q_\delta, V_\delta \in \mathcal{P}$. In particular the action difference is explicitly

$$Q_\delta = \sum_{x' \sim x} \bar{\psi}(x) w(x,x') U_A(x,x') (e^{q\delta(\pi(x),\pi(x'))} - 1)\psi(x').$$

The sum is over ordered hopping terms. If their photon ends coincide the parenthesis is zero; for a link in S its two orientations have exponents $q\delta_b$ and $-q\delta_b$. The mass term cancels. This fixes the sign of V_δ. The action difference and $e^{-S(A-i\delta)} = e^{-S(A)} V_\delta$ follow directly from the hopping formula, since the even polynomials commute.

We justify the contour shift on the actual flux sum. Take first the site plane $x_1 = j$. For $b = (x, \nu) \in S$, $\nu = 2, 3$, let $\sigma(b) = (x; 1\nu)$ be the plaquette leaving that plane. Its only side in S is b and

$$(du)_{\sigma(b)} = -u_b \quad (\operatorname{supp} u \subset S).$$

The plaquettes are distinct modulo periods because B embeds. Thus the action of $\mathbb{Z}^S$ on closed integer fluxes by $z \mapsto z + dk$ is free. Choose orbit representatives z_0. Here k is extended by zero to all other links. If $dk = 0$, the distinguished plaquette equations force each $k_b = 0$; this proves freeness without a cohomology assumption. Every closed flux belongs to exactly one representative orbit,

since the set of closed integer cochains is countable. The equivalence $(A, z_0 + dk) \sim (A + (2\pi/q)k, z_0)$ unfolds the compact link domains in S, giving, for an integrable configuration function H,

$$\int H = \sum_{z_0} \int_{[0,2\pi/q)^{\mathbb{T}^{(1)}\setminus S}} dA' \int_{\mathbb{R}^S} d\xi\, H((A', \xi), z_0).$$

To verify it directly, write each orbit as $z_0 + dk$ and change the integration variable on its S coordinates to $\xi = A|_S + (2\pi/q)k$. The half-open cubes $[0, 2\pi/q)^S + (2\pi/q)k$ partition $\mathbb{R}^S$. The field strength is unchanged by this simultaneous change, all hopping phases are unchanged, and the actual local integrand has the same value because its local lift has not changed. Thus it is an identity for functions on configurations. Tonelli proves it for nonnegative functions; apply that identity to the absolute value before using Fubini for an integrable one.

For fixed z_0, A', choose an integer one-cochain p_B with $dp_B = z_0$ on the cells of B, using the integer local lift lemma. This is a local choice on a contractible box and does not assert a primitive for the flux on the torus. Write $A(\zeta) = (A', \zeta)$ and $\hat{A}(\zeta) = A(\zeta) + (2\pi/q)p_B$ on B, $F(\zeta) = dA(\zeta) + (2\pi/q)z_0$. After the Berezin integral, the integrand

$$f(\zeta) = e^{-\langle F(\zeta), F(\zeta)\rangle/2} \Phi(F(\zeta)) J_{\mathbb{T}}[P](A(\zeta); z_0)$$

is entire. Here $J_{\mathbb{T}}[P](A; z_0)$ means the actual Berezin coefficient of $P(\hat{A})e^{-S(A)}$ with this fixed local primitive; it requires no inverse fermion matrix. The scalar pairing in the Maxwell factor is bilinear. Expand P into its finite local phase-monomial sum and expand e^{-S} in the Grassmann algebra. Each coefficient is a finite product of $e^{\pm iqA(b)}$ and $e^{i\langle \delta_P, \hat{A}\rangle}$ with real δ_P. The integer part of $\hat{A}$ contributes modulus one. On a fixed imaginary strip these coefficients are bounded independently of z_0 and the real part of A. The bound may depend on the finite torus, which is harmless for this exact identity. The imaginary field strength obeys $|\operatorname{Im} F_\sigma| \le 4\|\delta\|_\infty$, so the tube hypothesis bounds Φ there as well.

For each distinguished plaquette the real field strength is $-\operatorname{Re}\zeta_b + \xi_b^0$, with no other free S coordinate. Retaining these real Gaussian factors and discarding the others gives, on $|\operatorname{Im}\zeta_b| \le \|\delta\|_\infty$,

$$|f(\zeta)| \le C \prod_{b\in S} e^{-(\operatorname{Re}\zeta_b - \xi_b^0)^2/2},$$

where the centres depend on z_0, A' and the constant is finite. The imaginary Maxwell contribution is bounded and supported on at most $4|S|$ plaquettes, proving the estimate. For one variable integrate around the rectangle joining $[-R, R]$ to $[-R, R] - i\delta_b$. Each vertical integral is bounded by its finite height times a constant times $e^{-(R-|\xi_b^0|)^2/2}$, times the integrable Gaussian in the remaining real variables. It tends to zero. The horizontal integrals converge by the same majorant. Repeat in every coordinate, using Fubini with the fixed-strip bound. Thus $\int_{\mathbb{R}^S} f(\xi)\, d\xi = \int_{\mathbb{R}^S} f(\xi - i\delta)\, d\xi$ for this fixed flux representative. The vertical sides have been removed before the representatives are summed.

On the shifted line,

$$e^{-\langle F - id\delta, F - id\delta\rangle/2} = e^{-\|F\|^2/2} e^{i\langle F, d\delta\rangle + \|d\delta\|^2/2}.$$

Moreover $\langle F, d\delta\rangle = \langle \hat{A}, d^*d\delta\rangle$ on the local lift, and the fermion factor is $V_\delta P^{(\delta)}$. The orbitwise Gaussian estimate is not a majorant to be summed over z_0 after the other Maxwell factors have been discarded. We check endpoint integrability separately. On the original real domain the entire factor and every local polynomial coefficient are bounded. The finite Berezin action coefficients are bounded independently of the integer flux. The original endpoint is therefore at most $C_{\mathbb{T},P,\Phi} e^{-\|F\|^2/2}$ in absolute value. At the shifted endpoint use the tube norm of $\Phi(F - id\delta)$; the local polynomial $V_\delta P^{(\delta)}$ still has uniformly bounded real-potential coefficients, and $e^{i\langle F, d\delta\rangle}$ has modulus one. It has the same Maxwell majorant with another finite constant. The closed-flux Maxwell sum is integrable by the finite-volume convergence proof. Unfolding absolute values therefore gives absolute integrability of both endpoints with every flux sector retained. We may now sum the orbitwise equality and integrate over A', obtaining the claimed formula. All its plaquettes and sites lie in B by the assumed

neighbourhood condition. For another coordinate plane, orient the distinguished plaquette with its normal first; its coefficient of u_b is ± 1. The identical Gaussian and free-orbit argument applies. □

12.3. **A finite elimination step.** By a local integral identity we mean

$$Z_{\mathbb{T}}[\Phi P] = \sum_a Z_{\mathbb{T}}[\Phi^{[\eta_a]} P_a] \tag{$\mathcal{I}$}$$

for every entire Φ and every torus in which one fixed box containing all the local terms and shifts embeds. The index set is finite. The reflection identity has terms $(\Theta_\Pi P_a, -\theta_\Pi^* \eta_a)$; this follows from finite functional covariance and $((\Phi^\theta)^{[\eta]})^\theta = \Phi^{[-\theta^*\eta]}$.

Proposition 12.6 (Removing an outer photon plane)**.** *If the lowest plane read by P is at least j, there is a local integral identity whose terms P_a have lowest plane at least $j+1$ and highest plane at most the larger of $j+2$ and the old highest plane. If P is read in B, all new terms and shifts are read in B^{+8}. They depend on P, j alone. The analogous operation removes a highest plane j, giving highest plane at most $j-1$ and lowest plane at least the smaller of $j-2$ and the old lowest plane.*

Proof. Apply the six upward fermion replacements in thick layer j. They are performed in the order of its six thin layers. Each emptied layer stays empty, since a replacement creates fields only in the next two layers. The last map, from thin layer six, creates fields in thin layers one or two of thick layer $j+1$. The resulting polynomial has no fields in thick layer j; its new fields lie in layer $j+1$ and its new links lie at j or between j and $j+1$. It is read in B^{+6}, and its integral equals that of P with every bounded bosonic factor. Expand it as $\sum_a \lambda_a e^{i\langle \delta'_a, \hat{A}\rangle} \Psi_a$. At the lower plane the gauge condition gives

$$(\delta'_a)^\perp_j = d_{\mathrm{s}}^* (\delta'_a)_j,$$

since there are no fields there and no links below it. Let δ_a be supported on the slice links at $j+1$, with $(\delta_a)_{j+1} = (\delta'_a)_j$. The contour identity replaces the term by the shifted factor with shift $d\delta_a$ and by

$$P_a = \lambda_a e^{\|d\delta_a\|^2/2 + \langle \delta'_a, \delta_a\rangle} e^{i\langle \delta'_a + d^* d\delta_a, \hat{A}\rangle} V_{\delta_a} \Psi_a.$$

Its required neighbourhood is contained in B^{+8}. The scalar $e^{\langle \delta'_a, \delta_a\rangle}$ comes from substituting $\hat{A} - i\delta_a$ in its phase, and the other real scalar comes from completing the Maxwell square. The shifted bosonic factor is retained in this exact identity. For the support bound, the six recursions enlarge B at most by six. Moving the links supporting u from plane j to $j+1$ adds at most one. Their adjacent plaquettes, hopping ends and photon-site cubes are covered by one further unit enlargement. That same box contains the adjoint stencil.

Here is the discrete calculation which removes the unwanted links. Set $u = (\delta'_a)_j$. The only nonzero components of $d\delta_a$ are

$$(d\delta_a)^{\mathrm{el}}_j = u, \qquad (d\delta_a)^{\mathrm{el}}_{j+1} = -u, \qquad (d\delta_a)^{\mathrm{mag}}_{j+1} = d_{\mathrm{s}} u.$$

Pairing with a spatial test cochain v_l gives the electric sum $\sum_l \langle v_{l+1} - v_l, (d\delta_a)^{\mathrm{el}}_l \rangle$. Thus its spatial adjoint is the incoming electric copy minus the outgoing copy, plus the magnetic adjoint in that plane. Pairing with a test time cochain a_l gives $-\langle d_{\mathrm{s}} a_l, (d\delta_a)^{\mathrm{el}}_l \rangle$, so its time adjoint is $-d_{\mathrm{s}}^* (d\delta_a)^{\mathrm{el}}_l$. Consequently

$$\begin{aligned} (d^* d\delta_a)_j &= -u, & (d^* d\delta_a)^\perp_j &= -d_{\mathrm{s}}^* u, \\ (d^* d\delta_a)_{j+1} &= (2 + d_{\mathrm{s}}^* d_{\mathrm{s}}) u, & (d^* d\delta_a)^\perp_{j+1} &= d_{\mathrm{s}}^* u, \\ (d^* d\delta_a)_{j+2} &= -u, \end{aligned}$$

and all other components vanish. The two electric copies and the magnetic copy have disjoint plaquette supports, giving $\|d\delta_a\|^2 = 2\|u\|^2 + \|d_{\mathrm{s}} u\|^2$. The new exponent therefore vanishes at j and on the time links leaving j. The polynomial V_{δ_a} uses only interior thin layers of thick layer $j+1$: affected spatial type 3 bonds have both ends in thin layers two through five, and no time

hopping phase is shifted. The old polynomial has already lost all fermions at j, and substituting its potential changes only scalar coefficients. No new dependence remains at or below j, and no new dependence occurs above $j+2$. This proves all upward assertions. Apply a time reflection to this construction to obtain the downward version, with the shift sign already specified above. Pull the identity back from the reflected torus using finite covariance. Its factor becomes $\Phi^{[-\theta^* d\delta_a]}$ and its term becomes ΘP_a. The reflected stencil gives the reversed plane bounds without reflection invariance of the original torus or source. □

Lemma 12.7 (Estimating a local identity in a folded source)**.** *For a fixed local integral identity (*$\mathcal{I}$*), write* $P^\flat = \sum_a P_a$*. There are a fixed box* B_1*, a finite* C *and a threshold such that*

$$|\mu_i^t(P) - \mu_i^t(P^\flat)| \le (e^{Ct} - 1) \sum_a \sup_{\operatorname{supp}\beta \subset B_1} |\mu_i^t(\varphi_\beta P_a)|.$$

They are independent of i, t*. The same assertion holds for the reflected identity on the original perturbed torus.*

Proof. Write $O = \sum_b a_b \varphi_{\beta_b}$. For a compact shift η, write the chosen expression of its copy in cell κ as $O^{[\kappa]} = \sum_b a_{\kappa b} \varphi_{\beta_{\kappa b}}$. If g_κ is the wall-group element carrying the base cell to that cell, then

$$a_{\kappa b} = \begin{cases} a_b & g_\kappa \text{ even}, \\ \overline{a_b} & g_\kappa \text{ odd}, \end{cases} \qquad \beta_{\kappa b} = (-1)^{\operatorname{parity}(g_\kappa)} (g_\kappa^{-1})^* \beta_b.$$

These are real cochain exponents with $\|\beta_{\kappa b}\|_1 = \|\beta_b\|_1$ and $\sum_b |a_{\kappa b}| = |O|_1$. Thus

$$O_\mathfrak{d}(F - i\eta) - O_\mathfrak{d}(F) = \sum_{\kappa, b} a_{\kappa b} (e^{\langle \beta_{\kappa b}, \eta \rangle} - 1) \varphi_{\beta_{\kappa b}}(F).$$

Only cells containing a plaquette of $\operatorname{supp}\eta$ contribute. There are at most $|\operatorname{supp}\eta|$ such cells. Indeed a plaquette on which the pairing is nonzero is contained in the strict support of a source copy and hence in exactly one open cell. For each contributing cell choose one such plaquette. These choices are distinct. A diagonal cell has coordinate diameter at most h, so the entire support of every contributing copy lies in the support box of η enlarged by h. For real v use $|e^v - 1| \le e^{|v|} - 1$ and $|\langle \beta_{\kappa b}, \eta \rangle| \le \|\beta_b\|_1 \|\eta\|_\infty$. The coefficient sum is at most

$$C(\eta) = |\operatorname{supp}\eta| \cdot |O|_1 (e^{\max_b \|\beta_b\|_1 \|\eta\|_\infty} - 1).$$

For a sufficiently large torus that enlarged box embeds, so its periodisation creates no additional contributing cell. More explicitly, choose representatives of its intersecting period cells in that enlarged infinite-lattice box. It embeds once the volume is sufficiently large, so no two relevant plaquettes or cells are identified. Every torus source term meeting the periodized shift is one of these representatives. The bound and support are therefore independent of the volume. Let $R_\eta = O_\mathfrak{d}^{[\eta]} - O_\mathfrak{d}$. The mth power of its finite character expression has coefficient sum at most $C(\eta)^m$, since multiplying characters adds exponents and grouping cannot increase total variation. The exponential series hence gives

$$e^{t(O_\mathfrak{d}^{[\eta]} - O_\mathfrak{d})} = 1 + \sum_\alpha b_\alpha(t) \varphi_{\beta_\alpha}, \quad \sum_\alpha |b_\alpha(t)| \le e^{tC(\eta)} - 1.$$

All β_α are supported in the fixed enlarged box: addition of exponents introduces no new supporting plaquette. The sum of the coefficient bounds is $\sum_{m \ge 1} t^m C(\eta)^m / m! = e^{tC(\eta)} - 1$. Since characters have modulus one, convergence is absolute and uniform on real configurations. Apply the local identity with $\Phi = e^{tO_\mathfrak{d}}$, divide by the positive denominator. The exact identity is

$$\mu_i^t(P) = \sum_a \mu_i^t\Big(e^{tR_{\eta_a}} P_a\Big).$$

Subtract $\sum_a \mu_i^t(P_a)$ only now. Termwise integration is justified by the uniformly absolutely convergent character series and the finite-volume absolute integrability of the local polynomial. Thus its ath difference is at most

$$(e^{tC(\eta_a)} - 1) \sup_{\operatorname{supp}\beta \subset B_1} |\mu_i^t(\varphi_\beta P_a)|.$$

Choosing $C = \max_a C(\eta_a)$ and one box for all shifts gives the result. The reflected shifts are equally compact and have the same argument on D_i; neither D_i nor its source has been assumed invariant under that reflection. □

Theorem 12.8 (Reduction to a fixed link plane)**.** *For $P \in \mathcal{P}$ and $c = j_0 + \frac{1}{2}$, there is $Y_P \in \mathcal{P}_1(c)$ defined on $\mathbb{Z}^3$, independent of all volume sequences, folded data and source strengths, such that for each datum there are $C < \infty$ and i_0 with*

$$|\mu_i^t(P) - \mu_i^t(Y_P)| + |\mu_i^t(\Theta_c P) - \mu_i^t(\Theta_c Y_P)| \le Ct \quad (0 \le t \le 1,\ i \ge i_0).$$

The assertion holds for every coordinate link plane.

Proof. Remove successive lowest photon planes until the lowest is at least j_0, then successive highest planes until the highest is at most $j_0 + 1$. To make the termination and locality explicit, write l, h for the least and greatest photon planes of a nonconstant term. The upward map sends (l, h) to plane bounds $(l + 1, \max(h, l + 2))$. After $N_+ = \max(j_0 - l, 0)$ such maps the lowest bound is at least j_0, and the highest is at most $\max(h, j_0 + 1)$. If there were no upward map, retain the original bounds instead. The downward map sends a greatest plane h' to bounds $(\min(l, h' - 2), h' - 1)$. While $h' \ge j_0 + 2$, the new lowest bound stays at least j_0; the last possible value removed is $j_0 + 2$. Thus the downward phase terminates after finitely many maps and creates no dependence below j_0. When the original lowest plane exceeded the target plane, the downward maps may first create the two-plane strip; the same bound applies. Constants require no map. For a finite sum use common plane bounds and apply each map linearly, acting as the identity on terms which do not read that plane. The number of operations is then finite before any volume, source, or strength is chosen.

Each map produces a finite sum and enlarges a support box by at most eight. Thus, after the two finite phases, every term and every imaginary field shift lies in one fixed enlargement of the original box. Let the successive polynomials be $P^{(0)} = P, P^{(1)}, \dots, P^{(N)}$, where $P^{(a+1)}$ is the sum of the terms in the local identity for $P^{(a)}$. These polynomials and their exact local identities depend only on P and the target plane. They can be defined once on the infinite lattice and then periodized into any sufficiently large torus. All remaining dependence of $P^{(N)}$ lies on the two photon planes and the links between them.

Now apply upward fermion replacements in thin layers one through five of thick layer j_0, and downward replacements in layers six through two of thick layer $j_0 + 1$. The last operations create fields no lower than $\mathbb{X}_{j_0,6}$ and no higher than $\mathbb{X}_{j_0+1,1}$. Their new links are confined to the two photon planes and the links between them, by the layer lemma. The result Y_P is therefore a two-layer polynomial. These ten final operations are exact with the source present. The fermion maps at a thin layer with no fields act as the identity. Hence the two opposite lists may be carried out for a common polynomial, with fixed support, even when its terms have different thin-layer extremes. The last upward map from thin layer five can create fields only in thin layer six of j_0 or thin layer one of $j_0 + 1$; the last downward map from layer two has the reverse bound. This verifies precisely the two-layer support condition, including the ends of the intervening time links.

The earlier operations form a finite list of local integral identities. Apply the preceding error estimate to each identity and to its reflection, and use the uniform local bound on their finitely many terms. In detail, the source-shift lemma bounds the ath defect by $(e^{C_a t} - 1) \sum_b \sup_{\operatorname{supp}\beta \subset B_a} |\mu_i^t(\varphi_\beta P_{ab})|$. The terms P_{ab} are fixed gauge-invariant polynomials. The normal-form and insertion bounds already proved give finite constants L_{ab} and geometric integers k_{ab} for these suprema; their boxes contain B_a as well as the polynomial supports. The reflected

identity has its own finite list of such constants. The finitely many geometric integers and thresholds can be fixed before choosing i, t. Once $\Delta_i \le 1$ and $0 \le t \le 1$,

$$\gamma_{t,i}(k) \le e^{(2|O|_1+|Q|)k^3}, \qquad e^{Ct} - 1 \le (e^C - 1)t.$$

The sum of the finitely many resulting bounds is a constant times t. Indeed one may take the sum of $(e^{C_a} - 1)L_{ab}e^{(2|O|_1+|Q|)k_{ab}^3}$ over both lists as C. The exact final fermion replacements add no error. Telescoping the defects of $P^{(a)}, P^{(a+1)}$ proves the two estimates. In this estimate the shifted sources have already been treated by their exact local identities; no change of the original source or torus is made. Coordinate covariance transports the local identities, not the perturbed torus; the same error estimates on that torus therefore prove the statement in the other directions. No exact symmetry of the source is used. □

13. Translation invariance of the comparison states

The two preceding estimates now have a single use. The elimination theorem replaces a general local character by a two-layer element, and the reflection identity compares that element with its image in a coordinate link plane. Their errors vanish as the perturbation strength tends to zero. This proves reflection invariance of the comparison state. Composing two neighboring link reflections then gives the even lattice translations needed in the diagonal positivity argument.

Theorem 13.1. *Every comparison state is invariant under reflection in each coordinate link plane. It is consequently invariant under every translation in* $2\mathbb{Z}^3$.

Proof. Fix a coordinate link plane and a local character $P = \varphi_\beta$. It belongs to $\mathcal{P}$, since its local-lift exponent is $d^*\beta$ and $d^*d^*\beta = 0$. Choose the fixed polynomial Y_P of the reduction theorem. The triangle inequality gives

$$\begin{aligned}|\mu_i^t(\Theta_c P) - \overline{\mu_i^t(P)}| \le{}& |\mu_i^t(\Theta_c P) - \mu_i^t(\Theta_c Y_P)| \\ &+ |\mu_i^t(\Theta_c Y_P) - \overline{\mu_i^t(Y_P)}| + |\mu_i^t(Y_P) - \mu_i^t(P)|.\end{aligned}$$

The first and last terms sum to at most Ct. The middle term has the ordered limiting estimate of Theorem 11.8. Hence

$$\lim_{t\downarrow 0} \limsup_{i\to\infty} |\mu_i^t(\Theta_c P) - \overline{\mu_i^t(P)}| = 0.$$

For a comparison state, take its volume limits at each fixed strength along its defining subsequence, then its strength limit. Both P and $\Theta_c P$ are bounded continuous cylinder functions, so the volume limit of their difference is its difference of expectations under the corresponding law ν_t. Its modulus is bounded by the full upper volume limit above. The subsequent strength limit gives $\nu_0(\Theta_c P) = \overline{\nu_0(P)}$. This passage never asserts weak convergence of a fermionic expectation or an unbounded Y_P: those quantities were eliminated from the finite-volume defect before either limit. On bosonic characters $\Theta_c P = \varphi_{-\theta_c^*\beta}$, while positivity of the probability law gives $\overline{\nu_0(P)} = \nu_0(\varphi_{-\beta})$. Replace β by $-\beta$ and obtain $\nu_0(\varphi_{\theta_c^*\beta}) = \nu_0(\varphi_\beta)$. Finite-dimensional characteristic functions determine probability measures, and cylinder sets generate the Borel sets of Ω. Thus the law is invariant under θ_c^*, with the ordinary tensor pullback. Finally, composing reflections in the two link planes at c and $c+1$ gives translation by $2e_\mu$. The three coordinate directions generate $2\mathbb{Z}^3$. □

14. Diagonal reflection positivity in infinite volume

The comparison states constructed above are reflection positive at the walls of their cells. They depend on the observable used to construct them. We now transfer this property to each limit of the cubic torus laws. There are two steps. The layer elimination argument restores enough translations to move any local observable into a prescribed cell. A convex separation argument then recovers the single cubic limit from the observable-dependent comparison states.

Lemma 14.1 (Domination by a reflection-positive law)**.** *Fix $0 < \rho < 1$, $w_i > 0$, $M \in \mathbb{R}$ and $q > 0$. Suppose that $n_i \to \infty$ satisfies* (Div)*, and that $\mu_{C(n_i)}$ converges weakly on Ω to ν. For every $c_0 \in \mathbb{Z}$ and every real trigonometric polynomial $O \in \mathcal{T}$, there is $\nu_0 \in \mathcal{R}(c_0)$ such that*

$$\nu(O) \le \nu_0(O).$$

Here $\mathcal{R}(c_0)$ consists of the laws satisfying Gaussian domination that are invariant and reflection positive at $\{x_1 - x_2 = c_0\}$.

Proof. Choose $r \ge 1$ and $y_0 \in \mathbb{R}^3$ with $\operatorname{supp} O \subset y_0 + [-r, r]^3$. The divisibility condition gives an even $h > 8r + 8$ with $4h \in \mathcal{D}(\mathfrak{n})$. Indeed, the infinite divisor-closed set $\mathcal{D}(\mathfrak{n})$ is unbounded, and taking the least common multiple of a sufficiently large member with 8 gives a member of the form $4h$ with h even.

Choose an even integer a in

$$(c_0 + 2r - \ell_-(y_0),\ c_0 + h - 2r - \ell_-(y_0)).$$

The interval has length $h - 4r > 2$. Put $v = ae_1$ and $y = y_0 + v$. Then $c_0 < \ell_-(y) - 2r$ and $\ell_-(y) + 2r < c_0 + h$. Choose $c_+ \in \mathbb{Z}$ and $c_3 \in \mathbb{Z} + \frac{1}{2}$ so that

$$c_+ < \ell_+(y) - 2r < \ell_+(y) + 2r < c_+ + h, \qquad c_3 < y_3 - r < y_3 + r < c_3 + h.$$

Their permissible intervals have length greater than one. Since $|\ell_\pm(x) - \ell_\pm(y)| \le 2r$ on $y + [-r, r]^3$, the support of $\tau_v O$ lies in the open cell

$$Q_0 = \{c_0 < \ell_- < c_0 + h,\ c_+ < \ell_+ < c_+ + h,\ c_3 < x_3 < c_3 + h\}.$$

Thus $(c_0, c_+, c_3; h; \tau_v O)$ is admissible data for the original sequence $\mathfrak{n}$.

Apply Proposition 10.7 to these data and the subsequence giving ν. It supplies a comparison state ν_0 that satisfies Gaussian domination and is invariant and reflection positive at every cell wall, including $\ell_- = c_0$. Moreover,

$$\nu(\tau_v O) \le \nu_0(\tau_v O).$$

Each cubic law is translation invariant, so the bounded continuous functions O and $\tau_v O$ have equal integrals under ν. Theorem 13.1 gives invariance of ν_0 under $2\mathbb{Z}^3$, and $v \in 2\mathbb{Z}^3$. Consequently

$$\nu(O) = \nu(\tau_v O) \le \nu_0(\tau_v O) = \nu_0(O).$$

No uniform comparison state as O varies is asserted or needed. □

Theorem 14.2 (Infinite-volume reflection positivity)**.** *Fix $0 < \rho < 1$, $w_i > 0$, $M \in \mathbb{R}$ and $q > 0$. Let $n_i \to \infty$ satisfy* (Div)*. The laws $\mu_{C(n_i)}$ form a tight family on Ω. Every weak limit point satisfies Gaussian domination, is invariant under the lattice translations and cubic group, and is invariant and reflection positive at every mirror of the nine families.*

Proof. Choose an index i_0 such that every side n_i is at least four for $i \ge i_0$. A single plaquette embeds in $C(n_i)$ at every lattice position, with no identified edges, for all these indices. Periodization of $s1_\sigma$ has squared norm s^2, independently of σ and i. Coordinate domination therefore gives $\mu_{C(n_i)}(e^{\pm sF_\sigma}) \le e^{s^2/2}$ simultaneously for all σ. Markov's inequality with $s = R$ gives

$$\sup_{i \ge i_0} \sup_{\sigma} \mu_{C(n_i)}\{|F_\sigma| > R\} \le 2e^{-R^2/2}.$$

The threshold is common to all coordinates. In particular, no interchange of a coordinate-dependent threshold with a countable union is involved. Enumerate the plaquettes as $\sigma_1, \sigma_2, \ldots$. Given $\varepsilon > 0$, choose R_j so that $2e^{-R_j^2/2} \le \varepsilon 2^{-j}$. The product $\prod_j [-R_j, R_j]$ is compact for the product topology, and its intersection with the closed set Ω is compact. The union bound gives measure at least $1 - \varepsilon$ for the tail of the sequence. Finitely many initial laws can be included by adjoining finitely many compact sets. Prokhorov's theorem gives weak limit points.

Let ν be one of them. The plaquettes of a fixed finite cochain fit in one finite box. That box embeds once n_i exceeds its diameter, and then periodization preserves the cochain's squared norm. Coordinate domination applies with its original norm. For $L > 0$ its truncated exponential $\min(e^{\langle \beta, F\rangle}, L)$ is bounded continuous, so weak convergence bounds its expectation under ν by $e^{\|\beta\|^2/2}$. Monotone convergence in L proves Gaussian domination. Lattice translations and cubic transformations act continuously on Ω by signed coordinate permutations. Finite covariance gives invariance of the cubic torus laws under these maps; weak convergence passes each identity to ν. This proves invariance of the limit law, since bounded continuous cylinders determine it. Closedness of the field is built into Ω.

At a coordinate mirror, any bounded continuous cylinder observable on the positive side eventually fits inside a half of the cubic torus. More explicitly, for a mirror $x_\mu = c$, its finitely many supporting plaquettes lie in an interval $c < x_\mu < c + d$ for some fixed d. If $n_i > 2d$ and the remaining finite support box embeds, they lie in the open torus half $c < x_\mu < c + n_i/2$. The opposite wall is admissible because n_i is eventually even under Div. The cube is adapted to that coordinate direction and its closing character is -1. Finite reflection positivity applies at both integer and half-integer positions allowed by Definition 2.2. The reflected product is bounded and continuous, so its nonnegative expectation passes to the limit. Density of cylinder observables in the positive side L^2 space gives reflection positivity in the stated sense, using the invariant-law isometry proved in Lemma 5.3.

Fix $c_0 \in \mathbb{Z}$. The previous lemma gives, for each $O \in \mathcal{T}$, a law $\nu_0 \in \mathcal{R}(c_0)$ with $\nu(O) \le \nu_0(O)$. The class $\mathcal{R}(c_0)$ is convex and weakly compact. Lemma 5.3 therefore gives $\nu \in \mathcal{R}(c_0)$. Since c_0 was arbitrary, ν is reflection positive at every mirror of the family $x_1 - x_2 = c_0$. The cubic group carries this family to the six diagonal families. Explicitly, if g is such a transformation and $\theta' = g\theta g^{-1}$, then pulling a positive-side cylinder function for θ' back by g gives one for θ. Invariance of ν identifies the two reflected quadratic forms, so the former is nonnegative. The same conjugation proves invariance under θ'. These six families together with the three coordinate families are precisely the nine asserted families. □

The side sequence used for the continuum cutoff is a subsequence of 8 times powers of the fixed odd integer L. It satisfies the divisibility condition, so the theorem applies at every fixed spacing and fixed set of lattice parameters. Nothing in the proof requires a unique volume limit or convergence without subsequences.

15. Random field strength tensors

The continuum observable is a real antisymmetric tensor $F = (F_{12}, F_{13}, F_{23})$, interpreted as a tempered distribution. We identify it with a distributional two-form. This identification fixes the transformation rule of its indices; it does not introduce a different observable. Let

$$\mathcal{S} = \mathcal{S}(\mathbb{R}^3; \Lambda^2), \qquad \mathcal{S}' = \mathcal{S}'(\mathbb{R}^3; \Lambda^2).$$

The basis $dx_1 \wedge dx_2, dx_1 \wedge dx_3, dx_2 \wedge dx_3$ is orthonormal. Test functions are real unless their complexification is explicitly used. A distribution acts on a complex test by complex linearity. Write $F(f)$ for this action, and let Σ be the σ-algebra generated by the functions $F \mapsto F(f)$. For an open set $O \subset \mathbb{R}^3$, let Σ_O be generated by these functions with $f \in C_c^\infty(O; \Lambda^2)$. Completions of these σ-algebras are understood whenever they specify a subspace of $L^2(\mu)$. The complex L^2 scalar product is antilinear in its first argument.

For $g(x) = Rx + a$, where $R \in \mathrm{O}(3)$, put

$$(g \cdot f)(x) = (\wedge^2 R) f(g^{-1}x), \qquad (gF)(f) = F(g^{-1} \cdot f), \qquad (\widehat{g}\Phi)(F) = \Phi(g^{-1}F).$$

Thus $\widehat{g}F(f) = F(g \cdot f)$, and $\widehat{gh} = \widehat{g}\widehat{h}$. Write $U(a) = \widehat{\tau_a}$, where $\tau_a(x) = x + a$. If $g_*\mu = \mu$, then $\widehat{g}$ is unitary on $L^2(\mu)$. Let $r_{n,c}$ denote reflection in $n \cdot x = c$, with $|n| = 1$, and put

$$\Theta_{n,c}\Phi = \overline{\widehat{r}_{n,c}\Phi}.$$

In particular, $\theta_j = r_{e_j,0}$ and $\Theta_j = \Theta_{e_j,0}$. The pullback $\widehat{\theta}_j$ is complex linear; Θ_j is antilinear. We keep this distinction throughout the proof.

15.1. **A Hilbert space containing the limit laws.** Let h_k, $k \in \mathbb{N}_0^3$, be the normalized real Hermite functions, and let $h_\kappa = h_k e_c$, with $\kappa = (k, c) \in \mathcal{K} = \mathbb{N}_0^3 \times \{12, 13, 23\}$. The operator

$$\mathsf{A} = 1 + |x|^2 - \Delta$$

acts componentwise and satisfies $\mathsf{A}h_\kappa = \lambda_\kappa h_\kappa$, $\lambda_\kappa = 2|k|_1 + 4$. For real s, define

$$\|F\|_s^2 = \sum_\kappa \lambda_\kappa^{2s} |F(h_\kappa)|^2, \qquad H_s = \{F \in \mathcal{S}' : \|F\|_s < \infty\}.$$

The space H_s is the weighted sequence Hilbert space with this norm. Its inclusion in $\mathcal{S}'$ is specified by

$$F(f) = \sum_\kappa F(h_\kappa)\langle h_\kappa, f\rangle_2, \qquad |F(f)| \le \|F\|_{-s}\|f\|_s.$$

For later use set

$$Z_b = \sum_\kappa \lambda_\kappa^{-b} = 3\sum_{l \ge 0} \frac{(l+1)(l+2)}{2}(2l+4)^{-b}. \tag{15.1}$$

The series converges precisely when $b > 3$.

Lemma 15.1. *The Schwartz topology is defined by the norms $\|f\|_m$, $m = 0, 1, \ldots$. Hermite expansions converge in that topology, and $\mathcal{S}' = \bigcup_{m \ge 0} H_{-m}$. The space $(\mathcal{S}', \Sigma)$ is standard Borel. The trace of Σ on H_{-m} is its Borel σ-algebra. Every closed bounded ball of H_{-3} is compact in H_{-4}.*

Proof. Integration by parts shows $\langle h_\kappa, \mathsf{A}^m f\rangle = \lambda_\kappa^m \langle h_\kappa, f\rangle$. Thus a Schwartz function has Hermite coefficients decreasing faster than every power of λ_κ. Conversely, multiplication by x_i and differentiation by x_i are sums of the Hermite raising and lowering operators, whose coefficients are bounded by a constant times $\lambda_\kappa^{1/2}$. More explicitly, in one dimension,

$$x h_l = \sqrt{l/2}\, h_{l-1} + \sqrt{(l+1)/2}\, h_{l+1}, \qquad \partial_x h_l = \sqrt{l/2}\, h_{l-1} - \sqrt{(l+1)/2}\, h_{l+1}.$$

For the tensor product basis these formulas act on one index at a time. A composition of r multiplications and derivatives has at most 2^r terms, shifts indices by at most r, and has coefficients bounded by $C_r \lambda_\kappa^{r/2}$. Orthogonality, with the finite multiplicity of the shifted indices, consequently bounds its L^2 norm by $C_r \|f\|_{\lceil r/2 \rceil}$ on finite Hermite sums. The same estimates pass to their Hilbert limits. Every expression $x^\alpha \partial^\beta f$ therefore has an L^2 norm bounded by $C_{\alpha\beta}\|f\|_m$ for some m. Apply the Sobolev inequality in dimension three to its derivatives of order at most two. This bounds each Schwartz seminorm by one of the norms $\|f\|_m$. The opposite bounds follow by expanding A^m as a differential operator with polynomial coefficients. These estimates also prove convergence in the Schwartz topology of the finite Hermite sums: their tails tend to zero in every $\|\cdot\|_m$ by Parseval, and each Schwartz seminorm is bounded by one such norm.

If F is continuous on $\mathcal{S}$, then $|F(f)| \le C\|f\|_m$ for some m. Applying this bound to finite Hermite sums shows, by Hilbert-space duality, that $F \in H_{-m}$. Each weighted square-summable coefficient sequence defines a distribution by the displayed Cauchy–Schwarz bound, proving the converse assertion. The coordinate image of $\mathcal{S}'$ is therefore

$$\bigcup_{m \ge 0} \left\{ (t_\kappa) : \sum_\kappa \lambda_\kappa^{-2m} |t_\kappa|^2 < \infty \right\},$$

a Borel subset of the countable product $\mathbb{R}^{\mathcal{K}}$. The coordinate map and its inverse are measurable: evaluation at an arbitrary test is the pointwise limit of its finite Hermite sums. This proves the standard Borel assertion and the assertion about traces on Hilbert spaces.

For compactness, take a sequence with $\|F_l\|_{-3} \le M$. A diagonal subsequence has convergent individual Hermite coefficients. Fatou's lemma gives a limiting sequence with H_{-3} norm at most M. For every $B > 0$ the H_{-4} norm squared of the difference outside $\lambda_\kappa \le B$ is at most $4M^2B^{-2}$; the finite part converges. The subsequence consequently converges in H_{-4}. □

15.2. **Gaussian bounds and supported observables.** The estimate used throughout the continuum argument is

$$\mu(e^{F(f)}) \le e^{\|f\|_2^2/2} \qquad (f \in \mathcal{S}). \tag{G}$$

It is an exponential upper bound, not a statement that the field is Gaussian.

Lemma 15.2 (Consequences of Gaussian domination)**.** *Suppose that μ satisfies (G). The map $f \mapsto F(f)$ extends uniquely to a complex-linear contraction from $L^2(\mathbb{R}^3;\Lambda^2)$ to $L^2(\mu)$. Its values have mean zero and lie in every $L^p(\mu)$, $p < \infty$. For real f,*

$$\mu(|F(f)|^p) \le p\,2^{p/2}\Gamma(p/2)\|f\|_2^p, \qquad \mu(F(f)^2) \le \|f\|_2^2. \tag{15.2}$$

For complex f the L^p bound holds with an additional factor $2^{p/2}$ on the right. The extended field is real on real tests and measurable with respect to Σ_O when the test vanishes almost everywhere outside an open set O.

For $n \ge 1$, the moments

$$S_n(f_1,\dots,f_n) = \mu(F(f_1)\cdots F(f_n))$$

are continuous symmetric multilinear forms on L^2 tests. A law satisfying (G) is determined by these moments. It is invariant under a Euclidean map g if and only if all these moment forms are invariant under g.

Proof. For real f put $X = F(f)$ and $v = \|f\|_2^2$. The bound for tf gives $\mu(e^{tX}) \le e^{t^2v/2}$ for all real t. Jensen's inequality, followed by $t \to 0$ from either side, gives $\mu(X) = 0$. The inequality $\mu(\cosh(tX)) \le e^{t^2v/2}$, divided by $t^2/2$ and followed by Fatou's lemma as $t \to 0$, gives $\mu(X^2) \le v$. Chernoff's inequality, optimized at $t = u/v$, gives $\mu(|X| > u) \le 2e^{-u^2/(2v)}$; the case $v = 0$ follows from the second moment bound. Integration of $p\int_0^\infty u^{p-1}\mu(|X| > u)\,du$ proves (15.2).

The second moment estimate extends the field by L^2 density. For an approximating sequence a subsequence converges almost surely; Fatou's lemma preserves the exponential bound and all the moment bounds. Apply this also to differences to obtain convergence in every finite L^p. Complexification gives $\|F(f)\|_p \le C_p(\|\operatorname{Re} f\|_2 + \|\operatorname{Im} f\|_2) \le \sqrt{2}C_p\|f\|_2$ and the asserted reality rule. At $p = 2$ the sharper contraction follows from reality:

$$\mu(|F(f)|^2) = \mu(F(\operatorname{Re} f)^2) + \mu(F(\operatorname{Im} f)^2) \le \|\operatorname{Re} f\|_2^2 + \|\operatorname{Im} f\|_2^2 = \|f\|_2^2.$$

Smooth compactly supported functions in O are dense in $L^2(O)$. Since $L^2(\Sigma_O,\mu)$ is closed, the support assertion follows. The covariance rule $\widehat{g}F(f) = F(g\cdot f)$ extends to L^2 tests whenever $g_*\mu = \mu$, by continuity on both sides.

Hölder's inequality proves the multilinear bounds and continuity of S_n. For a real test the partial sums of its exponential series are dominated by $e^{|F(f)|} \le e^{F(f)} + e^{-F(f)}$, an integrable function. Hence

$$\mu(e^{iF(f)}) = \sum_{n\ge0} \frac{i^n}{n!}\mu(F(f)^n).$$

Moments therefore determine the characteristic functional. Its values on linear combinations determine each finite-dimensional joint law by uniqueness of Fourier transforms of probability measures. These laws determine the cylinder measure. Applying this to $g_*\mu$ proves the last assertion. □

Lemma 15.3 (Density of exponentials)**.** *For every open O there is a countable rational vector space $D_O \subset C_c^\infty(O;\Lambda^2)$, dense on each compact support, such that the finite linear combinations of $e^{iF(f)}$, $f \in D_O$, are dense in $L^2(\Sigma_O,\mu)$ for every law μ. Two laws having the same characteristic functional on $D_{\mathbb{R}^3}$ are equal. Under translation invariance, $U(a)$ is a strongly continuous unitary representation on $L^2(\mu)$. The analogous assertion holds for rotations under rotation invariance.*

Proof. Choose a compact exhaustion of O and countable dense families of smooth tests with supports in those compact sets, and take their rational span. Evaluation at the limiting tests is a pointwise limit, so these tests generate Σ_O. If an L^2 function G is orthogonal to all their exponentials, the finite complex measure $G\,d\mu$ has vanishing Fourier transform on every finite collection of the generating coordinates. Continuity extends the rational values to real values. Fourier uniqueness and the monotone class theorem give $G = 0$. The same argument proves uniqueness of laws. Finally, $U(a)e^{iF(f)} = e^{iF(\tau_a \cdot f)}$ converges pointwise as $a \to 0$, since translations are continuous on $\mathcal{S}$. Bounded convergence gives convergence in L^2, and density gives strong continuity on all of L^2. Rotations are treated identically. □

15.3. **Schwinger functions on tensor tests.** Let $\mathscr{S}_n = \mathcal{S}((\mathbb{R}^3)^n; (\Lambda^2)_{\mathbb{C}}^{\otimes n})$, $n \geq 1$, and $\mathscr{S}_0 = \mathbb{C}$. Use the seminorms

$$q_m(f) = \max_{|\alpha| \leq m} \sup_y (1+|y|^2)^{m/2} |\partial^\alpha f(y)|,$$

where the maximum also runs over tensor components. Let $a_{\boldsymbol{\kappa}}(f) = \langle h_{\kappa_1} \otimes \cdots \otimes h_{\kappa_n}, f \rangle_2$.

Lemma 15.4 (Hermite coefficients)**.** *There is a numerical constant C such that*

$$\sum_{\boldsymbol{\kappa}} |a_{\boldsymbol{\kappa}}(f)| \leq Z_4^{n/2} \|(\mathsf{A}^2)^{\otimes n} f\|_2 \leq C^n q_{6n}(f). \tag{15.3}$$

Proof. Integration by parts multiplies a Hermite coefficient by $\prod_p \lambda_{\kappa_p}^2$ when $(\mathsf{A}^2)^{\otimes n}$ is applied. Cauchy–Schwarz, with weights $\prod_p \lambda_{\kappa_p}^{-2}$, and Parseval's identity prove the first inequality. To prove the second, put $M(x) = 1 + |x|^2$. Leibniz's rule gives

$$\mathsf{A}^2 = M^2 - 2M\Delta - 4x \cdot \nabla - 6 + \Delta^2.$$

This is a sum of differential monomials of order at most four whose coefficient bounds, in units of M^2, have sum at most 34. Its n-fold tensor product is consequently bounded componentwise by

$$34^n \prod_p (1+|y_p|^2)^2 \max_{|\alpha| \leq 4n} |\partial^\alpha f(y)| \leq 34^n q_{6n}(f) \prod_p (1+|y_p|^2)^{-1}.$$

Here $(1+|y|^2)^n \geq \prod_p (1+|y_p|^2)$. There are 3^n components and $\int_{\mathbb{R}^3} (1+|x|^2)^{-2}\,dx = \pi^2$. The resulting L^2 bound is exponential in n, proving the claim. In particular one may take $C = 34\pi\sqrt{3Z_4}$. The coefficient total 34 comes from the five displayed groups in A^2: their bounds are respectively $1, 6, 12, 6, 9$ in units of M^2. The derivative order of their n-fold product is at most $4n$; the weight in q_{6n} is $(1+\sum_p |y_p|^2)^{3n}$. Thus both the differentiation and the weight used above are controlled by the stated seminorm, with no superexponential constant in n. □

Lemma 15.5 (Tensor tests)**.** *Suppose that μ satisfies* (G)*. For $f \in \mathscr{S}_n$ define*

$$F^{(n)}(f) = \sum_{\boldsymbol{\kappa}} a_{\boldsymbol{\kappa}}(f) \prod_{p=1}^{n} F(h_{\kappa_p}), \qquad F^{(0)}(c) = c. \tag{15.4}$$

The series converges absolutely for every $F \in \mathcal{S}'$ and in $L^2(\mu)$. It is the unique continuous linear map $\mathscr{S}_n \to L^2(\mu)$ equal to $F(f_1)\cdots F(f_n)$ on tensor products. For a numerical C,

$$\|F^{(n)}(f)\|_{L^2(\mu)} \leq C^n \sqrt{n!}\, q_{6n}(f). \tag{15.5}$$

It obeys multiplication, conjugation and Euclidean transformation rules:

$$F^{(m)}(f)F^{(n)}(f') = F^{(m+n)}(f \otimes f'),$$
$$\overline{F^{(n)}(f)} = F^{(n)}(\overline{f}),$$
$$\widehat{g}F^{(n)}(f) = F^{(n)}((g\cdot)^{\otimes n} f).$$

If $\operatorname{supp} f \subset O^n$*, then* $F^{(n)}(f) \in L^2(\Sigma_O, \mu)$*. The same conclusion holds with* $O = \{x_j > 0\}$ *when the support is contained in the closed positive half-space in each variable.*

Proof. For real unit tests, Hölder's inequality and (15.2) with exponent $2n$ give

$$\left\| \prod_{p=1}^{n} F(f_p) \right\|_2 \leq \sqrt{2^{n+1} n!} \prod_p \|f_p\|_2.$$

Minkowski's inequality and (15.3) prove L^2 convergence and (15.5). For a fixed distribution choose s with $F \in H_{-s}$. The sum of absolute values in (15.4) is at most

$$\|F\|_{-s}^n Z_4^{n/2} \|(\mathsf{A}^{s+2})^{\otimes n} f\|_2.$$

It therefore converges pointwise and defines a continuous functional on $\mathscr{S}_n$. On products it is the product of the individual Hermite series for $F(f_p)$. Finite tensor products are dense in $\mathscr{S}_n$ by Hermite expansion, which proves uniqueness and the three algebraic rules by continuity. For multiplication this last continuity is a statement about distributions before taking expectations: for each fixed F, the two sides are continuous in f, f' in their Schwartz topologies and agree on finite tensor sums. Hence they agree pointwise for every distribution, even though multiplication need not be continuous on a probability space's L^2 space. The moment estimates already proved place their common value in $L^2(\mu)$.

For compact support in O^n, choose $\chi \in C_c^\infty(O)$ equal to one near every projection of that support. Multiply the finite Hermite sums by $\chi^{\otimes n}$. They converge in $\mathscr{S}_n$ and are sums of products of tests supported in O. Their random variables belong to the closed subspace $L^2(\Sigma_O, \mu)$, so does the limit. For noncompact support in O^n, first multiply by expanding compact cutoffs; the Schwartz estimates give convergence in every seminorm. To spell out this approximation, choose $\chi \in C_c^\infty(\mathbb{R}^3)$ equal to one on the unit ball and put $\chi_R(x) = \chi(x/R)$. The derivatives of $1 - \chi_R^{\otimes n}$ are supported where some variable has size at least R. Leibniz's rule, followed by a higher Schwartz weight on f, makes each prescribed seminorm of $(1 - \chi_R^{\otimes n})f$ tend to zero. Each resulting compact support projects to compact subsets of O, to which the preceding argument applies. For support in the closed positive half-space translate each variable by εe_j. The translated test has support in the open positive half-space and converges in $\mathscr{S}_n$ as $\varepsilon \downarrow 0$. Thus positive-side measurability does not require a restriction excluding the time-zero boundary or the diagonals $y_p = y_q$. □

The Schwinger functions are now defined on all tensor tests by

$$S_n(f) = \mu(F^{(n)}(f)), \qquad S_0(c) = c. \tag{15.6}$$

In particular, their definition includes coincident arguments. The bound (15.5) is the linear growth condition used in the OS axioms.

15.4. **The charge, mass and cutoff sequence.**

Definition 15.6. Fix an odd integer $L \geq 3$, $e > 0$, $0 < \bar{m} \leq 1$, $0 < \rho < 1$, a point $v_0 \in C_0$, and the geometric weights $w_i = 48|\Pi_i \cap \overline{C}_0|$. Choose any real sequence $(m_{0,N})_{N \geq 0}$. For $N, T \geq 0$ put

$$\eta_N = L^{-N}, \qquad q_N = e\eta_N^{1/2}, \qquad m_N = \bar{m}\eta_N + m_{0,N}, \qquad M_N = m_N + \frac{\rho}{2}\sum_i w_i,$$

$$b = \lfloor \log_L \bar{m}^{-1} \rfloor, \qquad n_{N,T} = 8L^{b+N+T}, \qquad R_T = \eta_N n_{N,T} = 8L^{b+T}.$$

Use the normalized chamber measure on the cubic torus $C(n_{N,T})$ with charge q_N and mass parameter M_N. If $u = dA + 2\pi z/q_N$ is its field strength in lattice units, its physical value on the plaquette $\eta_N\sigma$ is $\eta_N^{-3/2}u(\sigma)$. Its periodic step interpolation, defined in (16.2), has law $P^{\mathrm{ch}}_{N,T}$ on tempered distributions.

At fixed N, let $\mathcal{V}_N$ be the set of weak limit points of these laws as $T \to \infty$, in the Hilbert space H_{-4} defined above. A volume-first limit is a weak limit of a sequence $\mu_N \in \mathcal{V}_N$ along some $N \to \infty$. The following sections prove that both choices of limit are nonempty. All references to volume-first limits in the main theorem use this specified cutoff sequence.

At fixed N the charge and mass stay fixed while the volume tends to infinity. The integer sides have the divisibility property required in the finite comparison: every divisor of $8L^{b+N+T_0}$ divides all later sides, and these eventual divisors are unbounded. The continuum normalization makes the lattice ℓ^2 norm of a test cochain bounded by the continuum L^2 norm; this is why the Gaussian estimate retains constant one as $\eta_N \to 0$.

16. The volume-first continuum limit

We use the infinite-volume reflection-positivity theorem for the chamber lattice law proved in the preceding sections. At each fixed cutoff index N it provides a subsequential infinite-volume law ν_N on the real lattice field strength variables $u(\sigma)$, σ a plaquette of $\mathbb{Z}^3$. Its properties needed here are precisely these:

(a) $du = 0$ almost surely, with the lattice coboundary;

(b) for every real finite two-cochain b,

$$\nu_N\Big(e^{\sum_\sigma b(\sigma)u(\sigma)}\Big) \le e^{\frac{1}{2}\sum_\sigma b(\sigma)^2}; \tag{16.1}$$

(c) invariance under integer translations and signed permutations;

(d) invariance and reflection positivity at all admitted translates of the coordinate and diagonal lattice planes.

The mass and charge may depend on N as specified in the lattice construction. The bounds in this list do not depend on them. We shall prove all subsequent limit passages from this finite list.

16.1. **The step interpolation.** Put $\eta = L^{-N}$ and, for $\sigma = (x;\mu\nu)$, set

$$Q_\eta(\sigma) = \eta\left(x + \tfrac{1}{2}(e_\mu + e_\nu) + [-\tfrac{1}{2},\tfrac{1}{2})^3\right).$$

For a fixed tensor component these cubes partition $\mathbb{R}^3$. Define

$$(\sharp_\eta u)_{\mu\nu}(y) = \eta^{-3/2}u(x;\mu\nu) \quad (y \in Q_\eta(x;\mu\nu)), \qquad f_{[\eta]}(\sigma) = \eta^{-3/2}\int_{Q_\eta(\sigma)} f_{\mu\nu}(y)\,dy. \tag{16.2}$$

Jensen's inequality gives

$$\sum_\sigma |f_{[\eta]}(\sigma)|^2 \le \|f\|_2^2. \tag{16.3}$$

Lemma 16.1. *The step interpolation is a measurable tempered distribution for ν_N-almost every u. Its law μ_N is a Borel probability measure on H_{-3} satisfying (G) and*

$$\int \|F\|_{-3}^2\, d\mu_N(F) \le Z_6. \tag{16.4}$$

It is invariant under translations by $\eta\mathbb{Z}^3$ and the cubic group.

Proof. The one-coordinate version of (16.1) gives $\nu_N(|u(\sigma)| > t) \le 2e^{-t^2/2}$. There are $O(m^2)$ plaquettes at distance m from the origin. The sum of the probabilities of $|u(\sigma)| > 1 + |x(\sigma)|_\infty$ is finite. Borel–Cantelli implies that almost surely $|u(\sigma)| \le C(u)(1 + |x(\sigma)|_\infty)$ for every plaquette.

The corresponding step function has at most linear growth and is a tempered distribution. For a Schwartz test,

$$(\sharp_\eta u)(f) = \sum_\sigma f_{[\eta]}(\sigma) u(\sigma),$$

with absolute convergence under that growth bound. Finite sums also converge in $L^2(\nu_N)$, by the second moment consequence of (16.1) and (16.3). Fatou's lemma preserves the exponential bound, giving (G). Evaluation is a limit of measurable finite sums. Its second moment on h_κ is at most one. Tonelli's theorem now gives (16.4). Membership in H_{-3} follows almost surely; setting the step map equal to zero on the null exceptional set makes it a Borel map into H_{-4}. The averaging cubes are carried into averaging cubes, with the tensor orientation signs, by lattice translations and signed permutations. This proves the stated invariances. □

We next check that these are the volume-first laws in the topology used for the continuum limit, rather than merely product-topology laws on lattice coordinates.

Lemma 16.2. *At a fixed spacing, if the periodic lattice laws converge on every finite collection of plaquettes to ν_N, their step-interpolated laws converge weakly on H_{-4} to μ_N.*

Proof. For a periodic box of physical side R, evaluation at a test f uses its periodization $f^R(y) = \sum_{v\in\mathbb{Z}^3} f(y + Rv)$. The lattice Gaussian bound gives

$$\mathbb{E}e^{tF^\sharp(f)} \le \exp(t^2 \|f^R\|^2_{L^2(\mathbb{R}^3/R\mathbb{Z}^3)}/2).$$

We need one bound uniform in $R \ge 1$. With $Q_R = [-R/2, R/2)^3$, Minkowski's and Cauchy–Schwarz inequalities give

$$\|f^R\|_2 \le \|f\|_2 + 32R^{-2}\||y|^2 f\|_{L^2(\mathbb{R}^3\setminus Q_R)} \le 65\|\mathsf{A}f\|_2. \tag{16.5}$$

Indeed, for $v \ne 0$ and $y \in Q_R$, $|y + Rv| \ge (R/2)|v|_\infty$; apply Cauchy–Schwarz to the sum of $\|f(\cdot + Rv)\|_{L^2(Q_R)}$ with weights $|v|_\infty^{-2}$. The sum of the fourth inverse powers is $\sum_{m\ge1}(24m^2+2)m^{-4} < 42$. This proves the first inequality with 32. To see the second, integration by parts gives

$$\|\mathsf{A}f\|_2^2 = \|(1 + |x|^2)f\|_2^2 + \|\Delta f\|_2^2 + 2\int (1 + |x|^2)|\nabla f|^2 - 6\|f\|_2^2.$$

Since $\mathsf{A} \ge 4$, this implies $\||x|^2 f\|_2 \le 2\|\mathsf{A}f\|_2$ and $\|f\|_2 \le \|\mathsf{A}f\|_2/4$, which suffice.

Thus the periodic step laws have $\mathbb{E}\|F\|^2_{-3} \le 65^2 Z_4$, uniformly in the volume. They are tight on H_{-4} by Lemma 15.1. Indeed, applying (16.5) to h_κ gives $\mathbb{E}|F(h_\kappa)|^2 \le 65^2\lambda_\kappa^2$; summing with weights λ_κ^{-6} gives the stated Z_4 bound. Markov's inequality bounds the mass outside the compact H_{-3} ball of radius M by $65^2 Z_4/M^2$. Such a ball is compact, hence closed, in the metric space H_{-4}. This is the tightness criterion used here. For compactly supported tests and sufficiently large boxes the characteristic function is a characteristic function of finitely many lattice variables, and hence converges to that of $\sharp_\eta u$. Approximate any Schwartz test by compactly supported tests in the norm $\|\mathsf{A}f\|_2$; the second moment bound (16.5) controls the difference uniformly in the volume. Every weak limit therefore has the characteristic functional of μ_N. Characteristic uniqueness shows that the entire selected volume subsequence converges to μ_N on H_{-4}. To justify convergence of the entire selected subsequence, every subsequence has a convergent further subsequence by tightness, and the preceding characteristic calculation identifies its limit with μ_N. Failure of convergence would give a bounded continuous function and a subsequence of its integrals staying away from the integral under μ_N, contradicting that identification. □

16.2. **Passing to zero spacing.**

Proposition 16.3. *Every sequence $N \to \infty$ has a subsequence along which μ_N converges weakly on H_{-4} to a law μ. Every such law satisfies (G), continuous translation invariance and cubic invariance. It obeys*

$$\mu(dF = 0) = 1. \tag{dF}$$

It is invariant and reflection positive at every translated plane in the nine families

$$x_j = c \quad (1 \le j \le 3), \qquad x_i + x_j = c, \quad x_i - x_j = c \quad (1 \le i < j \le 3).$$

Proof. The uniform estimate (16.4), compactness of H_{-3} balls in H_{-4} and Prokhorov's theorem give subsequences. Evaluation at a Schwartz test is continuous on H_{-4}. Apply weak convergence to truncated positive exponentials and then monotone convergence to preserve (G).

For $a \in \mathbb{R}^3$ choose $a_N \in \eta_N \mathbb{Z}^3$ with $a_N \to a$. The field estimate gives, for real f,

$$\left|\mu_N(e^{iF(\tau_a \cdot f)}) - \mu_N(e^{iF(\tau_{a_N} \cdot f)})\right| \le \|\tau_a \cdot f - \tau_{a_N} \cdot f\|_2 \to 0.$$

Lattice translation invariance and weak convergence give translation invariance of the limit characteristic functional. Cubic invariance passes directly, because signed permutations carry each averaging cube to the appropriate averaging cube. Characteristic uniqueness turns these identities into invariance of the law.

For the Bianchi identity let ξ be a real scalar Schwartz function, and form the test whose dual vector is $-\nabla\xi$: $d^*\xi = (-\partial_1\xi, \partial_2\xi, -\partial_3\xi)$ in the component order $(23, 13, 12)$. Replace each derivative by the centered difference

$$\partial_j^\eta \xi(y) = \eta^{-1}(\xi(y + \eta e_j/2) - \xi(y - \eta e_j/2))$$

to obtain $u^{(\eta)}$. Summation by parts gives $(\sharp_\eta u)(u^{(\eta)}) = 0$ when $du = 0$. To check it explicitly, integrate ξ over the cube $\eta(x + [0,1)^3)$, with factor $\eta^{-5/2}$, and call the result $\xi_\eta(x)$. The three coefficients of $u^{(\eta)}_{[\eta]}$ are the signed differences $\xi_\eta(x) - \xi_\eta(x - e_j)$; their pairing with u is $\sum_x \xi_\eta(x)(du)(x) = 0$. Here the coboundary convention is

$$(du)(x) = u_{23}(x + e_1) - u_{23}(x) - u_{13}(x + e_2) + u_{13}(x) + u_{12}(x + e_3) - u_{12}(x).$$

For example, the 23 coefficient of $u^{(\eta)}_{[\eta]}$ is $-\xi_\eta(x) + \xi_\eta(x - e_1)$. Reindexing its second summand gives the first two terms of this coboundary; the 13 and 12 coefficients give the other four terms with their signs. All sums converge absolutely by the almost sure linear growth proved in Lemma 16.1. The multiplier inequality $|\sin t - t| \le |t|^3/6$ gives

$$\|u^{(\eta)} - d^*\xi\|_2 \le \frac{\eta^2}{24} \sum_j \|\partial_j^3 \xi\|_2.$$

Consequently $F(d^*\xi)$ tends to zero in $L^2(\mu_N)$. Weak convergence gives $F(d^*\xi) = 0$ under μ. This passage needs no uniform integrability assumption: for each $K > 0$, the bounded continuous function $\min\{|F(d^*\xi)|^2, K\}$ has limiting expectation zero. Letting $K \to \infty$ proves the almost sure assertion. Choose a countable dense family of scalar Schwartz tests and use continuity of each distribution; this makes the Bianchi identity a single almost sure assertion, proving (dF).

For reflection positivity it suffices first to consider finite linear combinations $\Phi = \sum_l z_l e^{iF(f_l)}$ with compact tests strictly on one side of a plane through the origin. The support has a positive distance from the plane. For sufficiently small η, every lattice plaquette contributing to its test cochains lies strictly on that side: the averaging cubes and plaquettes have diameter at most a fixed multiple of η. The finite lattice positivity assertion applies. The reflected test map is exactly equivariant, because each of the nine reflections is a signed permutation of the coordinates. Passing the bounded characteristic-function quadratic form to the limit gives

$$\mu((\Theta_{n,0}\Phi)\Phi) \ge 0.$$

Reflection invariance passes by the same equivariance. Exponential density and the Cauchy–Schwarz estimate in $L^2(\mu)$ extend the inequality to every observable in the open half-space. Finally, translation invariance transports it to every offset c. □

Definition 16.4 (Reflection-positive laws)**.** Write $\mathcal{C}_9$ for the translation-invariant laws that are invariant and reflection positive at every plane of the nine families in Proposition 16.3. A law satisfies

(G), translation and cubic invariance, and coordinate reflection positivity. (H)

All laws constructed in Proposition 16.3 satisfy (H), (dF), and belong to $\mathcal{C}_9$.

Remark 16.5. The construction takes an infinite-volume subsequence at each fixed spacing, and then a subsequence of these laws as the spacing tends to zero. Lemma 16.2 shows that both limits are weak limits on the same Hilbert space. If desired, choose a periodic volume approximating each μ_N to within $1/N$ in a metric for weak convergence, while making that volume at least N. The triangle inequality gives a cofinal sequence of periodic cutoff laws with the same limit. This observation does not prove the converse assertion for arbitrary cofinal limits.

17. Reflection positivity and reconstruction

Let μ be translation invariant and invariant under a reflection r in a plane through the origin. Reflection positivity means

$$\mu((\Theta\Phi)\Phi) \geq 0 \quad \text{for every } \Phi \in L^2(\Sigma_+, \mu), \tag{17.1}$$

where Σ_+ is the σ-algebra of its open positive half-space. It is enough to check this on finite sums of supported characteristic exponentials: their density and $|\mu((\Theta\Phi)\Psi)| \leq \|\Phi\|_2\|\Psi\|_2$ give the extension. In this section the normal may be any unit vector. We write the proof for e_j to agree with the later spectral notation.

Definition 17.1. Put $\Sigma_j^+ = \Sigma_{\{x_j>0\}}$ and $\mathcal{E}_j^+ = L^2(\Sigma_j^+, \mu)$. On this space use the form

$$b_j(\Phi, \Psi) = \mu((\Theta_j\Phi)\Psi) = \langle \widehat{\theta}_j\Phi, \Psi\rangle_{L^2(\mu)}.$$

Its null quotient, completed in this form, is $\mathcal{H}_j$. The class of Φ is $[\Phi]_j$, and $\Omega_j = [1]_j$. This quotient construction is the OS reconstruction used here.

Lemma 17.2 (Translation averages)**.** *For a nonzero linear subspace $E \subset \mathbb{R}^3$, suppose that μ is invariant under its translations. The averages*

$$A_T^E\Phi = T^{-d}\int_{[0,T]^d} U(t_1v_1 + \cdots + t_dv_d)\Phi\, dt, \qquad d = \dim E,$$

with an orthonormal basis $v_1, \ldots, v_d$ of E, converge strongly in $L^2(\mu)$ to the orthogonal projection P^E onto the invariant vectors. This projection is conditional expectation onto the invariant σ-algebra $\mathcal{I}^E$. Invariant vectors admit invariant measurable representatives. The averages are contractions in every L^p, $1 \leq p \leq \infty$, when defined there.

Proof. The averages fix invariant vectors. For a vector of the form $\Psi - U(a)\Psi$, $a \in E$, changing variables in the averaging cube shows that its average tends to zero: the relative measure of the symmetric difference of the cube and its translate tends to zero. The orthogonal complement of the invariant vectors is the closed span of these differences, since a vector orthogonal to every difference is fixed by every $U(a)$. Uniform boundedness of the averages proves the strong limit. Minkowski's inequality proves the L^p contraction.

For bounded Φ, the averages have pointwise Borel versions, by joint measurability of the translation action. A subsequence of the L^2 convergent averages converges almost surely. The elementary boundary estimate $|A_T^E\Phi(F) - A_T^E\Phi(\tau_aF)| \leq 2\|\Phi\|_\infty|([0,T]^d + a)\triangle[0,T]^d|/T^d$ shows that the set of convergence and its limiting function are invariant under every $a \in E$. The limiting function is consequently $\mathcal{I}^E$-measurable. Joint measurability and Fubini identify the pointwise orbit integrals with the Bochner averages. The boundary estimate is simultaneous for every F and every translation,

so the limiting representative is invariant under all real translations, not merely under a countable subgroup.

For an arbitrary invariant L^2 vector Ψ, take bounded approximants Ψ_l in ordinary L^2. Their projections $P^E\Psi_l$ converge to Ψ and have the bounded invariant representatives just constructed. Choose a subsequence converging almost surely. Its convergence set is invariant, since every member of the sequence is exactly invariant. Assigning zero off that set gives an invariant Borel representative of Ψ. Conversely, an $\mathcal{I}^E$-measurable function is invariant. The orthogonal projection onto this subspace is conditional expectation, as follows from its defining integral identity. □

Lemma 17.3 (Invariant weights)**.** *Suppose that μ is invariant and reflection positive at $x_j = 0$, and invariant under translations along e_j. Every vector invariant under these translations is measurable with respect to either open half-space, at any offset, and is fixed by $\widehat{\theta}_j$. For every bounded invariant h,*

$$\mu(h\widehat{\theta}_j G) = \mu(hG) \qquad (G \in L^1(\mu));$$

if $h \geq 0$, then

$$\mu(h(\Theta_j G)G) \geq 0 \qquad (G \in \mathcal{E}_j^+). \tag{17.2}$$

These statements hold for an arbitrary normal after replacing e_j by that normal.

Proof. Approximate an invariant vector Φ by compactly supported characteristic polynomials G_l. Translate G_l far enough into the desired half-space. The error remains $\|G_l - \Phi\|_2$, because Φ is invariant and translations are unitary. The closed half-space-measurable subspace therefore contains Φ. The complex-linear reflection restricts to a self-adjoint involution on the invariant subspace, since it reverses translations along its normal. Reflection positivity makes that restriction nonnegative, by the just-proved half-space measurability. A nonnegative self-adjoint involution is the identity. Explicitly, write $J = \widehat{\theta}_j$ on the invariant subspace. The operators $(1 \pm J)/2$ are orthogonal projections. If $\Phi = (1 - J)\Psi/2$, then $b_j(\Phi, \Phi) = \langle J\Phi, \Phi\rangle_\mu = -\|\Phi\|_2^2$, so positivity forces $\Phi = 0$. Reflection fixation is consequently an equality in ordinary L^2, not merely an equality of OS classes.

Reflection invariance now gives the first integral identity. If $h \geq 0$, its square root is bounded, invariant and positive-side measurable, and is fixed by the reflection. Apply reflection positivity to $\sqrt{h}\,G$ to obtain (17.2). Every step uses only the normal direction, so the proof applies to every plane normal. □

Proposition 17.4 (Reconstruction)**.** *Suppose that μ is translation invariant, invariant under θ_j and reflection positive at $x_j = 0$. The class map $\mathcal{E}_j^+ \to \mathcal{H}_j$ is a contraction with dense range, $\|\Omega_j\| = 1$, and there are*

$$T_j(t) = e^{-t\mathsf{H}_j}, \quad t \geq 0, \qquad V_j(y), \quad y \perp e_j,$$

with $\mathsf{H}_j \geq 0$ self-adjoint and V_j a strongly continuous unitary representation, such that

$$T_j(t)[\Phi]_j = [U(te_j)\Phi]_j, \qquad V_j(y)[\Phi]_j = [U(y)\Phi]_j. \tag{17.3}$$

The semigroup and transverse representation commute and fix Ω_j. If P^j is the invariant projection on $L^2(\mu)$ for translations along e_j, and $P_{0,j}$ the projection onto $\ker \mathsf{H}_j$*, then*

$$P_{0,j}[\Phi]_j = [P^j\Phi]_j. \tag{17.4}$$

The class map restricts to an isometry of the ordinary invariant subspace onto $\ker \mathsf{H}_j$*.*

Proof. Reflection invariance makes b_j Hermitian, and positivity gives Cauchy–Schwarz for it. Its quadratic form is bounded by the ordinary L^2 norm squared. This proves the quotient and contraction assertions. Also $b_j(1, 1) = 1$.

For $t \geq 0$ translation preserves the positive-side subspace and

$$b_j(U(te_j)\Phi, U(te_j)\Phi) = b_j(\Phi, U(2te_j)\Phi).$$

Writing $|\Phi|_b = b_j(\Phi,\Phi)^{1/2}$ and iterating Cauchy–Schwarz gives

$$|U(te_j)\Phi|_b \le |\Phi|_b^{1-2^{-l}} |U(2^l te_j)\Phi|_b^{2^{-l}} \le |\Phi|_b^{1-2^{-l}} \|\Phi\|_2^{2^{-l}}.$$

Let $l \to \infty$. Thus translation induces a contraction on the quotient. The same reflection identity makes it self-adjoint. This also treats null vectors: the iterated bound gives $|U(te_j)\Phi|_b = 0$ whenever $|\Phi|_b = 0$. For self-adjointness, translation invariance and $\widehat{\theta}_j U(te_j) = U(-te_j)\widehat{\theta}_j$ give

$$b_j(U(te_j)\Phi,\Psi) = \langle U(-te_j)\widehat{\theta}_j\Phi,\Psi\rangle_\mu = b_j(\Phi,U(te_j)\Psi).$$

The semigroup law and strong continuity follow first on classes from ordinary translation continuity, then by density. Since $T_j(t) = T_j(t/2)^2$, the operators are nonnegative. They are injective: if $T_j(t)\psi = 0$, repeatedly taking square roots gives $T_j(2^{-l}t)\psi = 0$, whence $\psi = 0$ by strong continuity. Define $\mathsf{H}_j = -\log T_j(1)$ by the spectral calculus. It is nonnegative and self-adjoint. Uniqueness of nonnegative square roots proves $T_j(t) = e^{-t\mathsf{H}_j}$ first for nonnegative dyadic t, then for every t by continuity. This also proves uniqueness of the generator. More precisely, $T_j(1)$ has spectrum in $[0,1]$ and no spectral mass at zero, since its kernel is zero. If $E^{(1)}$ is its spectral measure, the domain of $-\log T_j(1)$ is

$$\left\{\psi : \int_{(0,1]} |\log u|^2 \, d\langle\psi, E^{(1)}(u)\psi\rangle < \infty\right\}.$$

It is dense by truncation to $u \ge \varepsilon$. The exponential of this generator at time 2^{-l} is the unique nonnegative 2^lth root of $T_j(1)$, and hence equals $T_j(2^{-l})$. Products give all nonnegative dyadic times; strong continuity gives all real nonnegative times. No derivative on an unproved dense domain is needed for this construction.

Transverse translations preserve the form exactly, because they commute with the reflection. Their inverse translations do the same, so they induce unitaries. Commutation, the group law and strong continuity follow on classes and then by density.

It remains to identify the zero-energy space. For $\Phi \in \mathcal{E}_j^+$ the ordinary averages of $U(te_j)\Phi$ converge to $P^j\Phi$. The class map is continuous, so their classes converge to $[P^j\Phi]_j$. The same averages in the OS space are averages of $e^{-t\mathsf{H}_j}$, which converge by the spectral theorem to $P_{0,j}[\Phi]_j$. This proves (17.4). Each finite positive-time average belongs to $\mathcal{E}_j^+$. Since this subspace is closed in ordinary L^2, its ordinary limit $P^j\Phi$ also belongs to it. The bounded class map commutes with the Bochner integral. The multiplier for the OS average is

$$\frac{1}{T}\int_0^T e^{-t\omega}\,dt = \begin{cases} (1-e^{-T\omega})/(T\omega), & \omega > 0, \\ 1, & \omega = 0. \end{cases}$$

It is bounded by one and tends to $1_{\{0\}}(\omega)$. Dominated convergence for each vector spectral measure proves the strong convergence to $P_{0,j}$ used above. Lemma 17.3 shows that invariant Φ belongs to $\mathcal{E}_j^+$ and is fixed by the complex-linear reflection. Hence $\|[\Phi]_j\| = \|\Phi\|_2$, and its class has zero energy. The image is closed because the ordinary invariant subspace is complete and the class map is an isometry on it. It contains $P_{0,j}[\mathcal{E}_j^+]$ by (17.4). This set is dense in $\ker \mathsf{H}_j$: approximate any vector there by classes and apply the bounded projection $P_{0,j}$. Thus the image is exactly $\ker \mathsf{H}_j$. This is the step which permits an OS zero-energy statement to imply ordinary translation invariance later. □

Lemma 17.5 (Joint spectral measure)**.** *There is a projection-valued measure $\mathfrak{E}_j$ on $[0,\infty)\times e_j^\perp$ such that*

$$e^{-t\mathsf{H}_j}V_j(y) = \int e^{-t\omega + ip\cdot y}\, d\mathfrak{E}_j(\omega,p). \tag{17.5}$$

For $i \ne j$, the transverse generator P_i is multiplication by p_i in this spectral representation. Its joint spectral measure with H_j is the image of $\mathfrak{E}_j$ under $(\omega,p)\mapsto(\omega,p_i)$.

Proof. Stone's theorem gives the two transverse self-adjoint generators. The transverse unitaries commute with each other and with every $e^{-t\mathsf{H}_j}$. Commutation with $e^{-\mathsf{H}_j}$ implies commutation with its spectral projections, and hence with those of H_j, since $\omega \mapsto e^{-\omega}$ is injective on $[0,\infty)$. The commuting spectral projections define their joint projection-valued measure; the joint spectral theorem gives the formula and the image assertion. The natural domain of each unbounded generator consists of the vectors for which the integral of its squared coordinate against the vector spectral measure is finite. □

18. Continuous rotations

The additional diagonal reflections have two consequences. First, they restrict the joint spectrum of time and space translations in the OS Hilbert space. Second, this restriction permits the analytic comparison of moments written with different coordinate directions as time. We prove these statements in that order. The higher moments require a positivity argument for their angular coefficients; the two-point function alone does not determine the law.

Throughout this section μ satisfies Gaussian domination, is carried by $dF = 0$, and is invariant under translations and signed coordinate permutations. Coordinate reflection positivity and its reconstruction are as in Proposition 17.4. Additional reflection or spectral assumptions will be stated where they are used.

18.1. **The tensor action and the spectral bound.** Identify a test two-form with its Hodge dual

$$b = \star f = (f_{23}, -f_{13}, f_{12}), \qquad \mathbb{B}(b) = F(f).$$

This is an isometry of the test spaces. For $g(x) = Rx + a$,

$$(g \cdot b)(x) = \det(R) R b(g^{-1}x), \qquad \widehat{g}\mathbb{B}(b) = \mathbb{B}(g \cdot b). \tag{18.1}$$

The determinant factor is the usual transformation of an axial vector; the observable remains the field strength tensor. Its closedness says

$$\mathbb{B}(\nabla u) = 0 \quad \text{in } L^2(\mu) \qquad (u \in \mathcal{S}(\mathbb{R}^3;\mathbb{C})). \tag{18.2}$$

The action on a gradient is $g \cdot \nabla u = \det(R)\nabla(u \circ g^{-1})$, so this identity is preserved by all orthogonal changes of frame, without assuming that those frames preserve μ.

Let $\mathcal{H}_j$ be the reconstructed Hilbert space, $[\Phi]_j$ its class map, $\Omega_j = [1]_j$ the vacuum, and $\mathfrak{E}_j$ the joint spectral measure of H_j and the transverse momenta. We use

$$e^{-t\mathsf{H}_j} V_j(y) = \int e^{-t\omega + ip\cdot y}\, d\mathfrak{E}_j(\omega, p), \qquad t \geq 0, \quad y \in e_j^{\perp}.$$

These formulas fix the sign of the momenta. The spectral condition needed below is

$$\mathfrak{E}_j\{(\omega,p) : \omega < |p_i|\} = 0 \qquad (i \neq j). \tag{18.3}$$

It is a statement on the whole Hilbert space. A bound only on vectors obtained from a single field insertion would not suffice for the higher-moment proof.

Proposition 18.1 (A diagonal reflection gives the spectral bound)**.** *Suppose a translation-invariant law is invariant and reflection positive at $x_j = 0$ and $x_j + x_i = 0$, $i \neq j$. Then its coordinate reconstruction satisfies (18.3) for this ordered pair (j,i).*

Proof. The proof compares two spectral representations of the same correlation. One representation is a positive measure on the complex half-plane; the other bounds products of Laplace transforms on its boundary axes. Suitable transforms grow on the part of the half-plane outside the required sector.

Fix $D > 0$ and an observable $\Phi \in L^2(\mu)$ supported in $B = \{0 < x_j < D,\ |x_i| < D\}$. Put $\psi = [\Phi]_j$, and let ϱ be its positive spectral measure in (ω, p_i). For $a, c \geq 0$ define

$$G(a,c) = \mu((\Theta_j\Phi)U(2De_j + a(e_j + e_i) + c(e_j - e_i))\Phi).$$

In the coordinate reconstruction, with $z = \omega - ip_i$,

$$G(a,c) = \int e^{-2D\omega} e^{-az - c\bar{z}}\, d\varrho(\omega, p_i). \tag{18.4}$$

For the diagonal reconstruction set $n_\pm = (e_j \pm e_i)/\sqrt{2}$ and $y = D(e_j + e_i)/2$. The observables

$$\Phi_1 = \widehat{r}_{n_+,0} U(-y) \widehat{\theta}_j \Phi, \qquad \Phi_2 = U(2De_j - y)\Phi$$

are supported on $x_j + x_i > 0$: the first arises by shifting $\theta_j B$ to $x_j + x_i < 0$ and reflecting it, and the second by shifting B to the positive side. They have L^2 norm $\|\Phi\|_2$. An orthogonal map taking e_j, e_i to n_+, n_- transports this reconstruction to the coordinate reconstruction of a rotated law. We do not identify that law with μ. The class map is a contraction, so its spectral representation gives a finite complex measure ϱ' of total variation at most $\|\Phi\|_2^2$, with

$$G(a,c) = \int e^{-a\omega' + icp'}\, d\varrho'(\omega', p'), \qquad \omega' \geq 0. \tag{18.5}$$

Indeed, translating the defining correlation by $-y$ makes its first factor $\overline{\widehat{r}_{n_+,0}\Phi_1}$, while its remaining translation is $\sqrt{2}an_+ + \sqrt{2}cn_-$. The factors $\sqrt{2}$ are absorbed into the spectral variables. The total-variation estimate follows by applying Cauchy–Schwarz to the mutually orthogonal spectral projections of any finite Borel partition. More explicitly, if E' is the joint spectral measure in the diagonal reconstruction and ψ_k are the classes of Φ_k, then the complex measure before rescaling is $\langle \psi_1, E'(\cdot)\psi_2 \rangle$. For a finite partition (A_l),

$$\begin{aligned} \sum_l |\langle \psi_1, E'(A_l)\psi_2 \rangle| &\leq \sum_l \|E'(A_l)\psi_1\|\, \|E'(A_l)\psi_2\| \\ &\leq \|\psi_1\|\, \|\psi_2\| \leq \|\Phi\|_2^2. \end{aligned}$$

The image measure has no larger total variation. This estimate requires neither positivity of the mixed spectral measure nor rotation invariance of the law.

For $h \in L^1(0,\infty)$ let $\widetilde{h}(z) = \int_0^\infty h(a) e^{-az}\, da$. Integrate both representations against $h(a)\overline{h(c)}$. Finite total variation and $|e^{-az}| \leq 1$ justify Fubini and yield

$$\int e^{-2D\omega} |\widetilde{h}(\omega - ip_i)|^2\, d\varrho \leq \|\Phi\|_2^2 \sup_{t \geq 0} |\widetilde{h}(t)| \sup_{s \in \mathbb{R}} |\widetilde{h}(is)|. \tag{18.6}$$

For either sign $\varepsilon \in \{\pm 1\}$ and $n \geq 1$, use the principal powers

$$\widetilde{h}_n(z) = e^{-n} z^{1 - 4i\varepsilon n/\pi} (1+z)^{-3}, \qquad \widetilde{h}_n(0) = 0.$$

On $\operatorname{Re} z \geq 0$, $|z|\,|1+z|^{-3} \leq 1$. The suprema on the positive and imaginary axes are at most e^{-n} and e^n, respectively, so their product is at most one.

These functions are Laplace transforms of actual L^1 functions. To check this, put $\nu = 1 - 4i\varepsilon n/\pi$ and factor

$$z^\nu (1+z)^{-3} = \big(1 - (1+z)^{-1}\big)^\nu (1+z)^{-(3-\nu)}.$$

The principal logarithms agree in this factorization on the right half-plane. The second factor is the transform of $a^{2-\nu} e^{-a}/\Gamma(3-\nu)$, an integrable function since $\operatorname{Re}\nu = 1$. The binomial coefficients of the first factor satisfy $|\binom{\nu}{k}| = O(k^{-2})$ by the Gamma formula and Stirling's estimate. Its series therefore converges in the total-variation norm as a sum of the point mass at zero and the Gamma probability densities whose transforms are $(1+z)^{-k}$. This identity holds on the closed half-plane by continuity, including $z = 0$. Convolution with the integrable second factor gives the desired h_n. Here is the density and branch check used in that claim. On the closed right half-plane, away from zero,

$$\log \frac{z}{1+z} = \log z - \log(1+z),$$

because z and $1+z$ have arguments of the same sign and $|\arg(1+z)| \le |\arg z| \le \pi/2$. Thus the factorization uses the principal powers on both sides. For $k \ge 1$ put $\gamma_k(a) = a^{k-1}e^{-a}/(k-1)!$. The measure

$$m_\nu = \delta_0 + \sum_{k\ge 1} (-1)^k \binom{\nu}{k} \gamma_k(a)\, da$$

has finite total variation, since $\|\gamma_k\|_1 = 1$ and

$$\binom{\nu}{k} = (-1)^k \frac{\Gamma(k-\nu)}{\Gamma(-\nu)\Gamma(k+1)} = O(k^{-2}).$$

Its transform is $(1-(1+z)^{-1})^\nu$. The binomial series converges uniformly on $|u| \le 1$. At $u = 1$ both its sum and the continuous extension of $(1-u)^\nu$ equal zero, since $\operatorname{Re}\nu = 1$. The function

$$q_\nu(a) = \frac{a^{2-\nu}e^{-a}}{\Gamma(3-\nu)}$$

is integrable and has transform $(1+z)^{-(3-\nu)}$, by the Gamma integral, first on the positive axis and then by analytic continuation. Consequently $h_n = e^{-n}(m_\nu * q_\nu)$ is an integrable function, with $\|h_n\|_1 \le e^{-n}\|m_\nu\|_{\mathrm{TV}}\|q_\nu\|_1$. Fubini proves its claimed transform on the closed half-plane.

On a sector $\varepsilon \arg z \ge \pi/4 + \delta$, $\delta > 0$,

$$|\widetilde{h}_n(z)|^2 \ge e^{8\delta n/\pi}|z|^2|1+z|^{-6}.$$

Equation (18.6), followed by $n \to \infty$, forces ϱ to vanish there. Both signs and a countable sequence of δ exclude precisely $|p_i| > \omega$. Indeed, integration of the lower bound over that sector gives

$$\int_{\varepsilon \arg z \ge \pi/4+\delta} e^{-2D\omega}|z|^2|1+z|^{-6}\, d\varrho \le e^{-8\delta n/\pi}\|\Phi\|_2^2.$$

The integral is zero as $n \to \infty$. Its weight is strictly positive on the sector, so the sector itself has zero measure. The point $z = 0$ already satisfies the cone inequality.

Finally, finite sums of $e^{iF(f)}$ with tests compactly supported in $x_j > 0$ are dense in the positive-side L^2 space by Lemma 15.3. Each such sum is supported in one of the sets B above. Their classes are dense in $\mathcal{H}_j$, and the bounded spectral projection in (18.3) vanishes on those classes. It therefore vanishes on the whole space. In detail, on any such class ψ the squared norm of that projection is $\langle \psi, E(A)\psi\rangle = \varrho_\psi(A) = 0$, where $A = \{\omega < |p_i|\}$. The projection $E(A)$ is bounded. The contraction class map carries the dense supported exponential span to a dense subspace of $\mathcal{H}_j$, so continuity forces $E(A) = 0$ on every vector. This supplies the cone for all the multi-insertion vectors used below. □

18.2. **Planar test functions and complex translations.** For the planar argument we now assume $\mathfrak{E}_1\{(\omega, p) : \omega < |p_2|\} = 0$. The preceding proposition supplies this hypothesis for a law with the stated diagonal reflection positivity. The quarter-turn symmetry transports it to the reconstruction with x_2 as time.

We now work in the (x_1, x_2) plane, writing $x = (y, x_3)$, $y = (y^0, y^1)$. Let R_t rotate this plane through t, with $R_t e_1 = \cos t\, e_1 + \sin t\, e_2$. The third coordinate is unchanged. Use the polarization vectors

$$e^+ = e_1 + ie_2, \quad e^- = e_1 - ie_2, \quad e^3 = e_3,$$

with angular weights $w(+) = 1, w(-) = -1, w(3) = 0$ and reflection signs $s(+) = s(-) = 1, s(3) = -1$. A label $\lambda = (\sigma, \chi)$ consists of one polarization and a real profile $\chi \in L^2(\mathbb{R})$. For $\alpha \in L^2(\mathbb{R}^2)$ define

$$(\alpha \otimes \lambda)(y, x_3) = \alpha(y)\chi(x_3)e^\sigma, \qquad \mathbb{B}_\lambda(\alpha) = \mathbb{B}(\alpha \otimes \lambda).$$

Their norm is $|e^{\sigma}|\|\alpha\|_2\|\chi\|_2$, and finite sums are dense in $L^2(\mathbb{R}^3;\mathbb{C}^3)$, also with any prescribed planar support and smooth compactly supported profiles. This follows from the density of product simple functions and the displayed basis.

For a tuple of labels write $w(\lambda)=\sum_a w(\sigma_a)$, $s(\lambda)=\prod_a s(\sigma_a)$, and $\bar\lambda$ for the tuple obtained by exchanging $+$ and $-$. Direct matrix multiplication gives

$$\begin{aligned} R_t\cdot(\alpha\otimes\lambda) &= e^{-iw(\sigma)t}(\alpha\circ R_{-t})\otimes\lambda,\\ \Theta_1\mathbb{B}_\lambda(\alpha) &= s(\sigma)\mathbb{B}_\lambda(\bar\alpha\circ\theta_1),\\ \Theta_2\mathbb{B}_\lambda(\alpha) &= -\mathbb{B}_\lambda(\bar\alpha\circ\theta_2),\\ \overline{\mathbb{B}_\lambda(\alpha)} &= \mathbb{B}_{\bar\lambda}(\bar\alpha). \end{aligned} \tag{18.7}$$

The antilinear reflection preserves the label. This distinction is needed when a Gram matrix is formed below. The axial-vector matrices for the reflections are $\det(\theta_1)\theta_1=\operatorname{diag}(1,-1,-1)$ and $\det(\theta_2)\theta_2=\operatorname{diag}(-1,1,-1)$. The first sends e^+ to e^- and the second sends it to $-e^-$. Complex conjugation in Θ_j sends e^- back to e^+. This gives the unchanged polarization labels and the different signs in (18.7). For a scalar product test $u(y,x_3)=\eta(y)\chi_0(x_3)$, with $\partial_z=(\partial_0-i\partial_1)/2$, $\partial_{\bar z}=(\partial_0+i\partial_1)/2$, one also has

$$\nabla u=(\partial_z\eta)\otimes(+,\chi_0)+(\partial_{\bar z}\eta)\otimes(-,\chi_0)+\eta\otimes(3,\chi_0'). \tag{18.8}$$

All identities extend from smooth tests to L^2 tests by Lemma 15.2.

For $\xi\in\mathbb{C}^2$ put

$$\xi^{\pm}=\xi^0\pm i\xi^1,\qquad d(\xi)=\operatorname{Re}\xi^0-|\operatorname{Im}\xi^1|=\min(\operatorname{Re}\xi^+,\operatorname{Re}\xi^-),\qquad \mathcal{W}=\{d(\xi)>0\}.$$

The function d is superadditive. By the spectral bound the operator

$$T(\xi)=\int e^{-\xi^0\omega+i\xi^1p_2}\,d\mathfrak{E}_1(\omega,p)\qquad(\xi\in\overline{\mathcal{W}}) \tag{18.9}$$

is a contraction. It is strongly continuous on the closed tube, holomorphic in operator norm on its interior, multiplicative, and $T(\xi)^*=T(-\theta_1\bar\xi)$. On a compact interior set each derivative of its spectral integrand is dominated by $C\omega^k e^{-\delta\omega}$, since $|p_2|\le\omega$. The bounded functional calculus therefore proves all these assertions, including norm holomorphy. In detail, on $d(\xi)\ge\delta>0$ the spectral multiplier of any derivative of total order k is bounded by $\omega^k e^{-\delta\omega}\le k!\delta^{-k}$. The Taylor series in a sufficiently small complex polydisc therefore converges in operator norm, by summing its geometric majorants. On the closed tube only strong continuity is asserted: for each ψ, the squared norm of the difference of two spectral multipliers is the integral of their squared difference against the finite measure $\langle\psi,\mathfrak{E}_1(\cdot)\psi\rangle$. Bounded convergence proves continuity there.

For later use put $\xi^*=(\bar\xi^0,-\bar\xi^1)$ and define complex rotations by $(R_\zeta\xi)^{\pm}=e^{\pm i\zeta}\xi^{\pm}$. The maps $\Lambda_\beta=R_{i\beta}$ preserve every rotated tube $\mathcal{W}_\upsilon=R_\upsilon\mathcal{W}$ and commute with $*$. Indeed, they multiply the two light coordinates by the positive numbers $e^{\mp\beta}$. In particular

$$d(\Lambda_\beta\xi)\ge e^{-|\beta|}d(\xi)\quad(\xi\in\overline{\mathcal{W}}). \tag{18.10}$$

Every rotated tube is convex and contains the real part of each of its points. Its real points are the half-plane $n_\upsilon\cdot y>0$, where $n_\upsilon=(\cos\upsilon,\sin\upsilon)$. These facts follow either from the light coordinates or from applying the real matrix $R_{-\upsilon}$ to the defining inequalities.

18.3. **Extension across coincident planar supports.** The next lemma explains why an identity proved for separated planar supports suffices. It is independent of a probability measure.

Proposition 18.2 (No contact terms)**.** *Let $O\subset\mathbb{R}^2$ be open and let D be a bounded N-linear form on $L^2(O\times\mathbb{R};\mathbb{C}^3)$. Suppose that it vanishes when its entries have pairwise disjoint compact planar supports, and also whenever one entry is ∇u, $u\in C_c^\infty(O\times\mathbb{R})$. Then $D=0$.*

Proof. We argue by induction on N, for all open O simultaneously. Compact planar truncations are dense in the indicated L^2 space. For $N = 1$ the first assumption already gives the conclusion.

For $N = 2$, represent the bounded form as $D(f, g) = \int f \cdot Tg$ for a bounded operator T; the pairing here is bilinear. The separated-support assumption gives $1_K T 1_{K'} = 0$ for disjoint compact $K, K' \subset O$. Inner regularity and strong convergence of compact truncations give the same identity for disjoint Borel sets. Applying it to A and $O \setminus A$ shows that T commutes with every planar indicator, hence with every bounded Borel multiplier $h(y)$. To justify the Borel-set passage, approximate the two disjoint Borel sets from within by compact sets for the finite measures $\int_A \int |f(y, x_3)|^2 \, dx_3 \, dy$ associated with the two entries. Passing to the L^2 limit gives the multiplier identity. Consequently

$$D(h\nabla u, g) = D(\nabla u, hg) = 0.$$

Fix a compact $K \subset O$ and choose $u_0 \in C_c^\infty(O)$ equal to one near K. For $\chi \in C_c^\infty(\mathbb{R})$, the gradients of $u_0\chi$, $u_0 y^0 \chi$, $u_0 y^1 \chi$, restricted to $K \times \mathbb{R}$, are respectively

$$\chi' e_3, \quad \chi e_1 + y^0 \chi' e_3, \quad \chi e_2 + y^1 \chi' e_3.$$

Multiplying them by $h1_K$ first kills the third component and then the first and second components in $D(\cdot, g)$. The profiles χ' are dense in $L^2(\mathbb{R})$: a vector orthogonal to them has distributional derivative zero and is an L^2 constant, hence zero. Every planar L^2 function supported in K is an L^2 limit of bounded planar multipliers times 1_K. The identities for $\chi' e_3$ annihilate their tensor products with a dense family of third-coordinate profiles. Thus the entire third-component subspace over K is annihilated. Subtracting the third components from the other two gradients annihilates the first and second component product tests. Finite sums of such products are dense, and compact planar truncations exhaust O. Hence $D(\cdot, g) = 0$.

Now take $N \geq 3$. The induction hypothesis first shows vanishing when one entry is supported in a compact set K and all others in $O \setminus K$. Fix the compactly supported entry and apply the $(N-1)$-variable assertion on that open complement. It next shows vanishing for any two nonempty groups supported in disjoint compact planar sets: fix the entries of one group and apply the lower-order assertion to the other; its separated-support hypothesis follows from the just-proved one-entry case. The gradient hypothesis is inherited in both applications by extension by zero. To verify the first induction, fix f_1 over $K \Subset O$. On $O \setminus K$ the remaining $(N-1)$-linear form is bounded and annihilates gradients. If its entries have pairwise disjoint compact planar supports, those supports are also disjoint from K, and the original separated-support hypothesis applies. The lower-order conclusion therefore proves the one-entry assertion.

For the second induction, fix one group over K and consider the lower-order form in the other group on $O \setminus K$. When the latter entries have pairwise disjoint compact supports, choose one of those supports. Every other entry, including the fixed group, is outside it, so the one-entry result applies. This verifies the separated-support hypothesis for that lower-order form, whose gradient hypothesis is unchanged. Its conclusion gives the two-group assertion for overlapping supports within each group.

It remains to consider entries with a common compact planar support. Partition the plane into half-open squares Q of side a. Expand all entries in $1_Q f_i$. Terms whose squares are not all equal vanish by the two-group assertion, first shrinking the squares to disjoint compact subsets and then passing to the L^2 limit. Thus

$$|D(f_1, \ldots, f_N)| \leq \|D\| \sum_Q \prod_i \|1_Q f_i\| \leq \|D\| \|f_1\| \|f_2\| \sup_Q \|1_Q f_3\| \prod_{i \geq 4} \|f_i\|.$$

The last supremum tends to zero as $a \to 0$ by absolute continuity of the integral of $y \mapsto \int |f_3(y, x_3)|^2 \, dx_3$. Density of compact truncations completes the induction. □

18.4. **Bounded insertions between two time slices.** We require a bound independent of the number of fields on either side of an insertion. An ordinary moment estimate does not have that uniformity. Reflection positivity provides it by repeatedly doubling a source in adjacent time slabs.

Write $U_l = U(le_1)$ and let $L^2_{[a,b]}$ denote tests supported in $a \le x_1 \le b$. Let $\mathcal{D}_+$ be the positive-side observables belonging to every finite $L^p(\mu)$. It contains the field polynomials and exponentials of linearly smeared fields.

Lemma 18.3 (Exponential source bound)**.** *For $l > 0$, $h \in L^2_{[0,l]}$ and bounded positive-side Φ,*

$$\|[e^{\mathbb{B}(h)} U_l \Phi]_1\| \le e^{\|\operatorname{Re} h\|^2/2} \|[\Phi]_1\|.$$

Proof. Put $M_l(h)\Phi = e^{\mathbb{B}(h)} U_l \Phi$ and $h^{\#} = \tau_l(\overline{\theta_1 \cdot h} + h)$. Reflection and translation of the integrand give exactly

$$\begin{aligned} \|[M_l(h)\Phi]_1\|^2 &= \mu\big(e^{\mathbb{B}(\overline{\theta_1 \cdot h})} (U_{-l}\Theta_1\Phi) e^{\mathbb{B}(h)} U_l \Phi\big) \\ &= \mu\big((\Theta_1\Phi) e^{\mathbb{B}(h^{\#})} U_{2l}\Phi\big) \\ &= \langle [\Phi]_1, [M_{2l}(h^{\#})\Phi]_1 \rangle. \end{aligned} \tag{18.11}$$

The second equality applies U_l to the whole integrand. It uses multiplicativity of translations and reflections, $\Theta_1 U_l = U_{-l}\Theta_1$, and invariance of μ. The two terms of $h^{\#}$ occupy the two adjacent slabs, and $\tau_{2l}\overline{\theta_1 \cdot h^{\#}} = h^{\#}$. This identity follows by applying the involution $h \mapsto \overline{\theta_1 \cdot h}$ twice and using $\theta_1 \tau_l = \tau_{-l}\theta_1$:

$$\tau_{2l}\overline{\theta_1 \cdot h^{\#}} = \tau_l(h + \overline{\theta_1 \cdot h}) = h^{\#}.$$

For any symmetric source k of width l' satisfying the analogous identity, its repeated source $k^{(n)} = \sum_{r=0}^{n-1} \tau_{rl'} k$ is symmetric and doubles to $k^{(2n)}$. Explicitly,

$$\tau_{nl'}\overline{\theta_1 \cdot k^{(n)}} = \sum_{r=0}^{n-1} \tau_{(n-1-r)l'} k = k^{(n)}, \qquad (k^{(n)})^{\#} = k^{(n)} + \tau_{nl'} k^{(n)} = k^{(2n)}.$$

Set $a_r = \|[M_{2^r l'}(k^{(2^r)})\Phi]_1\|$. Equation (18.11) gives $a_r^2 \le \|[\Phi]_1\| a_{r+1}$, hence

$$a_0 \le \|[\Phi]_1\|^{1-2^{-r}} a_r^{2^{-r}}.$$

Applying Gaussian domination to the doubled correlation gives $a_r^2 \le \|\Phi\|_\infty^2 e^{2^r \|\operatorname{Re} k\|^2}$: the 2^{r+1} real sources have disjoint slab interiors, so their squared norms add. More precisely,

$$\begin{aligned} a_r^2 &\le \|\Phi\|_\infty^2 \mu(e^{\mathbb{B}(\operatorname{Re} k^{(2^{r+1})})}) \\ &\le \|\Phi\|_\infty^2 \exp(2^r \|\operatorname{Re} k\|^2). \end{aligned}$$

Thus the iterated bound reads

$$a_0 \le \|[\Phi]_1\|^{1-2^{-r}} \|\Phi\|_\infty^{2^{-r}} e^{\|\operatorname{Re} k\|^2/2}.$$

If $\|[\Phi]_1\| = 0$, the first Cauchy–Schwarz inequality already gives $a_0 = 0$. Otherwise let $r \to \infty$ to obtain $a_0 \le e^{\|\operatorname{Re} k\|^2/2} \|[\Phi]_1\|$, also if that class is zero. Apply this to $k = h^{\#}$, for which $\|\operatorname{Re} h^{\#}\|^2 = 2\|\operatorname{Re} h\|^2$, and use (18.11) once more. All products are integrable because the Gaussian bound places every exponential in every finite L^p space. □

Proposition 18.4 (Insertion operators)**.** *For $f \in L^2_{[0,l]}$ there is a bounded, complex-linear operator $\mathcal{B}_l(f)$ on $\mathcal{H}_1$ with*

$$\mathcal{B}_l(f)[\Phi]_1 = [\mathbb{B}(f) U_l \Phi]_1 \quad (\Phi \in \mathcal{D}_+), \qquad \|\mathcal{B}_l(f)\| \le k_1 \|f\|, \quad k_1 = \sqrt{2}\exp(1/2).$$

For $f \in L^2_{[-d,d]}$ put $\mathcal{B}^{\mathrm{c}}_d(f) = \mathcal{B}_{2d}(\tau_d f)$. Then

$$\langle [\Phi']_1, \mathcal{B}^{\mathrm{c}}_d(f)[\Phi]_1 \rangle = \mu((\Theta_1 U_d \Phi')\mathbb{B}(f) U_d \Phi), \tag{18.12}$$

and its adjoint is $\mathcal{B}_d^{\mathrm{c}}(\overline{\theta_1 \cdot f})$. *If* $r \geq |t|$, $a \in \mathbb{R}$, *then*

$$T(r+t,a)\mathcal{B}_d^{\mathrm{c}}(f)T(r-t,-a) = \mathcal{B}_{d+r}^{\mathrm{c}}(\tau_{te_1+ae_2}f). \tag{18.13}$$

Proof. For a real f, the Hilbert-valued entire function $z \mapsto [e^{z\mathbb{B}(f)}U_l\Phi]_1$ is bounded by $e^{(\operatorname{Re} z)^2\|f\|^2/2}\|[\Phi]_1\|$. Its power series converges locally in L^2 by Gaussian domination. Cauchy's estimate on $|z| = \|f\|^{-1}$ bounds its first coefficient by $\exp(1/2)\|f\|\|[\Phi]_1\|$. Splitting real and imaginary parts uses $\|\operatorname{Re} f\| + \|\operatorname{Im} f\| \leq \sqrt{2}\|f\|$ and the complex linearity of the field to give the stated k_1. The entire power series is justified in ordinary L^2 before passing to the OS class: on a compact disc $|z| \leq R$, the series and its derivatives are dominated by a polynomial in $|\mathbb{B}(f)|$ times $e^{R|\mathbb{B}(f)|}$. Its square is integrable by Gaussian domination. Equivalently the nth coefficient has L^2 norm at most a constant times $C^n\|f\|^n/\sqrt{n!}$ for bounded Φ. The estimate kills null classes, so it defines the operator on the dense classes of bounded observations and then on all of $\mathcal{H}_1$. Truncate $\Phi \in \mathcal{D}_+$ in L^4; Hölder's inequality and $\mathbb{B}(f) \in L^4$ extend the defining identity to those observations.

For tests supported in $c \leq x_1 \leq c'$ within $[0,l]$, their translated defining classes give the localization identity

$$\mathcal{B}_l(f) = e^{-c\mathsf{H}_1}\mathcal{B}_{c'-c}(\tau_{-c}f)e^{-(l-c')\mathsf{H}_1}.$$

On a dense class $[\Phi]_1$, the right side equals

$$[U_c(\mathbb{B}(\tau_{-c}f)U_{c'-c}U_{l-c'}\Phi)]_1 = [\mathbb{B}(f)U_l\Phi]_1.$$

Thus the localization formula is an equality of bounded operators, not a formal rearrangement of unbounded fields. Translating its scalar-product formula by $-d$ gives (18.12). Conjugating that formula and reflecting its integrand proves the adjoint identity. Finally apply the reconstruction identities to both outer factors in (18.13). They translate the inserted test by $te_1 + ae_2$, while their time translations together with the central $2d$ add to $2(d+r)$. The translated test lies in $[-d-r, d+r]$. Both bounded operators therefore agree on the same dense classes, proving the formula. For the signs in that calculation, its scalar-product integrand before the final common translation is

$$(\Theta_1 U((d+r+t)e_1 - ae_2)\Phi')\,\mathbb{B}(f)\,U((d+r-t)e_1 - ae_2)\Phi.$$

Applying $U(te_1+ae_2)$ to the entire product changes the two outer factors to $\Theta_1 U_{d+r}\Phi'$ and $U_{d+r}\Phi$, and changes the inserted test to $\tau_{te_1+ae_2}f$. This is exactly the matrix formula for the right side of (18.13). □

18.5. **Kernels for separated planar points.** A point mass in the two planar variables is not an L^2 test. We first put semigroup factors on both sides of its approximation. The insertion bound and a holomorphic mean-value estimate then make the point limit a bounded operator. Products of these limiting operators will define the moment kernels.

Proposition 18.5 (Point insertions)**.** *For every label* $\lambda = (\sigma, \chi)$ *and* $\xi, \xi' \in \mathcal{W}$ *there is a bounded operator* $B_\lambda(\xi, \xi')$, *holomorphic in both variables, such that*

$$\|B_\lambda(\xi,\xi')\| \leq \frac{k_2\|\chi\|_2}{\min(d(\xi), d(\xi'))}, \qquad k_2 = \frac{8\exp(1/2)}{\pi}. \tag{18.14}$$

It is real-linear in χ *and obeys*

$$\begin{aligned} T(\kappa)B_\lambda(\xi,\xi')T(\kappa') &= B_\lambda(\kappa+\xi, \xi'+\kappa'), \\ B_\lambda(\xi,\xi')^* &= s(\sigma)B_\lambda(\xi'^*, \xi^*). \end{aligned} \tag{18.15}$$

Here $\kappa, \kappa' \in \overline{\mathcal{W}}$. *For compactly supported* $\alpha \in L^2(\mathbb{R}^2)$ *with* $|y^0| < d$ *on its support,*

$$\mathcal{B}_d^{\mathrm{c}}(\alpha \otimes \lambda) = \int \alpha(y)B_\lambda(dn_0 + y, dn_0 - y)\,dy, \qquad n_0 = (1,0), \tag{18.16}$$

as an operator-norm Bochner integral.

Proof. Let $j_\delta = (\pi\delta^2)^{-1}1_{|y|<\delta}$, set $A_\delta = \mathcal{B}^{\mathrm{c}}_\delta(j_\delta \otimes \lambda)$, and define

$$B^\delta_\lambda(\xi,\xi') = T(\xi - \delta n_0)A_\delta T(\xi' - \delta n_0)$$

when both tube distances exceed δ. The insertion covariance and the invariance of j_δ under θ_1 give the two identities (18.15) for this regularization and, for real y and $r \geq |y^0|$,

$$B^\delta_\lambda((r+\delta)n_0 + y, (r+\delta)n_0 - y) = \mathcal{B}^{\mathrm{c}}_{r+\delta}((\tau_y j_\delta)\otimes\lambda). \tag{18.17}$$

Put $d_0 = \min(d(\xi), d(\xi'))$ and $r = (d_0 - \delta)/2$. The unsandwiched insertion has bound $\|A_\delta\| \leq k_1\sqrt{2}\|\chi\|_2/(\sqrt{\pi}\delta)$. The next estimate uses the two outer semigroups to obtain a bound independent of this point-approximation divergence. For unit vectors ψ, ψ' the function $h(z) = \langle\psi, B^\delta_\lambda(\xi+z, \xi'-z)\psi'\rangle$ is holomorphic where $|\operatorname{Re} z^0| < r$, $|\operatorname{Im} z^1| < r$. Write $z = y + i(t,t')$. The outer factors

$$T(\xi - (r+\delta)n_0 + i(t,t')), \quad T(\xi' - (r+\delta)n_0 - i(t,t'))$$

are contractions for $|t'| < r$, and carry the imaginary spatial translation. The remaining central factor is the right side of (18.17). Thus, for $\eta \in L^2((-r,r)^2)$,

$$\left|\int \eta(y)h(y + i(t,t'))\,dy\right| \leq \sqrt{2}k_1\|\chi\|_2\|\eta * j_\delta\|_2 \leq 2\exp(1/2)\|\chi\|_2\|\eta\|_2.$$

The vectors left after the outer contractions do not depend on y, so this use of the insertion bound is legitimate. The central arguments are $(r+\delta)n_0 + y$ and $(r+\delta)n_0 - y$; adding the outer arguments gives $\xi + y + i(t,t')$ and $\xi' - y - i(t,t')$. Integrating the central operator against $\eta(y)$ gives $\mathcal{B}^{\mathrm{c}}_{r+\delta}((\eta * j_\delta)\otimes\lambda)$. The inserted test has support in $|y^0| < r + \delta$ and norm at most $\sqrt{2}\|\chi\|_2\|\eta * j_\delta\|_2$. Young's inequality uses $\|j_\delta\|_1 = 1$, giving the bound independent of δ. Taking the L^2 dual norm gives an integral of $|h|^2$ bounded by $4\exp(1)\|\chi\|_2^2$ on the real square for each t, t'. The mean-value inequality on a complex polydisc of radius $r' < r$, followed by integration over $|t|, |t'| < r$, gives

$$|h(0)|^2 \leq (\pi r'^2)^{-2}(2r)^2\,4\exp(1)\|\chi\|_2^2.$$

Let $r' \uparrow r$. Consequently

$$\|B^\delta_\lambda(\xi,\xi')\| \leq k_2\|\chi\|_2/(d_0 - \delta). \tag{18.18}$$

To verify the mean-value calculation, apply the scalar disc mean-value inequality in each variable:

$$|h(0)|^2 \leq (\pi r'^2)^{-2}\int_{|z^0|,|z^1|<r'} |h(z)|^2\,d^4z.$$

The polydisc is contained in the product of the real and imaginary squares of side $2r$. For every imaginary slice, the real-square integral is at most $4e\|\chi\|_2^2$. Integrating the imaginary square gives $16er^2\|\chi\|_2^2$. Taking square roots and substituting $2r = d_0 - \delta$ yields $8\sqrt{e}\|\chi\|_2/(\pi(d_0 - \delta))$.

We next identify the weak limit rather than multiply weakly convergent regularizations. Set $r_0 = d_0/4$. On the polydisc $|z^0|, |z^1| < r_0$ the matrix elements $h_\delta(z)$ above are uniformly bounded for $\delta < r_0$. For real $y \in (-r_0, r_0)^2$, integration of (18.17), and then insertion covariance, give

$$\int \eta(y)h_\delta(y)\,dy = \langle\psi, T(\xi - 2r_0n_0)\mathcal{B}^{\mathrm{c}}_{2r_0}((\eta * j_\delta)\otimes\lambda)T(\xi' - 2r_0n_0)\psi'\rangle.$$

This converges because $\eta * j_\delta \to \eta$ in L^2 and the insertion map is bounded. In this formula take $r = 2r_0 - \delta$ in (18.17). For $\delta < r_0$, the support of $\eta * j_\delta$ lies in $|y^0| < 2r_0$, and the two outer arguments remain in the tube. Thus the integrated matrix elements have a fixed norm limit. Every normally convergent subsequence of h_δ therefore has the same integrals against real square tests. Its continuous limit is unique on that square, hence unique on the polydisc by the holomorphic identity theorem. Montel's theorem now gives convergence of all these matrix elements. Riesz representation defines $B_\lambda(\xi,\xi')$; the bound and the identities follow by weak convergence. More explicitly, local boundedness gives a normally convergent subsequence of any sequence of the scalar functions h_δ. Two such limits have the same integrals against every real-square L^2 test by the preceding calculation. Their continuous restrictions agree on that square; the identity theorem

makes them agree on the polydisc. Therefore every sequence $\delta \to 0$ has the same scalar limit. The limits are sesquilinear in ψ, ψ' and bounded by $k_2\|\chi\|_2\|\psi\|\|\psi'\|/d_0$; Riesz representation gives a single bounded operator. Linearity in the profile, the shift rule and the adjoint rule pass through these matrix-element limits. Products of point operators are formed only after this bounded operator has been defined.

For small $\varepsilon > 0$, the shift identity writes this operator, near a fixed pair (ξ_0, ξ_0'), as

$$T(\xi - \xi_0 + \varepsilon n_0)B_\lambda(\xi_0 - \varepsilon n_0, \xi_0' - \varepsilon n_0)T(\xi' - \xi_0' + \varepsilon n_0).$$

The arguments of both semigroups are interior there. This proves operator-norm holomorphy. Finally integrate (18.17) at $r = d-\delta$ against α. Its right side converges in norm by the approximate-identity property. The uniform bound (18.18) on the compact support permits dominated convergence of matrix elements on the left. Norm continuity and (18.14) make the limiting integral a Bochner integral, proving (18.16). The right side of the regularized identity is $\mathcal{B}_d^c((\alpha * j_\delta) \otimes \lambda)$, which converges in norm by the L^2 insertion bound. The left side converges in matrix elements by dominated convergence and the positive gap $\inf_{\operatorname{supp}\alpha}(d - |y^0|) > 0$. The limiting integral exists in norm because its operator-valued integrand is norm continuous and bounded on that compact support, and $\alpha \in L^1$. Equality of all matrix elements identifies this integral with the norm limit of the regularized right side. □

The semigroup can be redistributed between adjacent point insertions. If $\kappa_1 + \kappa_2 = \kappa_1' + \kappa_2'$ and all four lie in $\mathcal{W}$, then

$$B_\lambda(\xi, \kappa_1)B_{\lambda'}(\kappa_2, \xi') = B_\lambda(\xi, \kappa_1')B_{\lambda'}(\kappa_2', \xi'). \tag{18.19}$$

To prove it, subtract the same sufficiently small εn_0 from each inner argument. Both sides become $B_\lambda(\xi, \varepsilon n_0)T(\kappa_1 + \kappa_2 - 2\varepsilon n_0)B_{\lambda'}(\varepsilon n_0, \xi')$ by (18.15).

Fix any $\kappa \in \mathcal{W}$. For $N \geq 2$ and $\xi_1, \dots, \xi_{N-1} \in \mathcal{W}$ define

$$G_N(\xi; \lambda) = \langle \Omega_1, B_{\lambda_1}(\kappa, \xi_1/2)B_{\lambda_2}(\xi_1/2, \xi_2/2) \cdots B_{\lambda_N}(\xi_{N-1}/2, \kappa)\Omega_1 \rangle. \tag{18.20}$$

Define $G_1 = \langle \Omega_1, B_{\lambda_1}(\kappa, \kappa)\Omega_1 \rangle$. For an ordered configuration

$$\mathcal{W}^{(n)} = \{(x_1, \dots, x_n) : x_1 \in \mathcal{W},\ x_{a+1} - x_a \in \mathcal{W}\},$$

define $\Psi(x; \lambda)$ by replacing the first outer argument κ in the corresponding product with x_1, then applying the product to Ω_1 without the left vacuum. The empty vector is Ω_1. All these definitions are independent of their outer κ: the shift identity and $T(\xi)\Omega_1 = T(\xi)^*\Omega_1 = \Omega_1$ reduce any two choices to the same small εn_0. The vacuum identities follow from reconstruction: every real time and transverse translation fixes $[1]_1$. Its joint spectral measure is carried by $(\omega, p) = (0, 0)$, so every complex-tube multiplier also equals one on it. With $\Delta_a = x_{a+1} - x_a$, the vector is

$$\Psi(x; \lambda) = B_{\lambda_1}(x_1, \Delta_1/2)B_{\lambda_2}(\Delta_1/2, \Delta_2/2) \cdots B_{\lambda_n}(\Delta_{n-1}/2, \kappa)\Omega_1 \quad (n \geq 2),$$

and for $n = 1$ it is $B_{\lambda_1}(x_1, \kappa)\Omega_1$.

Proposition 18.6 (Moment kernels and their Gram formula)**.** *The kernels are holomorphic and real-multilinear in the profiles. For $N \geq 2$,*

$$|G_N(\xi; \lambda)| \leq \prod_{a=1}^{N} (2k_2\|\chi_a\|_2)\big(\min_a d(\xi_a)\big)^{-N}. \tag{18.21}$$

If compact planar supports are strictly ordered in y^0, then

$$S_N(\alpha_1 \otimes \lambda_1, \dots, \alpha_N \otimes \lambda_N) = \int \prod_a \alpha_a(y_a) G_N(y_2 - y_1, \dots, y_N - y_{N-1}; \lambda)\, dY. \tag{18.22}$$

If the supports also lie at positive time, their product's class is the Bochner integral of $\prod_a \alpha_a(y_a)\Psi(Y; \lambda)$. In particular $G_1 = 0$.

The vectors Ψ are holomorphic and satisfy $T(\kappa')\Psi(x) = \Psi(x_1+\kappa',\dots,x_n+\kappa')$. Their Gram formula is

$$\langle \Psi(x;\lambda), \Psi(z;\lambda')\rangle = s(\lambda) G_{n+n'}((x_n - x_{n-1})^*, \dots, (x_2-x_1)^*, \\ x_1^* + z_1, z_2 - z_1, \dots, z_{n'} - z_{n'-1}; \lambda_n, \dots, \lambda_1, \lambda'_1, \dots, \lambda'_{n'}). \tag{18.23}$$

When one vector is empty, omit its labels and the central difference; the vacuum-vacuum value is one. Finally,

$$\overline{G_N(\xi;\lambda)} = G_N(\bar\xi;\bar\lambda). \tag{18.24}$$

Proof. Holomorphy follows from that of each bounded operator factor. For the estimate choose $\kappa = tn_0$ with $t \geq \min_a d(\xi_a)/2$; every factor in the product is bounded by its corresponding factor in (18.21).

To recover the moments, enclose each ordered support in a time interval $c_a - d_a < y^0 < c_a + d_a$, with successive intervals disjoint. For positive supports choose $c_1 - d_1 \geq 0$. The insertion identity, starting at the first interval and translating the remaining product to the positive side, gives successively

$$[\prod_a \mathbb{B}(f_a)]_1 = e^{-(c_1-d_1)\mathsf{H}_1} \mathcal{B}^{\mathrm{c}}_{d_1}(\tau_{-c_1} f_1) \prod_{a=2}^{N} \left[e^{-(c_a - d_a - c_{a-1} - d_{a-1})\mathsf{H}_1} \mathcal{B}^{\mathrm{c}}_{d_a}(\tau_{-c_a} f_a)\right] \Omega_1.$$

This formula is an induction using localization; all remaining products belong to $\mathcal{D}_+$. Substitute (18.16) into every factor. All supports are compact and have positive interval gaps, so the operator norms are uniformly bounded there and the products of integrals are Bochner integrable. Absorb the intervening semigroups into the adjacent arguments. Each pair of inner arguments has sum $y_{a+1} - y_a$ and may be split equally by (18.19). The first argument becomes y_1, and the last acts on the vacuum. The result is exactly the claimed integral of Ψ. Pairing with Ω_1 gives (18.22). A common real time translation puts arbitrary ordered supports in the positive half-plane without changing their moments or their successive differences. For $N = 1$ the identity gives $S_1 = G_1 \int \alpha$, so the mean-zero property implies $G_1 = 0$.

Taking adjoints of the factors defining the first vector reverses their order and introduces $s(\lambda)$. The two central arguments then add to $x_1^* + z_1$; redistributing them equally proves (18.23). For real arguments $x^* = -\theta_1 x$, so the ordered points in this combined kernel are $(\theta_1 x_n, \dots, \theta_1 x_1, z_1, \dots, z_{n'})$. Each reflected point keeps its polarization label; this accounts for the weights as well as the signs in the Gram formula. The vector translation formula follows by absorbing $T(\kappa')$ into its first argument. Finally conjugation of the real moment identity, together with (18.7), proves (18.24) at real arguments. Both sides are holomorphic after conjugating the variables on the left; the identity theorem on the convex tube extends it to all arguments. □

18.6. **The two-point function.** The first application of the analytic kernels is particularly simple. At two points there is only one displacement. The four coordinate half-planes therefore cover every nonzero displacement. Closedness of the field strength tensor then excludes the two angular frequencies which the quarter-turn symmetry alone would still permit.

Lemma 18.7 (An elementary Fourier bound)**.** *Suppose h is entire, $h(\zeta + \pi/2) = i^w h(\zeta)$ for an integer w, and $|h(t+i\beta)| \leq Ce^{A|\beta|}$ for all real t, β. Then*

$$h(\zeta) = \sum_{\substack{k \in w + 4\mathbb{Z} \\ |k| \leq A}} a_k e^{ik\zeta}.$$

Moreover $|a_k| \leq C$.

Proof. The function $e^{-iw\zeta}h(\zeta)$ has period $\pi/2$. It consequently defines a holomorphic function of $u = e^{4i\zeta}$ on $\mathbb{C} \setminus \{0\}$: two inverse choices differ by that period. Its Laurent coefficients give a locally normally convergent series with frequencies $k \in w + 4\mathbb{Z}$. Equivalently,

$$a_k = \frac{2}{\pi} \int_0^{\pi/2} h(t + i\beta) e^{-ik(t+i\beta)}\, dt.$$

The integral does not depend on β, by Cauchy's theorem on a rectangle; the vertical sides cancel because the integrand is periodic. Its absolute value is at most $Ce^{A|\beta|+k\beta}$. Letting $\beta \to -\infty$ if $k > A$, or $\beta \to +\infty$ if $k < -A$, proves the claimed vanishing. Taking $\beta = 0$ gives the remaining bound. □

Proposition 18.8 (Two-point covariance)**.** *For every $t \in \mathbb{R}$ and $b, b' \in L^2(\mathbb{R}^3; \mathbb{C}^3)$,*

$$S_2(R_t \cdot b, R_t \cdot b') = S_2(b, b').$$

Proof. Fix labels $\lambda = (\sigma, \chi)$ and $\lambda' = (\sigma', \chi')$, and write $w = w(\sigma) + w(\sigma')$. For $v \in (\pi/2)\mathbb{Z}$ put

$$G^{(v)}(y) = e^{iwv} G_2(R_{-v}y; \lambda, \lambda'), \qquad n_v \cdot y > 0.$$

The quarter-turn invariance of the law and the product-test transformation rule imply that this function is a density of the two-point moment whenever the difference of the two planar supports lies in that half-plane. On an overlap the two densities agree: integrate their difference against products of test functions with sufficiently small supports and then use continuity. Thus they define a smooth function G on $\mathbb{R}^2 \setminus \{0\}$. It is a density on every product of two disjoint compact planar supports. Indeed, subdivide such supports into small squares. For a sufficiently fine subdivision, each pair of squares is strictly ordered in one of the two coordinate directions; apply the density identity there and sum. Boundaries have measure zero, and the kernels are bounded on the compact set of occurring differences.

For $r > 0$, the functions

$$h^{(v)}(\zeta) = e^{iwv} G_2(r n_{\zeta - v}; \lambda, \lambda'), \qquad |\operatorname{Re}\zeta - v| < \pi/2,$$

are holomorphic. Here $n_\zeta = R_\zeta n_0$. On real overlaps they agree with $G(rn_t)$, and hence agree on the whole overlap by the identity theorem. Their strips cover $\mathbb{C}$, so they give one entire function h_r. Choose v within $\pi/4$ of t. Since

$$d(r n_{t+i\beta-v}) = r\cos(t - v)e^{-|\beta|},$$

the kernel estimate gives

$$|h_r(t + i\beta)| \le 2(2k_2)^2 \|\chi\|_2 \|\chi'\|_2 r^{-2} e^{2|\beta|}.$$

The quarter-turn rule gives $h_r(\zeta + \pi/2) = i^w h_r(\zeta)$. Lemma 18.7 therefore yields

$$G(rn_t) = \sum_{\substack{k \in w + 4\mathbb{Z} \\ |k| \le 2}} a^k_{\lambda\lambda'}(r) e^{ikt}, \qquad |a^k_{\lambda\lambda'}(r)| \le C\|\chi\|_2 \|\chi'\|_2 r^{-2}. \tag{18.25}$$

The coefficients are smooth in r, since they are angular integrals of the smooth function G.

Except for the pairs $(+,+)$ and $(-,-)$, the only permitted frequency is $k = w$. Those two pairs could also have, respectively, $k = -2$ and $k = 2$. To exclude them, take $\chi_0 \in C_c^\infty(\mathbb{R}; \mathbb{R})$. The gradient identity, integration by parts in the first planar variable, and the disjoint-support density formula imply

$$\partial_z G_{(+,\chi_0),\lambda'} + \partial_{\bar z} G_{(-,\chi_0),\lambda'} + G_{(3,\chi_0'),\lambda'} = 0 \quad \text{on } \mathbb{R}^2 \setminus \{0\}. \tag{18.26}$$

For clarity, the sign of the first two derivatives is positive: integration by parts contributes a minus sign, whereas differentiation of $G(y' - y)$ in y contributes a second minus sign. Products of small

test supports detect every point of the displayed punctured plane, so the distributional identity holds there and is a smooth identity. In polar coordinates,

$$\partial_z(a(r)e^{ikt}) = \tfrac{1}{2}e^{i(k-1)t}(a'(r) + ka(r)/r), \qquad \partial_{\bar z}(a(r)e^{ikt}) = \tfrac{1}{2}e^{i(k+1)t}(a'(r) - ka(r)/r).$$

With $\sigma' = +$, frequency -3 in (18.26) can come only from the $k = -2$ term of $G_{(+,\chi_0),(+,\chi')}$. Its coefficient satisfies $a' = 2a/r$, so $a(r) = cr^2$. The bound in (18.25), valid for every $r > 0$, forces $c = 0$. With $\sigma' = -$, frequency 3 gives the same equation for the $k = 2$ term of $G_{(-,\chi_0),(-,\chi')}$. This term also vanishes. Density and the L^2 moment bound extend these conclusions from χ_0 to every real L^2 profile. Hence

$$G(R_t y; \lambda, \lambda') = e^{iwt}G(y; \lambda, \lambda').$$

Together with the product-test transformation rule this proves the asserted moment identity for disjoint compact planar supports. Finite sums of product tests are dense in the vector-valued L^2 spaces with those supports, so it holds there for arbitrary vector-valued fields.

Finally the difference $D(b, b') = S_2(R_t \cdot b, R_t \cdot b') - S_2(b, b')$ is a bounded bilinear form, vanishes on disjoint compact planar supports, and vanishes if either entry is a gradient. Rotations preserve gradients, by (18.2). The no-contact Proposition 18.2 proves $D = 0$ on all L^2 fields, including fields whose supports meet. □

18.7. **Angular frequencies of higher moments.** At higher orders a real configuration need not be strictly ordered in either coordinate direction. There are two distinct steps to address this. We first glue the kernels wherever a coordinate ordering exists. We then use their angular expansion to continue each coefficient to every configuration of distinct planar points. This continuation concerns the densities; it does not yet assert rotation invariance of the moments.

For a permutation π and a frame $v \in (\pi/2)\mathbb{Z}$ define

$$\mathcal{Z}(\pi, v) = \{Z = (z_1, \dots, z_N) : z_{\pi(a+1)} - z_{\pi(a)} \in R_v\mathcal{W},\ 1 \le a < N\},$$

and, on this tube, define

$$G_v^\pi(Z; \lambda) = e^{iw(\lambda)v}G_N((R_{-v}(z_{\pi(a+1)} - z_{\pi(a)}))_{a<N}; \pi\lambda). \tag{18.27}$$

Here $\pi Z = (z_{\pi(1)}, \dots, z_{\pi(N)})$, and the same convention applies to labels. Write $\mathcal{Z}_N$ for the union of these tubes and $\mathcal{O}_N$ for its real points.

Lemma 18.9 (Gluing the ordered kernels)**.** *The functions* (18.27) *agree on overlaps and define a holomorphic function $\mathcal{G}_N$ on $\mathcal{Z}_N$. For compact planar supports whose product lies in $\mathcal{Z}(\pi, v)$,*

$$S_N(\alpha_1 \otimes \lambda_1, \dots, \alpha_N \otimes \lambda_N) = \int \prod_a \alpha_a(y_a)\, \mathcal{G}_N(Y; \lambda)\, dY. \tag{18.28}$$

The tube union is invariant under permutations, complex common translations, conjugation, $R_{\pi/2}$, and every Λ_β, $\beta \in \mathbb{R}$. On it,

$$\begin{aligned} \mathcal{G}_N(\pi Z; \pi\lambda) &= \mathcal{G}_N(Z; \lambda), & \mathcal{G}_N(Z + (u, \dots, u); \lambda) &= \mathcal{G}_N(Z; \lambda), \\ \mathcal{G}_N(R_{\pi/2}Z; \lambda) &= i^{w(\lambda)}\mathcal{G}_N(Z; \lambda), & \overline{\mathcal{G}_N(Z; \lambda)} &= \mathcal{G}_N(\bar Z; \bar\lambda). \end{aligned} \tag{18.29}$$

Moreover $Y \in \mathcal{O}_N$ exactly when either all its first coordinates or all its second coordinates are distinct.

Proof. Each tube is open and convex, is invariant under real part and conjugation, and has the asserted translation and Λ_β invariances, because these statements hold for its successive differences in $R_v\mathcal{W}$. Its real points satisfy $n_v \cdot y_{\pi(1)} < \cdots < n_v \cdot y_{\pi(N)}$. For real arguments the density formula follows from the ordered kernel formula by permuting the entries and rotating by the quarter turn R_{-v}. Symmetry of the moment form permits the permutation; the product-test rule supplies the factor e^{iwv} in (18.27).

An intersection of two tubes is convex. If nonempty, its real points form a nonempty real open set, since taking real part stays in the intersection. The two functions are densities of the same moments on every small product of real open sets in that intersection. They are continuous, so they agree at every real point there. The holomorphic identity theorem then proves equality on the entire intersection. One version of that theorem used here follows by applying the one-variable identity theorem successively on a small polydisc whose real part lies in the zero set; the resulting open complex zero set extends through the connected domain. This proves the gluing and (18.28).

The first two symmetries follow by relabelling the tubes and by their dependence on differences. The quarter turn sends a tube with frame v to the one with frame $v+\pi/2$, giving its stated phase directly. The last identity follows from the reality identity for G_N and $w(\bar\lambda)=-w(\lambda)$. The characterization of real points follows because the four normals are the two coordinate vectors and their negatives. No convexity of the entire tube union is needed here. Convexity is used for each intersection. Its nonempty real product of balls comes from taking the real part of a point in the intersection and using the strict defining inequalities. The two ordered-density formulas identify their kernels on this product by testing with products of smooth functions. Continuity gives equality at each real configuration there, and the identity theorem then gives equality throughout the connected complex intersection. □

Lemma 18.10 (Finite angular expansion)**.** *For a tuple λ of $N\ge 2$ labels, put $w=w(\lambda)$. There are unique holomorphic functions $c_m(\cdot;\lambda)$ on $\mathcal{Z}_N$, zero unless $|w+4m|\le N$, such that*

$$\mathcal{G}_N(\Lambda_\beta Z;\lambda)=\sum_m c_m(Z;\lambda)e^{-(w+4m)\beta},\qquad \beta\in\mathbb{R}. \tag{18.30}$$

They have the permutation and translation symmetries of $\mathcal{G}_N$, and satisfy

$$\begin{aligned} c_m(\Lambda_\beta Z;\lambda)&=e^{-(w+4m)\beta}c_m(Z;\lambda),\\ c_m(R_{\pi/2}Z;\lambda)&=i^w c_m(Z;\lambda), \qquad \sum_m c_m(Z;\lambda)=\mathcal{G}_N(Z;\lambda),\\ \overline{c_m(Y;\lambda)}&=c_{-m}(Y;\bar\lambda)\quad (Y\in\mathcal{O}_N). \end{aligned} \tag{18.31}$$

Proof. First consider real configurations whose consecutive differences have directions $\vartheta_a\in(-\vartheta_0,\vartheta_0)$ for a fixed $0<\vartheta_0<\pi/4$. Write their lengths as $r_a>0$ and $r_*=\min_a r_a$. For $\zeta=t+i\beta$ and a frame v,

$$\mathrm{Re}(R_{\zeta-v}(y_{a+1}-y_a))^{\pm}=r_a e^{\mp\beta}\cos(\vartheta_a+t-v).$$

The strips $|t-v|<\pi/2-\vartheta_0$ cover $\mathbb{C}$ and show that $h_Y(\zeta)=\mathcal{G}_N(R_\zeta Y;\lambda)$ is entire. Choosing $|t-v|\le\pi/4$, the kernel bound gives

$$|h_Y(t+i\beta)|\le\prod_a(2k_2\|\chi_a\|_2)[r_*\cos(\vartheta_0+\pi/4)]^{-N}e^{N|\beta|}.$$

It has quasi-period $\pi/2$ and weight w. The elementary Fourier lemma therefore gives the asserted finite expansion for these configurations, evaluated at $\zeta=i\beta$.

To extend the conclusion, let $\mathcal{M}=\{m:|w+4m|\le N\}$ and $P(u)=\prod_{m\in\mathcal{M}}(u+w+4m)$. On a fixed tube $\mathcal{Z}(\pi,v)$ the function $(\beta,Z)\mapsto G^\pi_v(\Lambda_\beta Z;\lambda)$ is holomorphic on a complex neighbourhood of $\mathbb{R}\times\mathcal{Z}(\pi,v)$. For each real β, its derivatives in β are holomorphic in Z. The configurations which, after permutation and rotation by R_{-v}, have the directions just considered form a nonempty real open subset of the tube. On that subset the function is annihilated by $P(\partial_\beta)$. The identity theorem in Z proves this differential equation throughout the tube, for every real β.

The roots of P are distinct. The solutions of this constant coefficient ordinary differential equation are linear combinations of $e^{-(w+4m)\beta}$, $m\in\mathcal{M}$. For completeness, factor the differential operator into its distinct first-order factors: successive solution of $(\partial_\beta-r)u=f$ gives this statement by induction. The coefficients are fixed linear combinations of the first $|\mathcal{M}|$ derivatives at $\beta=0$, since

the matrix $(r_m^l)_{l,m}$ is a Vandermonde matrix with distinct entries. They are thus holomorphic in Z. An explicit formula is

$$c_m(Z;\lambda) = \prod_{l\in\mathcal{M}\setminus\{m\}} \frac{\partial_\beta + w + 4l}{4(l-m)} \mathcal{G}_N(\Lambda_\beta Z;\lambda)\Big|_{\beta=0} . \tag{18.32}$$

On the exponential with index m every factor equals one; on any other allowed exponential one factor vanishes. Thus the formula gives the unique coefficient using finitely many derivatives of a holomorphic function. The differential equation extends from the narrow configurations because, for every fixed real β, its left side is holomorphic in Z and vanishes on a nonempty real open subset of the connected tube. The identity theorem supplies all the other configurations. On overlapping tubes uniqueness of these exponential coefficients identifies them, because every $\Lambda_\beta Z$ remains in the overlap. This gives the global functions c_m.

All the listed symmetries follow by comparing coefficients in (18.30). In particular $\overline{\Lambda_\beta Y} = \Lambda_{-\beta} Y$ for real Y; conjugation changes w to $-w$, so the coefficient at index m becomes the one at index $-m$. □

We measure the possible failure of angular covariance by the nonnegative integer

$$A(0) = A(1) = 0, \qquad A(N) = \max\{|m| : c_m(\cdot;\lambda) \not\equiv 0 \text{ for some } N \text{ labels}\} \quad (N \geq 2), \tag{18.33}$$

where the maximum is zero if all coefficients vanish. Since $|w| \leq N$ and $|w + 4m| \leq N$, we have $A(N) \leq N/2$. If a coefficient is nonzero somewhere, it is nonzero at some strictly time-ordered real configuration after a permutation of the labels: first rotate its tube by a quarter turn to frame zero, then permute the entries, and use the identity theorem. Conjugation in (18.31) shows that both indices $A(N)$ and $-A(N)$ occur whenever $A(N) > 0$.

Lemma 18.11 (Continuation to distinct configurations)**.** *Each c_m extends uniquely to a real analytic function $\widetilde{c}_m$ on the configurations of distinct planar points. The extension has the permutation and common translation symmetries, and for every real t,*

$$\widetilde{c}_m(R_t Y;\lambda) = e^{i(w+4m)t} \widetilde{c}_m(Y;\lambda). \tag{18.34}$$

Consequently $\widetilde{G}_N = \sum_m \widetilde{c}_m$ is real analytic there and agrees with $\mathcal{G}_N$ on $\mathcal{O}_N$.

Proof. For a real point Y of one tube, the angles near zero which keep $R_t Y$ in that tube form an interval. Indeed each successive difference, written after rotation into that frame as $r_a n_{\vartheta_a}$ with $|\vartheta_a| < \pi/2$, imposes $-\pi/2 - \vartheta_a < t < \pi/2 - \vartheta_a$ near zero. On the resulting strip in $t + i\beta$, analytic continuation of (18.30) gives

$$\mathcal{G}_N(R_\zeta Y;\lambda) = \sum_m c_m(Y;\lambda) e^{i(w+4m)\zeta}.$$

Comparison of coefficients at $\zeta = t + i\beta$ gives the phase rule locally whenever the rotation stays in that tube.

Consider configurations of distinct points no two different chords of which are perpendicular. They form an open dense set: the exceptional equations are the zero sets of finitely many nonzero polynomials. A nonzero real polynomial cannot vanish on a real open set, by induction in its variables using the one-variable statement. This set is invariant under rotation and is contained in $\mathcal{O}_N$. In fact failure of both coordinate orderings would give a vertical chord and a horizontal chord; they are perpendicular, or, if the two chords have the same endpoints, those endpoints coincide. On this generic set the local phase rule continues along every real angle: the phase-corrected coefficient is locally constant along the orbit. Approximating a configuration Y for which both Y and $R_t Y$ belong to $\mathcal{O}_N$ by generic ones proves the same phase identity there by continuity.

For any distinct configuration Y, all but finitely many angles modulo π make its first coordinates distinct. Choose such an angle t and set

$$\widetilde{c}_m(Y;\lambda) = e^{-i(w+4m)t} c_m(R_t Y;\lambda).$$

The preceding phase identity proves independence of the choice. The same t works near Y, so this is a real analytic local formula. It also proves (18.34) and all the stated symmetries. Uniqueness follows locally from that phase rule and agreement on $\mathcal{O}_N$.

We shall use the connectedness of the distinct-configuration domain. Its complement is a finite union of affine subspaces of codimension two. The complement of any such finite union is path connected: given two points outside it, choose a third point such that both joining segments avoid all those subspaces. The forbidden third points lie in finitely many sets of measure zero, since for each codimension two subspace they are swept out by rays through that subspace, a set of dimension at most one less than the ambient dimension. This argument also applies when finitely many point positions are fixed below. □

18.8. **Reflection positivity excludes the remaining frequencies.** The angular index $A(N)$ is bounded linearly in N. Positivity provides two additional constraints. Splitting a configuration across a reflection plane makes these indices convex. Taking the leading coefficient of a positive Gram kernel gives a positive kernel of its own. The latter allows a vanishing identity on a regular polygon to be propagated to configurations with fewer prescribed vertices.

Lemma 18.12 (Convexity of the angular index)**.** *For integers $a, b \geq 0$,*

$$2A(a+b) \leq A(2a) + A(2b). \tag{18.35}$$

Proof. Put $N = a + b$ and $M = A(N)$. If $N \leq 1$ or $M = 0$, the assertion is immediate. Otherwise choose a strictly time-ordered real configuration Y and labels λ with $c_{-M}(Y;\lambda) \neq 0$. Translate so that the first a points have negative time and the last b positive time. If a group is empty, put all points on the appropriate side. Reflect the first group and reverse its order, obtaining $x_k = \theta_1 y_{a+1-k}$ with labels $\lambda^x_k = \lambda_{a+1-k}$. Put $z_l = y_{a+l}$ with labels $\lambda^z_l = \lambda_{a+l}$. Both are positively time-ordered; write their weights as w_x, w_z, so $w = w_x + w_z$.

The Gram formula and $(\Lambda_\beta u)^* = \Lambda_\beta(u^*)$ give

$$\langle \Psi(\Lambda_\beta x;\lambda^x), \Psi(\Lambda_\beta z;\lambda^z)\rangle = s(\lambda^x)\mathcal{G}_N(\Lambda_\beta Y;\lambda). \tag{18.36}$$

For an empty group its vector is the vacuum. The two squared norms are, up to their signs, the kernels on the reflected doubled configurations

$$X' = (\theta_1 x_a, \dots, \theta_1 x_1, x_1, \dots, x_a), \qquad X'' = (\theta_1 z_b, \dots, \theta_1 z_1, z_1, \dots, z_b),$$

with the same label on each original point and its reflected copy. They have weights $2w_x, 2w_z$. This is exactly where the antilinear reflection rule matters: Θ_1 preserves the labels, whereas reflection without complex conjugation would exchange $+$ and $-$.

Apply Cauchy–Schwarz to (18.36), expand the three kernels, and multiply by $e^{2w\beta}$. The result is

$$\left|\sum_m c_m(Y;\lambda)e^{-4m\beta}\right|^2 \leq \left|\sum_{|m|\leq A(2a)} c_m(X';\lambda')e^{-4m\beta}\right| \left|\sum_{|m|\leq A(2b)} c_m(X'';\lambda'')e^{-4m\beta}\right|.$$

An empty factor equals one. For large positive β the left side grows as $|c_{-M}(Y;\lambda)|^2 e^{8M\beta}$; the right side is at most a constant times $e^{4(A(2a)+A(2b))\beta}$. Comparison of these rates proves (18.35). □

Lemma 18.13 (A positive kernel from the leading coefficient)**.** *Let $a \geq 1$, $M = A(2a)$, and $c_* = \widetilde{c}^{(2a)}_{-M}$. On labelled configurations of a distinct points in the positive half-plane, the kernel*

$$K_a((x;\lambda),(z;\lambda')) = s(\lambda)c_*(\theta_1 x, z;\lambda,\lambda')$$

is Hermitian and positive semidefinite. For any line L and any two such configurations on the same open side of L,

$$|c_*(\theta_L x, z; \lambda, \lambda')|^2 \le |c_*(\theta_L x, x; \lambda, \lambda)||c_*(\theta_L z, z; \lambda', \lambda')|. \tag{18.37}$$

Here reflection moves the points and keeps their labels.

Proof. First assume that the times in each configuration are distinct and put them in increasing order. For every real β,

$$\langle \Psi(\Lambda_\beta x; \lambda), \Psi(\Lambda_\beta z; \lambda')\rangle = s(\lambda) \sum_{|m|\le M} \widetilde{c}_m(\theta_1 x, z; \lambda, \lambda') e^{-(w(\lambda)+w(\lambda')+4m)\beta}.$$

For any finite family of labelled configurations the left side is a positive semidefinite Gram matrix. Multiplying its row and column at $(x;\lambda)$ by the positive number $e^{w(\lambda)\beta}$, and the whole matrix by $e^{-4M\beta}$, preserves positivity. If C_β is the Gram matrix and D_β has diagonal entries $e^{w(\lambda)\beta}$, the rescaled matrix is $e^{-4M\beta} D_\beta C_\beta D_\beta$. Its entry at $(x;\lambda), (z;\lambda')$ is

$$s(\lambda) \sum_{m\ge -M} \widetilde{c}_m(\theta_1 x, z; \lambda, \lambda') e^{-4(M+m)\beta}.$$

The finite sum tends to K_a because every term with $m > -M$ vanishes. For a finite column vector v, its limiting quadratic form is the limit of the nonnegative quadratic forms of these rescaled Gram matrices. Thus K_a is positive semidefinite. Configurations with distinct times are dense in the configurations of distinct points of the open half-plane. Continuity of c_* therefore extends this conclusion to time ties as well. Positive semidefinite matrices are Hermitian; their two-by-two principal minors give the usual kernel Cauchy–Schwarz inequality.

For a general line, choose $v \in L$ and an angle u such that n_u points into its chosen side. Set $x' = R_{-u}(x - v)$ and $z' = R_{-u}(z - v)$. Then

$$(\theta_L x, z) = (v, \dots, v) + R_u(\theta_1 x', z'),$$

and the exact phase identity gives

$$c_*(\theta_L x, z; \lambda, \lambda') = e^{i(w(\lambda)+w(\lambda')-4M)u} s(\lambda) K_a((x';\lambda), (z';\lambda')).$$

The prefactor has modulus one. Applying kernel Cauchy–Schwarz proves (18.37). This uses angular continuation of the coefficient, rather than reflection positivity of the original law at L. □

Because the coefficients are symmetric, it is convenient to evaluate c_* at a finite labelled set. Its points are distinct, and each point carries one label. Say that such a set S has property $\mathcal{N}(S)$ if c_* vanishes on every labelled configuration of $2a$ distinct points which contains S.

Lemma 18.14 (Removing prescribed mirror pairs)**.** *Suppose S' consists of $j \le a$ distinct labelled points on one open side of a line L. If $S \subset \theta_L S' \cup S'$ and $\mathcal{N}(S)$ holds, then both $\mathcal{N}(S')$ and $\mathcal{N}(\theta_L S')$ hold.*

Proof. Complete S' to a labelled set x of a points on its side of L, and let z be any a labelled points on the same side. The diagonal set $\theta_L x \cup x$ contains S, so c_* vanishes there. The leading-coefficient inequality implies that it also vanishes on $\theta_L x \cup z$.

Fix arbitrary labels for the $a - j$ additional points of x and the a points of z. Let y be the reflections of those additional points. The assertion just proved says that

$$c_*(\theta_L S' \cup y \cup z) = 0$$

on the open set where the y points lie on the opposite side and the z points on the chosen side, with all points distinct. The domain allowing arbitrary positions of y, z, with collisions among them and with $\theta_L S'$ excluded, is the complement of finitely many affine subspaces of codimension two. It is open and connected by the argument in the preceding lemma. The displayed function is real analytic on that domain. A real analytic function zero on a nonempty open subset of a connected domain is

zero throughout: its local power series propagates the zero set along overlapping balls on any path. Thus it vanishes for all allowed positions. Every completion of $\theta_L S'$ is of this form, and the labels were arbitrary, proving $\mathcal{N}(\theta_L S')$. For the other conclusion, run the same argument on the opposite side with $\theta_L S'$ in the role of S'. Its reflected union is still $\theta_L S' \cup S'$, so the original hypothesis $S \subset \theta_L S' \cup S'$ and $\mathcal{N}(S)$ is unchanged. The conclusion of that second argument is $\mathcal{N}(S')$. □

Proposition 18.15 (The polygon divisibility constraint)**.** *For every integer $a \geq 1$, a divides $2A(2a)$.*

Proof. For $a = 1$ divisibility is automatic, so assume $a \geq 2$. If a rotation through t about any point permutes a labelled configuration and preserves its labels, symmetry and the exact phase imply

$$(1 - e^{i(w+4m)t})\widetilde{c}_m(Y;\lambda) = 0.$$

In particular, for the vertices of a regular $2a$-gon carrying one common label, $t = \pi/a$ and $w = 2aw(\sigma)$. The phase is $e^{4\pi i m/a}$, so this coefficient vanishes unless a divides $2m$.

Suppose a does not divide $2M$, where $M = A(2a)$. Then $M > 0$. Fix one label, and call j consecutive vertices of a regular $2a$-gon, carrying that label, an arc of length j. The preceding selection rule shows $\mathcal{N}(S)$ for every arc of length $2a$. We claim that the same assertion for all arcs of length $j \geq 2$ implies it for all arcs of length $j' = \lceil j/2 \rceil$.

Let S' be the vertices $P_{k_0+1}, \dots, P_{k_0+j'}$ of such an arc, in cyclic order, and let L be the perpendicular bisector of the edge joining P_{k_0} and P_{k_0+1}. It passes through the centre and contains no vertex. Reflection maps P_{k_0+k} to P_{k_0+1-k}. Since $j' \leq a$, all of S' lies on one open side of L. Its reflected union is an arc of length $2j' \geq j$, and therefore contains an arc S of length j. The induction hypothesis gives $\mathcal{N}(S)$, and the mirror-pair lemma gives $\mathcal{N}(S')$.

Repeated replacement by $\lceil j/2 \rceil$ reaches $j = 1$. Every planar point is a vertex of a regular $2a$-gon, so c_* vanishes on every configuration containing a point with the fixed label. The label was arbitrary. Thus c_* vanishes for all labels and all distinct configurations, contradicting occurrence of the index $-M$. This contradiction proves the divisibility. □

Lemma 18.16 (A convex integer sequence)**.** *Let $\nu(a)$ be nonnegative integers, $\nu(0) = 0$, with $2\nu(a) \leq \nu(a-1) + \nu(a+1)$, $\nu(a) \leq Ca$, and $a \mid 2\nu(a)$ for every $a \geq 1$. Then $\nu(a) = a\nu(1)$.*

Proof. The integer increments $d_a = \nu(a) - \nu(a-1)$ are nondecreasing. They are bounded above by C, since an increment larger than C would force the eventual slope above C. They are consequently eventually constant, with value d. Put $c = \sum_{a<a_0}(d - d_a) \geq 0$, where $d_a = d$ for $a \geq a_0$. Then $\nu(a) = da - c$ for $a \geq a_0$. The divisibility says that every sufficiently large a divides $2c$, forcing $c = 0$. Every increment is therefore d, including the first, which is $\nu(1)$. □

Proposition 18.17 (Only the tensor weight remains)**.** *For every N, $A(N) = 0$. Thus, for distinct real configurations,*

$$\widetilde{G}_N(R_t Y;\lambda) = e^{iw(\lambda)t}\widetilde{G}_N(Y;\lambda). \tag{18.38}$$

Proof. The sequence $\nu(a) = A(2a)$ satisfies the integer-sequence lemma. Its midpoint convexity follows from (18.35) with the pair $a-1, a+1$, its bound is $\nu(a) \leq a$, and its divisibility was just proved. Hence $A(2a) = aA(2)$. Splitting an arbitrary N into $a = \lfloor N/2 \rfloor$ and $b = N - a$ gives $A(N) \leq NA(2)/2$.

The two-point proof already showed that its density has only frequency w. On a strictly time-ordered real open set it is the kernel $\mathcal{G}_2$. Comparing its angular expansion on a small interval with that identity, and using linear independence of the finitely many exponentials, gives $c_m = 0$ there for $m \neq 0$. The identity theorem on every ordered tube, together with permutation and quarter-turn symmetry, proves $A(2) = 0$. Consequently all $A(N)$ vanish. The exact phase identity for $\widetilde{c}_0$ now gives (18.38). □

18.9. **From densities to all moments.** The angular argument has established covariance of smooth densities away from coincident planar points. Two extensions remain. First we must show that the continued density represents the moments even where neither coordinate ordering is strict. Second we must allow the supports of the fields to meet. The spectral cone gives the first extension; closedness and L^2 continuity give the second.

Lemma 18.18 (Moving one separated cluster)**.** *Suppose Φ_- and Φ_+ are polynomials in fields with arguments supported, respectively, in $x_1 < c - \delta$ and $x_1 > c + \delta$, where $\delta > 0$. The function $s \mapsto \mu(\Phi_- U(se_2)\Phi_+)$ has a holomorphic extension to $|\operatorname{Im} s| < 2\delta$.*

Proof. Common translation reduces to $c = 0$. The polynomials $U(-\delta e_1)\Theta_1\Phi_-$ and $U(-\delta e_1)\Phi_+$ are positive-side L^2 observables; the moment bounds guarantee this integrability. Let their classes be ψ_-, ψ_+. Reconstruction and commutation of the translations give, for real s,

$$\mu(\Phi_- U(se_2)\Phi_+) = \langle \psi_-, e^{-2\delta \mathsf{H}_1} V_1(se_2)\psi_+\rangle = \langle \psi_-, T(2\delta, s)\psi_+\rangle.$$

The last expression is holomorphic for $2\delta - |\operatorname{Im} s| > 0$ by the complex semigroup construction. □

Proposition 18.19 (The density at every distinct configuration)**.** *For compact, pairwise disjoint planar supports of $\alpha_a \in L^2$,*

$$S_N(\alpha_1 \otimes \lambda_1, \dots, \alpha_N \otimes \lambda_N) = \int \prod_a \alpha_a(y_a)\widetilde{G}_N(Y;\lambda)\, dY. \tag{18.39}$$

Proof. The product of the supports is a compact subset of the distinct-point domain. Its density is bounded there, and each α_a is integrable, so the right side is absolutely convergent. Denote the difference between the two sides by $D(\alpha_1, \dots, \alpha_N)$. Say that D vanishes near a distinct configuration Y if it vanishes whenever the individual supports lie in sufficiently small discs about its points. The ordered density formula proves this for $Y \in \mathcal{O}_N$.

Induct on the number $k(Y)$ of pairs with equal second coordinate. When $k(Y) = 0$ the configuration belongs to $\mathcal{O}_N$. Otherwise choose a pair with equal second coordinate and different first coordinates. Cut between their first coordinates at a level different from every first coordinate. This splits the points into nonempty groups I, J, with every point of J later than every point of I. Let the time gap be $\delta > 0$. Move the group J by se_2 and denote the resulting configuration by $Y(s)$. Its points remain distinct for all real s: the groups have a time gap, and distances within each group do not change. Choose $s_0 > 0$ smaller than the absolute value of every nonzero difference of second coordinates in the original configuration; if there is no such difference, any sufficiently small positive s_0 will do. Then all cross-group coincidences of second coordinates disappear, while no new ones arise. In particular $k(Y(s_0)) < k(Y)$.

By induction D vanishes for tests in discs of radius $2r$ about $Y(s_0)$; decrease r so that $4r < \delta$ and $4r < \min_{a \neq b} |y_a - y_b|$. Fix tests supported in discs of radius r about Y, and translate those with index in J by se_2. The supports stay pairwise disjoint for all real s. The resulting difference $F(s)$ is real analytic on $\mathbb{R}$. Indeed its moment term is a correlation of two clusters separated by a time gap of at least $\delta - 2r$, so the preceding lemma gives a holomorphic extension to $|\operatorname{Im} s| < \delta - 2r$.

For its integral term change variables back to the fixed supports. It becomes

$$\int \prod_a \alpha_a(y_a)\widetilde{G}_N(Y + sv_J;\lambda)\, dY,$$

where $(v_J)_a = e_2$ for $a \in J$ and zero otherwise. On a compact real interval in s, the arguments of this integral range over a compact subset of the distinct-point domain. Real analyticity of $\widetilde{G}_N$ supplies local holomorphic extensions near this compact set. A finite covering by smaller polydiscs gives a uniform radius $\varepsilon > 0$ and bound C for extensions at every point of it. Cauchy's inequalities bound the lth derivative of the integral by $Cl!\varepsilon^{-l}\prod_a \|\alpha_a\|_1$. Differentiation under the integral is justified by those bounds. Taylor's formula then proves real analyticity of this term on the interval.

Every real s lies in such an interval. To check that the strip for the moment term has this width, place the separating time plane halfway between the two support groups. Their distance is at least $\delta - 2r$. The cluster lemma applies with any half-gap smaller than $(\delta - 2r)/2$; its compatible extensions give the open strip $|\operatorname{Im} s| < \delta - 2r$. For the integral term, local complex extensions are unique on overlaps adjoining the real set by the identity theorem. At a fixed real s_*, compactness of the fixed support product gives a common positive radius on which $\widetilde{G}_N(Y + (s_* + z)v_J; \lambda)$ is holomorphic in z and uniformly bounded. Since $\prod_a |\alpha_a|$ is integrable, Cauchy's formula justifies differentiation of its integral and the claimed real analyticity. This verifies both analytic terms before applying the identity theorem to F.

For $|s - s_0| < r$, the translated supports lie in the discs where the induction hypothesis applies, so $F(s) = 0$. The real analytic identity theorem yields $F(0) = 0$. This proves local vanishing near every distinct configuration.

Finally cover the compact product of the original supports by finitely many products of smaller discs on which that local vanishing holds. Subdivide each support into a sufficiently fine square grid. For each nonzero tuple of square pieces, choose one point from each piece in the original supports. The resulting configuration belongs to one of those products of smaller discs. Choose the mesh so that the closure of each square still lies in the corresponding full disc. Local vanishing applies to that tuple. Multilinearity sums the finitely many terms to zero, proving (18.39). □

Theorem 18.20 (Rotations in a coordinate plane)**.** *Let a law satisfy Gaussian domination, $dF = 0$, translation invariance, invariance and reflection positivity at $x_j = 0$, invariance under the quarter turn in the (j, i) plane, and the whole-Hilbert-space spectral bound $\mathsf{H}_j \geq |\mathsf{P}_i|$, where $i \neq j$. Then for every $N \geq 1$, real t, and all L^2 vector-valued fields,*

$$S_N(R_t^{ji} \cdot b_1, \dots, R_t^{ji} \cdot b_N) = S_N(b_1, \dots, b_N). \tag{18.40}$$

The law itself is invariant under these rotations.

Proof. Consider first $(j, i) = (1, 2)$. Fix $N \geq 2$ and t, and form the difference of the two sides of (18.40). It is a bounded N-linear form D on L^2, since rotations are L^2 isometries and the moment bound is multilinear. For product tests with disjoint compact planar supports, the full density formula, change of variables $Y \mapsto R_t Y$, and its tensor weight give

$$\begin{aligned} S_N(R_t \cdot (\alpha_1 \otimes \lambda_1), \dots, R_t \cdot (\alpha_N \otimes \lambda_N)) &= e^{-iwt} \int \prod_a \alpha_a(y_a) \widetilde{G}_N(R_t Y; \lambda)\, dY \\ &= S_N(\alpha_1 \otimes \lambda_1, \dots, \alpha_N \otimes \lambda_N). \end{aligned}$$

Finite sums of product tests are dense in the L^2 fields supported over each compact planar set, so D vanishes for all fields over pairwise disjoint such sets. If an entry is a compactly supported smooth gradient, both its original and rotated fields vanish almost surely. Thus D also vanishes on gradients. The no-contact proposition now proves $D = 0$ on all L^2 fields. For $N = 1$ the same assertion follows from the mean-zero property. Gaussian domination makes the exponential moment series absolutely convergent, so invariance of all moments gives equality of characteristic functions, and hence of the laws. This is the determinacy statement in Lemma 15.2.

For another ordered pair, choose an orthogonal coordinate permutation g taking e_1, e_2 to e_j, e_i and apply the preceding argument to $(g^{-1})_* \mu$. Gaussian domination and closedness are preserved because the test action is orthogonal and commutes with the exterior derivative. The reflection form is transported to the one at $x_1 = 0$. Its reconstructed class map $[\Phi]_1 \mapsto [\widehat{g}\Phi]_j$ is unitary, since the two reflection inner products are equal by change of variables. It intertwines time semigroups and transverse translations, and hence their joint spectral measures. The assumed spectral bound is therefore transported to $\mathsf{H}_1 \geq |\mathsf{P}_2|$. All other hypotheses transform in the same way, proving the assertion for (j, i). □

Theorem 18.21 (Euclidean invariance from the nine reflections)**.** *Let μ satisfy Gaussian domination, $dF = 0$, invariance under $\mathbb{R}^3 \rtimes O_{\mathrm{cub}}$, and reflection positivity at every translated coordinate or diagonal plane*

$$\begin{aligned} x_j &= c \quad (1 \le j \le 3), \\ x_j - x_i &= c, \qquad x_j + x_i = c \quad (1 \le j < i \le 3), \end{aligned} \qquad c \in \mathbb{R}.$$

Then μ is invariant under $\mathbb{R}^3 \rtimes O(3)$. All its L^2 field moments obey the tensor transformation law at every order, and it is reflection positive at every affine plane.

Proof. Proposition 18.1 gives $\mathsf{H}_j \ge |\mathsf{P}_i|$ for each pair. The preceding theorem proves invariance under every rotation in a coordinate plane. These rotations generate $SO(3)$. One elementary verification is to rotate the vector Re_3 into e_3 by successive rotations in the $(1,3)$ and $(2,3)$ planes. The remaining rotation fixes e_3, and hence is a rotation in the $(1,2)$ plane. Reversing these operations expresses R as a product of three coordinate-plane rotations. Cubic invariance includes $-I$; since $O(3) = SO(3) \cup (-SO(3))$, the full orthogonal group follows. The moment identities follow either by the same composition or by change of variables in the invariant law, with the L^2 extension of the fields.

For a plane $n \cdot x = c$, choose an affine orthogonal map g carrying $x_1 = 0$ to it and its positive side to $n \cdot x > c$. The observable pullback is unitary because the law is invariant, maps the two positive-side sigma fields to each other, and intertwines the two antilinear reflections. Consequently

$$\mu((\Theta_{n,c}\Phi)\Phi) = \mu((\Theta_1 \widehat{g}^{-1}\Phi)\widehat{g}^{-1}\Phi) \ge 0.$$

This proves reflection positivity at the chosen plane. □

Orthogonal changes of coordinates can be used in this argument before rotation invariance has been proved. For completeness, if $\mu' = (g^{-1})_*\mu$, the map $\widehat{g}$ is a unitary $L^2(\mu') \to L^2(\mu)$ taking observations in O to those in gO. It conjugates translations by $U(a) \mapsto U(ga)$ and reflection by $\theta \mapsto g\theta g^{-1}$. In the corresponding reconstructed spaces, $[\Phi] \mapsto [\widehat{g}\Phi]$ preserves the positive form, is onto, and hence is unitary. Applying it to the reconstruction identities transports the generators and their joint spectral projections. Thus cubic symmetry transports (18.3) among all ordered coordinate pairs.

The diagonal-reflection argument has proved, for every law in $\mathcal{C}_9$, the exact spectral inequalities

$$\mathfrak{E}_j\{(\omega, p) : \omega < |p_i|\} = 0 \qquad (i \ne j). \tag{C}$$

The constant in this cone is one. The following consequence is useful both for the ergodic decomposition and for clustering.

Proposition 18.22 (Zero energy and invariant vectors)**.** *Suppose that a translation-invariant law is invariant and reflection positive at all coordinate planes, and satisfies (C) for both transverse coordinates in each direction. Then*

$$\ker \mathsf{H}_j \subset \ker \mathsf{P}_i \quad (i \ne j), \qquad V_j(y)P_{0,j} = P_{0,j}, \qquad P^j = P^{123}.$$

Here P^{123} is the invariant projection for all translations. For a translation-ergodic law, $\ker \mathsf{H}_j = \mathbb{C}\Omega_j$.

Proof. The joint spectral measure is supported on $\omega \ge \max_{i \ne j} |p_i|$. Its part with $\omega = 0$ is therefore supported at $(0,0)$. This gives the first two assertions by the joint spectral theorem, including the generator domains. If Φ is invariant along e_j, its class has zero energy, by Proposition 17.4. Thus $[U(y)\Phi - \Phi]_j = 0$ for $y \perp e_j$. This difference is still invariant along e_j, since translations commute. The class map is an isometry on that subspace; consequently $U(y)\Phi = \Phi$ in ordinary L^2. The directional invariant subspace equals the full invariant subspace, which gives $P^j = P^{123}$. Under ergodicity the latter space is the constants, and the isometry onto the zero-energy space gives the last assertion. □

19. Ergodic components

An invariant probability measure can be a mixture of invariant ergodic measures. Reflection positivity of a mixture does not by itself give positivity of its components. The invariant-weight lemma supplies the extra argument needed here. We also need the same Gaussian bound in the components, rather than a bound with component-dependent constants.

Proposition 19.1 (Ergodic decomposition)**.** *Every translation-invariant law μ has a measurable decomposition*

$$\mu = \int \mu_\xi \, d\nu(\xi)$$

into translation-invariant ergodic laws. The probability measure on the space of laws is unique. For every integrable G, $\mu_\xi(G)$ represents conditional expectation given the invariant σ-algebra. If $\mu(A) = 1$, then $\mu_\xi(A) = 1$ for almost every ξ.

Proof. The state space is standard Borel by Lemma 15.1. The regular conditional distribution theorem is applied to a countably generated version of the invariant σ-algebra. To construct that version, let G_p, $p \geq 1$, enumerate the characteristic polynomials with rational complex coefficients from a countable generating test family. Their span is dense in $L^2(\mu)$. The bounded functions $g_p = P^{123}G_p$ have invariant Borel versions by Lemma 17.2. Their span is dense in the invariant L^2 subspace. Consequently $\mathcal{J} = \sigma(g_p : p \geq 1)$ and $\mathcal{I}$ agree modulo μ-null sets: approximate any bounded invariant function by that span in L^2 and select an almost surely convergent subsequence. Its limit is a $\mathcal{J}$-measurable version of the function.

Put $\beta(F) = (g_p(F))_{p\geq 1} \in \mathbb{C}^{\mathbb{N}}$. Both the state space and this countable product are standard Borel. The regular conditional distribution theorem for the Borel map β gives a probability kernel $F \mapsto \mu_F$ such that, for every measurable A, $\mu_F(A) = \mathbb{E}_\mu(1_A \mid \mathcal{I})(F)$ almost surely, where $\mathcal{I} = \mathcal{I}^{\mathbb{R}^3}$. The same identity holds for integrable functions by the monotone class theorem. The kernel is $\mathcal{J}$-measurable; for every measurable A and every $B \in \mathcal{J}$, its defining identity is

$$\int_B \mu_F(A) \, d\mu(F) = \mu(A \cap B).$$

Thus conditional expectations have invariant versions. This use of conditional distributions does not require a pointwise quotient of the state space by translation orbits. Let ν be the distribution of the kernel as a random law. The map is measurable for the σ-algebra generated by evaluation of laws on measurable sets, and $\mu = \int \rho \, d\nu(\rho)$. The set of invariant ergodic laws is measurable in this space of probability measures. Invariance is checked by countably many characteristic tests and rational translations. For an invariant law ρ, ergodicity is equivalent to

$$\lim_{l\to\infty} \rho(|A_l G_p - \rho(G_p)|^2) = 0 \qquad \text{for every } p.$$

The mean ergodic theorem and density prove this equivalence; each integral here is a Borel function of ρ. Thus the property is a countable Borel condition. In particular the assertion that ν is carried by ergodic laws has its ordinary measurable meaning.

We verify invariance and ergodicity of these laws, so that no such property is being assumed of a conditional kernel. If h is bounded and invariant, invariance of μ gives $\mu(hU(a)G) = \mu(hG)$. Hence the conditional expectations of $U(a)G$ and G are equal. Apply this to the countable characteristic family in Lemma 15.3 and rational translations. Outside one null set, the conditional characteristic functionals are invariant under all rational translations; continuity of tests and bounded convergence extend this to every translation. Characteristic uniqueness proves invariance of each such μ_F.

Use the countable characteristic algebra $G_1, G_2, \ldots$ chosen above. Put $g_p = \mathbb{E}_\mu(G_p \mid \mathcal{I})$. Choose $T_l \to \infty$ so that $\|A^{\mathbb{R}^3}_{T_l} G_p - g_p\|_2^2 \leq 2^{-l}$ for $p \leq l$. Tonelli's theorem and the kernel identity show that, outside a common null set,

$$\mu_F(|A_{T_l} G_p - g_p|^2) \longrightarrow 0 \quad \text{for every } p.$$

The conditional identities for g_p and $|g_p|^2$ give $\mu_F(|g_p - g_p(F)|^2) = 0$ almost surely. Thus the averages converge in $L^2(\mu_F)$ to the constant $\mu_F(G_p) = g_p(F)$. Translation averages in the invariant law μ_F also converge to its invariant projection. Exponential density proves that projection is the projection onto constants. The components are ergodic. All exceptional sets in this argument can be chosen simultaneously. For each fixed p, Tonelli gives

$$\int \sum_{l \geq p} \mu_F(|A_{T_l} G_p - g_p|^2)\, d\mu(F) = \sum_{l \geq p} \|A_{T_l} G_p - g_p\|_2^2 < \infty.$$

The inner series is finite for almost every F, so its terms tend to zero. Since g_p and $|g_p|^2$ are bounded and $\mathcal{J}$-measurable, their conditional integrals equal their values at F. Expanding a square gives $\mu_F(|g_p - g_p(F)|^2) = 0$. A countable intersection handles every p. Characteristic-polynomial density holds for every law on the state space, so checking the component's invariant projection on this family checks it on all of $L^2(\mu_F)$.

For uniqueness choose the same averaging sequence so that $A_{T_l} G_p \to g_p$ pointwise outside a common μ-null set; summability of the squared errors permits this. Every other mixture of ergodic laws representing μ assigns measure one to that set in almost every component. In an ergodic component ρ, the averages converge in $L^2(\rho)$ to $\rho(G_p)$, so their pointwise limit equals that number almost surely. It follows that $\mu_F(G_p) = \rho(G_p)$ for ρ-almost every F, for all p, and characteristic uniqueness gives $\mu_F = \rho$ there. The distribution of the conditional kernel is therefore the same for every such mixture, proving uniqueness of ν. Finally $1 = \mu(A) = \int \rho(A)\, d\nu(\rho)$ and $0 \leq \rho(A) \leq 1$ prove the full-measure assertion. □

Lemma 19.2 (The Gaussian bound in components)**.** *If a translation-invariant law satisfies (G), almost every translation-ergodic component satisfies (G) with exactly the same right-hand side.*

Proof. Fix a real test f, write $G = e^{F(f)}$ and $c = e^{\|f\|_2^2/2}$, and let $g = \mathbb{E}_\mu(G \mid \mathcal{I})$. Every finite L^p norm of G is finite. For an integer $n \geq 2$, Tonelli's theorem and (G) give

$$\mu((A_T G)^n) \leq T^{-3n} \int_{([0,T]^3)^n} \exp\left(\tfrac{n}{2}\|f\|_2^2 + \tfrac{1}{2} \sum_{p \neq q} \langle f, \tau_{x_p - x_q} \cdot f \rangle_2 \right) dx.$$

The correlation $\langle f, \tau_y \cdot f \rangle_2$ is bounded by $\|f\|_2^2$ and tends to zero as $|y| \to \infty$. Indeed, split the scalar product where $|x| \geq |y|/2$ and where $|x - y| \geq |y|/2$, and use the L^2 tail of f. Given $\varepsilon > 0$, choose D so that every pair farther apart than D has correlation at most ε in absolute value. The part of the averaging cube where some pair has distance less than D has relative volume at most $\binom{n}{2}(2D)^3 T^{-3}$, in the maximum norm. The integrand is bounded everywhere by $e^{n^2\|f\|_2^2/2}$. Consequently

$$\limsup_{T \to \infty} \mu((A_T G)^n) \leq c^n e^{n(n-1)\varepsilon/2}.$$

The averages converge to g in L^2 and are uniformly bounded in L^{2n}. Interpolation gives convergence in L^n, so $\mu(g^n) \leq c^n$ after $\varepsilon \downarrow 0$. Letting $n \to \infty$ proves $0 \leq g \leq c$ almost surely. The interpolation and this last pointwise conclusion can be made explicit. Conditional Jensen and the contraction of the averages give $\|A_T G - g\|_{2n} \leq 2\|G\|_{2n}$. For $n > 2$,

$$\|A_T G - g\|_n \leq \|A_T G - g\|_2^{1/(n-1)} (2\|G\|_{2n})^{(n-2)/(n-1)} \longrightarrow 0.$$

For $n = 2$ use the mean ergodic theorem. The inequality $|a^n - b^n| \leq n|a - b|(|a| + |b|)^{n-1}$ and Hölder give convergence of the nth moments. If $\mu(g > c + \delta) > 0$, then $\mu(g^n) \geq (c + \delta)^n \mu(g > c + \delta)$, contradicting $\mu(g^n) \leq c^n$ for large n. Hence $g \leq c$ almost surely. Since g equals the conditional component expectation of G, this is the required bound for this test.

Use a countable dense real test family to take one exceptional null set. For each component, evaluation is continuous at every distribution. Fatou's lemma and convergence of the test norms

extend the bound to every real Schwartz test. This proves (G) simultaneously for all tests in almost every component. □

Proposition 19.3 (Properties of the components)**.** *Suppose that μ satisfies (G), (dF) and belongs to $\mathcal{C}_9$. Almost every translation-ergodic component satisfies these same properties, including cubic invariance and the same Gaussian bound. Each such component has $P^j = P^{123}$ and $\ker \mathsf{H}_j = \mathbb{C}\Omega_j$.*

Proof. Translation invariance and ergodicity are supplied by Proposition 19.1, and (G) by the preceding lemma. The measurable set $\{dF = 0\}$ has full measure: it is the intersection of the zero sets of $F(d^*\xi)$ over a countable dense scalar test family. It therefore has full measure in almost every component.

Fix one of the nine reflection normals n. A bounded $\mathcal{I}$-measurable function is invariant along n, so Lemma 17.3 gives $\mu(h\widehat{r}_n G) = \mu(hG)$ for every bounded invariant h. The conditional expectations are equal. Applying this to countably many characteristic tests shows reflection invariance of the components by characteristic uniqueness. For a supported characteristic polynomial G put $K = (\Theta_n G)G$. If $h \geq 0$ is bounded and invariant, $\mu(hK) \geq 0$ by the same lemma. Thus $\mathbb{E}(K \mid \mathcal{I}) \geq 0$ almost surely: first test its positive and negative imaginary parts to see that it is real, then test the set where its real part is negative. Its conditional value is the component reflection quadratic form. Take the countable algebra of supported characteristic polynomials, then use density in each component to obtain full L^2 positivity. Choose rational complex coefficients in that supported algebra. Countably many conditional inequalities give a single exceptional null set. Having first established reflection invariance of the component, its reflection pullback is an ordinary L^2 isometry, and

$$\begin{aligned}|\mu_F((\Theta_n G)G) - \mu_F((\Theta_n G')G')| \\ \leq (\|G\|_{L^2(\mu_F)} + \|G'\|_{L^2(\mu_F)})\|G - G'\|_{L^2(\mu_F)}.\end{aligned}$$

This extends the inequalities first to arbitrary coefficients and then to every positive-side $L^2(\mu_F)$ observable. Reflection invariance is essential for this estimate, which explains the order of the two component arguments. There are only nine normals, so one null set suffices. Translation invariance transports these conclusions to all offsets.

The coordinate reflections and the reflections in $x_i = x_j$ generate the signed permutation group. Thus the component laws are cubic invariant as well. Apply the same-law diagonal cone theorem to each of them and Proposition 18.22 for the final spectral assertions. □

20. Clustering and the OS axioms

Proposition 20.1 (Uniform directional mixing)**.** *For a translation-invariant law that is invariant and reflection positive at $x_j = 0$,*

$$\sup_{y \perp e_j} \left| \langle \Phi, U(se_j + y)(1 - P^j)\Psi \rangle_\mu \right| \longrightarrow 0 \qquad (|s| \to \infty)$$

for all $\Phi, \Psi \in L^2(\mu)$.

Proof. First take $\Phi = \widehat{\theta}_j \Phi'$ with $\Phi', \Psi \in \mathcal{E}_j^+$ and $s \geq 0$. Reconstruction and (17.4) give

$$\left| \langle \Phi, U(se_j + y)(1 - P^j)\Psi \rangle_\mu \right| \leq \|[\Phi']_j\| \, \|e^{-s\mathsf{H}_j}(1 - P_{0,j})[\Psi]_j\|.$$

The bound is independent of y. The second factor tends to zero by the spectral theorem: $e^{-s\omega} 1_{\{\omega > 0\}}$ tends pointwise to zero and is bounded by one.

For arbitrary Φ, Ψ approximate by compactly supported characteristic polynomials Φ_1, Ψ_1. Choose r larger than their supports in the normal direction. Then $\Psi_2 = U(re_j)\Psi_1 \in \mathcal{E}_j^+$ and $\Phi_2 = \widehat{\theta}_j U(-re_j)\Phi_1 \in \mathcal{E}_j^+$, and unitarity gives

$$\langle \Phi_1, U(se_j + y)(1 - P^j)\Psi_1 \rangle = \langle \widehat{\theta}_j \Phi_2, U((s - 2r)e_j + y)(1 - P^j)\Psi_2 \rangle.$$

The first case applies for $s \geq 2r$. The approximation errors are bounded independently of s, y by Cauchy–Schwarz and the contraction $U(se_j + y)(1 - P^j)$. Let their norms tend to zero. For negative s exchange Φ, Ψ and take adjoints; $y \mapsto -y$ preserves the supremum. □

Corollary 20.2 (Clustering)**.** *Suppose that the cone hypothesis of Proposition 18.22 holds. Then for all $\Phi, \Psi \in L^2(\mu)$,*

$$\mu(\overline{\Phi}\, U(a)\Psi) \longrightarrow \langle \Phi, P^{123}\Psi\rangle_\mu \qquad (|a| \to \infty).$$

If μ is translation ergodic, the limit equals $\overline{\mu(\Phi)}\mu(\Psi)$.

Proof. Put $P = P^{123} = P^j$, for each j. The invariant part of the correlation equals $\langle \Phi, P\Psi\rangle$. Choose j with $|a_j| = |a|_\infty$ and write $a = a_j e_j + y$. The three uniform directional bounds imply that the remaining part tends to zero, since there are only three possible choices of j. Ergodicity makes P the projection onto constants. □

For a unit normal n define the tensor reflection

$$\Theta_k^n f = \overline{(r_n \cdot)^{\otimes k} f}, \qquad \Theta_0^n c = \overline{c}.$$

A permutation of the arguments also permutes the tensor indices; write f^π for this operation. These conventions agree on simple products with reflection or permutation of their individual factors.

Theorem 20.3 (The OS axioms for a law)**.** *Let μ satisfy (G), be invariant under the Euclidean group, and be reflection positive at coordinate planes. Suppose also that its directional spectral measures obey (C). The Schwinger functions (15.6) satisfy:*

OS0. For a numerical C and every $n \geq 1$, $|S_n(f)| \leq C^n \sqrt{n!}\, q_{6n}(f)$.

OS1. $S_n((g\cdot)^{\otimes n} f) = S_n(f)$ for every Euclidean transformation g, with the two-tensor action on the indices.

OS2. For every unit n, every finite sequence of tensor tests $f_k \in \mathscr{S}_k$ supported where $n \cdot y_l \geq 0$ for each argument, and $f_0 \in \mathbb{C}$,

$$\sum_{k,l} S_{k+l}(\Theta_k^n f_k \otimes f_l) \geq 0.$$

OS3. $S_n(f^\pi) = S_n(f)$ for every permutation π.

If μ is translation ergodic, it also satisfies:

OS4. For $f \in \mathscr{S}_m$, $f' \in \mathscr{S}_n$,

$$S_{m+n}(f \otimes \tau_a^{\otimes n} f') \longrightarrow S_m(f) S_n(f') \qquad (|a| \to \infty).$$

These statements impose no restriction at coincident arguments.

Proof. OS0 follows from (15.5). The tensor transformation identity and invariance of the law give OS1. Full invariance transports coordinate reflection positivity to every plane: if $ge_1 = n$, the map $\widehat{g}$ takes positive-side observables in direction e_1 to those in direction n, and $g\theta_1 g^{-1} = r_n$. Lemma 15.5 therefore places $\Phi = \sum_k F^{(k)}(f_k)$ in the appropriate positive-side L^2 space, including for tests touching the boundary. Its multiplication and conjugation rules give

$$\mu((\Theta_n \Phi)\Phi) = \sum_{k,l} S_{k+l}(\Theta_k^n f_k \otimes f_l) \geq 0,$$

which is OS2. Products of scalar random variables commute, so OS3 holds on simple products and then on all tensor tests by density and continuity. Finally apply Corollary 20.2 to $\overline{F^{(m)}(f)}$ and $F^{(n)}(f')$. The tensor multiplication and translation rules give exactly OS4. □

Corollary 20.4 (The chamber limit)**.** *Let μ be any volume-first continuum law of Proposition 16.3. Its Schwinger functions satisfy OS0–OS3. Almost every translation-ergodic component satisfies all five OS axioms.*

Proof. The diagonal-reflection theorem gives the same-law cone (C), and the rotation theorem gives full Euclidean invariance. Theorem 20.3 applies to μ. Proposition 19.3 gives (H), (dF) and the nine reflection families in almost every component. Apply the same cone and rotation theorems to those laws, then the ergodic clause of Theorem 20.3. Each application uses the hypotheses of that component law itself. □

Notation

Symbol	Meaning
C^k, C^k_{c}	Real lattice k-cochains and their finitely supported subspace; d is the coboundary and d^* its adjoint. Definition 2.1.
$\mathcal{L}, \Pi = (\ell, c)$	The nine coordinate and diagonal reflection families and an admissible lattice reflection plane. Definition 2.2.
W, O_{cub}	The integer translation group extended by signed coordinate permutations, and its orthogonal subgroup. Lemma 2.3.
$C_0, v_0, \mathbb{X}$	The fundamental chamber, its chosen interior point, and the orbit of fermion sites. Definition 2.4.
$\pi(x), y(x), x^{(i)}$	The nearest photon site, displacement from it, and the neighbor of bond type i. Definition 2.4.
A, z, F	A link potential, closed integer flux, and field strength $F = dA + 2\pi z/q$. Definition 2.6.
q, ρ, M, w_i	The lattice charge, Wilson parameter, scalar diagonal mass parameter, and four hopping weights. Definition 2.8.
$\mathbb{T} = (\Gamma, \chi)$	A photon period lattice together with the fermion boundary character. $\lvert\mathbb{T}\rvert$ is its number of photon sites. Definition 2.8.
$C(n_1, n_2, n_3)$, $D(a, b, c)$, $D'(a, b, c)$	The coordinate and two diagonal tori, with their specified boundary characters. Definition 2.9.
$Z_{\mathbb{T}}[G], \mathbb{E}_{\mathbb{T}}[G]$	The unnormalized and normalized finite functionals. Definition 2.11.
$\mathcal{G}(B), \mathcal{A}(U), \mathcal{P}$	Local gauge-invariant integrands of Gaussian growth, products supported in U, and the polynomial algebra used in elimination. Definitions 2.13 and 2.20.
$\Theta_\Pi, \zeta_I, \omega_I$	Antilinear Grassmann reflection, ordered crossing-mode monomials, and their positive weights. Definitions 2.16 and 6.11.
$\Omega, \mu_{\mathbb{T}}, \varphi_\beta$	Closed infinite-lattice field strengths, the periodic finite law, and its characteristic exponential. Definition 5.1.
$\eta_N, n_{N,T}, R_T$	Spacing, lattice side and physical side of the specified cutoff tori. Definition 15.6.
ν_N, μ_N, μ	A fixed-spacing infinite-volume lattice law, its step interpolation, and a zero-spacing limit. Section 16.
$\mathcal{S}, \mathcal{S}', \mathscr{S}_n$	Schwartz tensor tests, tempered tensor distributions, and n-variable tensor tests. Section 15.
H_s, A	Weighted Hermite Hilbert spaces and the operator $1 + \lvert x\rvert^2 - \Delta$ defining their weights. Section 15.
Σ_O	The sigma-algebra of observations supported in O. Section 15.
$U(a), \widehat{g}, \Theta_{n,c}$	Ordinary translations, linear Euclidean pullback on observations, and its antilinear reflection. Section 15.
$[\Phi]_j, \mathcal{H}_j, \Omega_j$	An OS equivalence class, its completed Hilbert space, and the class of the constant one. Definition 17.1.
$\mathsf{H}_j, \mathsf{P}_i, V_j$	Reconstructed nonnegative Hamiltonian, transverse momentum, and transverse translations. Proposition 17.4.
$P^j, P^{123}, P_{0,j}$	Ordinary directional and full invariant projections, and the OS zero-energy projection. Proposition 17.4.
$S_n, F^{(n)}, q_m$	Schwinger functions, tensor polynomial observations, and the Schwartz seminorm in the growth estimate. Lemma 15.5.

Symbol	Meaning
$\mathbb{B}(b), \lambda = (\sigma, \chi)$	The field evaluated on the two-form Hodge dual to b, and a product-test polarization with a real transverse profile. Section 18.
$w(\lambda), s(\lambda)$	The sum of angular weights and the product of antilinear-reflection signs of a tuple of labels. Equation 18.7.
$\mathcal{W}, d(\xi), \xi^*, T(\xi)$	The complex translation tube, its distance function, reflected conjugation, and its contraction semigroup. Equation 18.9.
$G_N, \Psi, \mathcal{G}_N, \widetilde{G}_N$	Ordered moment kernels and class vectors, the kernel on the union of ordering tubes, and its density on distinct real configurations. Proposition 18.6 and Lemma 18.11.
$c_m, \widetilde{c}_m, A(N)$	Angular coefficients on the complex tubes and their real extensions; the maximal excess frequency index at order N. Equations 18.30 and 18.33.

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